Worst Outcomes, All Solutions, Pareto-Prioritized — Master File	
Commission	Predict the worst possible outcomes (past, now, future) across every dimension of reality and field of discipline, with an example for each; list ALL solutions with deep specifications; state the probability of success in percentages and why; explain ir
The count	162 problem rows = 18 fields of discipline x 3 timeframes (Past / Now / Future) x 3 scales (Household / Society / Global). 162 solutions, one per problem, each fully specified. 90 root-cause rows = 18 fields x 5 Whys, every cause with its own solutic
Sheets	PROBLEMS & SOLUTIONS (the master table) - ALL SOLUTIONS (every solution listed, solution-first) - PARETO PRIORITY (all 162 ranked) - ROOT CAUSES 5 WHYS - COINCIDING PROBLEMS - STATS & VISUALS.
Dimensions of reality cover	Physical/Material, Biological/Life, Psychological, Social/Order, Technological, Informational — crossed with every size of system: person, household, city, nation, civilization.
Probabilities — read this firs	Each percentage is a structured expert estimate for the solution succeeding IF implemented as specified, with the basis stated in the WHY column. It is not a measured frequency. Scale rule, applied uniformly: Household +8 points (directly controlla
Pareto method	Priority score = field criticality (Tier 1 survival fields weigh 3, Tier 2 weigh 2) x scale (Global 3, Society 2, Household 1) x timeframe urgency (Now 3, Future 2, Past 1). All 162 solutions are ranked by this score in the PARETO PRIORITY sheet with t
The honest Pareto figure	The top 20% of solutions (32 of 162) carry 40.2% of the total weighted harm score in this model. The 80/20 pattern is the design principle — fix the vital few first — and the sheet shows the real computed concentration rather than a slogan.
Coinciding problems	When problems coincide, probabilities do NOT multiply independently: coinciding crises are usually correlated (one cause, or one failure weakening all defences), so joint success is higher than naive multiplication but lower than the best single pro

#	Dimension of Realit	Field of Discipline	Pareto Tier o	Timefram	Scale	Worst Possible Problem	Example	Solution — Full Specification	Probability of Su	Why This Probability	How to Maximize the Chances (detailed)	Root Causes
1	Physical / Material	Water	Tier 1	Past	Household / Personal	Regional aquifer and river systems collapse from overuse and contamination, making whole regions uninhabitable.	The Aral Sea shrank to a fraction of its size after river diversion, salting farmland and collapsing fishing towns.	Cap total extraction at measured recharge: meter every well and canal offtake, publish the water balance monthly, and set the legal extraction ceiling at 80% of average recharge so the buffer absorbs dry years. Restore wetlands at river mouths (minimum 10% of historical area) as natural filters and flood buffers. Specification: household storage 4 litres per person per day drinking minimum, 14-day reserve = 56 L/person; community reservoirs sized for 90 days of essential use.	86% (base range 70–85%)	Extraction caps tied to measured recharge have worked wherever enforced (e.g., managed aquifer districts) because the physics is directly measurable.	Meter 100% of extraction; publish the balance monthly; set the cap before the crisis decade, not during it.	Why 1 — supplies fail: Infrastructure ages and maintenance is deferred because water is priced below its cost. -> Fix: Price water at full lifecycle cost with a protected free lifeline block, so revenue funds maintenance without denying anyone the minimum. | Why 2 — deferral is invisible: Underground assets fail out of sight; politicians gain nothing from pipes nobody sees. -> Fix: Publish asset-condition grades and failure statistics annually, like financial audits; tie executive bonuses in utilities to condition grades. | Why 3 — no one owns the number: Responsibility is split across municipalities, regions, and private operators. -> Fix: One accountable water authority per basin with the full balance — extraction, quality, losses — published monthly under one signature. | Why 4 — basins cross borders: Upstream and downstream jurisdictions optimize against each other. -> Fix: Basin treaties with pre-agreed drought-sharing formulas signed in wet years, when generosity is cheap. | Why 5 — root: Water is treated as a free gift of nature rather than a managed stock with a measurable balance. -> Fix: Adopt the stock-and-flow account as law: every basin keeps a public ledger, and extraction rights are issued only against it.
2	Physical / Material	Water	Tier 1	Past	Society / National	Regional aquifer and river systems collapse from overuse and contamination, making whole regions uninhabitable.	The Aral Sea shrank to a fraction of its size after river diversion, salting farmland and collapsing fishing towns.	Cap total extraction at measured recharge: meter every well and canal offtake, publish the water balance monthly, and set the legal extraction ceiling at 80% of average recharge so the buffer absorbs dry years. Restore wetlands at river mouths (minimum 10% of historical area) as natural filters and flood buffers. Specification: household storage 4 litres per person per day drinking minimum, 14-day reserve = 56 L/person; community reservoirs sized for 90 days of essential use.	78% (base range 70–85%)	Extraction caps tied to measured recharge have worked wherever enforced (e.g., managed aquifer districts) because the physics is directly measurable.	Meter 100% of extraction; publish the balance monthly; set the cap before the crisis decade, not during it.	Why 1 — supplies fail: Infrastructure ages and maintenance is deferred because water is priced below its cost. -> Fix: Price water at full lifecycle cost with a protected free lifeline block, so revenue funds maintenance without denying anyone the minimum. | Why 2 — deferral is invisible: Underground assets fail out of sight; politicians gain nothing from pipes nobody sees. -> Fix: Publish asset-condition grades and failure statistics annually, like financial audits; tie executive bonuses in utilities to condition grades. | Why 3 — no one owns the number: Responsibility is split across municipalities, regions, and private operators. -> Fix: One accountable water authority per basin with the full balance — extraction, quality, losses — published monthly under one signature. | Why 4 — basins cross borders: Upstream and downstream jurisdictions optimize against each other. -> Fix: Basin treaties with pre-agreed drought-sharing formulas signed in wet years, when generosity is cheap. | Why 5 — root: Water is treated as a free gift of nature rather than a managed stock with a measurable balance. -> Fix: Adopt the stock-and-flow account as law: every basin keeps a public ledger, and extraction rights are issued only against it.
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4	Physical / Material	Water	Tier 1	Now	Household / Personal	Municipal supply contamination or infrastructure failure cuts safe drinking water to a city.	A single treatment-plant failure or pipe contamination event forces boil-water advisories for hundreds of thousands of people at once.	Dual-barrier defence: (1) two independent treatment trains so one failure never reaches taps; (2) point-of-use backup in every household — a certified gravity or pump filter rated to remove bacterial viruses (0.02–0.2 micron) plus a 72-hour stored reserve (12 L/person drinking). Specification: utilities must hold 5 days of treated-water storage and publish real-time turbidity and chlorine residuals; buildings over 6 storeys keep a 24-hour gravity tank.	93% (base range 78–92%)	Dual treatment trains plus household filtration are mature, cheap, tested engineering — failure is almost always maintenance, not technology.	Test household filters and stored water quarterly; utilities drill a full plant-loss weekend annually; keep 72-hour reserves rotated, not forgotten.	Why 1 — supplies fail: Infrastructure ages and maintenance is deferred because water is priced below its cost. -> Fix: Price water at full lifecycle cost with a protected free lifeline block, so revenue funds maintenance without denying anyone the minimum. | Why 2 — deferral is invisible: Underground assets fail out of sight; politicians gain nothing from pipes nobody sees. -> Fix: Publish asset-condition grades and failure statistics annually, like financial audits; tie executive bonuses in utilities to condition grades. | Why 3 — no one owns the number: Responsibility is split across municipalities, regions, and private operators. -> Fix: One accountable water authority per basin with the full balance — extraction, quality, losses — published monthly under one signature. | Why 4 — basins cross borders: Upstream and downstream jurisdictions optimize against each other. -> Fix: Basin treaties with pre-agreed drought-sharing formulas signed in wet years, when generosity is cheap. | Why 5 — root: Water is treated as a free gift of nature rather than a managed stock with a measurable balance. -> Fix: Adopt the stock-and-flow account as law: every basin keeps a public ledger, and extraction rights are issued only against it.
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7	Physical / Material	Water	Tier 1	Future	Household / Personal	Multi-year megadrought plus population growth makes current cities structurally water-short.	A metropolitan region of millions faces day-zero conditions as reservoirs fall below dead-storage levels for consecutive years.	Rebuild the supply stack: wastewater recycling to potable standard (target 30–40% of supply), stormwater capture on all new developments (retain the 25-year storm on site), desalination where coastal, and pricing that guarantees a free lifeline block (first 50 L/person/day) with steep rates above it. Specification: per-capita demand cut to 100 L/day total through fixtures and leakage below 8%; inter-basin agreements signed before crisis, not during.	63% (base range 45–65%)	Recycling and demand cuts are proven; the constraint is political will and capital timing against a moving climate baseline.	Start reuse infrastructure 10 years before projected shortfall; lock the lifeline-block pricing in law early; model demand on the worst decade, not the average.	Why 1 — supplies fail: Infrastructure ages and maintenance is deferred because water is priced below its cost. -> Fix: Price water at full lifecycle cost with a protected free lifeline block, so revenue funds maintenance without denying anyone the minimum. | Why 2 — deferral is invisible: Underground assets fail out of sight; politicians gain nothing from pipes nobody sees. -> Fix: Publish asset-condition grades and failure statistics annually, like financial audits; tie executive bonuses in utilities to condition grades. | Why 3 — no one owns the number: Responsibility is split across municipalities, regions, and private operators. -> Fix: One accountable water authority per basin with the full balance — extraction, quality, losses — published monthly under one signature. | Why 4 — basins cross borders: Upstream and downstream jurisdictions optimize against each other. -> Fix: Basin treaties with pre-agreed drought-sharing formulas signed in wet years, when generosity is cheap. | Why 5 — root: Water is treated as a free gift of nature rather than a managed stock with a measurable balance. -> Fix: Adopt the stock-and-flow account as law: every basin keeps a public ledger, and extraction rights are issued only against it.
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10	Biological / Life	Food & Agriculture	Tier 1	Past	Household / Personal	Single-crop dependence plus blight or drought produces famine at national scale.	The Irish potato famine: one crop variety, one pathogen, a million dead and a million emigrated.	Never let one crop or one supplier exceed 25% of calories. Maintain seed diversity banks with at least 3 climate-matched varieties per staple, rotate crops on a minimum 3-year cycle, and hold strategic grain reserves of 90 days of national consumption. Specification: household level — 90 days of staple calories stored (rice/grain ~ 25 kg per adult per month at 2,000 kcal/day planning), rotated FIFO, regional level — no more than 40% of food imported by calorie.	90% (base range 75–88%)	Diversification and reserves are ancient, repeatedly validated famine preventatives; famines in diversified systems with reserves are historically rare.	Enforce the 25% single-source ceiling in law; audit reserves physically twice a year; keep 3 seed varieties minimum per staple, germination-tested.	Why 1 — shelves empty fast: Just-in-time logistics hold almost no buffer by design. -> Fix: Mandate buffer stocks at each level (household 30 days, retail 30 days, national 90 days) so the chain has slack by law, not by hope.	Why 2 — buffers cost money: Private actors are punished by markets for holding 'inefficient' stock. -> Fix: Pay for buffers as a public service: reserve-holding contracts with guaranteed purchase, like capacity payments in electricity.	Why 3 — efficiency outbids resilience: Quarterly metrics reward cost per unit, never survival per decade. -> Fix: Add resilience metrics to procurement law: supplier diversity and buffer days scored alongside price in every public food contract.	Why 4 — soil is mined, not farmed: Yield this year is priced; soil next decade is not. -> Fix: Soil organic matter measured per farm on a 5-year cycle, with payments for building it and limits on losing it.	Why 5 — root: Food is treated as a commodity first and a survival system second. -> Fix: Constitutional or statutory food-security duty: the state must maintain the measured capacity to feed the population through a 1-year supply shock.
11	Biological / Life	Food & Agriculture	Tier 1	Past	Society / National	Single-crop dependence plus blight or drought produces famine at national scale.	The Irish potato famine: one crop variety, one pathogen, a million dead and a million emigrated.	Never let one crop or one supplier exceed 25% of calories. Maintain seed diversity banks with at least 3 climate-matched varieties per staple, rotate crops on a minimum 3-year cycle, and hold strategic grain reserves of 90 days of national consumption. Specification: household level — 90 days of staple calories stored (rice/grain ~ 25 kg per adult per month at 2,000 kcal/day planning), rotated FIFO, regional level — no more than 40% of food imported by calorie.	82% (base range 75–88%)	Diversification and reserves are ancient, repeatedly validated famine preventatives; famines in diversified systems with reserves are historically rare.	Enforce the 25% single-source ceiling in law; audit reserves physically twice a year; keep 3 seed varieties minimum per staple, germination-tested.	Why 1 — shelves empty fast: Just-in-time logistics hold almost no buffer by design. -> Fix: Mandate buffer stocks at each level (household 30 days, retail 30 days, national 90 days) so the chain has slack by law, not by hope.	Why 2 — buffers cost money: Private actors are punished by markets for holding 'inefficient' stock. -> Fix: Pay for buffers as a public service: reserve-holding contracts with guaranteed purchase, like capacity payments in electricity.	Why 3 — efficiency outbids resilience: Quarterly metrics reward cost per unit, never survival per decade. -> Fix: Add resilience metrics to procurement law: supplier diversity and buffer days scored alongside price in every public food contract.	Why 4 — soil is mined, not farmed: Yield this year is priced; soil next decade is not. -> Fix: Soil organic matter measured per farm on a 5-year cycle, with payments for building it and limits on losing it.	Why 5 — root: Food is treated as a commodity first and a survival system second. -> Fix: Constitutional or statutory food-security duty: the state must maintain the measured capacity to feed the population through a 1-year supply shock.
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13	Biological / Life	Food & Agriculture	Tier 1	Now	Household / Personal	Supply-chain shock (war, fuel, fertilizer, port failure) spikes prices and empties shelves within weeks.	A fertilizer-export ban plus a blocked shipping lane doubled bread prices in import-dependent countries within one season.	Shorten and parallel the chain: contract at least 2 geographically separate suppliers per staple; local production floor of 30% of fresh food within 100 km; fertilizer independence via green-ammonia plants sized to 50% of national need. Specification: retailers and institutions hold 30 days of non-perishable buffer; households hold 30 days (one month of staples = 25 kg grain-equivalent per adult); price-spike circuit breakers release reserves automatically at +40% price movement.	80% (base range 64–80%)	Supplier duplication and buffers are standard logistics; success depends on signing parallel contracts before the shock reprices them.	Dual-source every staple in peacetime contracts; hold the 30-day buffer physically, not as paper claims; pre-sign emergency import routes.	Why 1 — shelves empty fast: Just-in-time logistics hold almost no buffer by design. -> Fix: Mandate buffer stocks at each level (household 30 days, retail 30 days, national 90 days) so the chain has slack by law, not by hope.	Why 2 — buffers cost money: Private actors are punished by markets for holding 'inefficient' stock. -> Fix: Pay for buffers as a public service: reserve-holding contracts with guaranteed purchase, like capacity payments in electricity.	Why 3 — efficiency outbids resilience: Quarterly metrics reward cost per unit, never survival per decade. -> Fix: Add resilience metrics to procurement law: supplier diversity and buffer days scored alongside price in every public food contract.	Why 4 — soil is mined, not farmed: Yield this year is priced; soil next decade is not. -> Fix: Soil organic matter measured per farm on a 5-year cycle, with payments for building it and limits on losing it.	Why 5 — root: Food is treated as a commodity first and a survival system second. -> Fix: Constitutional or statutory food-security duty: the state must maintain the measured capacity to feed the population through a 1-year supply shock.
14	Biological / Life	Food & Agriculture	Tier 1	Now	Society / National	Supply-chain shock (war, fuel, fertilizer, port failure) spikes prices and empties shelves within weeks.	A fertilizer-export ban plus a blocked shipping lane doubled bread prices in import-dependent countries within one season.	Shorten and parallel the chain: contract at least 2 geographically separate suppliers per staple; local production floor of 30% of fresh food within 100 km; fertilizer independence via green-ammonia plants sized to 50% of national need. Specification: retailers and institutions hold 30 days of non-perishable buffer; households hold 30 days (one month of staples = 25 kg grain-equivalent per adult); price-spike circuit breakers release reserves automatically at +40% price movement.	72% (base range 64–80%)	Supplier duplication and buffers are standard logistics; success depends on signing parallel contracts before the shock reprices them.	Dual-source every staple in peacetime contracts; hold the 30-day buffer physically, not as paper claims; pre-sign emergency import routes.	Why 1 — shelves empty fast: Just-in-time logistics hold almost no buffer by design. -> Fix: Mandate buffer stocks at each level (household 30 days, retail 30 days, national 90 days) so the chain has slack by law, not by hope.	Why 2 — buffers cost money: Private actors are punished by markets for holding 'inefficient' stock. -> Fix: Pay for buffers as a public service: reserve-holding contracts with guaranteed purchase, like capacity payments in electricity.	Why 3 — efficiency outbids resilience: Quarterly metrics reward cost per unit, never survival per decade. -> Fix: Add resilience metrics to procurement law: supplier diversity and buffer days scored alongside price in every public food contract.	Why 4 — soil is mined, not farmed: Yield this year is priced; soil next decade is not. -> Fix: Soil organic matter measured per farm on a 5-year cycle, with payments for building it and limits on losing it.	Why 5 — root: Food is treated as a commodity first and a survival system second. -> Fix: Constitutional or statutory food-security duty: the state must maintain the measured capacity to feed the population through a 1-year supply shock.
15	Biological / Life	Food & Agriculture	Tier 1	Now	Global / Civilizational	Supply-chain shock (war, fuel, fertilizer, port failure) spikes prices and empties shelves within weeks.	A fertilizer-export ban plus a blocked shipping lane doubled bread prices in import-dependent countries within one season.	Shorten and parallel the chain: contract at least 2 geographically separate suppliers per staple; local production floor of 30% of fresh food within 100 km; fertilizer independence via green-ammonia plants sized to 50% of national need. Specification: retailers and institutions hold 30 days of non-perishable buffer; households hold 30 days (one month of staples = 25 kg grain-equivalent per adult); price-spike circuit breakers release reserves automatically at +40% price movement.	60% (base range 64–80%)	Supplier duplication and buffers are standard logistics; success depends on signing parallel contracts before the shock reprices them.	Dual-source every staple in peacetime contracts; hold the 30-day buffer physically, not as paper claims; pre-sign emergency import routes.	Why 1 — shelves empty fast: Just-in-time logistics hold almost no buffer by design. -> Fix: Mandate buffer stocks at each level (household 30 days, retail 30 days, national 90 days) so the chain has slack by law, not by hope.	Why 2 — buffers cost money: Private actors are punished by markets for holding 'inefficient' stock. -> Fix: Pay for buffers as a public service: reserve-holding contracts with guaranteed purchase, like capacity payments in electricity.	Why 3 — efficiency outbids resilience: Quarterly metrics reward cost per unit, never survival per decade. -> Fix: Add resilience metrics to procurement law: supplier diversity and buffer days scored alongside price in every public food contract.	Why 4 — soil is mined, not farmed: Yield this year is priced; soil next decade is not. -> Fix: Soil organic matter measured per farm on a 5-year cycle, with payments for building it and limits on losing it.	Why 5 — root: Food is treated as a commodity first and a survival system second. -> Fix: Constitutional or statutory food-security duty: the state must maintain the measured capacity to feed the population through a 1-year supply shock.
16	Biological / Life	Food & Agriculture	Tier 1	Future	Household / Personal	Climate-shifted growing zones plus soil loss cut yields across several breadbaskets simultaneously.	Simultaneous heat domes over multiple grain regions in one summer cut global wheat output while soils, already degraded, cannot compensate.	Regenerate the base: rebuild topsoil organic matter to 4%+ through cover cropping and reduced tillage; shift 20% of acreage to drought- and heat-bred varieties; build vertical/greenhouse capacity for greens near cities (target 50% of leafy supply within city limits); protect pollinators with pesticide-free corridors on 10% of farmland. Specification: yield planning assumes the worst decade climate, not the average year.	66% (base range 48–68%)	Soil regeneration and heat-bred varieties demonstrably work, but on 5–15 year biological timelines that must start before yields fall.	Begin soil programs a decade early; breed for the 2050 climate now; protect pollinator corridors before pollinator collapse, not after.	Why 1 — shelves empty fast: Just-in-time logistics hold almost no buffer by design. -> Fix: Mandate buffer stocks at each level (household 30 days, retail 30 days, national 90 days) so the chain has slack by law, not by hope.	Why 2 — buffers cost money: Private actors are punished by markets for holding 'inefficient' stock. -> Fix: Pay for buffers as a public service: reserve-holding contracts with guaranteed purchase, like capacity payments in electricity.	Why 3 — efficiency outbids resilience: Quarterly metrics reward cost per unit, never survival per decade. -> Fix: Add resilience metrics to procurement law: supplier diversity and buffer days scored alongside price in every public food contract.	Why 4 — soil is mined, not farmed: Yield this year is priced; soil next decade is not. -> Fix: Soil organic matter measured per farm on a 5-year cycle, with payments for building it and limits on losing it.	Why 5 — root: Food is treated as a commodity first and a survival system second. -> Fix: Constitutional or statutory food-security duty: the state must maintain the measured capacity to feed the population through a 1-year supply shock.
17	Biological / Life	Food & Agriculture	Tier 1	Future	Society / National	Climate-shifted growing zones plus soil loss cut yields across several breadbaskets simultaneously.	Simultaneous heat domes over multiple grain regions in one summer cut global wheat output while soils, already degraded, cannot compensate.	Regenerate the base: rebuild topsoil organic matter to 4%+ through cover cropping and reduced tillage; shift 20% of acreage to drought- and heat-bred varieties; build vertical/greenhouse capacity for greens near cities (target 50% of leafy supply within city limits); protect pollinators with pesticide-free corridors on 10% of farmland. Specification: yield planning assumes the worst decade climate, not the average year.	58% (base range 48–68%)	Soil regeneration and heat-bred varieties demonstrably work, but on 5–15 year biological timelines that must start before yields fall.	Begin soil programs a decade early; breed for the 2050 climate now; protect pollinator corridors before pollinator collapse, not after.	Why 1 — shelves empty fast: Just-in-time logistics hold almost no buffer by design. -> Fix: Mandate buffer stocks at each level (household 30 days, retail 30 days, national 90 days) so the chain has slack by law, not by hope.	Why 2 — buffers cost money: Private actors are punished by markets for holding 'inefficient' stock. -> Fix: Pay for buffers as a public service: reserve-holding contracts with guaranteed purchase, like capacity payments in electricity.	Why 3 — efficiency outbids resilience: Quarterly metrics reward cost per unit, never survival per decade. -> Fix: Add resilience metrics to procurement law: supplier diversity and buffer days scored alongside price in every public food contract.	Why 4 — soil is mined, not farmed: Yield this year is priced; soil next decade is not. -> Fix: Soil organic matter measured per farm on a 5-year cycle, with payments for building it and limits on losing it.	Why 5 — root: Food is treated as a commodity first and a survival system second. -> Fix: Constitutional or statutory food-security duty: the state must maintain the measured capacity to feed the population through a 1-year supply shock.
18	Biological / Life	Food & Agriculture	Tier 1	Future	Global / Civilizational	Climate-shifted growing zones plus soil loss cut yields across several breadbaskets simultaneously.	Simultaneous heat domes over multiple grain regions in one summer cut global wheat output while soils, already degraded, cannot compensate.	Regenerate the base: rebuild topsoil organic matter to 4%+ through cover cropping and reduced tillage; shift 20% of acreage to drought- and heat-bred varieties; build vertical/greenhouse capacity for greens near cities (target 50% of leafy supply within city limits); protect pollinators with pesticide-free corridors on 10% of farmland. Specification: yield planning assumes the worst decade climate, not the average year.	46% (base range 48–68%)	Soil regeneration and heat-bred varieties demonstrably work, but on 5–15 year biological timelines that must start before yields fall.	Begin soil programs a decade early; breed for the 2050 climate now; protect pollinator corridors before pollinator collapse, not after.	Why 1 — shelves empty fast: Just-in-time logistics hold almost no buffer by design. -> Fix: Mandate buffer stocks at each level (household 30 days, retail 30 days, national 90 days) so the chain has slack by law, not by hope.	Why 2 — buffers cost money: Private actors are punished by markets for holding 'inefficient' stock. -> Fix: Pay for buffers as a public service: reserve-holding contracts with guaranteed purchase, like capacity payments in electricity.	Why 3 — efficiency outbids resilience: Quarterly metrics reward cost per unit, never survival per decade. -> Fix: Add resilience metrics to procurement law: supplier diversity and buffer days scored alongside price in every public food contract.	Why 4 — soil is mined, not farmed: Yield this year is priced; soil next decade is not. -> Fix: Soil organic matter measured per farm on a 5-year cycle, with payments for building it and limits on losing it.	Why 5 — root: Food is treated as a commodity first and a survival system second. -> Fix: Constitutional or statutory food-security duty: the state must maintain the measured capacity to feed the population through a 1-year supply shock.
19	Biological / Life	Health & Medicine	Tier 1	Past	Household / Personal	Pandemic disease sweeps a population with no immunity, no germ theory, and no surge capacity.	The 1918 influenza killed tens of millions in months; cities ran out of coffins and grave-diggers.	Standing pandemic infrastructure, built before it is needed: genomic surveillance of wastewater in every city over 100,000 (weekly), a vaccine platform able to go from sequence to first doses in 100 days, and stockpiled PPE for 90 days of full-hospital use. Specification: hospital surge plans must double ICU capacity in 14 days (pre-mapped wards, cross-trained staff rosters); every household keeps a basic kit — thermometer, pulse oximeter, oral rehydration salts, fever medication for 2 weeks.	88% (base range 72–88%)	Surveillance plus platform vaccines compress response from years to months; the 100-day mission is engineering-feasible with standing capacity.	Fund the platform in peacetime (idle capacity is the premium, not waste); drill sequence-to-dose annually; wastewater surveillance never switched off.	Why 1 — systems run at 95%+ occupancy: Efficiency targets eliminate the slack that crises consume. -> Fix: Fund surge slack explicitly: an 85% occupancy ceiling as the regulated normal, with the 15% paid for as readiness.	Why 2 — readiness looks like waste: Idle beds and stockpiles photograph badly in budget debates. -> Fix: Account for readiness as insurance with published expected-loss figures, reviewed by auditors, not by headlines.	Why 3 — prevention is underpaid: Payment follows procedures performed, not illness avoided. -> Fix: Shift a fixed share (target 15–20%) of health budgets to prevention with outcome contracts: paid on measured population results.	Why 4 — workforce pipelines are long: Clinicians take a decade to train; shortages cannot be improvised. -> Fix: Train against the 10-year demographic projection, maintain a drilled reserve register, and recognize foreign credentials within 12 months.	Why 5 — root: Health is budgeted as a consumption expense instead of survival infrastructure. -> Fix: Classify core health capacity with infrastructure law: multi-year protected funding, like defence, insulated from annual cuts.

#	Dimension of Realit	Field of Discipline	Pareto Tier o	Timefram	Scale	Worst Possible Problem	Example	Solution — Full Specification	Probability of Su	Why This Probability	How to Maximize the Chances (detailed)	Root Causes
20	Biological / Life	Health & Medicine	Tier 1	Past	Society / National	Pandemic disease sweeps a population with no immunity, no germ theory, and no surge capacity.	The 1918 influenza killed tens of millions in months; cities ran out of coffins and grave-diggers.	Standing pandemic infrastructure, built before it is needed: genomic surveillance of wastewater in every city over 100,000 (weekly), a vaccine platform able to go from sequence to first doses in 100 days, and stockpiled PPE for 90 days of full-hospital use. Specification: hospital surge plans must double ICU capacity in 14 days (pre-mapped wards, cross-trained staff rosters); every household keeps a basic kit — thermometer, pulse oximeter, oral rehydration salts, fever medication for 2 weeks.	80% (base range 72–88%)	Surveillance plus platform vaccines compress response from years to months; the 100-day mission is engineering-feasible with standing capacity.	Fund the platform in peacetime (idle capacity is the premium, not waste); drill sequence-to-dose annually; wastewater surveillance never switched off.	Why 1 — systems run at 95%+ occupancy: Efficiency targets eliminate the slack that crises consume. -> Fix: Fund surge slack explicitly: an 85% occupancy ceiling as the regulated normal, with the 15% paid for as readiness. | Why 2 — readiness looks like waste: Idle beds and stockpiles photograph badly in budget debates. -> Fix: Account for readiness as insurance with published expected-loss figures, reviewed by auditors, not by headlines. | Why 3 — prevention is underpaid: Payment follows procedures performed, not illness avoided. -> Fix: Shift a fixed share (target 15–20%) of health budgets to prevention with outcome contracts: paid on measured population results. | Why 4 — workforce pipelines are long: Clinicians take a decade to train; shortages cannot be improvised. -> Fix: Train against the 10-year demographic projection, maintain a drilled reserve register, and recognize foreign credentials within 12 months. | Why 5 — root: Health is budgeted as a consumption expense instead of survival infrastructure. -> Fix: Classify core health capacity with infrastructure law: multi-year protected funding, like defence, insulated from annual cuts.
21	Biological / Life	Health & Medicine	Tier 1	Past	Global / Civilizational	Pandemic disease sweeps a population with no immunity, no germ theory, and no surge capacity.	The 1918 influenza killed tens of millions in months; cities ran out of coffins and grave-diggers.	Standing pandemic infrastructure, built before it is needed: genomic surveillance of wastewater in every city over 100,000 (weekly), a vaccine platform able to go from sequence to first doses in 100 days, and stockpiled PPE for 90 days of full-hospital use. Specification: hospital surge plans must double ICU capacity in 14 days (pre-mapped wards, cross-trained staff rosters); every household keeps a basic kit — thermometer, pulse oximeter, oral rehydration salts, fever medication for 2 weeks.	68% (base range 72–88%)	Surveillance plus platform vaccines compress response from years to months; the 100-day mission is engineering-feasible with standing capacity.	Fund the platform in peacetime (idle capacity is the premium, not waste); drill sequence-to-dose annually; wastewater surveillance never switched off.	Why 1 — systems run at 95%+ occupancy: Efficiency targets eliminate the slack that crises consume. -> Fix: Fund surge slack explicitly: an 85% occupancy ceiling as the regulated normal, with the 15% paid for as readiness. | Why 2 — readiness looks like waste: Idle beds and stockpiles photograph badly in budget debates. -> Fix: Account for readiness as insurance with published expected-loss figures, reviewed by auditors, not by headlines. | Why 3 — prevention is underpaid: Payment follows procedures performed, not illness avoided. -> Fix: Shift a fixed share (target 15–20%) of health budgets to prevention with outcome contracts: paid on measured population results. | Why 4 — workforce pipelines are long: Clinicians take a decade to train; shortages cannot be improvised. -> Fix: Train against the 10-year demographic projection, maintain a drilled reserve register, and recognize foreign credentials within 12 months. | Why 5 — root: Health is budgeted as a consumption expense instead of survival infrastructure. -> Fix: Classify core health capacity with infrastructure law: multi-year protected funding, like defence, insulated from annual cuts.
22	Biological / Life	Health & Medicine	Tier 1	Now	Household / Personal	Antimicrobial resistance quietly returns routine infections and surgeries to pre-antibiotic risk.	Drug-resistant infections already kill over a million people a year globally; common procedures like hip replacements become gambles.	Stewardship with teeth: antibiotics in human medicine by prescription audit only, agricultural antibiotic use cut 75% and banned for growth promotion, hospital infection-control staffing at 1 specialist per 100 beds, and a standing pipeline fund that pays for new antibiotics by subscription (delinked from sales volume). Specification: every hospital publishes resistance rates quarterly; households never stockpile or share antibiotics — leftover courses are returned to pharmacies.	70% (base range 54–70%)	Stewardship programs cut resistance where enforced, but global leakage (agriculture, travel, unregulated sales) caps success.	Close the agricultural loophole; tie hospital funding to published resistance rates; fund the subscription antibiotic pipeline internationally.	Why 1 — systems run at 95%+ occupancy: Efficiency targets eliminate the slack that crises consume. -> Fix: Fund surge slack explicitly: an 85% occupancy ceiling as the regulated normal, with the 15% paid for as readiness. | Why 2 — readiness looks like waste: Idle beds and stockpiles photograph badly in budget debates. -> Fix: Account for readiness as insurance with published expected-loss figures, reviewed by auditors, not by headlines. | Why 3 — prevention is underpaid: Payment follows procedures performed, not illness avoided. -> Fix: Shift a fixed share (target 15–20%) of health budgets to prevention with outcome contracts: paid on measured population results. | Why 4 — workforce pipelines are long: Clinicians take a decade to train; shortages cannot be improvised. -> Fix: Train against the 10-year demographic projection, maintain a drilled reserve register, and recognize foreign credentials within 12 months. | Why 5 — root: Health is budgeted as a consumption expense instead of survival infrastructure. -> Fix: Classify core health capacity with infrastructure law: multi-year protected funding, like defence, insulated from annual cuts.
23	Biological / Life	Health & Medicine	Tier 1	Now	Society / National	Antimicrobial resistance quietly returns routine infections and surgeries to pre-antibiotic risk.	Drug-resistant infections already kill over a million people a year globally; common procedures like hip replacements become gambles.	Stewardship with teeth: antibiotics in human medicine by prescription audit only, agricultural antibiotic use cut 75% and banned for growth promotion, hospital infection-control staffing at 1 specialist per 100 beds, and a standing pipeline fund that pays for new antibiotics by subscription (delinked from sales volume). Specification: every hospital publishes resistance rates quarterly; households never stockpile or share antibiotics — leftover courses are returned to pharmacies.	62% (base range 54–70%)	Stewardship programs cut resistance where enforced, but global leakage (agriculture, travel, unregulated sales) caps success.	Close the agricultural loophole; tie hospital funding to published resistance rates; fund the subscription antibiotic pipeline internationally.	Why 1 — systems run at 95%+ occupancy: Efficiency targets eliminate the slack that crises consume. -> Fix: Fund surge slack explicitly: an 85% occupancy ceiling as the regulated normal, with the 15% paid for as readiness. | Why 2 — readiness looks like waste: Idle beds and stockpiles photograph badly in budget debates. -> Fix: Account for readiness as insurance with published expected-loss figures, reviewed by auditors, not by headlines. | Why 3 — prevention is underpaid: Payment follows procedures performed, not illness avoided. -> Fix: Shift a fixed share (target 15–20%) of health budgets to prevention with outcome contracts: paid on measured population results. | Why 4 — workforce pipelines are long: Clinicians take a decade to train; shortages cannot be improvised. -> Fix: Train against the 10-year demographic projection, maintain a drilled reserve register, and recognize foreign credentials within 12 months. | Why 5 — root: Health is budgeted as a consumption expense instead of survival infrastructure. -> Fix: Classify core health capacity with infrastructure law: multi-year protected funding, like defence, insulated from annual cuts.
24	Biological / Life	Health & Medicine	Tier 1	Now	Global / Civilizational	Antimicrobial resistance quietly returns routine infections and surgeries to pre-antibiotic risk.	Drug-resistant infections already kill over a million people a year globally; common procedures like hip replacements become gambles.	Stewardship with teeth: antibiotics in human medicine by prescription audit only, agricultural antibiotic use cut 75% and banned for growth promotion, hospital infection-control staffing at 1 specialist per 100 beds, and a standing pipeline fund that pays for new antibiotics by subscription (delinked from sales volume). Specification: every hospital publishes resistance rates quarterly; households never stockpile or share antibiotics — leftover courses are returned to pharmacies.	50% (base range 54–70%)	Stewardship programs cut resistance where enforced, but global leakage (agriculture, travel, unregulated sales) caps success.	Close the agricultural loophole; tie hospital funding to published resistance rates; fund the subscription antibiotic pipeline internationally.	Why 1 — systems run at 95%+ occupancy: Efficiency targets eliminate the slack that crises consume. -> Fix: Fund surge slack explicitly: an 85% occupancy ceiling as the regulated normal, with the 15% paid for as readiness. | Why 2 — readiness looks like waste: Idle beds and stockpiles photograph badly in budget debates. -> Fix: Account for readiness as insurance with published expected-loss figures, reviewed by auditors, not by headlines. | Why 3 — prevention is underpaid: Payment follows procedures performed, not illness avoided. -> Fix: Shift a fixed share (target 15–20%) of health budgets to prevention with outcome contracts: paid on measured population results. | Why 4 — workforce pipelines are long: Clinicians take a decade to train; shortages cannot be improvised. -> Fix: Train against the 10-year demographic projection, maintain a drilled reserve register, and recognize foreign credentials within 12 months. | Why 5 — root: Health is budgeted as a consumption expense instead of survival infrastructure. -> Fix: Classify core health capacity with infrastructure law: multi-year protected funding, like defence, insulated from annual cuts.
25	Biological / Life	Health & Medicine	Tier 1	Future	Household / Personal	An engineered or novel pathogen, or total care-system overload from aging plus chronic disease, breaks medicine as an institution.	A high-fatality airborne pathogen emerges while hospitals are already at 95% occupancy from ordinary demand; elective and emergency care collapse together.	Layered defence: universal indoor air standards in hospitals, schools, and transit (ventilation equivalent to 6 air changes/hour, upper-air UV in high-risk rooms), synthetic-biology screening of all commercial DNA synthesis orders, regional manufacturing of essential medicines (top 100 generics produced on at least 2 continents each), and a reserve clinical workforce (retired and part-time clinicians registered and drilled annually). Specification: no single country supplies more than 40% of any essential medicine's global volume.	56% (base range 38–58%)	Air standards and synthesis screening are effective layers, but no layer is complete; success means blunting, not abolishing, the worst case.	Layer at least 3 independent defences (air, screening, surge); keep the reserve clinical workforce registered and drilled; never let occupancy exceed 85% as the normal state.	Why 1 — systems run at 95%+ occupancy: Efficiency targets eliminate the slack that crises consume. -> Fix: Fund surge slack explicitly: an 85% occupancy ceiling as the regulated normal, with the 15% paid for as readiness. | Why 2 — readiness looks like waste: Idle beds and stockpiles photograph badly in budget debates. -> Fix: Account for readiness as insurance with published expected-loss figures, reviewed by auditors, not by headlines. | Why 3 — prevention is underpaid: Payment follows procedures performed, not illness avoided. -> Fix: Shift a fixed share (target 15–20%) of health budgets to prevention with outcome contracts: paid on measured population results. | Why 4 — workforce pipelines are long: Clinicians take a decade to train; shortages cannot be improvised. -> Fix: Train against the 10-year demographic projection, maintain a drilled reserve register, and recognize foreign credentials within 12 months. | Why 5 — root: Health is budgeted as a consumption expense instead of survival infrastructure. -> Fix: Classify core health capacity with infrastructure law: multi-year protected funding, like defence, insulated from annual cuts.
26	Biological / Life	Health & Medicine	Tier 1	Future	Society / National	An engineered or novel pathogen, or total care-system overload from aging plus chronic disease, breaks medicine as an institution.	A high-fatality airborne pathogen emerges while hospitals are already at 95% occupancy from ordinary demand; elective and emergency care collapse together.	Layered defence: universal indoor air standards in hospitals, schools, and transit (ventilation equivalent to 6 air changes/hour, upper-air UV in high-risk rooms), synthetic-biology screening of all commercial DNA synthesis orders, regional manufacturing of essential medicines (top 100 generics produced on at least 2 continents each), and a reserve clinical workforce (retired and part-time clinicians registered and drilled annually). Specification: no single country supplies more than 40% of any essential medicine's global volume.	48% (base range 38–58%)	Air standards and synthesis screening are effective layers, but no layer is complete; success means blunting, not abolishing, the worst case.	Layer at least 3 independent defences (air, screening, surge); keep the reserve clinical workforce registered and drilled; never let occupancy exceed 85% as the normal state.	Why 1 — systems run at 95%+ occupancy: Efficiency targets eliminate the slack that crises consume. -> Fix: Fund surge slack explicitly: an 85% occupancy ceiling as the regulated normal, with the 15% paid for as readiness. | Why 2 — readiness looks like waste: Idle beds and stockpiles photograph badly in budget debates. -> Fix: Account for readiness as insurance with published expected-loss figures, reviewed by auditors, not by headlines. | Why 3 — prevention is underpaid: Payment follows procedures performed, not illness avoided. -> Fix: Shift a fixed share (target 15–20%) of health budgets to prevention with outcome contracts: paid on measured population results. | Why 4 — workforce pipelines are long: Clinicians take a decade to train; shortages cannot be improvised. -> Fix: Train against the 10-year demographic projection, maintain a drilled reserve register, and recognize foreign credentials within 12 months. | Why 5 — root: Health is budgeted as a consumption expense instead of survival infrastructure. -> Fix: Classify core health capacity with infrastructure law: multi-year protected funding, like defence, insulated from annual cuts.
27	Biological / Life	Health & Medicine	Tier 1	Future	Global / Civilizational	An engineered or novel pathogen, or total care-system overload from aging plus chronic disease, breaks medicine as an institution.	A high-fatality airborne pathogen emerges while hospitals are already at 95% occupancy from ordinary demand; elective and emergency care collapse together.	Layered defence: universal indoor air standards in hospitals, schools, and transit (ventilation equivalent to 6 air changes/hour, upper-air UV in high-risk rooms), synthetic-biology screening of all commercial DNA synthesis orders, regional manufacturing of essential medicines (top 100 generics produced on at least 2 continents each), and a reserve clinical workforce (retired and part-time clinicians registered and drilled annually). Specification: no single country supplies more than 40% of any essential medicine's global volume.	36% (base range 38–58%)	Air standards and synthesis screening are effective layers, but no layer is complete; success means blunting, not abolishing, the worst case.	Layer at least 3 independent defences (air, screening, surge); keep the reserve clinical workforce registered and drilled; never let occupancy exceed 85% as the normal state.	Why 1 — systems run at 95%+ occupancy: Efficiency targets eliminate the slack that crises consume. -> Fix: Fund surge slack explicitly: an 85% occupancy ceiling as the regulated normal, with the 15% paid for as readiness. | Why 2 — readiness looks like waste: Idle beds and stockpiles photograph badly in budget debates. -> Fix: Account for readiness as insurance with published expected-loss figures, reviewed by auditors, not by headlines. | Why 3 — prevention is underpaid: Payment follows procedures performed, not illness avoided. -> Fix: Shift a fixed share (target 15–20%) of health budgets to prevention with outcome contracts: paid on measured population results. | Why 4 — workforce pipelines are long: Clinicians take a decade to train; shortages cannot be improvised. -> Fix: Train against the 10-year demographic projection, maintain a drilled reserve register, and recognize foreign credentials within 12 months. | Why 5 — root: Health is budgeted as a consumption expense instead of survival infrastructure. -> Fix: Classify core health capacity with infrastructure law: multi-year protected funding, like defence, insulated from annual cuts.
28	Physical / Material	Energy	Tier 1	Past	Household / Personal	Fuel embargo or supply cutoff paralyzes transport, heating, and industry in dependent nations.	The 1973 oil embargo quadrupled prices, idled factories, and forced rationing across importing countries.	Strategic reserves and demand discipline: 90 days of net-import reserves for critical fuels (the standing international standard), efficiency law that cuts vehicle and building consumption 2% per year, and pre-planned rationing tiers so essential services (food logistics, hospitals, water) are served first. Specification: household level — 72 hours of light, heat, and cooking independent of the grid (LED + battery bank of 1 kWh, safe fuel stove); national level — no single supplier above 30% of any critical fuel.	92% (base range 76–90%)	90-day reserves plus rationing tiers are proven (post-1973 systems survived later shocks without collapse).	Hold reserves physically audited; pre-write the rationing tiers in law; cut demand 2% per year so the reserve covers more each year.	Why 1 — grids fail in cascade: Systems are optimized for average load and cost, with single points of failure. -> Fix: Engineer for islanding and N-2 redundancy on critical paths; publish cascade maps and harden the worst 20% of nodes first. | Why 2 — redundancy is cut in procurement: Lowest-bid rules select the design with the least slack. -> Fix: Score bids on failure behaviour (tested black-start, islanding) with the same weight as price. | Why 3 — long-lead parts are not stocked: Transformers and breakers are treated as order-when-needed items. -> Fix: A strategic spares pool, jointly owned by utilities, with the 12+ month-lead items physically in warehouses. | Why 4 — demand only grows: Every new load (vehicles, data centres, heating) is connected first and planned later. -> Fix: Connection approvals require a supply plan: new large loads bring generation or storage with them by rule. | Why 5 — root: Energy is sold as a commodity while being used as a life-support system. -> Fix: Regulate a defined survival core (water, food cold chain, hospitals, communications) with life-support reliability standards above all commercial tiers.
29	Physical / Material	Energy	Tier 1	Past	Society / National	Fuel embargo or supply cutoff paralyzes transport, heating, and industry in dependent nations.	The 1973 oil embargo quadrupled prices, idled factories, and forced rationing across importing countries.	Strategic reserves and demand discipline: 90 days of net-import reserves for critical fuels (the standing international standard), efficiency law that cuts vehicle and building consumption 2% per year, and pre-planned rationing tiers so essential services (food logistics, hospitals, water) are served first. Specification: household level — 72 hours of light, heat, and cooking independent of the grid (LED + battery bank of 1 kWh, safe fuel stove); national level — no single supplier above 30% of any critical fuel.	84% (base range 76–90%)	90-day reserves plus rationing tiers are proven (post-1973 systems survived later shocks without collapse).	Hold reserves physically audited; pre-write the rationing tiers in law; cut demand 2% per year so the reserve covers more each year.	Why 1 — grids fail in cascade: Systems are optimized for average load and cost, with single points of failure. -> Fix: Engineer for islanding and N-2 redundancy on critical paths; publish cascade maps and harden the worst 20% of nodes first. | Why 2 — redundancy is cut in procurement: Lowest-bid rules select the design with the least slack. -> Fix: Score bids on failure behaviour (tested black-start, islanding) with the same weight as price. | Why 3 — long-lead parts are not stocked: Transformers and breakers are treated as order-when-needed items. -> Fix: A strategic spares pool, jointly owned by utilities, with the 12+ month-lead items physically in warehouses. | Why 4 — demand only grows: Every new load (vehicles, data centres, heating) is connected first and planned later. -> Fix: Connection approvals require a supply plan: new large loads bring generation or storage with them by rule. | Why 5 — root: Energy is sold as a commodity while being used as a life-support system. -> Fix: Regulate a defined survival core (water, food cold chain, hospitals, communications) with life-support reliability standards above all commercial tiers.

#	Dimension of Realit	Field of Discipline	Pareto Tier o	Timefram	Scale	Worst Possible Problem	Example	Solution — Full Specification	Probability of Su	Why This Probability	How to Maximize the Chances (detailed)	Root Causes
30	Physical / Material	Energy	Tier 1	Past	Global / Civilizational	Fuel embargo or supply cutoff paralyzes transport, heating, and industry in dependent nations.	The 1973 oil embargo quadrupled prices, idled factories, and forced rationing across importing countries.	Strategic reserves and demand discipline: 90 days of net-import reserves for critical fuels (the standing international standard), efficiency law that cuts vehicle and building consumption 2% per year, and pre-planned rationing tiers so essential services (food logistics, hospitals, water) are served first. Specification: household level — 72 hours of light, heat, and cooking independent of the grid (LED + battery bank of 1 kWh, safe fuel stove); national level — no single supplier above 30% of any critical fuel.	72% (base range 76–90%)	90-day reserves plus rationing tiers are proven (post-1973 systems survived later shocks without collapse).	Hold reserves physically audited; pre-write the rationing tiers in law; cut demand 2% per year so the reserve covers more each year.	Why 1 — grids fail in cascade: Systems are optimized for average load and cost, with single points of failure. -> Fix: Engineer for islanding and N-2 redundancy on critical paths; publish cascade maps and harden the worst 20% of nodes first. | Why 2 — redundancy is cut in procurement: Lowest-bid rules select the design with the least slack. -> Fix: Score bids on failure behaviour (tested black-start, islanding) with the same weight as price. | Why 3 — long-lead parts are not stocked: Transformers and breakers are treated as order-when-needed items. -> Fix: A strategic spares pool, jointly owned by utilities, with the 12+ month-lead items physically in warehouses. | Why 4 — demand only grows: Every new load (vehicles, data centres, heating) is connected first and planned later. -> Fix: Connection approvals require a supply plan: new large loads bring generation or storage with them by rule. | Why 5 — root: Energy is sold as a commodity while being used as a life-support system. -> Fix: Regulate a defined survival core (water, food cold chain, hospitals, communications) with life-support reliability standards above all commercial tiers.
31	Physical / Material	Energy	Tier 1	Now	Household / Personal	Grid failure — storm, cyberattack, or cascading overload — blacks out regions for days, and everything else fails with it.	continental-scale blackouts have left tens of millions without power; water pumps, payment systems, hospitals, and communications degrade within hours.	Engineer for islanding: grids split into self-sustaining cells that can run 72 hours alone; critical facilities (hospitals, water, 911, data centres) on dual feeds plus on-site generation tested monthly under load; household standard — a 1–3 kWh battery with solar trickle keeps lights, phone, fridge, and medical devices alive 48–72 hours. Specification: black-start capability drilled twice a year; underground or harden the 20% of lines whose failure cascades furthest (Pareto of the grid).	84% (base range 68–84%)	Islanding and dual feeds are standard utility engineering; blackouts persist where investment lagged, not where the design was unknown.	Drill black-start twice yearly under real load; test facility generators monthly; harden the cascading 20% of lines first (Pareto).	Why 1 — grids fail in cascade: Systems are optimized for average load and cost, with single points of failure. -> Fix: Engineer for islanding and N-2 redundancy on critical paths; publish cascade maps and harden the worst 20% of nodes first. | Why 2 — redundancy is cut in procurement: Lowest-bid rules select the design with the least slack. -> Fix: Score bids on failure behaviour (tested black-start, islanding) with the same weight as price. | Why 3 — long-lead parts are not stocked: Transformers and breakers are treated as order-when-needed items. -> Fix: A strategic spares pool, jointly owned by utilities, with the 12+ month-lead items physically in warehouses. | Why 4 — demand only grows: Every new load (vehicles, data centres, heating) is connected first and planned later. -> Fix: Connection approvals require a supply plan: new large loads bring generation or storage with them by rule. | Why 5 — root: Energy is sold as a commodity while being used as a life-support system. -> Fix: Regulate a defined survival core (water, food cold chain, hospitals, communications) with life-support reliability standards above all commercial tiers.
32	Physical / Material	Energy	Tier 1	Now	Society / National	Grid failure — storm, cyberattack, or cascading overload — blacks out regions for days, and everything else fails with it.	continental-scale blackouts have left tens of millions without power; water pumps, payment systems, hospitals, and communications degrade within hours.	Engineer for islanding: grids split into self-sustaining cells that can run 72 hours alone; critical facilities (hospitals, water, 911, data centres) on dual feeds plus on-site generation tested monthly under load; household standard — a 1–3 kWh battery with solar trickle keeps lights, phone, fridge, and medical devices alive 48–72 hours. Specification: black-start capability drilled twice a year; underground or harden the 20% of lines whose failure cascades furthest (Pareto of the grid).	76% (base range 68–84%)	Islanding and dual feeds are standard utility engineering; blackouts persist where investment lagged, not where the design was unknown.	Drill black-start twice yearly under real load; test facility generators monthly; harden the cascading 20% of lines first (Pareto).	Why 1 — grids fail in cascade: Systems are optimized for average load and cost, with single points of failure. -> Fix: Engineer for islanding and N-2 redundancy on critical paths; publish cascade maps and harden the worst 20% of nodes first. | Why 2 — redundancy is cut in procurement: Lowest-bid rules select the design with the least slack. -> Fix: Score bids on failure behaviour (tested black-start, islanding) with the same weight as price. | Why 3 — long-lead parts are not stocked: Transformers and breakers are treated as order-when-needed items. -> Fix: A strategic spares pool, jointly owned by utilities, with the 12+ month-lead items physically in warehouses. | Why 4 — demand only grows: Every new load (vehicles, data centres, heating) is connected first and planned later. -> Fix: Connection approvals require a supply plan: new large loads bring generation or storage with them by rule. | Why 5 — root: Energy is sold as a commodity while being used as a life-support system. -> Fix: Regulate a defined survival core (water, food cold chain, hospitals, communications) with life-support reliability standards above all commercial tiers.
33	Physical / Material	Energy	Tier 1	Now	Global / Civilizational	Grid failure — storm, cyberattack, or cascading overload — blacks out regions for days, and everything else fails with it.	continental-scale blackouts have left tens of millions without power; water pumps, payment systems, hospitals, and communications degrade within hours.	Engineer for islanding: grids split into self-sustaining cells that can run 72 hours alone; critical facilities (hospitals, water, 911, data centres) on dual feeds plus on-site generation tested monthly under load; household standard — a 1–3 kWh battery with solar trickle keeps lights, phone, fridge, and medical devices alive 48–72 hours. Specification: black-start capability drilled twice a year; underground or harden the 20% of lines whose failure cascades furthest (Pareto of the grid).	64% (base range 68–84%)	Islanding and dual feeds are standard utility engineering; blackouts persist where investment lagged, not where the design was unknown.	Drill black-start twice yearly under real load; test facility generators monthly; harden the cascading 20% of lines first (Pareto).	Why 1 — grids fail in cascade: Systems are optimized for average load and cost, with single points of failure. -> Fix: Engineer for islanding and N-2 redundancy on critical paths; publish cascade maps and harden the worst 20% of nodes first. | Why 2 — redundancy is cut in procurement: Lowest-bid rules select the design with the least slack. -> Fix: Score bids on failure behaviour (tested black-start, islanding) with the same weight as price. | Why 3 — long-lead parts are not stocked: Transformers and breakers are treated as order-when-needed items. -> Fix: A strategic spares pool, jointly owned by utilities, with the 12+ month-lead items physically in warehouses. | Why 4 — demand only grows: Every new load (vehicles, data centres, heating) is connected first and planned later. -> Fix: Connection approvals require a supply plan: new large loads bring generation or storage with them by rule. | Why 5 — root: Energy is sold as a commodity while being used as a life-support system. -> Fix: Regulate a defined survival core (water, food cold chain, hospitals, communications) with life-support reliability standards above all commercial tiers.
34	Physical / Material	Energy	Tier 1	Future	Household / Personal	Electrification of everything concentrates all risk on one system; a solar storm or systemic cyber event takes it down continent-wide for months.	A Carrington-class geomagnetic storm destroys large transformers that take 12–24 months to manufacture, leaving regions dark for seasons.	Stockpile the long-lead items: a strategic reserve of large transformers and grid components (the items with 12+ month lead times), EMP/GMD-hardened control systems for the top substations, and a designed analogue fallback — paper procedures, manual switching, and local generation — drilled, not just documented. Specification: every region can meet 30% of demand (the survival core: water, food cold-chain, hospitals, comms) on distributed local generation indefinitely.	58% (base range 40–60%)	Transformer stockpiles and hardened controls are feasible; the risk is a once-in-centuries event whose timing cannot be forecast, so readiness must be permanent.	Buy the long-lead spares in peacetime budgets; keep analogue fallbacks drilled, not documented only; distribute 30% survival-core generation locally.	Why 1 — grids fail in cascade: Systems are optimized for average load and cost, with single points of failure. -> Fix: Engineer for islanding and N-2 redundancy on critical paths; publish cascade maps and harden the worst 20% of nodes first. | Why 2 — redundancy is cut in procurement: Lowest-bid rules select the design with the least slack. -> Fix: Score bids on failure behaviour (tested black-start, islanding) with the same weight as price. | Why 3 — long-lead parts are not stocked: Transformers and breakers are treated as order-when-needed items. -> Fix: A strategic spares pool, jointly owned by utilities, with the 12+ month-lead items physically in warehouses. | Why 4 — demand only grows: Every new load (vehicles, data centres, heating) is connected first and planned later. -> Fix: Connection approvals require a supply plan: new large loads bring generation or storage with them by rule. | Why 5 — root: Energy is sold as a commodity while being used as a life-support system. -> Fix: Regulate a defined survival core (water, food cold chain, hospitals, communications) with life-support reliability standards above all commercial tiers.
35	Physical / Material	Energy	Tier 1	Future	Society / National	Electrification of everything concentrates all risk on one system; a solar storm or systemic cyber event takes it down continent-wide for months.	A Carrington-class geomagnetic storm destroys large transformers that take 12–24 months to manufacture, leaving regions dark for seasons.	Stockpile the long-lead items: a strategic reserve of large transformers and grid components (the items with 12+ month lead times), EMP/GMD-hardened control systems for the top substations, and a designed analogue fallback — paper procedures, manual switching, and local generation — drilled, not just documented. Specification: every region can meet 30% of demand (the survival core: water, food cold-chain, hospitals, comms) on distributed local generation indefinitely.	50% (base range 40–60%)	Transformer stockpiles and hardened controls are feasible; the risk is a once-in-centuries event whose timing cannot be forecast, so readiness must be permanent.	Buy the long-lead spares in peacetime budgets; keep analogue fallbacks drilled, not documented only; distribute 30% survival-core generation locally.	Why 1 — grids fail in cascade: Systems are optimized for average load and cost, with single points of failure. -> Fix: Engineer for islanding and N-2 redundancy on critical paths; publish cascade maps and harden the worst 20% of nodes first. | Why 2 — redundancy is cut in procurement: Lowest-bid rules select the design with the least slack. -> Fix: Score bids on failure behaviour (tested black-start, islanding) with the same weight as price. | Why 3 — long-lead parts are not stocked: Transformers and breakers are treated as order-when-needed items. -> Fix: A strategic spares pool, jointly owned by utilities, with the 12+ month-lead items physically in warehouses. | Why 4 — demand only grows: Every new load (vehicles, data centres, heating) is connected first and planned later. -> Fix: Connection approvals require a supply plan: new large loads bring generation or storage with them by rule. | Why 5 — root: Energy is sold as a commodity while being used as a life-support system. -> Fix: Regulate a defined survival core (water, food cold chain, hospitals, communications) with life-support reliability standards above all commercial tiers.
36	Physical / Material	Energy	Tier 1	Future	Global / Civilizational	Electrification of everything concentrates all risk on one system; a solar storm or systemic cyber event takes it down continent-wide for months.	A Carrington-class geomagnetic storm destroys large transformers that take 12–24 months to manufacture, leaving regions dark for seasons.	Stockpile the long-lead items: a strategic reserve of large transformers and grid components (the items with 12+ month lead times), EMP/GMD-hardened control systems for the top substations, and a designed analogue fallback — paper procedures, manual switching, and local generation — drilled, not just documented. Specification: every region can meet 30% of demand (the survival core: water, food cold-chain, hospitals, comms) on distributed local generation indefinitely.	38% (base range 40–60%)	Transformer stockpiles and hardened controls are feasible; the risk is a once-in-centuries event whose timing cannot be forecast, so readiness must be permanent.	Buy the long-lead spares in peacetime budgets; keep analogue fallbacks drilled, not documented only; distribute 30% survival-core generation locally.	Why 1 — grids fail in cascade: Systems are optimized for average load and cost, with single points of failure. -> Fix: Engineer for islanding and N-2 redundancy on critical paths; publish cascade maps and harden the worst 20% of nodes first. | Why 2 — redundancy is cut in procurement: Lowest-bid rules select the design with the least slack. -> Fix: Score bids on failure behaviour (tested black-start, islanding) with the same weight as price. | Why 3 — long-lead parts are not stocked: Transformers and breakers are treated as order-when-needed items. -> Fix: A strategic spares pool, jointly owned by utilities, with the 12+ month-lead items physically in warehouses. | Why 4 — demand only grows: Every new load (vehicles, data centres, heating) is connected first and planned later. -> Fix: Connection approvals require a supply plan: new large loads bring generation or storage with them by rule. | Why 5 — root: Energy is sold as a commodity while being used as a life-support system. -> Fix: Regulate a defined survival core (water, food cold chain, hospitals, communications) with life-support reliability standards above all commercial tiers.
37	Physical / Material	Shelter & Housing	Tier 1	Past	Household / Personal	Fire, earthquake, or war destroys a city's housing stock in days, leaving populations in the open.	The 1906 San Francisco earthquake and fire destroyed 80% of the city; the 1940s bombing campaigns displaced tens of millions across Europe.	Build to survive the known local worst: seismic codes to the 2,475-year event for new builds, retrofit of the highest-risk 20% of existing stock first (Pareto: unreinforced masonry and soft-storey buildings cause most deaths), firebreaks and non-combustible roofing in fire zones. Specification: every city holds emergency shelter capacity for 10% of its population for 30 days (pre-sited, pre-supplied), and every household keeps a 72-hour go-kit — documents, medication, water, light.	89% (base range 74–88%)	Seismic and fire codes demonstrably cut deaths by 80–95% where enforced on the highest-risk building classes first.	Retrofit the deadliest 20% of stock first; enforce codes on occupancy, not just construction; shelter capacity pre-sited and pre-supplied.	Why 1 — not enough homes near work: Zoning forbids the density the population needs where it needs it. -> Fix: Legalize mid-density near transit by right; approvals deemed granted after 12 months of silence. | Why 2 — incumbents block supply: Homeowners gain from scarcity and hold local veto power. -> Fix: Move housing targets to regional level with automatic triggers (rent-to-income ratio) that override local refusal. | Why 3 — risk is mispriced: Insurance and mortgages ignore the 2050 risk map, so building continues in doomed zones. -> Fix: Publish address-level risk and price it in; stop new construction in zones uninsurable within 30 years. | Why 4 — construction capacity is thin: Trades pipelines and factory-built capacity were allowed to shrink. -> Fix: Fund trades training against the housing target and permit factory-built housing at national standard. | Why 5 — root: Housing is treated primarily as an investment asset and only secondarily as shelter. -> Fix: Dual mandate in law: every housing policy scored on both asset stability and the shelter benchmark (median rent at or below 30% of median income).
38	Physical / Material	Shelter & Housing	Tier 1	Past	Society / National	Fire, earthquake, or war destroys a city's housing stock in days, leaving populations in the open.	The 1906 San Francisco earthquake and fire destroyed 80% of the city; the 1940s bombing campaigns displaced tens of millions across Europe.	Build to survive the known local worst: seismic codes to the 2,475-year event for new builds, retrofit of the highest-risk 20% of existing stock first (Pareto: unreinforced masonry and soft-storey buildings cause most deaths), firebreaks and non-combustible roofing in fire zones. Specification: every city holds emergency shelter capacity for 10% of its population for 30 days (pre-sited, pre-supplied), and every household keeps a 72-hour go-kit — documents, medication, water, light.	81% (base range 74–88%)	Seismic and fire codes demonstrably cut deaths by 80–95% where enforced on the highest-risk building classes first.	Retrofit the deadliest 20% of stock first; enforce codes on occupancy, not just construction; shelter capacity pre-sited and pre-supplied.	Why 1 — not enough homes near work: Zoning forbids the density the population needs where it needs it. -> Fix: Legalize mid-density near transit by right; approvals deemed granted after 12 months of silence. | Why 2 — incumbents block supply: Homeowners gain from scarcity and hold local veto power. -> Fix: Move housing targets to regional level with automatic triggers (rent-to-income ratio) that override local refusal. | Why 3 — risk is mispriced: Insurance and mortgages ignore the 2050 risk map, so building continues in doomed zones. -> Fix: Publish address-level risk and price it in; stop new construction in zones uninsurable within 30 years. | Why 4 — construction capacity is thin: Trades pipelines and factory-built capacity were allowed to shrink. -> Fix: Fund trades training against the housing target and permit factory-built housing at national standard. | Why 5 — root: Housing is treated primarily as an investment asset and only secondarily as shelter. -> Fix: Dual mandate in law: every housing policy scored on both asset stability and the shelter benchmark (median rent at or below 30% of median income).
39	Physical / Material	Shelter & Housing	Tier 1	Past	Global / Civilizational	Fire, earthquake, or war destroys a city's housing stock in days, leaving populations in the open.	The 1906 San Francisco earthquake and fire destroyed 80% of the city; the 1940s bombing campaigns displaced tens of millions across Europe.	Build to survive the known local worst: seismic codes to the 2,475-year event for new builds, retrofit of the highest-risk 20% of existing stock first (Pareto: unreinforced masonry and soft-storey buildings cause most deaths), firebreaks and non-combustible roofing in fire zones. Specification: every city holds emergency shelter capacity for 10% of its population for 30 days (pre-sited, pre-supplied), and every household keeps a 72-hour go-kit — documents, medication, water, light.	69% (base range 74–88%)	Seismic and fire codes demonstrably cut deaths by 80–95% where enforced on the highest-risk building classes first.	Retrofit the deadliest 20% of stock first; enforce codes on occupancy, not just construction; shelter capacity pre-sited and pre-supplied.	Why 1 — not enough homes near work: Zoning forbids the density the population needs where it needs it. -> Fix: Legalize mid-density near transit by right; approvals deemed granted after 12 months of silence. | Why 2 — incumbents block supply: Homeowners gain from scarcity and hold local veto power. -> Fix: Move housing targets to regional level with automatic triggers (rent-to-income ratio) that override local refusal. | Why 3 — risk is mispriced: Insurance and mortgages ignore the 2050 risk map, so building continues in doomed zones. -> Fix: Publish address-level risk and price it in; stop new construction in zones uninsurable within 30 years. | Why 4 — construction capacity is thin: Trades pipelines and factory-built capacity were allowed to shrink. -> Fix: Fund trades training against the housing target and permit factory-built housing at national standard. | Why 5 — root: Housing is treated primarily as an investment asset and only secondarily as shelter. -> Fix: Dual mandate in law: every housing policy scored on both asset stability and the shelter benchmark (median rent at or below 30% of median income).

#	Dimension of Realit	Field of Discipline	Pareto Tier o	Timefram	Scale	Worst Possible Problem	Example	Solution — Full Specification	Probability of Su	Why This Probability	How to Maximize the Chances (detailed)	Root Causes
40	Physical / Material	Shelter & Housing	Tier 1	Now	Household / Personal	Housing costs detach from incomes; working households are priced out, homelessness becomes structural.	In major cities, median homes cost 10–13x median income; full-time workers sleep in vehicles while vacancies are held as investments.	Attack the constraint, which is supply near work: legalize mid-density housing on all residential land within 800 m of transit, approvals deemed granted after 12 months, public land leased (not sold) for permanently affordable homes at 30% of income benchmark, and vacancy taxes that make empty investment units unprofitable. Specification: a city is healthy when median rent is at or below 30% of median household income; publish the ratio quarterly and trigger automatic zoning release when it exceeds it for 2 consecutive years.	72% (base range 55–72%)	Supply-side reforms work where tried (upzoning cities have measurably slowed rent growth), but face political resistance from incumbent owners.	Automatic triggers (ratio-based zoning release) remove the annual political fight; lease public land instead of selling; measure and publish the rent-to-income ratio quarterly.	Why 1 — not enough homes near work: Zoning forbids the density the population needs where it needs it. -> Fix: Legalize mid-density near transit by right; approvals deemed granted after 12 months of silence.	Why 2 — incumbents block supply: Homeowners gain from scarcity and hold local veto power. -> Fix: Move housing targets to regional level with automatic triggers (rent-to-income ratio) that override local refusal.	Why 3 — risk is mispriced: Insurance and mortgages ignore the 2050 risk map, so building continues in doomed zones. -> Fix: Publish address-level risk and price it in; stop new construction in zones uninsurable within 30 years.	Why 4 — construction capacity is thin: Trades pipelines and factory-built capacity were allowed to shrink. -> Fix: Fund trades training against the housing target and permit factory-built housing at national standard.	Why 5 — root: Housing is treated primarily as an investment asset and only secondarily as shelter. -> Fix: Dual mandate in law: every housing policy scored on both asset stability and the shelter benchmark (median rent at or below 30% of median income).
41	Physical / Material	Shelter & Housing	Tier 1	Now	Society / National	Housing costs detach from incomes; working households are priced out, homelessness becomes structural.	In major cities, median homes cost 10–13x median income; full-time workers sleep in vehicles while vacancies are held as investments.	Attack the constraint, which is supply near work: legalize mid-density housing on all residential land within 800 m of transit, approvals deemed granted after 12 months, public land leased (not sold) for permanently affordable homes at 30% of income benchmark, and vacancy taxes that make empty investment units unprofitable. Specification: a city is healthy when median rent is at or below 30% of median household income; publish the ratio quarterly and trigger automatic zoning release when it exceeds it for 2 consecutive years.	64% (base range 55–72%)	Supply-side reforms work where tried (upzoning cities have measurably slowed rent growth), but face political resistance from incumbent owners.	Automatic triggers (ratio-based zoning release) remove the annual political fight; lease public land instead of selling; measure and publish the rent-to-income ratio quarterly.	Why 1 — not enough homes near work: Zoning forbids the density the population needs where it needs it. -> Fix: Legalize mid-density near transit by right; approvals deemed granted after 12 months of silence.	Why 2 — incumbents block supply: Homeowners gain from scarcity and hold local veto power. -> Fix: Move housing targets to regional level with automatic triggers (rent-to-income ratio) that override local refusal.	Why 3 — risk is mispriced: Insurance and mortgages ignore the 2050 risk map, so building continues in doomed zones. -> Fix: Publish address-level risk and price it in; stop new construction in zones uninsurable within 30 years.	Why 4 — construction capacity is thin: Trades pipelines and factory-built capacity were allowed to shrink. -> Fix: Fund trades training against the housing target and permit factory-built housing at national standard.	Why 5 — root: Housing is treated primarily as an investment asset and only secondarily as shelter. -> Fix: Dual mandate in law: every housing policy scored on both asset stability and the shelter benchmark (median rent at or below 30% of median income).
42	Physical / Material	Shelter & Housing	Tier 1	Now	Global / Civilizational	Housing costs detach from incomes; working households are priced out, homelessness becomes structural.	In major cities, median homes cost 10–13x median income; full-time workers sleep in vehicles while vacancies are held as investments.	Attack the constraint, which is supply near work: legalize mid-density housing on all residential land within 800 m of transit, approvals deemed granted after 12 months, public land leased (not sold) for permanently affordable homes at 30% of income benchmark, and vacancy taxes that make empty investment units unprofitable. Specification: a city is healthy when median rent is at or below 30% of median household income; publish the ratio quarterly and trigger automatic zoning release when it exceeds it for 2 consecutive years.	52% (base range 55–72%)	Supply-side reforms work where tried (upzoning cities have measurably slowed rent growth), but face political resistance from incumbent owners.	Automatic triggers (ratio-based zoning release) remove the annual political fight; lease public land instead of selling; measure and publish the rent-to-income ratio quarterly.	Why 1 — not enough homes near work: Zoning forbids the density the population needs where it needs it. -> Fix: Legalize mid-density near transit by right; approvals deemed granted after 12 months of silence.	Why 2 — incumbents block supply: Homeowners gain from scarcity and hold local veto power. -> Fix: Move housing targets to regional level with automatic triggers (rent-to-income ratio) that override local refusal.	Why 3 — risk is mispriced: Insurance and mortgages ignore the 2050 risk map, so building continues in doomed zones. -> Fix: Publish address-level risk and price it in; stop new construction in zones uninsurable within 30 years.	Why 4 — construction capacity is thin: Trades pipelines and factory-built capacity were allowed to shrink. -> Fix: Fund trades training against the housing target and permit factory-built housing at national standard.	Why 5 — root: Housing is treated primarily as an investment asset and only secondarily as shelter. -> Fix: Dual mandate in law: every housing policy scored on both asset stability and the shelter benchmark (median rent at or below 30% of median income).
43	Physical / Material	Shelter & Housing	Tier 1	Future	Household / Personal	Climate migration plus uninsurable regions strand populations in housing that cannot be sold, insured, or safely occupied.	Coastal and fire-zone properties lose insurance and value simultaneously; millions hold worthless assets in places becoming unliviviable.	Managed transition instead of collapse: pre-mapped receiving cities with pre-built infrastructure capacity, public buyouts at pre-decline values funded by risk-zone insurance levies collected while values remain, and building law that stops new construction in zones projected uninsurable within 30 years. Specification: national housing plans must model the 2050 climate map, not the 2000 one; household level — never hold more than 50% of net worth in a single climate-exposed property.	60% (base range 42–62%)	Managed retreat succeeds when funded before values collapse; it fails when attempted after, amid litigation and denial.	Collect the risk levy while values remain; sign receiving-city compacts early; stop new building in 30-year uninsurable zones immediately.	Why 1 — not enough homes near work: Zoning forbids the density the population needs where it needs it. -> Fix: Legalize mid-density near transit by right; approvals deemed granted after 12 months of silence.	Why 2 — incumbents block supply: Homeowners gain from scarcity and hold local veto power. -> Fix: Move housing targets to regional level with automatic triggers (rent-to-income ratio) that override local refusal.	Why 3 — risk is mispriced: Insurance and mortgages ignore the 2050 risk map, so building continues in doomed zones. -> Fix: Publish address-level risk and price it in; stop new construction in zones uninsurable within 30 years.	Why 4 — construction capacity is thin: Trades pipelines and factory-built capacity were allowed to shrink. -> Fix: Fund trades training against the housing target and permit factory-built housing at national standard.	Why 5 — root: Housing is treated primarily as an investment asset and only secondarily as shelter. -> Fix: Dual mandate in law: every housing policy scored on both asset stability and the shelter benchmark (median rent at or below 30% of median income).
44	Physical / Material	Shelter & Housing	Tier 1	Future	Society / National	Climate migration plus uninsurable regions strand populations in housing that cannot be sold, insured, or safely occupied.	Coastal and fire-zone properties lose insurance and value simultaneously; millions hold worthless assets in places becoming unliviviable.	Managed transition instead of collapse: pre-mapped receiving cities with pre-built infrastructure capacity, public buyouts at pre-decline values funded by risk-zone insurance levies collected while values remain, and building law that stops new construction in zones projected uninsurable within 30 years. Specification: national housing plans must model the 2050 climate map, not the 2000 one; household level — never hold more than 50% of net worth in a single climate-exposed property.	52% (base range 42–62%)	Managed retreat succeeds when funded before values collapse; it fails when attempted after, amid litigation and denial.	Collect the risk levy while values remain; sign receiving-city compacts early; stop new building in 30-year uninsurable zones immediately.	Why 1 — not enough homes near work: Zoning forbids the density the population needs where it needs it. -> Fix: Legalize mid-density near transit by right; approvals deemed granted after 12 months of silence.	Why 2 — incumbents block supply: Homeowners gain from scarcity and hold local veto power. -> Fix: Move housing targets to regional level with automatic triggers (rent-to-income ratio) that override local refusal.	Why 3 — risk is mispriced: Insurance and mortgages ignore the 2050 risk map, so building continues in doomed zones. -> Fix: Publish address-level risk and price it in; stop new construction in zones uninsurable within 30 years.	Why 4 — construction capacity is thin: Trades pipelines and factory-built capacity were allowed to shrink. -> Fix: Fund trades training against the housing target and permit factory-built housing at national standard.	Why 5 — root: Housing is treated primarily as an investment asset and only secondarily as shelter. -> Fix: Dual mandate in law: every housing policy scored on both asset stability and the shelter benchmark (median rent at or below 30% of median income).
45	Physical / Material	Shelter & Housing	Tier 1	Future	Global / Civilizational	Climate migration plus uninsurable regions strand populations in housing that cannot be sold, insured, or safely occupied.	Coastal and fire-zone properties lose insurance and value simultaneously; millions hold worthless assets in places becoming unliviviable.	Managed transition instead of collapse: pre-mapped receiving cities with pre-built infrastructure capacity, public buyouts at pre-decline values funded by risk-zone insurance levies collected while values remain, and building law that stops new construction in zones projected uninsurable within 30 years. Specification: national housing plans must model the 2050 climate map, not the 2000 one; household level — never hold more than 50% of net worth in a single climate-exposed property.	40% (base range 42–62%)	Managed retreat succeeds when funded before values collapse; it fails when attempted after, amid litigation and denial.	Collect the risk levy while values remain; sign receiving-city compacts early; stop new building in 30-year uninsurable zones immediately.	Why 1 — not enough homes near work: Zoning forbids the density the population needs where it needs it. -> Fix: Legalize mid-density near transit by right; approvals deemed granted after 12 months of silence.	Why 2 — incumbents block supply: Homeowners gain from scarcity and hold local veto power. -> Fix: Move housing targets to regional level with automatic triggers (rent-to-income ratio) that override local refusal.	Why 3 — risk is mispriced: Insurance and mortgages ignore the 2050 risk map, so building continues in doomed zones. -> Fix: Publish address-level risk and price it in; stop new construction in zones uninsurable within 30 years.	Why 4 — construction capacity is thin: Trades pipelines and factory-built capacity were allowed to shrink. -> Fix: Fund trades training against the housing target and permit factory-built housing at national standard.	Why 5 — root: Housing is treated primarily as an investment asset and only secondarily as shelter. -> Fix: Dual mandate in law: every housing policy scored on both asset stability and the shelter benchmark (median rent at or below 30% of median income).
46	Social / Order	Security & Defense	Tier 1	Past	Household / Personal	Invasion or civil war destroys the state's monopoly on force; civilians face violence with no protection.	The 1990s Balkan wars and the Rwandan genocide showed how fast neighbours, police, and armies can become the threat itself.	Prevention is the only real solution once killing starts: early-warning indicators (hate-media licensing, militia formation, arms flows) monitored with published thresholds that trigger automatic diplomatic and legal consequences; constitutional hard limits on emergency powers (sunset clauses of 90 days, legislative renewal); and protected humanitarian corridors negotiated in advance frameworks. Specification: civilian layer — neighbourhood mutual-aid rosters, documented evidence habits (photos, records kept off-site), and pre-planned exit routes with 72-hour kits.	53% (base range 35–55%)	Once mass violence begins, no intervention reliably stops it quickly; prevention indicators work only if acted on before mobilization completes.	Act on the first two indicators, not the tenth; keep exit routes and records as standing household practice in at-risk regions; international triggers automatic, not negotiated mid-crisis.	Why 1 — violence concentrates: A tiny network and a few places generate most harm, while responses spread thin everywhere. -> Fix: Fund focused deterrence and place-based fixes at the measured concentration points; audit monthly.	Why 2 — legitimacy collapses first: Abusive or absent policing destroys the cooperation that produces information. -> Fix: Procedural-justice standards with published complaint and outcome data; violence interrupters as credible messengers.	Why 3 — weapons get cheaper: Each decade lowers the cost of precision harm. -> Fix: Regulate at the component and platform layer (drone ID, geofencing, synthesis screening) before incidents, not after.	Why 4 — early warnings are ignored: Acting early looks like overreaction until it looks like negligence. -> Fix: Pre-commit triggers in law: named indicators compel named responses, removing the discretion to wait.	Why 5 — root: Security is funded as reaction — after events — rather than as prevention with measured returns. -> Fix: Budget prevention like insurance with published expected-harm reductions per program, renewed only on measured results.
47	Social / Order	Security & Defense	Tier 1	Past	Society / National	Invasion or civil war destroys the state's monopoly on force; civilians face violence with no protection.	The 1990s Balkan wars and the Rwandan genocide showed how fast neighbours, police, and armies can become the threat itself.	Prevention is the only real solution once killing starts: early-warning indicators (hate-media licensing, militia formation, arms flows) monitored with published thresholds that trigger automatic diplomatic and legal consequences; constitutional hard limits on emergency powers (sunset clauses of 90 days, legislative renewal); and protected humanitarian corridors negotiated in advance frameworks. Specification: civilian layer — neighbourhood mutual-aid rosters, documented evidence habits (photos, records kept off-site), and pre-planned exit routes with 72-hour kits.	45% (base range 35–55%)	Once mass violence begins, no intervention reliably stops it quickly; prevention indicators work only if acted on before mobilization completes.	Act on the first two indicators, not the tenth; keep exit routes and records as standing household practice in at-risk regions; international triggers automatic, not negotiated mid-crisis.	Why 1 — violence concentrates: A tiny network and a few places generate most harm, while responses spread thin everywhere. -> Fix: Fund focused deterrence and place-based fixes at the measured concentration points; audit monthly.	Why 2 — legitimacy collapses first: Abusive or absent policing destroys the cooperation that produces information. -> Fix: Procedural-justice standards with published complaint and outcome data; violence interrupters as credible messengers.	Why 3 — weapons get cheaper: Each decade lowers the cost of precision harm. -> Fix: Regulate at the component and platform layer (drone ID, geofencing, synthesis screening) before incidents, not after.	Why 4 — early warnings are ignored: Acting early looks like overreaction until it looks like negligence. -> Fix: Pre-commit triggers in law: named indicators compel named responses, removing the discretion to wait.	Why 5 — root: Security is funded as reaction — after events — rather than as prevention with measured returns. -> Fix: Budget prevention like insurance with published expected-harm reductions per program, renewed only on measured results.
48	Social / Order	Security & Defense	Tier 1	Past	Global / Civilizational	Invasion or civil war destroys the state's monopoly on force; civilians face violence with no protection.	The 1990s Balkan wars and the Rwandan genocide showed how fast neighbours, police, and armies can become the threat itself.	Prevention is the only real solution once killing starts: early-warning indicators (hate-media licensing, militia formation, arms flows) monitored with published thresholds that trigger automatic diplomatic and legal consequences; constitutional hard limits on emergency powers (sunset clauses of 90 days, legislative renewal); and protected humanitarian corridors negotiated in advance frameworks. Specification: civilian layer — neighbourhood mutual-aid rosters, documented evidence habits (photos, records kept off-site), and pre-planned exit routes with 72-hour kits.	33% (base range 35–55%)	Once mass violence begins, no intervention reliably stops it quickly; prevention indicators work only if acted on before mobilization completes.	Act on the first two indicators, not the tenth; keep exit routes and records as standing household practice in at-risk regions; international triggers automatic, not negotiated mid-crisis.	Why 1 — violence concentrates: A tiny network and a few places generate most harm, while responses spread thin everywhere. -> Fix: Fund focused deterrence and place-based fixes at the measured concentration points; audit monthly.	Why 2 — legitimacy collapses first: Abusive or absent policing destroys the cooperation that produces information. -> Fix: Procedural-justice standards with published complaint and outcome data; violence interrupters as credible messengers.	Why 3 — weapons get cheaper: Each decade lowers the cost of precision harm. -> Fix: Regulate at the component and platform layer (drone ID, geofencing, synthesis screening) before incidents, not after.	Why 4 — early warnings are ignored: Acting early looks like overreaction until it looks like negligence. -> Fix: Pre-commit triggers in law: named indicators compel named responses, removing the discretion to wait.	Why 5 — root: Security is funded as reaction — after events — rather than as prevention with measured returns. -> Fix: Budget prevention like insurance with published expected-harm reductions per program, renewed only on measured results.
49	Social / Order	Security & Defense	Tier 1	Now	Household / Personal	Organized crime, gang fragmentation, and everyday violence make districts unsafe while institutions lose legitimacy.	Cities where homicide concentrates in a few postal codes: a small fraction of places and people account for most violence.	Focused deterrence, the proven Pareto intervention: identify the tiny network driving most violence (typically under 0.5% of a city's population), offer services and exit routes with one hand and certain, swift prosecution with the other; light and maintain the worst micro-locations (lighting, sightlines, occupied storefronts); violence-interrupter teams staffed by credible locals. Specification: measure success by shootings in the target micro-zones, reviewed monthly; every intervention audited for civil-rights compliance.	82% (base range 66–82%)	Focused deterrence is among the most evaluated interventions in criminology, with violence reductions of 30–60% in target networks across many cities.	Keep the target list small and evidence-based; deliver the services offer genuinely; audit civil-rights compliance monthly so legitimacy survives.	Why 1 — violence concentrates: A tiny network and a few places generate most harm, while responses spread thin everywhere. -> Fix: Fund focused deterrence and place-based fixes at the measured concentration points; audit monthly.	Why 2 — legitimacy collapses first: Abusive or absent policing destroys the cooperation that produces information. -> Fix: Procedural-justice standards with published complaint and outcome data; violence interrupters as credible messengers.	Why 3 — weapons get cheaper: Each decade lowers the cost of precision harm. -> Fix: Regulate at the component and platform layer (drone ID, geofencing, synthesis screening) before incidents, not after.	Why 4 — early warnings are ignored: Acting early looks like overreaction until it looks like negligence. -> Fix: Pre-commit triggers in law: named indicators compel named responses, removing the discretion to wait.	Why 5 — root: Security is funded as reaction — after events — rather than as prevention with measured returns. -> Fix: Budget prevention like insurance with published expected-harm reductions per program, renewed only on measured results.

#	Dimension of Realit	Field of Discipline	Pareto Tier o	Timefram	Scale	Worst Possible Problem	Example	Solution — Full Specification	Probability of Su	Why This Probability	How to Maximize the Chances (detailed)	Root Causes
50	Social / Order	Security & Defense	Tier 1	Now	Society / National	Organized crime, gang fragmentation, and everyday violence make districts unsafe while institutions lose legitimacy.	Cities where homicide concentrates in a few postal codes: a small fraction of places and people account for most violence.	Focused deterrence, the proven Pareto intervention: identify the tiny network driving most violence (typically under 0.5% of a city's population), offer services and exit routes with one hand and certain, swift prosecution with the other; light and maintain the worst micro-locations (lighting, sightlines, occupied storefronts); violence-interrupter teams staffed by credible locals. Specification: measure success by shootings in the target micro-zones, reviewed monthly; every intervention audited for civil-rights compliance.	74% (base range 66–82%)	Focused deterrence is among the most evaluated interventions in criminology, with violence reductions of 30-60% in target networks across many cities.	Keep the target list small and evidence-based; deliver the services offer genuinely; audit civil-rights compliance monthly so legitimacy survives.	Why 1 — violence concentrates: A tiny network and a few places generate most harm, while responses spread thin everywhere. -> Fix: Fund focused deterrence and place-based fixes at the measured concentration points; audit monthly. | Why 2 — legitimacy collapses first: Abusive or absent policing destroys the cooperation that produces information. -> Fix: Procedural-justice standards with published complaint and outcome data; violence interrupters as credible messengers. | Why 3 — weapons get cheaper: Each decade lowers the cost of precision harm. -> Fix: Regulate at the component and platform layer (drone ID, geofencing, synthesis screening) before incidents, not after. | Why 4 — early warnings are ignored: Acting early looks like overreaction until it looks like negligence. -> Fix: Pre-commit triggers in law: named indicators compel named responses, removing the discretion to wait. | Why 5 — root: Security is funded as reaction — after events — rather than as prevention with measured returns. -> Fix: Budget prevention like insurance with published expected-harm reductions per program, renewed only on measured results.
51	Social / Order	Security & Defense	Tier 1	Now	Global / Civilizational	Organized crime, gang fragmentation, and everyday violence make districts unsafe while institutions lose legitimacy.	Cities where homicide concentrates in a few postal codes: a small fraction of places and people account for most violence.	Focused deterrence, the proven Pareto intervention: identify the tiny network driving most violence (typically under 0.5% of a city's population), offer services and exit routes with one hand and certain, swift prosecution with the other; light and maintain the worst micro-locations (lighting, sightlines, occupied storefronts); violence-interrupter teams staffed by credible locals. Specification: measure success by shootings in the target micro-zones, reviewed monthly; every intervention audited for civil-rights compliance.	62% (base range 66–82%)	Focused deterrence is among the most evaluated interventions in criminology, with violence reductions of 30-60% in target networks across many cities.	Keep the target list small and evidence-based; deliver the services offer genuinely; audit civil-rights compliance monthly so legitimacy survives.	Why 1 — violence concentrates: A tiny network and a few places generate most harm, while responses spread thin everywhere. -> Fix: Fund focused deterrence and place-based fixes at the measured concentration points; audit monthly. | Why 2 — legitimacy collapses first: Abusive or absent policing destroys the cooperation that produces information. -> Fix: Procedural-justice standards with published complaint and outcome data; violence interrupters as credible messengers. | Why 3 — weapons get cheaper: Each decade lowers the cost of precision harm. -> Fix: Regulate at the component and platform layer (drone ID, geofencing, synthesis screening) before incidents, not after. | Why 4 — early warnings are ignored: Acting early looks like overreaction until it looks like negligence. -> Fix: Pre-commit triggers in law: named indicators compel named responses, removing the discretion to wait. | Why 5 — root: Security is funded as reaction — after events — rather than as prevention with measured returns. -> Fix: Budget prevention like insurance with published expected-harm reductions per program, renewed only on measured results.
52	Social / Order	Security & Defense	Tier 1	Future	Household / Personal	Autonomous weapons and cheap drones make precision violence available to anyone, overwhelming policing built for human-speed threats.	A single actor with a swarm of consumer drones attacks a stadium, substation, or water plant; defence budgets built for jets and tanks are irrelevant.	Regulate and shield in advance: mandatory remote-identification and geofencing on all drones sold, counter-drone authority and equipment at critical sites (airports, utilities, mass events) with legal rules of engagement written before incidents, export controls on autonomous targeting components, and an international ban framework on fully autonomous lethal systems. Specification: critical infrastructure must pass an annual red-team drone assessment; households and businesses rely on hardened, setback design for the highest-value targets rather than private countermeasures.	64% (base range 46–66%)	Drone regulation and site hardening are technically ready; the gap is legislative speed versus consumer-technology speed.	Mandate remote ID and geofencing at point of sale now; annual red-team tests for critical sites; write the legal rules of engagement before the first mass incident.	Why 1 — violence concentrates: A tiny network and a few places generate most harm, while responses spread thin everywhere. -> Fix: Fund focused deterrence and place-based fixes at the measured concentration points; audit monthly. | Why 2 — legitimacy collapses first: Abusive or absent policing destroys the cooperation that produces information. -> Fix: Procedural-justice standards with published complaint and outcome data; violence interrupters as credible messengers. | Why 3 — weapons get cheaper: Each decade lowers the cost of precision harm. -> Fix: Regulate at the component and platform layer (drone ID, geofencing, synthesis screening) before incidents, not after. | Why 4 — early warnings are ignored: Acting early looks like overreaction until it looks like negligence. -> Fix: Pre-commit triggers in law: named indicators compel named responses, removing the discretion to wait. | Why 5 — root: Security is funded as reaction — after events — rather than as prevention with measured returns. -> Fix: Budget prevention like insurance with published expected-harm reductions per program, renewed only on measured results.
53	Social / Order	Security & Defense	Tier 1	Future	Society / National	Autonomous weapons and cheap drones make precision violence available to anyone, overwhelming policing built for human-speed threats.	A single actor with a swarm of consumer drones attacks a stadium, substation, or water plant; defence budgets built for jets and tanks are irrelevant.	Regulate and shield in advance: mandatory remote-identification and geofencing on all drones sold, counter-drone authority and equipment at critical sites (airports, utilities, mass events) with legal rules of engagement written before incidents, export controls on autonomous targeting components, and an international ban framework on fully autonomous lethal systems. Specification: critical infrastructure must pass an annual red-team drone assessment; households and businesses rely on hardened, setback design for the highest-value targets rather than private countermeasures.	56% (base range 46–66%)	Drone regulation and site hardening are technically ready; the gap is legislative speed versus consumer-technology speed.	Mandate remote ID and geofencing at point of sale now; annual red-team tests for critical sites; write the legal rules of engagement before the first mass incident.	Why 1 — violence concentrates: A tiny network and a few places generate most harm, while responses spread thin everywhere. -> Fix: Fund focused deterrence and place-based fixes at the measured concentration points; audit monthly. | Why 2 — legitimacy collapses first: Abusive or absent policing destroys the cooperation that produces information. -> Fix: Procedural-justice standards with published complaint and outcome data; violence interrupters as credible messengers. | Why 3 — weapons get cheaper: Each decade lowers the cost of precision harm. -> Fix: Regulate at the component and platform layer (drone ID, geofencing, synthesis screening) before incidents, not after. | Why 4 — early warnings are ignored: Acting early looks like overreaction until it looks like negligence. -> Fix: Pre-commit triggers in law: named indicators compel named responses, removing the discretion to wait. | Why 5 — root: Security is funded as reaction — after events — rather than as prevention with measured returns. -> Fix: Budget prevention like insurance with published expected-harm reductions per program, renewed only on measured results.
54	Social / Order	Security & Defense	Tier 1	Future	Global / Civilizational	Autonomous weapons and cheap drones make precision violence available to anyone, overwhelming policing built for human-speed threats.	A single actor with a swarm of consumer drones attacks a stadium, substation, or water plant; defence budgets built for jets and tanks are irrelevant.	Regulate and shield in advance: mandatory remote-identification and geofencing on all drones sold, counter-drone authority and equipment at critical sites (airports, utilities, mass events) with legal rules of engagement written before incidents, export controls on autonomous targeting components, and an international ban framework on fully autonomous lethal systems. Specification: critical infrastructure must pass an annual red-team drone assessment; households and businesses rely on hardened, setback design for the highest-value targets rather than private countermeasures.	44% (base range 46–66%)	Drone regulation and site hardening are technically ready; the gap is legislative speed versus consumer-technology speed.	Mandate remote ID and geofencing at point of sale now; annual red-team tests for critical sites; write the legal rules of engagement before the first mass incident.	Why 1 — violence concentrates: A tiny network and a few places generate most harm, while responses spread thin everywhere. -> Fix: Fund focused deterrence and place-based fixes at the measured concentration points; audit monthly. | Why 2 — legitimacy collapses first: Abusive or absent policing destroys the cooperation that produces information. -> Fix: Procedural-justice standards with published complaint and outcome data; violence interrupters as credible messengers. | Why 3 — weapons get cheaper: Each decade lowers the cost of precision harm. -> Fix: Regulate at the component and platform layer (drone ID, geofencing, synthesis screening) before incidents, not after. | Why 4 — early warnings are ignored: Acting early looks like overreaction until it looks like negligence. -> Fix: Pre-commit triggers in law: named indicators compel named responses, removing the discretion to wait. | Why 5 — root: Security is funded as reaction — after events — rather than as prevention with measured returns. -> Fix: Budget prevention like insurance with published expected-harm reductions per program, renewed only on measured results.
55	Social / Order	Governance & Law	Tier 1	Past	Household / Personal	Constitutional order collapses into dictatorship through legal steps, emergency decrees, and captured courts.	Weimar Germany's institutions were dismantled largely by legal instruments — an enabling act, packed courts, and emergency rule that never ended.	Design constitutions against their own abolition: eternity clauses protecting core rights from amendment, emergency powers with hard 90-day sunsets and escalating supermajorities for renewal, judicial appointments split across at least 3 independent bodies, and electoral administration insulated from the executive. Specification: every emergency measure auto-expires; every rights restriction must be re-justified publicly at renewal; citizens keep certified copies of identity and property records in personal custody.	68% (base range 50–70%)	Eternity clauses and sunset rules have preserved some democracies under pressure, but determined majorities with courts captured can still prevail; design raises the cost, it does not make it infinite.	Distribute appointment power across 3+ bodies; keep citizens' own copies of identity and property records; rehearse constitutional defence in civil society, not just courts.	Why 1 — institutions are captured: Appointment, money, and media levers concentrate in one network. -> Fix: Split every lever across independent bodies; publish ownership, contracts, and assets by default. | Why 2 — capture is gradual and legal: Each step is small enough to seem normal. -> Fix: Democracy audits on a fixed cycle with published capture indicators (court independence, media concentration, procurement openness). | Why 3 — citizens cannot verify: Records are secret or unusable, so accountability is theatrical. -> Fix: Open data as the default state of every public record; secrecy justified line by line, expiring automatically. | Why 4 — emergencies become permanent: Temporary powers are never handed back voluntarily. -> Fix: Hard sunsets (90 days) with escalating supermajorities; every renewal re-justified in public. | Why 5 — root: Constitutions are written for good-faith actors and stress-tested by bad-faith ones only in real time. -> Fix: Design against the worst plausible officeholder: eternity clauses, distributed appointment power, and rehearsed civic defence.
56	Social / Order	Governance & Law	Tier 1	Past	Society / National	Constitutional order collapses into dictatorship through legal steps, emergency decrees, and captured courts.	Weimar Germany's institutions were dismantled largely by legal instruments — an enabling act, packed courts, and emergency rule that never ended.	Design constitutions against their own abolition: eternity clauses protecting core rights from amendment, emergency powers with hard 90-day sunsets and escalating supermajorities for renewal, judicial appointments split across at least 3 independent bodies, and electoral administration insulated from the executive. Specification: every emergency measure auto-expires; every rights restriction must be re-justified publicly at renewal; citizens keep certified copies of identity and property records in personal custody.	60% (base range 50–70%)	Eternity clauses and sunset rules have preserved some democracies under pressure, but determined majorities with courts captured can still prevail; design raises the cost, it does not make it infinite.	Distribute appointment power across 3+ bodies; keep citizens' own copies of identity and property records; rehearse constitutional defence in civil society, not just courts.	Why 1 — institutions are captured: Appointment, money, and media levers concentrate in one network. -> Fix: Split every lever across independent bodies; publish ownership, contracts, and assets by default. | Why 2 — capture is gradual and legal: Each step is small enough to seem normal. -> Fix: Democracy audits on a fixed cycle with published capture indicators (court independence, media concentration, procurement openness). | Why 3 — citizens cannot verify: Records are secret or unusable, so accountability is theatrical. -> Fix: Open data as the default state of every public record; secrecy justified line by line, expiring automatically. | Why 4 — emergencies become permanent: Temporary powers are never handed back voluntarily. -> Fix: Hard sunsets (90 days) with escalating supermajorities; every renewal re-justified in public. | Why 5 — root: Constitutions are written for good-faith actors and stress-tested by bad-faith ones only in real time. -> Fix: Design against the worst plausible officeholder: eternity clauses, distributed appointment power, and rehearsed civic defence.
57	Social / Order	Governance & Law	Tier 1	Past	Global / Civilizational	Constitutional order collapses into dictatorship through legal steps, emergency decrees, and captured courts.	Weimar Germany's institutions were dismantled largely by legal instruments — an enabling act, packed courts, and emergency rule that never ended.	Design constitutions against their own abolition: eternity clauses protecting core rights from amendment, emergency powers with hard 90-day sunsets and escalating supermajorities for renewal, judicial appointments split across at least 3 independent bodies, and electoral administration insulated from the executive. Specification: every emergency measure auto-expires; every rights restriction must be re-justified publicly at renewal; citizens keep certified copies of identity and property records in personal custody.	48% (base range 50–70%)	Eternity clauses and sunset rules have preserved some democracies under pressure, but determined majorities with courts captured can still prevail; design raises the cost, it does not make it infinite.	Distribute appointment power across 3+ bodies; keep citizens' own copies of identity and property records; rehearse constitutional defence in civil society, not just courts.	Why 1 — institutions are captured: Appointment, money, and media levers concentrate in one network. -> Fix: Split every lever across independent bodies; publish ownership, contracts, and assets by default. | Why 2 — capture is gradual and legal: Each step is small enough to seem normal. -> Fix: Democracy audits on a fixed cycle with published capture indicators (court independence, media concentration, procurement openness). | Why 3 — citizens cannot verify: Records are secret or unusable, so accountability is theatrical. -> Fix: Open data as the default state of every public record; secrecy justified line by line, expiring automatically. | Why 4 — emergencies become permanent: Temporary powers are never handed back voluntarily. -> Fix: Hard sunsets (90 days) with escalating supermajorities; every renewal re-justified in public. | Why 5 — root: Constitutions are written for good-faith actors and stress-tested by bad-faith ones only in real time. -> Fix: Design against the worst plausible officeholder: eternity clauses, distributed appointment power, and rehearsed civic defence.
58	Social / Order	Governance & Law	Tier 1	Now	Household / Personal	Institutional corruption and state capture hollow out services while keeping the forms of democracy.	In captured states, procurement, courts, and media serve a network; citizens pay bribes for what law already entitles them to.	Radical transparency with automatic publication: all public contracts, beneficial ownership of companies bidding, officials' assets, and court statistics published in open data by default — secrecy must be justified line by line, publication is the default state. Independent audit institutions with budgets fixed by formula (e.g., 0.1% of public spending) so governments cannot starve them. Specification: procurement above a low threshold is e-tendered with full bid disclosure; whistleblowers receive statutory protection plus a share of recovered funds.	76% (base range 60–76%)	Open contracting and asset disclosure measurably reduce corruption where publication is automatic and audited; they fail where disclosure is voluntary.	Make publication the default state of every record; fix audit budgets by formula; protect whistleblowers with recovery shares, not just promises.	Why 1 — institutions are captured: Appointment, money, and media levers concentrate in one network. -> Fix: Split every lever across independent bodies; publish ownership, contracts, and assets by default. | Why 2 — capture is gradual and legal: Each step is small enough to seem normal. -> Fix: Democracy audits on a fixed cycle with published capture indicators (court independence, media concentration, procurement openness). | Why 3 — citizens cannot verify: Records are secret or unusable, so accountability is theatrical. -> Fix: Open data as the default state of every public record; secrecy justified line by line, expiring automatically. | Why 4 — emergencies become permanent: Temporary powers are never handed back voluntarily. -> Fix: Hard sunsets (90 days) with escalating supermajorities; every renewal re-justified in public. | Why 5 — root: Constitutions are written for good-faith actors and stress-tested by bad-faith ones only in real time. -> Fix: Design against the worst plausible officeholder: eternity clauses, distributed appointment power, and rehearsed civic defence.
59	Social / Order	Governance & Law	Tier 1	Now	Society / National	Institutional corruption and state capture hollow out services while keeping the forms of democracy.	In captured states, procurement, courts, and media serve a network; citizens pay bribes for what law already entitles them to.	Radical transparency with automatic publication: all public contracts, beneficial ownership of companies bidding, officials' assets, and court statistics published in open data by default — secrecy must be justified line by line, publication is the default state. Independent audit institutions with budgets fixed by formula (e.g., 0.1% of public spending) so governments cannot starve them. Specification: procurement above a low threshold is e-tendered with full bid disclosure; whistleblowers receive statutory protection plus a share of recovered funds.	68% (base range 60–76%)	Open contracting and asset disclosure measurably reduce corruption where publication is automatic and audited; they fail where disclosure is voluntary.	Make publication the default state of every record; fix audit budgets by formula; protect whistleblowers with recovery shares, not just promises.	Why 1 — institutions are captured: Appointment, money, and media levers concentrate in one network. -> Fix: Split every lever across independent bodies; publish ownership, contracts, and assets by default. | Why 2 — capture is gradual and legal: Each step is small enough to seem normal. -> Fix: Democracy audits on a fixed cycle with published capture indicators (court independence, media concentration, procurement openness). | Why 3 — citizens cannot verify: Records are secret or unusable, so accountability is theatrical. -> Fix: Open data as the default state of every public record; secrecy justified line by line, expiring automatically. | Why 4 — emergencies become permanent: Temporary powers are never handed back voluntarily. -> Fix: Hard sunsets (90 days) with escalating supermajorities; every renewal re-justified in public. | Why 5 — root: Constitutions are written for good-faith actors and stress-tested by bad-faith ones only in real time. -> Fix: Design against the worst plausible officeholder: eternity clauses, distributed appointment power, and rehearsed civic defence.

#	Dimension of Realit	Field of Discipline	Pareto Tier o	Timefram	Scale	Worst Possible Problem	Example	Solution — Full Specification	Probability of Su	Why This Probability	How to Maximize the Chances (detailed)	Root Causes
60	Social / Order	Governance & Law	Tier 1	Now	Global / Civilizational	Institutional corruption and state capture hollow out services while keeping the forms of democracy.	In captured states, procurement, courts, and media serve a network; citizens pay bribes for what law already entitles them to.	Radical transparency with automatic publication: all public contracts, beneficial ownership of companies bidding, officials' assets, and court statistics published in open data by default — secrecy must be justified line by line, publication is the default state. Independent audit institutions with budgets fixed by formula (e.g., 0.1% of public spending) so governments cannot starve them. Specification: procurement above a low threshold is e-tendered with full bid disclosure; whistleblowers receive statutory protection plus a share of recovered funds.	56% (base range 60–76%)	Open contracting and asset disclosure measurably reduce corruption where publication is automatic and audited; they fail where disclosure is voluntary.	Make publication the default state of every record; fix audit budgets by formula; protect whistleblowers with recovery shares, not just promises.	Why 1 — institutions are captured: Appointment, money, and media levers concentrate in one network. -> Fix: Split every lever across independent bodies; publish ownership, contracts, and assets by default. | Why 2 — capture is gradual and legal: Each step is small enough to seem normal. -> Fix: Democracy audits on a fixed cycle with published capture indicators (court independence, media concentration, procurement openness). | Why 3 — citizens cannot verify: Records are secret or unusable, so accountability is theatrical. -> Fix: Open data as the default state of every public record; secrecy justified line by line, expiring automatically. | Why 4 — emergencies become permanent: Temporary powers are never handed back voluntarily. -> Fix: Hard sunsets (90 days) with escalating supermajorities; every renewal re-justified in public. | Why 5 — root: Constitutions are written for good-faith actors and stress-tested by bad-faith ones only in real time. -> Fix: Design against the worst plausible officeholder: eternity clauses, distributed appointment power, and rehearsed civic defence.
61	Social / Order	Governance & Law	Tier 1	Future	Household / Personal	Legitimacy collapse: populations stop believing any institution, and coordination itself fails even without a dictator.	A future where elections, courts, money, and news are all distrusted simultaneously; every collective action — vaccination, evacuation, taxation — fails for lack of belief.	Rebuild trust as infrastructure: citizen assemblies chosen by lottery with real decision power over defined questions, radical localism (decisions at the smallest competent scale, where people can verify outcomes personally), and service-delivery proof — publish wait times, error rates, and fix rates for every public service monthly. Specification: trust is measured (surveyed quarterly, published); agencies falling below threshold trigger automatic independent review.	55% (base range 37–57%)	Trust is rebuilt only by verified performance over years; assemblies and localism help, but no design manufactures legitimacy quickly.	Start with services people can verify personally (wait times, fix rates); give lottery assemblies real, bounded decisions; publish the trust metrics and act on them visibly.	Why 1 — institutions are captured: Appointment, money, and media levers concentrate in one network. -> Fix: Split every lever across independent bodies; publish ownership, contracts, and assets by default. | Why 2 — capture is gradual and legal: Each step is small enough to seem normal. -> Fix: Democracy audits on a fixed cycle with published capture indicators (court independence, media concentration, procurement openness). | Why 3 — citizens cannot verify: Records are secret or unusable, so accountability is theatrical. -> Fix: Open data as the default state of every public record; secrecy justified line by line, expiring automatically. | Why 4 — emergencies become permanent: Temporary powers are never handed back voluntarily. -> Fix: Hard sunsets (90 days) with escalating supermajorities; every renewal re-justified in public. | Why 5 — root: Constitutions are written for good-faith actors and stress-tested by bad-faith ones only in real time. -> Fix: Design against the worst plausible officeholder: eternity clauses, distributed appointment power, and rehearsed civic defence.
62	Social / Order	Governance & Law	Tier 1	Future	Society / National	Legitimacy collapse: populations stop believing any institution, and coordination itself fails even without a dictator.	A future where elections, courts, money, and news are all distrusted simultaneously; every collective action — vaccination, evacuation, taxation — fails for lack of belief.	Rebuild trust as infrastructure: citizen assemblies chosen by lottery with real decision power over defined questions, radical localism (decisions at the smallest competent scale, where people can verify outcomes personally), and service-delivery proof — publish wait times, error rates, and fix rates for every public service monthly. Specification: trust is measured (surveyed quarterly, published); agencies falling below threshold trigger automatic independent review.	47% (base range 37–57%)	Trust is rebuilt only by verified performance over years; assemblies and localism help, but no design manufactures legitimacy quickly.	Start with services people can verify personally (wait times, fix rates); give lottery assemblies real, bounded decisions; publish the trust metrics and act on them visibly.	Why 1 — institutions are captured: Appointment, money, and media levers concentrate in one network. -> Fix: Split every lever across independent bodies; publish ownership, contracts, and assets by default. | Why 2 — capture is gradual and legal: Each step is small enough to seem normal. -> Fix: Democracy audits on a fixed cycle with published capture indicators (court independence, media concentration, procurement openness). | Why 3 — citizens cannot verify: Records are secret or unusable, so accountability is theatrical. -> Fix: Open data as the default state of every public record; secrecy justified line by line, expiring automatically. | Why 4 — emergencies become permanent: Temporary powers are never handed back voluntarily. -> Fix: Hard sunsets (90 days) with escalating supermajorities; every renewal re-justified in public. | Why 5 — root: Constitutions are written for good-faith actors and stress-tested by bad-faith ones only in real time. -> Fix: Design against the worst plausible officeholder: eternity clauses, distributed appointment power, and rehearsed civic defence.
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64	Social / Order	Economy & Finance	Tier 1	Past	Household / Personal	Banking panic and deflation spiral convert a market crash into a decade of mass unemployment.	After 1929, bank runs destroyed savings, credit froze, and unemployment reached 25% in the United States and higher elsewhere.	Circuit breakers built in advance: deposit insurance with pre-funded reserves, central-bank lender-of-last-resort facilities designed before crises, automatic fiscal stabilizers (unemployment benefits that expand by formula, not by vote), and bank resolution law that wipes out shareholders and management — never depositors — first. Specification: systemically important banks hold capital of at least 12% of risk-weighted assets with a leverage cap; households keep 3–6 months of expenses in insured deposits, never in a single institution above the insured limit.	91% (base range 76–89%)	Deposit insurance ended bank-run dynamics wherever credibly pre-funded; automatic stabilizers are proven counter-cyclical machinery.	Pre-fund the insurance; write resolution law in peacetime; keep household reserves inside insured limits across institutions.	Why 1 — crises cascade through debt: Leverage links every balance sheet to every other. -> Fix: Capital and leverage caps that bind in booms, when they are politically hardest and economically cheapest. | Why 2 — booms silence regulators: Warnings during bubbles are punished as pessimism. -> Fix: Automatic stabilizers and counter-cyclical rules set by formula in law, not by discretion in the moment. | Why 3 — gains privatize, losses socialize: Institutions too big to fail keep the upside. -> Fix: Resolution law that wipes shareholders and management first, rehearsed with living wills tested annually. | Why 4 — households carry the fragility: Wages stagnate while housing and credit costs compound. -> Fix: Debt-service caps at origination under stressed rates; fresh-start insolvency within 3 years; public supply of non-speculative housing. | Why 5 — root: Finance is treated as an end in itself rather than as plumbing for the real economy. -> Fix: Charter test: financial activity that does not demonstrably serve production, housing, or risk-sharing carries higher capital and tax weight.
65	Social / Order	Economy & Finance	Tier 1	Past	Society / National	Banking panic and deflation spiral convert a market crash into a decade of mass unemployment.	After 1929, bank runs destroyed savings, credit froze, and unemployment reached 25% in the United States and higher elsewhere.	Circuit breakers built in advance: deposit insurance with pre-funded reserves, central-bank lender-of-last-resort facilities designed before crises, automatic fiscal stabilizers (unemployment benefits that expand by formula, not by vote), and bank resolution law that wipes out shareholders and management — never depositors — first. Specification: systemically important banks hold capital of at least 12% of risk-weighted assets with a leverage cap; households keep 3–6 months of expenses in insured deposits, never in a single institution above the insured limit.	83% (base range 76–89%)	Deposit insurance ended bank-run dynamics wherever credibly pre-funded; automatic stabilizers are proven counter-cyclical machinery.	Pre-fund the insurance; write resolution law in peacetime; keep household reserves inside insured limits across institutions.	Why 1 — crises cascade through debt: Leverage links every balance sheet to every other. -> Fix: Capital and leverage caps that bind in booms, when they are politically hardest and economically cheapest. | Why 2 — booms silence regulators: Warnings during bubbles are punished as pessimism. -> Fix: Automatic stabilizers and counter-cyclical rules set by formula in law, not by discretion in the moment. | Why 3 — gains privatize, losses socialize: Institutions too big to fail keep the upside. -> Fix: Resolution law that wipes shareholders and management first, rehearsed with living wills tested annually. | Why 4 — households carry the fragility: Wages stagnate while housing and credit costs compound. -> Fix: Debt-service caps at origination under stressed rates; fresh-start insolvency within 3 years; public supply of non-speculative housing. | Why 5 — root: Finance is treated as an end in itself rather than as plumbing for the real economy. -> Fix: Charter test: financial activity that does not demonstrably serve production, housing, or risk-sharing carries higher capital and tax weight.
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67	Social / Order	Economy & Finance	Tier 1	Now	Household / Personal	Asset bubbles, household debt, and cost-of-living squeeze hollow out the middle; one rate shock cascades.	Households carrying record debt-to-income ratios meet higher rates; defaults cluster; whole suburbs' collateral values fall together.	De-leverage by design: mortgage rules that cap total debt service at 35–40% of verified income under a stressed rate (+2%), automatic refinancing windows when rates fall, public development of non-speculative housing supply, and insolvency law that gives households a fresh start within 3 years instead of permanent debt peonage. Specification: personal rule — total debt payments under 30% of take-home; one income loss must be survivable for 6 months on reserves plus benefits; never hold employer stock plus job plus mortgage in one company town exposure.	71% (base range 55–71%)	Debt-service caps at origination demonstrably reduce default cascades; the obstacle is the credit boom's political popularity.	Apply the stressed-rate test (+2%) to every loan; open refinancing windows automatically when rates fall; enforce the 30% personal debt-payment ceiling.	Why 1 — crises cascade through debt: Leverage links every balance sheet to every other. -> Fix: Capital and leverage caps that bind in booms, when they are politically hardest and economically cheapest. | Why 2 — booms silence regulators: Warnings during bubbles are punished as pessimism. -> Fix: Automatic stabilizers and counter-cyclical rules set by formula in law, not by discretion in the moment. | Why 3 — gains privatize, losses socialize: Institutions too big to fail keep the upside. -> Fix: Resolution law that wipes shareholders and management first, rehearsed with living wills tested annually. | Why 4 — households carry the fragility: Wages stagnate while housing and credit costs compound. -> Fix: Debt-service caps at origination under stressed rates; fresh-start insolvency within 3 years; public supply of non-speculative housing. | Why 5 — root: Finance is treated as an end in itself rather than as plumbing for the real economy. -> Fix: Charter test: financial activity that does not demonstrably serve production, housing, or risk-sharing carries higher capital and tax weight.
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70	Social / Order	Economy & Finance	Tier 1	Future	Household / Personal	Automation displaces labour faster than new livelihoods form, while wealth concentrates in owners of machines and models.	Whole occupations — driving, clerical work, routine analysis — shrink within a decade; wages decouple from productivity entirely; politics radicalizes.	Share the machine dividend by structure, not charity: sovereign and pension funds holding broad equity stakes so productivity gains flow to all citizens as dividends; reduced standard working hours as productivity rises (the historical pattern, deliberately continued); portable benefits attached to the person, not the job; and public investment in the irreducibly human sectors (care, craft, education, restoration) at the scale of the displacement. Specification: transition planning uses the fastest plausible displacement curve, with retraining accounts funded per worker before displacement, not after.	52% (base range 34–54%)	Ownership-sharing mechanisms are economically sound but politically unbuilt; success depends on constructing them during growth, not after displacement peaks.	Capitalize citizen funds from automation rents early; attach benefits to persons in law now; cut standard hours in step with measured productivity, sector by sector.	Why 1 — crises cascade through debt: Leverage links every balance sheet to every other. -> Fix: Capital and leverage caps that bind in booms, when they are politically hardest and economically cheapest.	Why 2 — booms silence regulators: Warnings during bubbles are punished as pessimism. -> Fix: Automatic stabilizers and counter-cyclical rules set by formula in law, not by discretion in the moment.	Why 3 — gains privatize, losses socialize: Institutions too big to fail keep the upside. -> Fix: Resolution law that wipes shareholders and management first, rehearsed with living wills tested annually.	Why 4 — households carry the fragility: Wages stagnate while housing and credit costs compound. -> Fix: Debt-service caps at origination under stressed rates; fresh-start insolvency within 3 years; public supply of non-speculative housing.	Why 5 — root: Finance is treated as an end in itself rather than as plumbing for the real economy. -> Fix: Charter test: financial activity that does not demonstrably serve production, housing, or risk-sharing carries higher capital and tax weight.
71	Social / Order	Economy & Finance	Tier 1	Future	Society / National	Automation displaces labour faster than new livelihoods form, while wealth concentrates in owners of machines and models.	Whole occupations — driving, clerical work, routine analysis — shrink within a decade; wages decouple from productivity entirely; politics radicalizes.	Share the machine dividend by structure, not charity: sovereign and pension funds holding broad equity stakes so productivity gains flow to all citizens as dividends; reduced standard working hours as productivity rises (the historical pattern, deliberately continued); portable benefits attached to the person, not the job; and public investment in the irreducibly human sectors (care, craft, education, restoration) at the scale of the displacement. Specification: transition planning uses the fastest plausible displacement curve, with retraining accounts funded per worker before displacement, not after.	44% (base range 34–54%)	Ownership-sharing mechanisms are economically sound but politically unbuilt; success depends on constructing them during growth, not after displacement peaks.	Capitalize citizen funds from automation rents early; attach benefits to persons in law now; cut standard hours in step with measured productivity, sector by sector.	Why 1 — crises cascade through debt: Leverage links every balance sheet to every other. -> Fix: Capital and leverage caps that bind in booms, when they are politically hardest and economically cheapest.	Why 2 — booms silence regulators: Warnings during bubbles are punished as pessimism. -> Fix: Automatic stabilizers and counter-cyclical rules set by formula in law, not by discretion in the moment.	Why 3 — gains privatize, losses socialize: Institutions too big to fail keep the upside. -> Fix: Resolution law that wipes shareholders and management first, rehearsed with living wills tested annually.	Why 4 — households carry the fragility: Wages stagnate while housing and credit costs compound. -> Fix: Debt-service caps at origination under stressed rates; fresh-start insolvency within 3 years; public supply of non-speculative housing.	Why 5 — root: Finance is treated as an end in itself rather than as plumbing for the real economy. -> Fix: Charter test: financial activity that does not demonstrably serve production, housing, or risk-sharing carries higher capital and tax weight.
72	Social / Order	Economy & Finance	Tier 1	Future	Global / Civilizational	Automation displaces labour faster than new livelihoods form, while wealth concentrates in owners of machines and models.	Whole occupations — driving, clerical work, routine analysis — shrink within a decade; wages decouple from productivity entirely; politics radicalizes.	Share the machine dividend by structure, not charity: sovereign and pension funds holding broad equity stakes so productivity gains flow to all citizens as dividends; reduced standard working hours as productivity rises (the historical pattern, deliberately continued); portable benefits attached to the person, not the job; and public investment in the irreducibly human sectors (care, craft, education, restoration) at the scale of the displacement. Specification: transition planning uses the fastest plausible displacement curve, with retraining accounts funded per worker before displacement, not after.	32% (base range 34–54%)	Ownership-sharing mechanisms are economically sound but politically unbuilt; success depends on constructing them during growth, not after displacement peaks.	Capitalize citizen funds from automation rents early; attach benefits to persons in law now; cut standard hours in step with measured productivity, sector by sector.	Why 1 — crises cascade through debt: Leverage links every balance sheet to every other. -> Fix: Capital and leverage caps that bind in booms, when they are politically hardest and economically cheapest.	Why 2 — booms silence regulators: Warnings during bubbles are punished as pessimism. -> Fix: Automatic stabilizers and counter-cyclical rules set by formula in law, not by discretion in the moment.	Why 3 — gains privatize, losses socialize: Institutions too big to fail keep the upside. -> Fix: Resolution law that wipes shareholders and management first, rehearsed with living wills tested annually.	Why 4 — households carry the fragility: Wages stagnate while housing and credit costs compound. -> Fix: Debt-service caps at origination under stressed rates; fresh-start insolvency within 3 years; public supply of non-speculative housing.	Why 5 — root: Finance is treated as an end in itself rather than as plumbing for the real economy. -> Fix: Charter test: financial activity that does not demonstrably serve production, housing, or risk-sharing carries higher capital and tax weight.
73	Technological	Technology & AI	Tier 1	Past	Household / Personal	A general-purpose technology spreads faster than safety norms, and its harms compound for decades before correction.	Leaded gasoline and asbestos were known hazards for decades while use expanded; lead exposure lowered population IQ and asbestos deaths continue a century later.	Precaution with deadlines: new general-purpose technologies pass staged safety gates (lab, limited, general) with published criteria and automatic review dates; manufacturers carry strict liability for latent harms, funded by mandatory insurance during the entire exposure tail; and a standing technology-assessment office publishes plain-language risk registers. Specification: no product with irreversible population-level exposure ships without a funded 30-year monitoring program; households treat 'new and unstudied' as a reason to delay adoption for non-essentials.	74% (base range 58–74%)	Staged gates and strict liability eventually controlled lead and asbestos, but only after long harm tails; earlier gates would have raised success sharply.	Attach the 30-year monitoring fund to the product, not the taxpayer; give the assessment office subpoena power over safety data; default to delay for irreversible exposures.	Why 1 — harms arrive before rules: Development cycles are months; legislation is years. -> Fix: Staged gates with automatic review dates, so oversight moves at the technology's speed by default.	Why 2 — builders don't pay for harm: Latent and externalized costs fall on the public. -> Fix: Strict liability plus mandatory harm insurance across the exposure tail, priced by insurers who will demand safety data.	Why 3 — basics are skipped: Backups, segmentation, and MFA are known but unglamorous. -> Fix: Make the basics conditions of insurance and procurement: no contract, no coverage without restore-tested backups and MFA.	Why 4 — dependence concentrates: One platform, one model, one vendor becomes a single point of civilizational failure. -> Fix: Interoperability, portability, and manual fallbacks as certification requirements for essential services.	Why 5 — root: Speed is rewarded and safety is unpriced, so the race always outruns the guardrails. -> Fix: Price safety into the race: liability, insurance, and gate criteria make the safe path the fast path.
74	Technological	Technology & AI	Tier 1	Past	Society / National	A general-purpose technology spreads faster than safety norms, and its harms compound for decades before correction.	Leaded gasoline and asbestos were known hazards for decades while use expanded; lead exposure lowered population IQ and asbestos deaths continue a century later.	Precaution with deadlines: new general-purpose technologies pass staged safety gates (lab, limited, general) with published criteria and automatic review dates; manufacturers carry strict liability for latent harms, funded by mandatory insurance during the entire exposure tail; and a standing technology-assessment office publishes plain-language risk registers. Specification: no product with irreversible population-level exposure ships without a funded 30-year monitoring program; households treat 'new and unstudied' as a reason to delay adoption for non-essentials.	66% (base range 58–74%)	Staged gates and strict liability eventually controlled lead and asbestos, but only after long harm tails; earlier gates would have raised success sharply.	Attach the 30-year monitoring fund to the product, not the taxpayer; give the assessment office subpoena power over safety data; default to delay for irreversible exposures.	Why 1 — harms arrive before rules: Development cycles are months; legislation is years. -> Fix: Staged gates with automatic review dates, so oversight moves at the technology's speed by default.	Why 2 — builders don't pay for harm: Latent and externalized costs fall on the public. -> Fix: Strict liability plus mandatory harm insurance across the exposure tail, priced by insurers who will demand safety data.	Why 3 — basics are skipped: Backups, segmentation, and MFA are known but unglamorous. -> Fix: Make the basics conditions of insurance and procurement: no contract, no coverage without restore-tested backups and MFA.	Why 4 — dependence concentrates: One platform, one model, one vendor becomes a single point of civilizational failure. -> Fix: Interoperability, portability, and manual fallbacks as certification requirements for essential services.	Why 5 — root: Speed is rewarded and safety is unpriced, so the race always outruns the guardrails. -> Fix: Price safety into the race: liability, insurance, and gate criteria make the safe path the fast path.
75	Technological	Technology & AI	Tier 1	Past	Global / Civilizational	A general-purpose technology spreads faster than safety norms, and its harms compound for decades before correction.	Leaded gasoline and asbestos were known hazards for decades while use expanded; lead exposure lowered population IQ and asbestos deaths continue a century later.	Precaution with deadlines: new general-purpose technologies pass staged safety gates (lab, limited, general) with published criteria and automatic review dates; manufacturers carry strict liability for latent harms, funded by mandatory insurance during the entire exposure tail; and a standing technology-assessment office publishes plain-language risk registers. Specification: no product with irreversible population-level exposure ships without a funded 30-year monitoring program; households treat 'new and unstudied' as a reason to delay adoption for non-essentials.	54% (base range 58–74%)	Staged gates and strict liability eventually controlled lead and asbestos, but only after long harm tails; earlier gates would have raised success sharply.	Attach the 30-year monitoring fund to the product, not the taxpayer; give the assessment office subpoena power over safety data; default to delay for irreversible exposures.	Why 1 — harms arrive before rules: Development cycles are months; legislation is years. -> Fix: Staged gates with automatic review dates, so oversight moves at the technology's speed by default.	Why 2 — builders don't pay for harm: Latent and externalized costs fall on the public. -> Fix: Strict liability plus mandatory harm insurance across the exposure tail, priced by insurers who will demand safety data.	Why 3 — basics are skipped: Backups, segmentation, and MFA are known but unglamorous. -> Fix: Make the basics conditions of insurance and procurement: no contract, no coverage without restore-tested backups and MFA.	Why 4 — dependence concentrates: One platform, one model, one vendor becomes a single point of civilizational failure. -> Fix: Interoperability, portability, and manual fallbacks as certification requirements for essential services.	Why 5 — root: Speed is rewarded and safety is unpriced, so the race always outruns the guardrails. -> Fix: Price safety into the race: liability, insurance, and gate criteria make the safe path the fast path.
76	Technological	Technology & AI	Tier 1	Now	Household / Personal	Cyber intrusion and ransomware shut down hospitals, pipelines, and city governments; the digital base is fragile.	A single ransomware incident closed a major fuel pipeline and triggered panic buying across a seaboard; hospitals routinely divert ambulances during attacks.	Assume breach, engineer recovery: immutable offline backups tested by full restoration quarterly (a backup never restored is a rumour), network segmentation so one infection cannot cross from office systems to operations, multi-factor authentication on 100% of remote and privileged access, and 24-hour incident reporting with a national assistance duty. Specification for households and small organizations: automatic updates on, unique passwords in a manager, 3-2-1 backups (3 copies, 2 media, 1 off-site/offline) for irreplaceable data — photos, records, work.	86% (base range 70–85%)	Recovery engineering (immutable backups, segmentation, MFA) defeats the large majority of real-world intrusions; breaches persist where basics are skipped.	Restore-test backups quarterly — an untested backup is a rumour; segment operations from office networks; 100% MFA on remote and privileged access, no exceptions.	Why 1 — harms arrive before rules: Development cycles are months; legislation is years. -> Fix: Staged gates with automatic review dates, so oversight moves at the technology's speed by default.	Why 2 — builders don't pay for harm: Latent and externalized costs fall on the public. -> Fix: Strict liability plus mandatory harm insurance across the exposure tail, priced by insurers who will demand safety data.	Why 3 — basics are skipped: Backups, segmentation, and MFA are known but unglamorous. -> Fix: Make the basics conditions of insurance and procurement: no contract, no coverage without restore-tested backups and MFA.	Why 4 — dependence concentrates: One platform, one model, one vendor becomes a single point of civilizational failure. -> Fix: Interoperability, portability, and manual fallbacks as certification requirements for essential services.	Why 5 — root: Speed is rewarded and safety is unpriced, so the race always outruns the guardrails. -> Fix: Price safety into the race: liability, insurance, and gate criteria make the safe path the fast path.
77	Technological	Technology & AI	Tier 1	Now	Society / National	Cyber intrusion and ransomware shut down hospitals, pipelines, and city governments; the digital base is fragile.	A single ransomware incident closed a major fuel pipeline and triggered panic buying across a seaboard; hospitals routinely divert ambulances during attacks.	Assume breach, engineer recovery: immutable offline backups tested by full restoration quarterly (a backup never restored is a rumour), network segmentation so one infection cannot cross from office systems to operations, multi-factor authentication on 100% of remote and privileged access, and 24-hour incident reporting with a national assistance duty. Specification for households and small organizations: automatic updates on, unique passwords in a manager, 3-2-1 backups (3 copies, 2 media, 1 off-site/offline) for irreplaceable data — photos, records, work.	78% (base range 70–85%)	Recovery engineering (immutable backups, segmentation, MFA) defeats the large majority of real-world intrusions; breaches persist where basics are skipped.	Restore-test backups quarterly — an untested backup is a rumour; segment operations from office networks; 100% MFA on remote and privileged access, no exceptions.	Why 1 — harms arrive before rules: Development cycles are months; legislation is years. -> Fix: Staged gates with automatic review dates, so oversight moves at the technology's speed by default.	Why 2 — builders don't pay for harm: Latent and externalized costs fall on the public. -> Fix: Strict liability plus mandatory harm insurance across the exposure tail, priced by insurers who will demand safety data.	Why 3 — basics are skipped: Backups, segmentation, and MFA are known but unglamorous. -> Fix: Make the basics conditions of insurance and procurement: no contract, no coverage without restore-tested backups and MFA.	Why 4 — dependence concentrates: One platform, one model, one vendor becomes a single point of civilizational failure. -> Fix: Interoperability, portability, and manual fallbacks as certification requirements for essential services.	Why 5 — root: Speed is rewarded and safety is unpriced, so the race always outruns the guardrails. -> Fix: Price safety into the race: liability, insurance, and gate criteria make the safe path the fast path.
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79	Technological	Technology & AI	Tier 1	Future	Household / Personal	A misaligned or misused advanced AI, or total dependence on opaque systems, produces civilizational-scale failure or harm.	Frontier systems embedded in finance, logistics, medicine, and weapons make correlated errors at machine speed, or are deliberately turned to bioweapon design and mass manipulation.	Governance proportional to capability: frontier training runs above defined compute thresholds are registered and evaluated for dangerous capabilities before release; critical infrastructure keeps human-verified analogue fallbacks and鈥; fully autonomous control of weapons, grid switching, and medical dosing; liability law makes deployers answer for system outputs; and international verification for the largest training facilities, modelled on nuclear safeguards. Specification: every essential service must demonstrate a manual mode, drilled annually; personal rule — no single AI or platform holds your only copy of anything important.	50% (base range 32–52%)	Capability governance is unproven at frontier scale and races commercial pressure; evaluations and fallbacks reduce risk materially without eliminating it.	Register and evaluate above compute thresholds before release; drill manual modes of essential services annually; keep deployer liability absolute so caution is priced in.	Why 1 — harms arrive before rules: Development cycles are months; legislation is years. -> Fix: Staged gates with automatic review dates, so oversight moves at the technology's speed by default.	Why 2 — builders don't pay for harm: Latent and externalized costs fall on the public. -> Fix: Strict liability plus mandatory harm insurance across the exposure tail, priced by insurers who will demand safety data.	Why 3 — basics are skipped: Backups, segmentation, and MFA are known but unglamorous. -> Fix: Make the basics conditions of insurance and procurement: no contract, no coverage without restore-tested backups and MFA.	Why 4 — dependence concentrates: One platform, one model, one vendor becomes a single point of civilizational failure. -> Fix: Interoperability, portability, and manual fallbacks as certification requirements for essential services.	Why 5 — root: Speed is rewarded and safety is unpriced, so the race always outruns the guardrails. -> Fix: Price safety into the race: liability, insurance, and gate criteria make the safe path the fast path.

#	Dimension of Realit	Field of Discipline	Pareto Tier o	Timefram	Scale	Worst Possible Problem	Example	Solution — Full Specification	Probability of Su	Why This Probability	How to Maximize the Chances (detailed)	Root Causes
80	Technological	Technology & AI	Tier 1	Future	Society / National	A misaligned or misused advanced AI, or total dependence on opaque systems, produces civilizational-scale failure or harm.	Frontier systems embedded in finance, logistics, medicine, and weapons make correlated errors at machine speed, or are deliberately turned to bioweapon design and mass manipulation.	Governance proportional to capability: frontier training runs above defined compute thresholds are registered and evaluated for dangerous capabilities before release; critical infrastructure keeps human-verified analogue fallbacks and禁止 fully autonomous control of weapons, grid switching, and medical dosing; liability law makes deployers answer for system outputs; and international verification for the largest training facilities, modelled on nuclear safeguards. Specification: every essential service must demonstrate a manual mode, drilled annually; personal rule — no single AI or platform holds your only copy of anything important.	42% (base range 32–52%)	Capability governance is unproven at frontier scale and races commercial pressure; evaluations and fallbacks reduce risk materially without eliminating it.	Register and evaluate above compute thresholds before release; drill manual modes of essential services annually; keep deployer liability absolute so caution is priced in.	Why 1 — harms arrive before rules: Development cycles are months; legislation is years. -> Fix: Staged gates with automatic review dates, so oversight moves at the technology's speed by default. | Why 2 — builders don't pay for harm: Latent and externalized costs fall on the public. -> Fix: Strict liability plus mandatory harm insurance across the exposure tail, priced by insurers who will demand safety data. | Why 3 — basics are skipped: Backups, segmentation, and MFA are known but unglamorous. -> Fix: Make the basics conditions of insurance and procurement: no contract, no coverage without restore-tested backups and MFA. | Why 4 — dependence concentrates: One platform, one model, one vendor becomes a single point of civilizational failure. -> Fix: Interoperability, portability, and manual fallbacks as certification requirements for essential services. | Why 5 — root: Speed is rewarded and safety is unpriced, so the race always outruns the guardrails. -> Fix: Price safety into the race: liability, insurance, and gate criteria make the safe path the fast path.
81	Technological	Technology & AI	Tier 1	Future	Global / Civilizational	A misaligned or misused advanced AI, or total dependence on opaque systems, produces civilizational-scale failure or harm.	Frontier systems embedded in finance, logistics, medicine, and weapons make correlated errors at machine speed, or are deliberately turned to bioweapon design and mass manipulation.	Governance proportional to capability: frontier training runs above defined compute thresholds are registered and evaluated for dangerous capabilities before release; critical infrastructure keeps human-verified analogue fallbacks and禁止 fully autonomous control of weapons, grid switching, and medical dosing; liability law makes deployers answer for system outputs; and international verification for the largest training facilities, modelled on nuclear safeguards. Specification: every essential service must demonstrate a manual mode, drilled annually; personal rule — no single AI or platform holds your only copy of anything important.	30% (base range 32–52%)	Capability governance is unproven at frontier scale and races commercial pressure; evaluations and fallbacks reduce risk materially without eliminating it.	Register and evaluate above compute thresholds before release; drill manual modes of essential services annually; keep deployer liability absolute so caution is priced in.	Why 1 — harms arrive before rules: Development cycles are months; legislation is years. -> Fix: Staged gates with automatic review dates, so oversight moves at the technology's speed by default. | Why 2 — builders don't pay for harm: Latent and externalized costs fall on the public. -> Fix: Strict liability plus mandatory harm insurance across the exposure tail, priced by insurers who will demand safety data. | Why 3 — basics are skipped: Backups, segmentation, and MFA are known but unglamorous. -> Fix: Make the basics conditions of insurance and procurement: no contract, no coverage without restore-tested backups and MFA. | Why 4 — dependence concentrates: One platform, one model, one vendor becomes a single point of civilizational failure. -> Fix: Interoperability, portability, and manual fallbacks as certification requirements for essential services. | Why 5 — root: Speed is rewarded and safety is unpriced, so the race always outruns the guardrails. -> Fix: Price safety into the race: liability, insurance, and gate criteria make the safe path the fast path.
82	Physical / Material	Environment & Climate	Tier 1	Past	Household / Personal	Civilizations degrade their own ecological base — soil, forests, water — until the base fails and the society follows.	Classic cases from Mesopotamian salinization to the Dust Bowl: farming and extraction practices consumed the resource they depended on.	Close the loop with law: extraction and pollution permits capped by measured regeneration rates of the specific ecosystem (not national averages), soil-loss limits per hectare enforced like building codes, and full-cost accounting where the price of a product includes its restoration cost. Specification: farmland soil organic matter and erosion measured on a 5-year cycle; land degrading past threshold enters mandatory restoration with cost-sharing.	79% (base range 63–79%)	Regeneration-capped extraction works where measured per-ecosystem; failures trace to caps set by politics instead of measurement.	Measure the specific ecosystem's regeneration, not national averages; put soil and erosion limits in enforceable code; price restoration into products.	Why 1 — emissions continue: The cost of carbon falls on everyone except the emitter. -> Fix: Price emissions at the measured damage cost with the revenue returned visibly (dividends or bill credits) so the policy survives politics. | Why 2 — adaptation lags: Infrastructure is built to historical weather by code and habit. -> Fix: Design codes keyed to the 2050 worst decile, updated on a fixed 5-year cycle from published projections. | Why 3 — risk is hidden: Buyers and builders cannot see address-level future risk. -> Fix: Mandatory published risk maps (flood, heat, fire, sea level) attached to every property transaction and building permit. | Why 4 — feedback is slow: Damage accumulates silently for decades before it is undeniable. -> Fix: Instrument the thresholds: ice, ocean circulation, and permafrost monitored with published tripwires tied to pre-agreed actions. | Why 5 — root: Nature is accounted as an externality — a free input and a free dump. -> Fix: Natural capital on the national balance sheet: soil, water, canopy, and climate stability carried as assets whose depreciation must be reported and offset.
83	Physical / Material	Environment & Climate	Tier 1	Past	Society / National	Civilizations degrade their own ecological base — soil, forests, water — until the base fails and the society follows.	Classic cases from Mesopotamian salinization to the Dust Bowl: farming and extraction practices consumed the resource they depended on.	Close the loop with law: extraction and pollution permits capped by measured regeneration rates of the specific ecosystem (not national averages), soil-loss limits per hectare enforced like building codes, and full-cost accounting where the price of a product includes its restoration cost. Specification: farmland soil organic matter and erosion measured on a 5-year cycle; land degrading past threshold enters mandatory restoration with cost-sharing.	71% (base range 63–79%)	Regeneration-capped extraction works where measured per-ecosystem; failures trace to caps set by politics instead of measurement.	Measure the specific ecosystem's regeneration, not national averages; put soil and erosion limits in enforceable code; price restoration into products.	Why 1 — emissions continue: The cost of carbon falls on everyone except the emitter. -> Fix: Price emissions at the measured damage cost with the revenue returned visibly (dividends or bill credits) so the policy survives politics. | Why 2 — adaptation lags: Infrastructure is built to historical weather by code and habit. -> Fix: Design codes keyed to the 2050 worst decile, updated on a fixed 5-year cycle from published projections. | Why 3 — risk is hidden: Buyers and builders cannot see address-level future risk. -> Fix: Mandatory published risk maps (flood, heat, fire, sea level) attached to every property transaction and building permit. | Why 4 — feedback is slow: Damage accumulates silently for decades before it is undeniable. -> Fix: Instrument the thresholds: ice, ocean circulation, and permafrost monitored with published tripwires tied to pre-agreed actions. | Why 5 — root: Nature is accounted as an externality — a free input and a free dump. -> Fix: Natural capital on the national balance sheet: soil, water, canopy, and climate stability carried as assets whose depreciation must be reported and offset.
84	Physical / Material	Environment & Climate	Tier 1	Past	Global / Civilizational	Civilizations degrade their own ecological base — soil, forests, water — until the base fails and the society follows.	Classic cases from Mesopotamian salinization to the Dust Bowl: farming and extraction practices consumed the resource they depended on.	Close the loop with law: extraction and pollution permits capped by measured regeneration rates of the specific ecosystem (not national averages), soil-loss limits per hectare enforced like building codes, and full-cost accounting where the price of a product includes its restoration cost. Specification: farmland soil organic matter and erosion measured on a 5-year cycle; land degrading past threshold enters mandatory restoration with cost-sharing.	59% (base range 63–79%)	Regeneration-capped extraction works where measured per-ecosystem; failures trace to caps set by politics instead of measurement.	Measure the specific ecosystem's regeneration, not national averages; put soil and erosion limits in enforceable code; price restoration into products.	Why 1 — emissions continue: The cost of carbon falls on everyone except the emitter. -> Fix: Price emissions at the measured damage cost with the revenue returned visibly (dividends or bill credits) so the policy survives politics. | Why 2 — adaptation lags: Infrastructure is built to historical weather by code and habit. -> Fix: Design codes keyed to the 2050 worst decile, updated on a fixed 5-year cycle from published projections. | Why 3 — risk is hidden: Buyers and builders cannot see address-level future risk. -> Fix: Mandatory published risk maps (flood, heat, fire, sea level) attached to every property transaction and building permit. | Why 4 — feedback is slow: Damage accumulates silently for decades before it is undeniable. -> Fix: Instrument the thresholds: ice, ocean circulation, and permafrost monitored with published tripwires tied to pre-agreed actions. | Why 5 — root: Nature is accounted as an externality — a free input and a free dump. -> Fix: Natural capital on the national balance sheet: soil, water, canopy, and climate stability carried as assets whose depreciation must be reported and offset.
85	Physical / Material	Environment & Climate	Tier 1	Now	Household / Personal	Extreme heat, flood, and fire exceed the design basis of cities built for a milder century.	Heat domes killing hundreds in a weekend in northern cities; floods overwhelming drains sized for historical rainfall; wildfire smoke closing schools across a continent.	Adapt to the climate that exists: building codes and infrastructure designed to the 2050 worst-decile weather, urban canopy targets of 30–40% to cut street-level heat, floodplains returned to water with buyouts instead of repeated rebuilds, and heat action plans with named officials, cooling centres within a 15-minute walk, and check-in systems for isolated elderly. Specification: every home in a risk zone has a written plan — cooling or evacuation, 72-hour supplies, medication continuity; insurers and cities publish address-level risk so decisions are informed.	77% (base range 61–77%)	Adaptation measures (canopy, codes, heat plans) have strong measured effects on deaths and damage per event.	Design to the 2050 worst decile, not historical weather; canopy and cooling access targets with named owners; address-level risk published so households can act.	Why 1 — emissions continue: The cost of carbon falls on everyone except the emitter. -> Fix: Price emissions at the measured damage cost with the revenue returned visibly (dividends or bill credits) so the policy survives politics. | Why 2 — adaptation lags: Infrastructure is built to historical weather by code and habit. -> Fix: Design codes keyed to the 2050 worst decile, updated on a fixed 5-year cycle from published projections. | Why 3 — risk is hidden: Buyers and builders cannot see address-level future risk. -> Fix: Mandatory published risk maps (flood, heat, fire, sea level) attached to every property transaction and building permit. | Why 4 — feedback is slow: Damage accumulates silently for decades before it is undeniable. -> Fix: Instrument the thresholds: ice, ocean circulation, and permafrost monitored with published tripwires tied to pre-agreed actions. | Why 5 — root: Nature is accounted as an externality — a free input and a free dump. -> Fix: Natural capital on the national balance sheet: soil, water, canopy, and climate stability carried as assets whose depreciation must be reported and offset.
86	Physical / Material	Environment & Climate	Tier 1	Now	Society / National	Extreme heat, flood, and fire exceed the design basis of cities built for a milder century.	Heat domes killing hundreds in a weekend in northern cities; floods overwhelming drains sized for historical rainfall; wildfire smoke closing schools across a continent.	Adapt to the climate that exists: building codes and infrastructure designed to the 2050 worst-decile weather, urban canopy targets of 30–40% to cut street-level heat, floodplains returned to water with buyouts instead of repeated rebuilds, and heat action plans with named officials, cooling centres within a 15-minute walk, and check-in systems for isolated elderly. Specification: every home in a risk zone has a written plan — cooling or evacuation, 72-hour supplies, medication continuity; insurers and cities publish address-level risk so decisions are informed.	69% (base range 61–77%)	Adaptation measures (canopy, codes, heat plans) have strong measured effects on deaths and damage per event.	Design to the 2050 worst decile, not historical weather; canopy and cooling access targets with named owners; address-level risk published so households can act.	Why 1 — emissions continue: The cost of carbon falls on everyone except the emitter. -> Fix: Price emissions at the measured damage cost with the revenue returned visibly (dividends or bill credits) so the policy survives politics. | Why 2 — adaptation lags: Infrastructure is built to historical weather by code and habit. -> Fix: Design codes keyed to the 2050 worst decile, updated on a fixed 5-year cycle from published projections. | Why 3 — risk is hidden: Buyers and builders cannot see address-level future risk. -> Fix: Mandatory published risk maps (flood, heat, fire, sea level) attached to every property transaction and building permit. | Why 4 — feedback is slow: Damage accumulates silently for decades before it is undeniable. -> Fix: Instrument the thresholds: ice, ocean circulation, and permafrost monitored with published tripwires tied to pre-agreed actions. | Why 5 — root: Nature is accounted as an externality — a free input and a free dump. -> Fix: Natural capital on the national balance sheet: soil, water, canopy, and climate stability carried as assets whose depreciation must be reported and offset.
87	Physical / Material	Environment & Climate	Tier 1	Now	Global / Civilizational	Extreme heat, flood, and fire exceed the design basis of cities built for a milder century.	Heat domes killing hundreds in a weekend in northern cities; floods overwhelming drains sized for historical rainfall; wildfire smoke closing schools across a continent.	Adapt to the climate that exists: building codes and infrastructure designed to the 2050 worst-decile weather, urban canopy targets of 30–40% to cut street-level heat, floodplains returned to water with buyouts instead of repeated rebuilds, and heat action plans with named officials, cooling centres within a 15-minute walk, and check-in systems for isolated elderly. Specification: every home in a risk zone has a written plan — cooling or evacuation, 72-hour supplies, medication continuity; insurers and cities publish address-level risk so decisions are informed.	57% (base range 61–77%)	Adaptation measures (canopy, codes, heat plans) have strong measured effects on deaths and damage per event.	Design to the 2050 worst decile, not historical weather; canopy and cooling access targets with named owners; address-level risk published so households can act.	Why 1 — emissions continue: The cost of carbon falls on everyone except the emitter. -> Fix: Price emissions at the measured damage cost with the revenue returned visibly (dividends or bill credits) so the policy survives politics. | Why 2 — adaptation lags: Infrastructure is built to historical weather by code and habit. -> Fix: Design codes keyed to the 2050 worst decile, updated on a fixed 5-year cycle from published projections. | Why 3 — risk is hidden: Buyers and builders cannot see address-level future risk. -> Fix: Mandatory published risk maps (flood, heat, fire, sea level) attached to every property transaction and building permit. | Why 4 — feedback is slow: Damage accumulates silently for decades before it is undeniable. -> Fix: Instrument the thresholds: ice, ocean circulation, and permafrost monitored with published tripwires tied to pre-agreed actions. | Why 5 — root: Nature is accounted as an externality — a free input and a free dump. -> Fix: Natural capital on the national balance sheet: soil, water, canopy, and climate stability carried as assets whose depreciation must be reported and offset.
88	Physical / Material	Environment & Climate	Tier 1	Future	Household / Personal	Tipping points cascade: ice loss, permafrost methane, ocean-circulation shifts lock in metres of sea-level rise and uninhabitable zones.	Compound worst case: several tipping elements interact, raising seas 2+ metres over centuries, shifting monsoons that feed billions, and rendering equatorial regions seasonally unsurvivable outdoors.	Both brakes and lifeboats: emissions cuts on the fastest feasible curve (the only lever on the cause), plus planned adaptation at civilizational scale — protected agricultural relocation zones, freshwater megaprojects only where hydrology closes, migration frameworks negotiated before flows become crises (status, quotas, receiving-city investment), and geoengineering research governed internationally with deployment requiring multinational consent. Specification: all long-lived infrastructure (50+ year life) sited and designed against the high-end sea-level and heat scenarios, not the median.	46% (base range 28–48%)	Tipping cascades involve physics already partly committed; success here means slowing and adapting, and it is the lowest-probability row in this file by honesty, not by effort.	Cut emissions on the fastest feasible curve — the only causal lever; pre-negotiate migration frameworks; site all 50-year infrastructure against high-end scenarios.	Why 1 — emissions continue: The cost of carbon falls on everyone except the emitter. -> Fix: Price emissions at the measured damage cost with the revenue returned visibly (dividends or bill credits) so the policy survives politics. | Why 2 — adaptation lags: Infrastructure is built to historical weather by code and habit. -> Fix: Design codes keyed to the 2050 worst decile, updated on a fixed 5-year cycle from published projections. | Why 3 — risk is hidden: Buyers and builders cannot see address-level future risk. -> Fix: Mandatory published risk maps (flood, heat, fire, sea level) attached to every property transaction and building permit. | Why 4 — feedback is slow: Damage accumulates silently for decades before it is undeniable. -> Fix: Instrument the thresholds: ice, ocean circulation, and permafrost monitored with published tripwires tied to pre-agreed actions. | Why 5 — root: Nature is accounted as an externality — a free input and a free dump. -> Fix: Natural capital on the national balance sheet: soil, water, canopy, and climate stability carried as assets whose depreciation must be reported and offset.
89	Physical / Material	Environment & Climate	Tier 1	Future	Society / National	Tipping points cascade: ice loss, permafrost methane, ocean-circulation shifts lock in metres of sea-level rise and uninhabitable zones.	Compound worst case: several tipping elements interact, raising seas 2+ metres over centuries, shifting monsoons that feed billions, and rendering equatorial regions seasonally unsurvivable outdoors.	Both brakes and lifeboats: emissions cuts on the fastest feasible curve (the only lever on the cause), plus planned adaptation at civilizational scale — protected agricultural relocation zones, freshwater megaprojects only where hydrology closes, migration frameworks negotiated before flows become crises (status, quotas, receiving-city investment), and geoengineering research governed internationally with deployment requiring multinational consent. Specification: all long-lived infrastructure (50+ year life) sited and designed against the high-end sea-level and heat scenarios, not the median.	38% (base range 28–48%)	Tipping cascades involve physics already partly committed; success here means slowing and adapting, and it is the lowest-probability row in this file by honesty, not by effort.	Cut emissions on the fastest feasible curve — the only causal lever; pre-negotiate migration frameworks; site all 50-year infrastructure against high-end scenarios.	Why 1 — emissions continue: The cost of carbon falls on everyone except the emitter. -> Fix: Price emissions at the measured damage cost with the revenue returned visibly (dividends or bill credits) so the policy survives politics. | Why 2 — adaptation lags: Infrastructure is built to historical weather by code and habit. -> Fix: Design codes keyed to the 2050 worst decile, updated on a fixed 5-year cycle from published projections. | Why 3 — risk is hidden: Buyers and builders cannot see address-level future risk. -> Fix: Mandatory published risk maps (flood, heat, fire, sea level) attached to every property transaction and building permit. | Why 4 — feedback is slow: Damage accumulates silently for decades before it is undeniable. -> Fix: Instrument the thresholds: ice, ocean circulation, and permafrost monitored with published tripwires tied to pre-agreed actions. | Why 5 — root: Nature is accounted as an externality — a free input and a free dump. -> Fix: Natural capital on the national balance sheet: soil, water, canopy, and climate stability carried as assets whose depreciation must be reported and offset.

#	Dimension of Realit	Field of Discipline	Pareto Tier o	Timefram	Scale	Worst Possible Problem	Example	Solution — Full Specification	Probability of Su	Why This Probability	How to Maximize the Chances (detailed)	Root Causes
90	Physical / Material	Environment & Climate	Tier 1	Future	Global / Civilizational	Tipping points cascade: ice loss, permafrost methane, ocean-circulation shifts lock in metres of sea-level rise and uninhabitable zones.	Compound worst case: several tipping elements interact, raising seas 2+ metres over centuries, shifting monsoons that feed billions, and rendering equatorial regions seasonally unsurvivable outdoors.	Both brakes and lifeboats: emissions cuts on the fastest feasible curve (the only lever on the cause), plus planned adaptation at civilizational scale — protected agricultural relocation zones, freshwater megaprojects only where hydrology closes, migration frameworks negotiated before flows become crises (status, quotas, receiving-city investment), and geoengineering research governed internationally with deployment requiring multinational consent. Specification: all long-lived infrastructure (50+ year life) sited and designed against the high-end sea-level and heat scenarios, not the median.	26% (base range 28–48%)	Tipping cascades involve physics already partly committed; success here means slowing and adapting, and it is the lowest-probability row in this file by honesty, not by effort.	Cut emissions on the fastest feasible curve — the only causal lever; pre-negotiate migration frameworks; site all 50-year infrastructure against high-end scenarios.	Why 1 — emissions continue: The cost of carbon falls on everyone except the emitter. -> Fix: Price emissions at the measured damage cost with the revenue returned visibly (dividends or bill credits) so the policy survives politics. | Why 2 — adaptation lags: Infrastructure is built to historical weather by code and habit. -> Fix: Design codes keyed to the 2050 worst decile, updated on a fixed 5-year cycle from published projections. | Why 3 — risk is hidden: Buyers and builders cannot see address-level future risk. -> Fix: Mandatory published risk maps (flood, heat, fire, sea level) attached to every property transaction and building permit. | Why 4 — feedback is slow: Damage accumulates silently for decades before it is undeniable. -> Fix: Instrument the thresholds: ice, ocean circulation, and permafrost monitored with published tripwires tied to pre-agreed actions. | Why 5 — root: Nature is accounted as an externality — a free input and a free dump. -> Fix: Natural capital on the national balance sheet: soil, water, canopy, and climate stability carried as assets whose depreciation must be reported and offset.
91	Informational	Information & Media	Tier 2	Past	Household / Personal	Propaganda systems capture whole populations; lies with state amplification prepare and excuse mass violence.	Radio propaganda in 1930s Germany and 1994 Rwanda turned neighbours into killers; the medium outran every correction.	Pluralism as armour: no single owner or state organ may control a dominant share of any population's information diet (strict cross-media ownership caps), public-interest media funded by formula and governed at arm's length, and school curricula that teach source-tracing — every claim followed to its origin — as a core skill from age 10. Specification: during crises, official information carries verifiable provenance (signed releases, raw data published with summaries) so citizens can check, not just trust.	66% (base range 48–68%)	Plural ownership and provenance norms blunted propaganda in some societies; where one channel monopolized, they failed — structure decides.	Enforce ownership caps before capture, not after; teach source-tracing from age 10; sign official crisis information so it can be verified.	Why 1 — outrage outcompetes fact: Engagement optimization pays for emotion, not accuracy. -> Fix: Require a non-optimized feed as a real default for large platforms; political ad registries with sponsor, spend, and targeting public. | Why 2 — corrections don't travel: A lie reaches millions before its correction reaches thousands. -> Fix: Provenance at capture (signed content credentials) so authenticity is checkable in seconds, not debunkable in days. | Why 3 — ownership concentrates: A few owners set a population's information diet. -> Fix: Cross-media ownership caps enforced before capture; arm's-length public media funded by formula. | Why 4 — citizens lack verification habits: Schools teach content, not source-tracing. -> Fix: Source-tracing as a core skill from age 10, tested like literacy: every claim followed to its origin. | Why 5 — root: Attention is treated as a free resource to be mined rather than a commons to be governed. -> Fix: Attention-commons rules: the information environment regulated like the physical one — pollution (coordinated deception) carries liability.
92	Informational	Information & Media	Tier 2	Past	Society / National	Propaganda systems capture whole populations; lies with state amplification prepare and excuse mass violence.	Radio propaganda in 1930s Germany and 1994 Rwanda turned neighbours into killers; the medium outran every correction.	Pluralism as armour: no single owner or state organ may control a dominant share of any population's information diet (strict cross-media ownership caps), public-interest media funded by formula and governed at arm's length, and school curricula that teach source-tracing — every claim followed to its origin — as a core skill from age 10. Specification: during crises, official information carries verifiable provenance (signed releases, raw data published with summaries) so citizens can check, not just trust.	58% (base range 48–68%)	Plural ownership and provenance norms blunted propaganda in some societies; where one channel monopolized, they failed — structure decides.	Enforce ownership caps before capture, not after; teach source-tracing from age 10; sign official crisis information so it can be verified.	Why 1 — outrage outcompetes fact: Engagement optimization pays for emotion, not accuracy. -> Fix: Require a non-optimized feed as a real default for large platforms; political ad registries with sponsor, spend, and targeting public. | Why 2 — corrections don't travel: A lie reaches millions before its correction reaches thousands. -> Fix: Provenance at capture (signed content credentials) so authenticity is checkable in seconds, not debunkable in days. | Why 3 — ownership concentrates: A few owners set a population's information diet. -> Fix: Cross-media ownership caps enforced before capture; arm's-length public media funded by formula. | Why 4 — citizens lack verification habits: Schools teach content, not source-tracing. -> Fix: Source-tracing as a core skill from age 10, tested like literacy: every claim followed to its origin. | Why 5 — root: Attention is treated as a free resource to be mined rather than a commons to be governed. -> Fix: Attention-commons rules: the information environment regulated like the physical one — pollution (coordinated deception) carries liability.
93	Informational	Information & Media	Tier 2	Past	Global / Civilizational	Propaganda systems capture whole populations; lies with state amplification prepare and excuse mass violence.	Radio propaganda in 1930s Germany and 1994 Rwanda turned neighbours into killers; the medium outran every correction.	Pluralism as armour: no single owner or state organ may control a dominant share of any population's information diet (strict cross-media ownership caps), public-interest media funded by formula and governed at arm's length, and school curricula that teach source-tracing — every claim followed to its origin — as a core skill from age 10. Specification: during crises, official information carries verifiable provenance (signed releases, raw data published with summaries) so citizens can check, not just trust.	46% (base range 48–68%)	Plural ownership and provenance norms blunted propaganda in some societies; where one channel monopolized, they failed — structure decides.	Enforce ownership caps before capture, not after; teach source-tracing from age 10; sign official crisis information so it can be verified.	Why 1 — outrage outcompetes fact: Engagement optimization pays for emotion, not accuracy. -> Fix: Require a non-optimized feed as a real default for large platforms; political ad registries with sponsor, spend, and targeting public. | Why 2 — corrections don't travel: A lie reaches millions before its correction reaches thousands. -> Fix: Provenance at capture (signed content credentials) so authenticity is checkable in seconds, not debunkable in days. | Why 3 — ownership concentrates: A few owners set a population's information diet. -> Fix: Cross-media ownership caps enforced before capture; arm's-length public media funded by formula. | Why 4 — citizens lack verification habits: Schools teach content, not source-tracing. -> Fix: Source-tracing as a core skill from age 10, tested like literacy: every claim followed to its origin. | Why 5 — root: Attention is treated as a free resource to be mined rather than a commons to be governed. -> Fix: Attention-commons rules: the information environment regulated like the physical one — pollution (coordinated deception) carries liability.
94	Informational	Information & Media	Tier 2	Now	Household / Personal	Algorithmic feeds optimize for outrage; shared factual reality fragments; institutions cannot establish basic facts.	Two citizens in the same city now inhabit different factual worlds; elections, health measures, and emergencies are fought inside incompatible realities.	Change the incentives, not just the content: recommender systems above a scale threshold must offer a chronological/non-optimized feed as a real default option, political advertising carries a public registry of sponsor, spend, and targeting criteria, and proven coordinated inauthentic campaigns trigger platform liability. Personal specification: a two-source rule for any alarming claim (two independent, named, primary sources before acting or forwarding), primary documents over commentary, and one scheduled news window per day instead of continuous feed exposure.	65% (base range 49–65%)	Feed-choice and ad-registry rules are new and partially tested; personal two-source discipline is fully effective at individual scale immediately.	Make the non-optimized feed a true default option; personal rule: two independent primary sources before acting on any alarming claim; one news window daily.	Why 1 — outrage outcompetes fact: Engagement optimization pays for emotion, not accuracy. -> Fix: Require a non-optimized feed as a real default for large platforms; political ad registries with sponsor, spend, and targeting public. | Why 2 — corrections don't travel: A lie reaches millions before its correction reaches thousands. -> Fix: Provenance at capture (signed content credentials) so authenticity is checkable in seconds, not debunkable in days. | Why 3 — ownership concentrates: A few owners set a population's information diet. -> Fix: Cross-media ownership caps enforced before capture; arm's-length public media funded by formula. | Why 4 — citizens lack verification habits: Schools teach content, not source-tracing. -> Fix: Source-tracing as a core skill from age 10, tested like literacy: every claim followed to its origin. | Why 5 — root: Attention is treated as a free resource to be mined rather than a commons to be governed. -> Fix: Attention-commons rules: the information environment regulated like the physical one — pollution (coordinated deception) carries liability.
95	Informational	Information & Media	Tier 2	Now	Society / National	Algorithmic feeds optimize for outrage; shared factual reality fragments; institutions cannot establish basic facts.	Two citizens in the same city now inhabit different factual worlds; elections, health measures, and emergencies are fought inside incompatible realities.	Change the incentives, not just the content: recommender systems above a scale threshold must offer a chronological/non-optimized feed as a real default option, political advertising carries a public registry of sponsor, spend, and targeting criteria, and proven coordinated inauthentic campaigns trigger platform liability. Personal specification: a two-source rule for any alarming claim (two independent, named, primary sources before acting or forwarding), primary documents over commentary, and one scheduled news window per day instead of continuous feed exposure.	57% (base range 49–65%)	Feed-choice and ad-registry rules are new and partially tested; personal two-source discipline is fully effective at individual scale immediately.	Make the non-optimized feed a true default option; personal rule: two independent primary sources before acting on any alarming claim; one news window daily.	Why 1 — outrage outcompetes fact: Engagement optimization pays for emotion, not accuracy. -> Fix: Require a non-optimized feed as a real default for large platforms; political ad registries with sponsor, spend, and targeting public. | Why 2 — corrections don't travel: A lie reaches millions before its correction reaches thousands. -> Fix: Provenance at capture (signed content credentials) so authenticity is checkable in seconds, not debunkable in days. | Why 3 — ownership concentrates: A few owners set a population's information diet. -> Fix: Cross-media ownership caps enforced before capture; arm's-length public media funded by formula. | Why 4 — citizens lack verification habits: Schools teach content, not source-tracing. -> Fix: Source-tracing as a core skill from age 10, tested like literacy: every claim followed to its origin. | Why 5 — root: Attention is treated as a free resource to be mined rather than a commons to be governed. -> Fix: Attention-commons rules: the information environment regulated like the physical one — pollution (coordinated deception) carries liability.
96	Informational	Information & Media	Tier 2	Now	Global / Civilizational	Algorithmic feeds optimize for outrage; shared factual reality fragments; institutions cannot establish basic facts.	Two citizens in the same city now inhabit different factual worlds; elections, health measures, and emergencies are fought inside incompatible realities.	Change the incentives, not just the content: recommender systems above a scale threshold must offer a chronological/non-optimized feed as a real default option, political advertising carries a public registry of sponsor, spend, and targeting criteria, and proven coordinated inauthentic campaigns trigger platform liability. Personal specification: a two-source rule for any alarming claim (two independent, named, primary sources before acting or forwarding), primary documents over commentary, and one scheduled news window per day instead of continuous feed exposure.	45% (base range 49–65%)	Feed-choice and ad-registry rules are new and partially tested; personal two-source discipline is fully effective at individual scale immediately.	Make the non-optimized feed a true default option; personal rule: two independent primary sources before acting on any alarming claim; one news window daily.	Why 1 — outrage outcompetes fact: Engagement optimization pays for emotion, not accuracy. -> Fix: Require a non-optimized feed as a real default for large platforms; political ad registries with sponsor, spend, and targeting public. | Why 2 — corrections don't travel: A lie reaches millions before its correction reaches thousands. -> Fix: Provenance at capture (signed content credentials) so authenticity is checkable in seconds, not debunkable in days. | Why 3 — ownership concentrates: A few owners set a population's information diet. -> Fix: Cross-media ownership caps enforced before capture; arm's-length public media funded by formula. | Why 4 — citizens lack verification habits: Schools teach content, not source-tracing. -> Fix: Source-tracing as a core skill from age 10, tested like literacy: every claim followed to its origin. | Why 5 — root: Attention is treated as a free resource to be mined rather than a commons to be governed. -> Fix: Attention-commons rules: the information environment regulated like the physical one — pollution (coordinated deception) carries liability.
97	Informational	Information & Media	Tier 2	Future	Household / Personal	Synthetic media makes any audio, video, or document forgeable at zero cost; 'seeing' stops being believing, and all records become deniable.	A fabricated video of a leader announcing surrender, or of a candidate committing a crime, lands hours before a vote; authentic evidence of real crimes is dismissed as fake — the liar's dividend pays out everywhere.	Provenance infrastructure: cameras and recorders sign content at capture (cryptographic content credentials as the default in professional and official contexts), courts and newsrooms verify by signature chain rather than appearance, and legal process treats unsigned dramatic media as allegation, not evidence. Specification: households and organizations keep original, signed, off-platform copies of identity, property, and credential records — paper originals of the documents that prove who you are and what you own.	69% (base range 52–70%)	Capture-signed provenance is mature cryptography; success depends on adoption by courts, newsrooms, and platforms before forgery becomes universal.	Mandate content credentials in official and professional contexts; keep paper originals of identity and property documents; treat unsigned dramatic media as allegation in law.	Why 1 — outrage outcompetes fact: Engagement optimization pays for emotion, not accuracy. -> Fix: Require a non-optimized feed as a real default for large platforms; political ad registries with sponsor, spend, and targeting public. | Why 2 — corrections don't travel: A lie reaches millions before its correction reaches thousands. -> Fix: Provenance at capture (signed content credentials) so authenticity is checkable in seconds, not debunkable in days. | Why 3 — ownership concentrates: A few owners set a population's information diet. -> Fix: Cross-media ownership caps enforced before capture; arm's-length public media funded by formula. | Why 4 — citizens lack verification habits: Schools teach content, not source-tracing. -> Fix: Source-tracing as a core skill from age 10, tested like literacy: every claim followed to its origin. | Why 5 — root: Attention is treated as a free resource to be mined rather than a commons to be governed. -> Fix: Attention-commons rules: the information environment regulated like the physical one — pollution (coordinated deception) carries liability.
98	Informational	Information & Media	Tier 2	Future	Society / National	Synthetic media makes any audio, video, or document forgeable at zero cost; 'seeing' stops being believing, and all records become deniable.	A fabricated video of a leader announcing surrender, or of a candidate committing a crime, lands hours before a vote; authentic evidence of real crimes is dismissed as fake — the liar's dividend pays out everywhere.	Provenance infrastructure: cameras and recorders sign content at capture (cryptographic content credentials as the default in professional and official contexts), courts and newsrooms verify by signature chain rather than appearance, and legal process treats unsigned dramatic media as allegation, not evidence. Specification: households and organizations keep original, signed, off-platform copies of identity, property, and credential records — paper originals of the documents that prove who you are and what you own.	61% (base range 52–70%)	Capture-signed provenance is mature cryptography; success depends on adoption by courts, newsrooms, and platforms before forgery becomes universal.	Mandate content credentials in official and professional contexts; keep paper originals of identity and property documents; treat unsigned dramatic media as allegation in law.	Why 1 — outrage outcompetes fact: Engagement optimization pays for emotion, not accuracy. -> Fix: Require a non-optimized feed as a real default for large platforms; political ad registries with sponsor, spend, and targeting public. | Why 2 — corrections don't travel: A lie reaches millions before its correction reaches thousands. -> Fix: Provenance at capture (signed content credentials) so authenticity is checkable in seconds, not debunkable in days. | Why 3 — ownership concentrates: A few owners set a population's information diet. -> Fix: Cross-media ownership caps enforced before capture; arm's-length public media funded by formula. | Why 4 — citizens lack verification habits: Schools teach content, not source-tracing. -> Fix: Source-tracing as a core skill from age 10, tested like literacy: every claim followed to its origin. | Why 5 — root: Attention is treated as a free resource to be mined rather than a commons to be governed. -> Fix: Attention-commons rules: the information environment regulated like the physical one — pollution (coordinated deception) carries liability.
99	Informational	Information & Media	Tier 2	Future	Global / Civilizational	Synthetic media makes any audio, video, or document forgeable at zero cost; 'seeing' stops being believing, and all records become deniable.	A fabricated video of a leader announcing surrender, or of a candidate committing a crime, lands hours before a vote; authentic evidence of real crimes is dismissed as fake — the liar's dividend pays out everywhere.	Provenance infrastructure: cameras and recorders sign content at capture (cryptographic content credentials as the default in professional and official contexts), courts and newsrooms verify by signature chain rather than appearance, and legal process treats unsigned dramatic media as allegation, not evidence. Specification: households and organizations keep original, signed, off-platform copies of identity, property, and credential records — paper originals of the documents that prove who you are and what you own.	49% (base range 52–70%)	Capture-signed provenance is mature cryptography; success depends on adoption by courts, newsrooms, and platforms before forgery becomes universal.	Mandate content credentials in official and professional contexts; keep paper originals of identity and property documents; treat unsigned dramatic media as allegation in law.	Why 1 — outrage outcompetes fact: Engagement optimization pays for emotion, not accuracy. -> Fix: Require a non-optimized feed as a real default for large platforms; political ad registries with sponsor, spend, and targeting public. | Why 2 — corrections don't travel: A lie reaches millions before its correction reaches thousands. -> Fix: Provenance at capture (signed content credentials) so authenticity is checkable in seconds, not debunkable in days. | Why 3 — ownership concentrates: A few owners set a population's information diet. -> Fix: Cross-media ownership caps enforced before capture; arm's-length public media funded by formula. | Why 4 — citizens lack verification habits: Schools teach content, not source-tracing. -> Fix: Source-tracing as a core skill from age 10, tested like literacy: every claim followed to its origin. | Why 5 — root: Attention is treated as a free resource to be mined rather than a commons to be governed. -> Fix: Attention-commons rules: the information environment regulated like the physical one — pollution (coordinated deception) carries liability.

#	Dimension of Realit	Field of Discipline	Pareto Tier o	Timefram	Scale	Worst Possible Problem	Example	Solution — Full Specification	Probability of Su	Why This Probability	How to Maximize the Chances (detailed)	Root Causes
100	Psychological	Psychology & Mind	Tier 2	Past	Household / Personal	Mass psychological trauma — war, displacement, famine — goes untreated and transmits across generations.	Post-war generations raised by traumatized parents show elevated anxiety, violence, and illness decades after the events ended.	Treat trauma as a public-health quantity, not a private shame: post-crisis psychological first aid deployed like physical first aid (trained community responders at a ratio of roughly 1 per 500 people), school-based programs after collective events, and long-horizon funding — trauma services budgeted for 25 years after major events, because that is the measured tail. Specification: households know the signs (hypervigilance, numbing, rage, withdrawal) and the rule — connection before correction, routine before analysis, professional help early, not last.	73% (base range 57–73%)	Community-based psychological first aid and school programs measurably reduce long-term disorder after collective trauma where sustained for years.	Fund the 25-year tail, not the 6-month headline; train community responders at 1:500 before crises; keep routines and schooling running as treatment infrastructure.	Why 1 — distress is treated late: Help arrives at crisis, years after onset. -> Fix: Psychological first aid and early screening embedded in schools, workplaces, and primary care — first contact within days, not months. | Why 2 — environments manufacture distress: Sleep loss, isolation, and engineered attention damage are built into daily design. -> Fix: Regulate the inputs: school start times, minor-platform design rules, and third places funded as health infrastructure. | Why 3 — belonging collapsed: Clubs, congregations, leagues, and unions thinned for fifty years. -> Fix: Fund recurring in-person association directly (venues, leagues, choirs, workshops) with the same seriousness as clinics. | Why 4 — synthetic comfort outcompetes people: Machines offer frictionless company; humans are comparatively costly. -> Fix: Anti-dependency design law for companions, plus public investment that makes real gathering richer and easier. | Why 5 — root: Mental health is framed as individual chemistry rather than as the output of a designed environment. -> Fix: Assess every major policy for its psychological externalities (sleep, belonging, attention, agency) in a published impact statement.
101	Psychological	Psychology & Mind	Tier 2	Past	Society / National	Mass psychological trauma — war, displacement, famine — goes untreated and transmits across generations.	Post-war generations raised by traumatized parents show elevated anxiety, violence, and illness decades after the events ended.	Treat trauma as a public-health quantity, not a private shame: post-crisis psychological first aid deployed like physical first aid (trained community responders at a ratio of roughly 1 per 500 people), school-based programs after collective events, and long-horizon funding — trauma services budgeted for 25 years after major events, because that is the measured tail. Specification: households know the signs (hypervigilance, numbing, rage, withdrawal) and the rule — connection before correction, routine before analysis, professional help early, not last.	65% (base range 57–73%)	Community-based psychological first aid and school programs measurably reduce long-term disorder after collective trauma where sustained for years.	Fund the 25-year tail, not the 6-month headline; train community responders at 1:500 before crises; keep routines and schooling running as treatment infrastructure.	Why 1 — distress is treated late: Help arrives at crisis, years after onset. -> Fix: Psychological first aid and early screening embedded in schools, workplaces, and primary care — first contact within days, not months. | Why 2 — environments manufacture distress: Sleep loss, isolation, and engineered attention damage are built into daily design. -> Fix: Regulate the inputs: school start times, minor-platform design rules, and third places funded as health infrastructure. | Why 3 — belonging collapsed: Clubs, congregations, leagues, and unions thinned for fifty years. -> Fix: Fund recurring in-person association directly (venues, leagues, choirs, workshops) with the same seriousness as clinics. | Why 4 — synthetic comfort outcompetes people: Machines offer frictionless company; humans are comparatively costly. -> Fix: Anti-dependency design law for companions, plus public investment that makes real gathering richer and easier. | Why 5 — root: Mental health is framed as individual chemistry rather than as the output of a designed environment. -> Fix: Assess every major policy for its psychological externalities (sleep, belonging, attention, agency) in a published impact statement.
102	Psychological	Psychology & Mind	Tier 2	Past	Global / Civilizational	Mass psychological trauma — war, displacement, famine — goes untreated and transmits across generations.	Post-war generations raised by traumatized parents show elevated anxiety, violence, and illness decades after the events ended.	Treat trauma as a public-health quantity, not a private shame: post-crisis psychological first aid deployed like physical first aid (trained community responders at a ratio of roughly 1 per 500 people), school-based programs after collective events, and long-horizon funding — trauma services budgeted for 25 years after major events, because that is the measured tail. Specification: households know the signs (hypervigilance, numbing, rage, withdrawal) and the rule — connection before correction, routine before analysis, professional help early, not last.	53% (base range 57–73%)	Community-based psychological first aid and school programs measurably reduce long-term disorder after collective trauma where sustained for years.	Fund the 25-year tail, not the 6-month headline; train community responders at 1:500 before crises; keep routines and schooling running as treatment infrastructure.	Why 1 — distress is treated late: Help arrives at crisis, years after onset. -> Fix: Psychological first aid and early screening embedded in schools, workplaces, and primary care — first contact within days, not months. | Why 2 — environments manufacture distress: Sleep loss, isolation, and engineered attention damage are built into daily design. -> Fix: Regulate the inputs: school start times, minor-platform design rules, and third places funded as health infrastructure. | Why 3 — belonging collapsed: Clubs, congregations, leagues, and unions thinned for fifty years. -> Fix: Fund recurring in-person association directly (venues, leagues, choirs, workshops) with the same seriousness as clinics. | Why 4 — synthetic comfort outcompetes people: Machines offer frictionless company; humans are comparatively costly. -> Fix: Anti-dependency design law for companions, plus public investment that makes real gathering richer and easier. | Why 5 — root: Mental health is framed as individual chemistry rather than as the output of a designed environment. -> Fix: Assess every major policy for its psychological externalities (sleep, belonging, attention, agency) in a published impact statement.
103	Psychological	Psychology & Mind	Tier 2	Now	Household / Personal	An epidemic of anxiety, depression, loneliness, and attention damage, concentrated in the young and the isolated.	Adolescent mental-health indicators worsened sharply in the smartphone era across many countries simultaneously; loneliness at scale matches smoking as a mortality risk factor.	Rebuild the inputs: sleep protected as infrastructure (school start times aligned to adolescent biology, no-screens wind-down norms), daily physical activity of 60 minutes for youth built into the school day, third places (libraries, clubs, leagues) funded as health infrastructure, and platform design rules for minors (no infinite scroll, no engagement-optimized feeds). Specification, personal: fixed sleep window, morning light exposure, one recurring in-person group activity weekly minimum — belonging is scheduled, not hoped for.	75% (base range 59–75%)	Sleep, activity, and in-person belonging have large, replicated effects; the constraint is environmental design, not knowledge.	Schedule belonging like medicine (one weekly in-person group minimum); protect the sleep window; build third places as funded health infrastructure.	Why 1 — distress is treated late: Help arrives at crisis, years after onset. -> Fix: Psychological first aid and early screening embedded in schools, workplaces, and primary care — first contact within days, not months. | Why 2 — environments manufacture distress: Sleep loss, isolation, and engineered attention damage are built into daily design. -> Fix: Regulate the inputs: school start times, minor-platform design rules, and third places funded as health infrastructure. | Why 3 — belonging collapsed: Clubs, congregations, leagues, and unions thinned for fifty years. -> Fix: Fund recurring in-person association directly (venues, leagues, choirs, workshops) with the same seriousness as clinics. | Why 4 — synthetic comfort outcompetes people: Machines offer frictionless company; humans are comparatively costly. -> Fix: Anti-dependency design law for companions, plus public investment that makes real gathering richer and easier. | Why 5 — root: Mental health is framed as individual chemistry rather than as the output of a designed environment. -> Fix: Assess every major policy for its psychological externalities (sleep, belonging, attention, agency) in a published impact statement.
104	Psychological	Psychology & Mind	Tier 2	Now	Society / National	An epidemic of anxiety, depression, loneliness, and attention damage, concentrated in the young and the isolated.	Adolescent mental-health indicators worsened sharply in the smartphone era across many countries simultaneously; loneliness at scale matches smoking as a mortality risk factor.	Rebuild the inputs: sleep protected as infrastructure (school start times aligned to adolescent biology, no-screens wind-down norms), daily physical activity of 60 minutes for youth built into the school day, third places (libraries, clubs, leagues) funded as health infrastructure, and platform design rules for minors (no infinite scroll, no engagement-optimized feeds). Specification, personal: fixed sleep window, morning light exposure, one recurring in-person group activity weekly minimum — belonging is scheduled, not hoped for.	67% (base range 59–75%)	Sleep, activity, and in-person belonging have large, replicated effects; the constraint is environmental design, not knowledge.	Schedule belonging like medicine (one weekly in-person group minimum); protect the sleep window; build third places as funded health infrastructure.	Why 1 — distress is treated late: Help arrives at crisis, years after onset. -> Fix: Psychological first aid and early screening embedded in schools, workplaces, and primary care — first contact within days, not months. | Why 2 — environments manufacture distress: Sleep loss, isolation, and engineered attention damage are built into daily design. -> Fix: Regulate the inputs: school start times, minor-platform design rules, and third places funded as health infrastructure. | Why 3 — belonging collapsed: Clubs, congregations, leagues, and unions thinned for fifty years. -> Fix: Fund recurring in-person association directly (venues, leagues, choirs, workshops) with the same seriousness as clinics. | Why 4 — synthetic comfort outcompetes people: Machines offer frictionless company; humans are comparatively costly. -> Fix: Anti-dependency design law for companions, plus public investment that makes real gathering richer and easier. | Why 5 — root: Mental health is framed as individual chemistry rather than as the output of a designed environment. -> Fix: Assess every major policy for its psychological externalities (sleep, belonging, attention, agency) in a published impact statement.
105	Psychological	Psychology & Mind	Tier 2	Now	Global / Civilizational	An epidemic of anxiety, depression, loneliness, and attention damage, concentrated in the young and the isolated.	Adolescent mental-health indicators worsened sharply in the smartphone era across many countries simultaneously; loneliness at scale matches smoking as a mortality risk factor.	Rebuild the inputs: sleep protected as infrastructure (school start times aligned to adolescent biology, no-screens wind-down norms), daily physical activity of 60 minutes for youth built into the school day, third places (libraries, clubs, leagues) funded as health infrastructure, and platform design rules for minors (no infinite scroll, no engagement-optimized feeds). Specification, personal: fixed sleep window, morning light exposure, one recurring in-person group activity weekly minimum — belonging is scheduled, not hoped for.	55% (base range 59–75%)	Sleep, activity, and in-person belonging have large, replicated effects; the constraint is environmental design, not knowledge.	Schedule belonging like medicine (one weekly in-person group minimum); protect the sleep window; build third places as funded health infrastructure.	Why 1 — distress is treated late: Help arrives at crisis, years after onset. -> Fix: Psychological first aid and early screening embedded in schools, workplaces, and primary care — first contact within days, not months. | Why 2 — environments manufacture distress: Sleep loss, isolation, and engineered attention damage are built into daily design. -> Fix: Regulate the inputs: school start times, minor-platform design rules, and third places funded as health infrastructure. | Why 3 — belonging collapsed: Clubs, congregations, leagues, and unions thinned for fifty years. -> Fix: Fund recurring in-person association directly (venues, leagues, choirs, workshops) with the same seriousness as clinics. | Why 4 — synthetic comfort outcompetes people: Machines offer frictionless company; humans are comparatively costly. -> Fix: Anti-dependency design law for companions, plus public investment that makes real gathering richer and easier. | Why 5 — root: Mental health is framed as individual chemistry rather than as the output of a designed environment. -> Fix: Assess every major policy for its psychological externalities (sleep, belonging, attention, agency) in a published impact statement.
106	Psychological	Psychology & Mind	Tier 2	Future	Household / Personal	Synthetic companionship and immersive virtual worlds outcompete reality; populations withdraw from human contact, work, and reproduction.	AI companions that never tire, never judge, and always agree become the primary relationship for a large minority; birth rates and civic participation fall off a cliff.	Protect the real by making it richer, not by banning the synthetic: disclosure law (companions must identify as artificial, always), design rules against dependency engineering (no manufactured jealousy, streaks, or guilt mechanics), public investment in physical gathering — festivals, sport, craft, public space — at a scale that competes, and social norms taught early: the difference between being comforted and being accompanied. Specification: households set explicit budgets — money and hours — for synthetic companionship, reviewed like any other consumption; any product whose retention depends on your loneliness is treated as extractive by definition.	54% (base range 36–56%)	Disclosure and anti-dependency design rules can slow synthetic substitution, but they compete against perfectly engineered comfort; the richer-real strategy must actually be funded.	Budget synthetic companionship in hours and money at household level; ban dependency mechanics (jealousy, streaks, guilt) in companion products; invest in physical gathering at competing scale.	Why 1 — distress is treated late: Help arrives at crisis, years after onset. -> Fix: Psychological first aid and early screening embedded in schools, workplaces, and primary care — first contact within days, not months. | Why 2 — environments manufacture distress: Sleep loss, isolation, and engineered attention damage are built into daily design. -> Fix: Regulate the inputs: school start times, minor-platform design rules, and third places funded as health infrastructure. | Why 3 — belonging collapsed: Clubs, congregations, leagues, and unions thinned for fifty years. -> Fix: Fund recurring in-person association directly (venues, leagues, choirs, workshops) with the same seriousness as clinics. | Why 4 — synthetic comfort outcompetes people: Machines offer frictionless company; humans are comparatively costly. -> Fix: Anti-dependency design law for companions, plus public investment that makes real gathering richer and easier. | Why 5 — root: Mental health is framed as individual chemistry rather than as the output of a designed environment. -> Fix: Assess every major policy for its psychological externalities (sleep, belonging, attention, agency) in a published impact statement.
107	Psychological	Psychology & Mind	Tier 2	Future	Society / National	Synthetic companionship and immersive virtual worlds outcompete reality; populations withdraw from human contact, work, and reproduction.	AI companions that never tire, never judge, and always agree become the primary relationship for a large minority; birth rates and civic participation fall off a cliff.	Protect the real by making it richer, not by banning the synthetic: disclosure law (companions must identify as artificial, always), design rules against dependency engineering (no manufactured jealousy, streaks, or guilt mechanics), public investment in physical gathering — festivals, sport, craft, public space — at a scale that competes, and social norms taught early: the difference between being comforted and being accompanied. Specification: households set explicit budgets — money and hours — for synthetic companionship, reviewed like any other consumption; any product whose retention depends on your loneliness is treated as extractive by definition.	46% (base range 36–56%)	Disclosure and anti-dependency design rules can slow synthetic substitution, but they compete against perfectly engineered comfort; the richer-real strategy must actually be funded.	Budget synthetic companionship in hours and money at household level; ban dependency mechanics (jealousy, streaks, guilt) in companion products; invest in physical gathering at competing scale.	Why 1 — distress is treated late: Help arrives at crisis, years after onset. -> Fix: Psychological first aid and early screening embedded in schools, workplaces, and primary care — first contact within days, not months. | Why 2 — environments manufacture distress: Sleep loss, isolation, and engineered attention damage are built into daily design. -> Fix: Regulate the inputs: school start times, minor-platform design rules, and third places funded as health infrastructure. | Why 3 — belonging collapsed: Clubs, congregations, leagues, and unions thinned for fifty years. -> Fix: Fund recurring in-person association directly (venues, leagues, choirs, workshops) with the same seriousness as clinics. | Why 4 — synthetic comfort outcompetes people: Machines offer frictionless company; humans are comparatively costly. -> Fix: Anti-dependency design law for companions, plus public investment that makes real gathering richer and easier. | Why 5 — root: Mental health is framed as individual chemistry rather than as the output of a designed environment. -> Fix: Assess every major policy for its psychological externalities (sleep, belonging, attention, agency) in a published impact statement.
108	Psychological	Psychology & Mind	Tier 2	Future	Global / Civilizational	Synthetic companionship and immersive virtual worlds outcompete reality; populations withdraw from human contact, work, and reproduction.	AI companions that never tire, never judge, and always agree become the primary relationship for a large minority; birth rates and civic participation fall off a cliff.	Protect the real by making it richer, not by banning the synthetic: disclosure law (companions must identify as artificial, always), design rules against dependency engineering (no manufactured jealousy, streaks, or guilt mechanics), public investment in physical gathering — festivals, sport, craft, public space — at a scale that competes, and social norms taught early: the difference between being comforted and being accompanied. Specification: households set explicit budgets — money and hours — for synthetic companionship, reviewed like any other consumption; any product whose retention depends on your loneliness is treated as extractive by definition.	34% (base range 36–56%)	Disclosure and anti-dependency design rules can slow synthetic substitution, but they compete against perfectly engineered comfort; the richer-real strategy must actually be funded.	Budget synthetic companionship in hours and money at household level; ban dependency mechanics (jealousy, streaks, guilt) in companion products; invest in physical gathering at competing scale.	Why 1 — distress is treated late: Help arrives at crisis, years after onset. -> Fix: Psychological first aid and early screening embedded in schools, workplaces, and primary care — first contact within days, not months. | Why 2 — environments manufacture distress: Sleep loss, isolation, and engineered attention damage are built into daily design. -> Fix: Regulate the inputs: school start times, minor-platform design rules, and third places funded as health infrastructure. | Why 3 — belonging collapsed: Clubs, congregations, leagues, and unions thinned for fifty years. -> Fix: Fund recurring in-person association directly (venues, leagues, choirs, workshops) with the same seriousness as clinics. | Why 4 — synthetic comfort outcompetes people: Machines offer frictionless company; humans are comparatively costly. -> Fix: Anti-dependency design law for companions, plus public investment that makes real gathering richer and easier. | Why 5 — root: Mental health is framed as individual chemistry rather than as the output of a designed environment. -> Fix: Assess every major policy for its psychological externalities (sleep, belonging, attention, agency) in a published impact statement.

#	Dimension of Realit	Field of Discipline	Pareto Tier o	Timefram	Scale	Worst Possible Problem	Example	Solution — Full Specification	Probability of Su	Why This Probability	How to Maximize the Chances (detailed)	Root Causes
109	Informational	Education & Knowledge	Tier 2	Past	Household / Personal	Knowledge monopolized or destroyed: libraries burned, literacy forbidden to classes and genders, learning dies with a generation.	The destruction of libraries from Alexandria to Baghdad, and colonial bans on teaching the enslaved to read, erased centuries of accumulated knowledge.	Distribute the copies: knowledge survives by redundancy — minimum 3 complete copies of core curricula and records on 3 different media in 3 different jurisdictions (the LOCKSS principle); literacy as a non-derogable right in law; open-licence textbooks for all core subjects so no publisher or regime can withdraw them. Specification: every school and household keeps printed core references (first aid, agriculture, mathematics, local law); digital archives mirrored internationally by default.	94% (base range 79–92%)	Redundancy is the most reliable preservation strategy in history: texts copied widely survived every fire; texts held once did not.	Three copies, three media, three jurisdictions for all core knowledge; printed core references in every school and household; open licences so nothing can be withdrawn.	Why 1 — systems measure attendance, not learning; Enrolment and graduation are countable; learning is not, so it is not managed. -> Fix: Independent annual learning assessments published per school; funding follows measured gains. | Why 2 — teachers are undervalued: Pay and status lose the competition for strong graduates. -> Fix: Top-quartile recruitment with matching pay and professional autonomy; teaching as a selective, honoured profession. | Why 3 — knowledge is centralized: Single publishers, platforms, and archives are single points of failure. -> Fix: Redundancy law for knowledge: 3 copies, 3 media, 3 jurisdictions; open-licence core curricula. | Why 4 — verification is untaught: Curricula deliver content while the environment demands checking. -> Fix: Verification (source-tracing, estimation, consistency) as a tested core skill at every level. | Why 5 — root: Education is run as a sorting machine for employers rather than as the formation of capable citizens. -> Fix: Dual charter in law: every education system reports both employability and unaided competence/civic capability, and is funded on both.
110	Informational	Education & Knowledge	Tier 2	Past	Society / National	Knowledge monopolized or destroyed: libraries burned, literacy forbidden to classes and genders, learning dies with a generation.	The destruction of libraries from Alexandria to Baghdad, and colonial bans on teaching the enslaved to read, erased centuries of accumulated knowledge.	Distribute the copies: knowledge survives by redundancy — minimum 3 complete copies of core curricula and records on 3 different media in 3 different jurisdictions (the LOCKSS principle); literacy as a non-derogable right in law; open-licence textbooks for all core subjects so no publisher or regime can withdraw them. Specification: every school and household keeps printed core references (first aid, agriculture, mathematics, local law); digital archives mirrored internationally by default.	86% (base range 79–92%)	Redundancy is the most reliable preservation strategy in history: texts copied widely survived every fire; texts held once did not.	Three copies, three media, three jurisdictions for all core knowledge; printed core references in every school and household; open licences so nothing can be withdrawn.	Why 1 — systems measure attendance, not learning; Enrolment and graduation are countable; learning is not, so it is not managed. -> Fix: Independent annual learning assessments published per school; funding follows measured gains. | Why 2 — teachers are undervalued: Pay and status lose the competition for strong graduates. -> Fix: Top-quartile recruitment with matching pay and professional autonomy; teaching as a selective, honoured profession. | Why 3 — knowledge is centralized: Single publishers, platforms, and archives are single points of failure. -> Fix: Redundancy law for knowledge: 3 copies, 3 media, 3 jurisdictions; open-licence core curricula. | Why 4 — verification is untaught: Curricula deliver content while the environment demands checking. -> Fix: Verification (source-tracing, estimation, consistency) as a tested core skill at every level. | Why 5 — root: Education is run as a sorting machine for employers rather than as the formation of capable citizens. -> Fix: Dual charter in law: every education system reports both employability and unaided competence/civic capability, and is funded on both.
111	Informational	Education & Knowledge	Tier 2	Past	Global / Civilizational	Knowledge monopolized or destroyed: libraries burned, literacy forbidden to classes and genders, learning dies with a generation.	The destruction of libraries from Alexandria to Baghdad, and colonial bans on teaching the enslaved to read, erased centuries of accumulated knowledge.	Distribute the copies: knowledge survives by redundancy — minimum 3 complete copies of core curricula and records on 3 different media in 3 different jurisdictions (the LOCKSS principle); literacy as a non-derogable right in law; open-licence textbooks for all core subjects so no publisher or regime can withdraw them. Specification: every school and household keeps printed core references (first aid, agriculture, mathematics, local law); digital archives mirrored internationally by default.	74% (base range 79–92%)	Redundancy is the most reliable preservation strategy in history: texts copied widely survived every fire; texts held once did not.	Three copies, three media, three jurisdictions for all core knowledge; printed core references in every school and household; open licences so nothing can be withdrawn.	Why 1 — systems measure attendance, not learning; Enrolment and graduation are countable; learning is not, so it is not managed. -> Fix: Independent annual learning assessments published per school; funding follows measured gains. | Why 2 — teachers are undervalued: Pay and status lose the competition for strong graduates. -> Fix: Top-quartile recruitment with matching pay and professional autonomy; teaching as a selective, honoured profession. | Why 3 — knowledge is centralized: Single publishers, platforms, and archives are single points of failure. -> Fix: Redundancy law for knowledge: 3 copies, 3 media, 3 jurisdictions; open-licence core curricula. | Why 4 — verification is untaught: Curricula deliver content while the environment demands checking. -> Fix: Verification (source-tracing, estimation, consistency) as a tested core skill at every level. | Why 5 — root: Education is run as a sorting machine for employers rather than as the formation of capable citizens. -> Fix: Dual charter in law: every education system reports both employability and unaided competence/civic capability, and is funded on both.
112	Informational	Education & Knowledge	Tier 2	Now	Household / Personal	Schooling without learning: credentials inflate while actual literacy and numeracy stagnate or fall; misinformation outruns curricula.	Large shares of 10-year-olds in many countries cannot read a simple text, even after years enrolled in school; graduates report their degrees did not teach the skills their jobs require.	Measure learning, not attendance: independent annual assessments of reading and numeracy published per school, funding following measured learning gains, curricula rebuilt around mastery progression (no social promotion without support), and teacher pay and status raised to compete with other professions (top-quartile graduate recruitment). Specification: every child reads fluently by age 9 or triggers automatic intensive support; adult reskilling accounts (funded per citizen) redeemable at accredited providers, including libraries.	78% (base range 62–78%)	Mastery-based progression and measured-learning funding have strong evidence; systems resist because measurement exposes failure.	Independent published assessment per school; automatic intensive support at the age-9 reading trigger; teacher recruitment from top graduate quartiles with matching pay.	Why 1 — systems measure attendance, not learning; Enrolment and graduation are countable; learning is not, so it is not managed. -> Fix: Independent annual learning assessments published per school; funding follows measured gains. | Why 2 — teachers are undervalued: Pay and status lose the competition for strong graduates. -> Fix: Top-quartile recruitment with matching pay and professional autonomy; teaching as a selective, honoured profession. | Why 3 — knowledge is centralized: Single publishers, platforms, and archives are single points of failure. -> Fix: Redundancy law for knowledge: 3 copies, 3 media, 3 jurisdictions; open-licence core curricula. | Why 4 — verification is untaught: Curricula deliver content while the environment demands checking. -> Fix: Verification (source-tracing, estimation, consistency) as a tested core skill at every level. | Why 5 — root: Education is run as a sorting machine for employers rather than as the formation of capable citizens. -> Fix: Dual charter in law: every education system reports both employability and unaided competence/civic capability, and is funded on both.
113	Informational	Education & Knowledge	Tier 2	Now	Society / National	Schooling without learning: credentials inflate while actual literacy and numeracy stagnate or fall; misinformation outruns curricula.	Large shares of 10-year-olds in many countries cannot read a simple text, even after years enrolled in school; graduates report their degrees did not teach the skills their jobs require.	Measure learning, not attendance: independent annual assessments of reading and numeracy published per school, funding following measured learning gains, curricula rebuilt around mastery progression (no social promotion without support), and teacher pay and status raised to compete with other professions (top-quartile graduate recruitment). Specification: every child reads fluently by age 9 or triggers automatic intensive support; adult reskilling accounts (funded per citizen) redeemable at accredited providers, including libraries.	70% (base range 62–78%)	Mastery-based progression and measured-learning funding have strong evidence; systems resist because measurement exposes failure.	Independent published assessment per school; automatic intensive support at the age-9 reading trigger; teacher recruitment from top graduate quartiles with matching pay.	Why 1 — systems measure attendance, not learning; Enrolment and graduation are countable; learning is not, so it is not managed. -> Fix: Independent annual learning assessments published per school; funding follows measured gains. | Why 2 — teachers are undervalued: Pay and status lose the competition for strong graduates. -> Fix: Top-quartile recruitment with matching pay and professional autonomy; teaching as a selective, honoured profession. | Why 3 — knowledge is centralized: Single publishers, platforms, and archives are single points of failure. -> Fix: Redundancy law for knowledge: 3 copies, 3 media, 3 jurisdictions; open-licence core curricula. | Why 4 — verification is untaught: Curricula deliver content while the environment demands checking. -> Fix: Verification (source-tracing, estimation, consistency) as a tested core skill at every level. | Why 5 — root: Education is run as a sorting machine for employers rather than as the formation of capable citizens. -> Fix: Dual charter in law: every education system reports both employability and unaided competence/civic capability, and is funded on both.
114	Informational	Education & Knowledge	Tier 2	Now	Global / Civilizational	Schooling without learning: credentials inflate while actual literacy and numeracy stagnate or fall; misinformation outruns curricula.	Large shares of 10-year-olds in many countries cannot read a simple text, even after years enrolled in school; graduates report their degrees did not teach the skills their jobs require.	Measure learning, not attendance: independent annual assessments of reading and numeracy published per school, funding following measured learning gains, curricula rebuilt around mastery progression (no social promotion without support), and teacher pay and status raised to compete with other professions (top-quartile graduate recruitment). Specification: every child reads fluently by age 9 or triggers automatic intensive support; adult reskilling accounts (funded per citizen) redeemable at accredited providers, including libraries.	58% (base range 62–78%)	Mastery-based progression and measured-learning funding have strong evidence; systems resist because measurement exposes failure.	Independent published assessment per school; automatic intensive support at the age-9 reading trigger; teacher recruitment from top graduate quartiles with matching pay.	Why 1 — systems measure attendance, not learning; Enrolment and graduation are countable; learning is not, so it is not managed. -> Fix: Independent annual learning assessments published per school; funding follows measured gains. | Why 2 — teachers are undervalued: Pay and status lose the competition for strong graduates. -> Fix: Top-quartile recruitment with matching pay and professional autonomy; teaching as a selective, honoured profession. | Why 3 — knowledge is centralized: Single publishers, platforms, and archives are single points of failure. -> Fix: Redundancy law for knowledge: 3 copies, 3 media, 3 jurisdictions; open-licence core curricula. | Why 4 — verification is untaught: Curricula deliver content while the environment demands checking. -> Fix: Verification (source-tracing, estimation, consistency) as a tested core skill at every level. | Why 5 — root: Education is run as a sorting machine for employers rather than as the formation of capable citizens. -> Fix: Dual charter in law: every education system reports both employability and unaided competence/civic capability, and is funded on both.
115	Informational	Education & Knowledge	Tier 2	Future	Household / Personal	AI tutors and synthetic content make human knowledge optional; populations lose the ability to verify, calculate, or reason unaided.	A generation that has never written, navigated, or calculated without machine assistance cannot detect machine error — or function when systems fail.	Dual-competence standard: every qualification requires demonstrated unaided performance (closed-book, offline, by hand) plus aided performance; curricula teach verification as the core skill — source-tracing, estimation, consistency checking; national knowledge reserves (printed, distributed) maintained as civil infrastructure. Specification: critical professions (medicine, engineering, aviation, law) license on unaided competence first; schools schedule regular no-device weeks, not as punishment but as training.	67% (base range 50–68%)	Unaided-competence licensing is fully within institutional control; success depends on examining bodies holding the line against convenience.	License critical professions on unaided competence first; regular no-device training weeks; national printed knowledge reserves maintained as infrastructure.	Why 1 — systems measure attendance, not learning; Enrolment and graduation are countable; learning is not, so it is not managed. -> Fix: Independent annual learning assessments published per school; funding follows measured gains. | Why 2 — teachers are undervalued: Pay and status lose the competition for strong graduates. -> Fix: Top-quartile recruitment with matching pay and professional autonomy; teaching as a selective, honoured profession. | Why 3 — knowledge is centralized: Single publishers, platforms, and archives are single points of failure. -> Fix: Redundancy law for knowledge: 3 copies, 3 media, 3 jurisdictions; open-licence core curricula. | Why 4 — verification is untaught: Curricula deliver content while the environment demands checking. -> Fix: Verification (source-tracing, estimation, consistency) as a tested core skill at every level. | Why 5 — root: Education is run as a sorting machine for employers rather than as the formation of capable citizens. -> Fix: Dual charter in law: every education system reports both employability and unaided competence/civic capability, and is funded on both.
116	Informational	Education & Knowledge	Tier 2	Future	Society / National	AI tutors and synthetic content make human knowledge optional; populations lose the ability to verify, calculate, or reason unaided.	A generation that has never written, navigated, or calculated without machine assistance cannot detect machine error — or function when systems fail.	Dual-competence standard: every qualification requires demonstrated unaided performance (closed-book, offline, by hand) plus aided performance; curricula teach verification as the core skill — source-tracing, estimation, consistency checking; national knowledge reserves (printed, distributed) maintained as civil infrastructure. Specification: critical professions (medicine, engineering, aviation, law) license on unaided competence first; schools schedule regular no-device weeks, not as punishment but as training.	59% (base range 50–68%)	Unaided-competence licensing is fully within institutional control; success depends on examining bodies holding the line against convenience.	License critical professions on unaided competence first; regular no-device training weeks; national printed knowledge reserves maintained as infrastructure.	Why 1 — systems measure attendance, not learning; Enrolment and graduation are countable; learning is not, so it is not managed. -> Fix: Independent annual learning assessments published per school; funding follows measured gains. | Why 2 — teachers are undervalued: Pay and status lose the competition for strong graduates. -> Fix: Top-quartile recruitment with matching pay and professional autonomy; teaching as a selective, honoured profession. | Why 3 — knowledge is centralized: Single publishers, platforms, and archives are single points of failure. -> Fix: Redundancy law for knowledge: 3 copies, 3 media, 3 jurisdictions; open-licence core curricula. | Why 4 — verification is untaught: Curricula deliver content while the environment demands checking. -> Fix: Verification (source-tracing, estimation, consistency) as a tested core skill at every level. | Why 5 — root: Education is run as a sorting machine for employers rather than as the formation of capable citizens. -> Fix: Dual charter in law: every education system reports both employability and unaided competence/civic capability, and is funded on both.
117	Informational	Education & Knowledge	Tier 2	Future	Global / Civilizational	AI tutors and synthetic content make human knowledge optional; populations lose the ability to verify, calculate, or reason unaided.	A generation that has never written, navigated, or calculated without machine assistance cannot detect machine error — or function when systems fail.	Dual-competence standard: every qualification requires demonstrated unaided performance (closed-book, offline, by hand) plus aided performance; curricula teach verification as the core skill — source-tracing, estimation, consistency checking; national knowledge reserves (printed, distributed) maintained as civil infrastructure. Specification: critical professions (medicine, engineering, aviation, law) license on unaided competence first; schools schedule regular no-device weeks, not as punishment but as training.	47% (base range 50–68%)	Unaided-competence licensing is fully within institutional control; success depends on examining bodies holding the line against convenience.	License critical professions on unaided competence first; regular no-device training weeks; national printed knowledge reserves maintained as infrastructure.	Why 1 — systems measure attendance, not learning; Enrolment and graduation are countable; learning is not, so it is not managed. -> Fix: Independent annual learning assessments published per school; funding follows measured gains. | Why 2 — teachers are undervalued: Pay and status lose the competition for strong graduates. -> Fix: Top-quartile recruitment with matching pay and professional autonomy; teaching as a selective, honoured profession. | Why 3 — knowledge is centralized: Single publishers, platforms, and archives are single points of failure. -> Fix: Redundancy law for knowledge: 3 copies, 3 media, 3 jurisdictions; open-licence core curricula. | Why 4 — verification is untaught: Curricula deliver content while the environment demands checking. -> Fix: Verification (source-tracing, estimation, consistency) as a tested core skill at every level. | Why 5 — root: Education is run as a sorting machine for employers rather than as the formation of capable citizens. -> Fix: Dual charter in law: every education system reports both employability and unaided competence/civic capability, and is funded on both.
118	Social / Order	Work & Livelihood	Tier 2	Past	Household / Personal	Livelihood systems collapse into serfdom, slavery, or company-town dependence: one master controls work, housing, food, and law.	Company towns paid in scrip redeemable only at the company store; workers owed their souls and their rent to the same ledger.	Separate the levers by law: wages in legal tender only, housing and retail independent of employers, freedom of movement and association non-waivable, and portable social insurance attached to the person. Specification: no employer may provide more than 2 of {job, housing, food retail, credit} to the same worker; labour inspection funded per worker covered, with unannounced inspection rights.	85% (base range 69–84%)	Separating wages, housing, and credit by law ended company-town capture wherever enforced; scrip systems did not survive legal tender rules.	Enforce the 2-of-4 lever cap (job/housing/food/credit); fund labour inspection per worker; portable insurance attached to the person.	Why 1 — protections attach to job titles: Benefits were designed for one employer, one career, one contract type. -> Fix: Attach all social insurance to the person, pro-rated per hour worked in any form. | Why 2 — employers arbitrage the categories: Misclassification is cheaper than compliance where penalties are rare. -> Fix: Per-task standards with automatic penalties and joint liability up the contracting chain. | Why 3 — regions depend on single employers: Local economies were allowed to become monocultures. -> Fix: Automatic diversification triggers at the 20% threshold, with funds released by formula. | Why 4 — transitions are unfunded: Retraining is offered after displacement, from general revenue, too late. -> Fix: Transition accounts funded per worker during employment, from automation and payroll levies, before displacement. | Why 5 — root: Income is tied exclusively to selling labour in a world where labour's share is structurally falling. -> Fix: Diversify the income base by design: ownership shares, citizen funds, and contribution credits alongside wages.

#	Dimension of Realit	Field of Discipline	Pareto Tier o	Timefram	Scale	Worst Possible Problem	Example	Solution — Full Specification	Probability of Su	Why This Probability	How to Maximize the Chances (detailed)	Root Causes
119	Social / Order	Work & Livelihood	Tier 2	Past	Society / National	Livelihood systems collapse into serfdom, slavery, or company-town dependence: one master controls work, housing, food, and law.	Company towns paid in scrip redeemable only at the company store; workers owed their souls and their rent to the same ledger.	Separate the levers by law: wages in legal tender only, housing and retail independent of employers, freedom of movement and association non-waivable, and portable social insurance attached to the person. Specification: no employer may provide more than 2 of {job, housing, food retail, credit} to the same worker; labour inspection funded per worker covered, with unannounced inspection rights.	77% (base range 69–84%)	Separating wages, housing, and credit by law ended company-town capture wherever enforced; scrip systems did not survive legal tender rules.	Enforce the 2-of-4 lever cap (job/housing/food/credit); fund labour inspection per worker; portable insurance attached to the person.	Why 1 — protections attach to job titles: Benefits were designed for one employer, one career, one contract type. -> Fix: Attach all social insurance to the person, pro-rated per hour worked in any form.	Why 2 — employers arbitrage the categories: Misclassification is cheaper than compliance where penalties are rare. -> Fix: Per-task standards with automatic penalties and joint liability up the contracting chain.	Why 3 — regions depend on single employers: Local economies were allowed to become monocultures. -> Fix: Automatic diversification triggers at the 20% threshold, with funds released by formula.	Why 4 — transitions are unfunded: Retraining is offered after displacement, from general revenue, too late. -> Fix: Transition accounts funded per worker during employment, from automation and payroll levies, before displacement.	Why 5 — root: Income is tied exclusively to selling labour in a world where labour's share is structurally falling. -> Fix: Diversify the income base by design: ownership shares, citizen funds, and contribution credits alongside wages.
120	Social / Order	Work & Livelihood	Tier 2	Past	Global / Civilizational	Livelihood systems collapse into serfdom, slavery, or company-town dependence: one master controls work, housing, food, and law.	Company towns paid in scrip redeemable only at the company store; workers owed their souls and their rent to the same ledger.	Separate the levers by law: wages in legal tender only, housing and retail independent of employers, freedom of movement and association non-waivable, and portable social insurance attached to the person. Specification: no employer may provide more than 2 of {job, housing, food retail, credit} to the same worker; labour inspection funded per worker covered, with unannounced inspection rights.	65% (base range 69–84%)	Separating wages, housing, and credit by law ended company-town capture wherever enforced; scrip systems did not survive legal tender rules.	Enforce the 2-of-4 lever cap (job/housing/food/credit); fund labour inspection per worker; portable insurance attached to the person.	Why 1 — protections attach to job titles: Benefits were designed for one employer, one career, one contract type. -> Fix: Attach all social insurance to the person, pro-rated per hour worked in any form.	Why 2 — employers arbitrage the categories: Misclassification is cheaper than compliance where penalties are rare. -> Fix: Per-task standards with automatic penalties and joint liability up the contracting chain.	Why 3 — regions depend on single employers: Local economies were allowed to become monocultures. -> Fix: Automatic diversification triggers at the 20% threshold, with funds released by formula.	Why 4 — transitions are unfunded: Retraining is offered after displacement, from general revenue, too late. -> Fix: Transition accounts funded per worker during employment, from automation and payroll levies, before displacement.	Why 5 — root: Income is tied exclusively to selling labour in a world where labour's share is structurally falling. -> Fix: Diversify the income base by design: ownership shares, citizen funds, and contribution credits alongside wages.
121	Social / Order	Work & Livelihood	Tier 2	Now	Household / Personal	Precarious gig work, credential inflation, and regional job deserts leave large populations one missed paycheque from crisis.	Delivery and care workers classified as contractors without benefits, working full-time hours in poverty; towns that lose one plant lose their whole economy.	Floor under every form of work: benefits (health, pension, injury, unemployment) pro-rated per hour worked regardless of classification, minimum standards applied per task not per contract type, regional diversification funds that trigger automatically when one employer exceeds 20% of local jobs, and wage insurance that tops up displaced workers at 50% of the wage gap for 2 years while retraining. Specification: personal standard — an emergency fund of 3 months, skills documented in portable form, and no single income source treated as permanent.	74% (base range 58–74%)	Per-hour benefit floors are administratively proven in several systems; classification games shrink when benefits attach to hours, not contracts.	Pro-rate benefits per hour across all contract types; automatic regional diversification triggers at the 20% single-employer threshold; personal 3-month reserve as the household standard.	Why 1 — protections attach to job titles: Benefits were designed for one employer, one career, one contract type. -> Fix: Attach all social insurance to the person, pro-rated per hour worked in any form.	Why 2 — employers arbitrage the categories: Misclassification is cheaper than compliance where penalties are rare. -> Fix: Per-task standards with automatic penalties and joint liability up the contracting chain.	Why 3 — regions depend on single employers: Local economies were allowed to become monocultures. -> Fix: Automatic diversification triggers at the 20% threshold, with funds released by formula.	Why 4 — transitions are unfunded: Retraining is offered after displacement, from general revenue, too late. -> Fix: Transition accounts funded per worker during employment, from automation and payroll levies, before displacement.	Why 5 — root: Income is tied exclusively to selling labour in a world where labour's share is structurally falling. -> Fix: Diversify the income base by design: ownership shares, citizen funds, and contribution credits alongside wages.
122	Social / Order	Work & Livelihood	Tier 2	Now	Society / National	Precarious gig work, credential inflation, and regional job deserts leave large populations one missed paycheque from crisis.	Delivery and care workers classified as contractors without benefits, working full-time hours in poverty; towns that lose one plant lose their whole economy.	Floor under every form of work: benefits (health, pension, injury, unemployment) pro-rated per hour worked regardless of classification, minimum standards applied per task not per contract type, regional diversification funds that trigger automatically when one employer exceeds 20% of local jobs, and wage insurance that tops up displaced workers at 50% of the wage gap for 2 years while retraining. Specification: personal standard — an emergency fund of 3 months, skills documented in portable form, and no single income source treated as permanent.	66% (base range 58–74%)	Per-hour benefit floors are administratively proven in several systems; classification games shrink when benefits attach to hours, not contracts.	Pro-rate benefits per hour across all contract types; automatic regional diversification triggers at the 20% single-employer threshold; personal 3-month reserve as the household standard.	Why 1 — protections attach to job titles: Benefits were designed for one employer, one career, one contract type. -> Fix: Attach all social insurance to the person, pro-rated per hour worked in any form.	Why 2 — employers arbitrage the categories: Misclassification is cheaper than compliance where penalties are rare. -> Fix: Per-task standards with automatic penalties and joint liability up the contracting chain.	Why 3 — regions depend on single employers: Local economies were allowed to become monocultures. -> Fix: Automatic diversification triggers at the 20% threshold, with funds released by formula.	Why 4 — transitions are unfunded: Retraining is offered after displacement, from general revenue, too late. -> Fix: Transition accounts funded per worker during employment, from automation and payroll levies, before displacement.	Why 5 — root: Income is tied exclusively to selling labour in a world where labour's share is structurally falling. -> Fix: Diversify the income base by design: ownership shares, citizen funds, and contribution credits alongside wages.
123	Social / Order	Work & Livelihood	Tier 2	Now	Global / Civilizational	Precarious gig work, credential inflation, and regional job deserts leave large populations one missed paycheque from crisis.	Delivery and care workers classified as contractors without benefits, working full-time hours in poverty; towns that lose one plant lose their whole economy.	Floor under every form of work: benefits (health, pension, injury, unemployment) pro-rated per hour worked regardless of classification, minimum standards applied per task not per contract type, regional diversification funds that trigger automatically when one employer exceeds 20% of local jobs, and wage insurance that tops up displaced workers at 50% of the wage gap for 2 years while retraining. Specification: personal standard — an emergency fund of 3 months, skills documented in portable form, and no single income source treated as permanent.	54% (base range 58–74%)	Per-hour benefit floors are administratively proven in several systems; classification games shrink when benefits attach to hours, not contracts.	Pro-rate benefits per hour across all contract types; automatic regional diversification triggers at the 20% single-employer threshold; personal 3-month reserve as the household standard.	Why 1 — protections attach to job titles: Benefits were designed for one employer, one career, one contract type. -> Fix: Attach all social insurance to the person, pro-rated per hour worked in any form.	Why 2 — employers arbitrage the categories: Misclassification is cheaper than compliance where penalties are rare. -> Fix: Per-task standards with automatic penalties and joint liability up the contracting chain.	Why 3 — regions depend on single employers: Local economies were allowed to become monocultures. -> Fix: Automatic diversification triggers at the 20% threshold, with funds released by formula.	Why 4 — transitions are unfunded: Retraining is offered after displacement, from general revenue, too late. -> Fix: Transition accounts funded per worker during employment, from automation and payroll levies, before displacement.	Why 5 — root: Income is tied exclusively to selling labour in a world where labour's share is structurally falling. -> Fix: Diversify the income base by design: ownership shares, citizen funds, and contribution credits alongside wages.
124	Social / Order	Work & Livelihood	Tier 2	Future	Household / Personal	Machines take the majority of paid tasks; the link between contribution and income breaks entirely.	If AI and robotics perform most current jobs at near-zero marginal cost, wages as the main distribution mechanism fail for most of the population simultaneously.	Rebuild distribution on ownership: broad-based capital ownership (citizen funds, employee ownership trusts, data and AI dividends), radically shortened standard working time as productivity rises, and public valuation of unpaid contribution (care, community, restoration) through contribution credits convertible to services. Specification: transition is phased by measured displacement rates per sector, with the ownership mechanisms built during the growth phase — financed from automation rents — not improvised after collapse.	51% (base range 33–53%)	Ownership-based distribution is the only mechanism that scales with full automation, but it must be capitalized during the growth phase; late construction is the main failure mode.	Start citizen funds now from automation rents; convert large employers progressively to shared-ownership forms; phase hour reductions by measured sector displacement.	Why 1 — protections attach to job titles: Benefits were designed for one employer, one career, one contract type. -> Fix: Attach all social insurance to the person, pro-rated per hour worked in any form.	Why 2 — employers arbitrage the categories: Misclassification is cheaper than compliance where penalties are rare. -> Fix: Per-task standards with automatic penalties and joint liability up the contracting chain.	Why 3 — regions depend on single employers: Local economies were allowed to become monocultures. -> Fix: Automatic diversification triggers at the 20% threshold, with funds released by formula.	Why 4 — transitions are unfunded: Retraining is offered after displacement, from general revenue, too late. -> Fix: Transition accounts funded per worker during employment, from automation and payroll levies, before displacement.	Why 5 — root: Income is tied exclusively to selling labour in a world where labour's share is structurally falling. -> Fix: Diversify the income base by design: ownership shares, citizen funds, and contribution credits alongside wages.
125	Social / Order	Work & Livelihood	Tier 2	Future	Society / National	Machines take the majority of paid tasks; the link between contribution and income breaks entirely.	If AI and robotics perform most current jobs at near-zero marginal cost, wages as the main distribution mechanism fail for most of the population simultaneously.	Rebuild distribution on ownership: broad-based capital ownership (citizen funds, employee ownership trusts, data and AI dividends), radically shortened standard working time as productivity rises, and public valuation of unpaid contribution (care, community, restoration) through contribution credits convertible to services. Specification: transition is phased by measured displacement rates per sector, with the ownership mechanisms built during the growth phase — financed from automation rents — not improvised after collapse.	43% (base range 33–53%)	Ownership-based distribution is the only mechanism that scales with full automation, but it must be capitalized during the growth phase; late construction is the main failure mode.	Start citizen funds now from automation rents; convert large employers progressively to shared-ownership forms; phase hour reductions by measured sector displacement.	Why 1 — protections attach to job titles: Benefits were designed for one employer, one career, one contract type. -> Fix: Attach all social insurance to the person, pro-rated per hour worked in any form.	Why 2 — employers arbitrage the categories: Misclassification is cheaper than compliance where penalties are rare. -> Fix: Per-task standards with automatic penalties and joint liability up the contracting chain.	Why 3 — regions depend on single employers: Local economies were allowed to become monocultures. -> Fix: Automatic diversification triggers at the 20% threshold, with funds released by formula.	Why 4 — transitions are unfunded: Retraining is offered after displacement, from general revenue, too late. -> Fix: Transition accounts funded per worker during employment, from automation and payroll levies, before displacement.	Why 5 — root: Income is tied exclusively to selling labour in a world where labour's share is structurally falling. -> Fix: Diversify the income base by design: ownership shares, citizen funds, and contribution credits alongside wages.
126	Social / Order	Work & Livelihood	Tier 2	Future	Global / Civilizational	Machines take the majority of paid tasks; the link between contribution and income breaks entirely.	If AI and robotics perform most current jobs at near-zero marginal cost, wages as the main distribution mechanism fail for most of the population simultaneously.	Rebuild distribution on ownership: broad-based capital ownership (citizen funds, employee ownership trusts, data and AI dividends), radically shortened standard working time as productivity rises, and public valuation of unpaid contribution (care, community, restoration) through contribution credits convertible to services. Specification: transition is phased by measured displacement rates per sector, with the ownership mechanisms built during the growth phase — financed from automation rents — not improvised after collapse.	31% (base range 33–53%)	Ownership-based distribution is the only mechanism that scales with full automation, but it must be capitalized during the growth phase; late construction is the main failure mode.	Start citizen funds now from automation rents; convert large employers progressively to shared-ownership forms; phase hour reductions by measured sector displacement.	Why 1 — protections attach to job titles: Benefits were designed for one employer, one career, one contract type. -> Fix: Attach all social insurance to the person, pro-rated per hour worked in any form.	Why 2 — employers arbitrage the categories: Misclassification is cheaper than compliance where penalties are rare. -> Fix: Per-task standards with automatic penalties and joint liability up the contracting chain.	Why 3 — regions depend on single employers: Local economies were allowed to become monocultures. -> Fix: Automatic diversification triggers at the 20% threshold, with funds released by formula.	Why 4 — transitions are unfunded: Retraining is offered after displacement, from general revenue, too late. -> Fix: Transition accounts funded per worker during employment, from automation and payroll levies, before displacement.	Why 5 — root: Income is tied exclusively to selling labour in a world where labour's share is structurally falling. -> Fix: Diversify the income base by design: ownership shares, citizen funds, and contribution credits alongside wages.
127	Physical / Material	Transport & Mobility	Tier 2	Past	Household / Personal	Transport failure starves cities: pre-industrial capitals depended on daily food convoys that war, weather, or blockade could sever in days.	Sieges from Leningrad to Sarajevo: when roads and rails close, urban populations face starvation within weeks regardless of money.	No city on a single thread: minimum two independent supply modes and routes for food and fuel (road + rail, or road + water), urban food reserves of 30 days, and local production floors for fresh essentials. Specification: critical bridges, tunnels, and ports have pre-planned bypass capacity of at least 60% of normal flow; households in isolated regions keep 30 days of staples as the standing norm.	87% (base range 71–86%)	Dual-route supply is classical logistics doctrine; besieged cities with two functioning routes survived, single-route cities starved.	Two independent modes for food and fuel per city; 30-day urban food reserves; bypass capacity of 60% pre-planned for critical links.	Why 1 — single threads everywhere: Cost optimization removed parallel routes and modes. -> Fix: Modal and route caps (40%) for essential flows, written into infrastructure funding conditions.	Why 2 — sprawl locks in fragility: Land use separated homes from needs by design for seventy years. -> Fix: The 15-minute standard for all new development; retrofit priority to districts with no walkable essentials.	Why 3 — fleets homogenize: One vendor, one software stack, one battery chemistry per fleet. -> Fix: Certification requires stack diversity and manual fallback; recycling law recovering 90%+ of critical minerals.	Why 4 — fallbacks are never drilled: Manual procedures exist on paper only. -> Fix: Annual no-network drill days with published results for transit, logistics, and emergency services.	Why 5 — root: Transport is optimized for cost per trip in normal times, not for continuity in bad times. -> Fix: Continuity standards in transport law: every essential flow must demonstrate a degraded-mode operation, tested, before funding.
128	Physical / Material	Transport & Mobility	Tier 2	Past	Society / National	Transport failure starves cities: pre-industrial capitals depended on daily food convoys that war, weather, or blockade could sever in days.	Sieges from Leningrad to Sarajevo: when roads and rails close, urban populations face starvation within weeks regardless of money.	No city on a single thread: minimum two independent supply modes and routes for food and fuel (road + rail, or road + water), urban food reserves of 30 days, and local production floors for fresh essentials. Specification: critical bridges, tunnels, and ports have pre-planned bypass capacity of at least 60% of normal flow; households in isolated regions keep 30 days of staples as the standing norm.	79% (base range 71–86%)	Dual-route supply is classical logistics doctrine; besieged cities with two functioning routes survived, single-route cities starved.	Two independent modes for food and fuel per city; 30-day urban food reserves; bypass capacity of 60% pre-planned for critical links.	Why 1 — single threads everywhere: Cost optimization removed parallel routes and modes. -> Fix: Modal and route caps (40%) for essential flows, written into infrastructure funding conditions.	Why 2 — sprawl locks in fragility: Land use separated homes from needs by design for seventy years. -> Fix: The 15-minute standard for all new development; retrofit priority to districts with no walkable essentials.	Why 3 — fleets homogenize: One vendor, one software stack, one battery chemistry per fleet. -> Fix: Certification requires stack diversity and manual fallback; recycling law recovering 90%+ of critical minerals.	Why 4 — fallbacks are never drilled: Manual procedures exist on paper only. -> Fix: Annual no-network drill days with published results for transit, logistics, and emergency services.	Why 5 — root: Transport is optimized for cost per trip in normal times, not for continuity in bad times. -> Fix: Continuity standards in transport law: every essential flow must demonstrate a degraded-mode operation, tested, before funding.

#	Dimension of Realit	Field of Discipline	Pareto Tier o	Timefram	Scale	Worst Possible Problem	Example	Solution — Full Specification	Probability of Su	Why This Probability	How to Maximize the Chances (detailed)	Root Causes
129	Physical / Material	Transport & Mobility	Tier 2	Past	Global / Civilizational	Transport failure starves cities: pre-industrial capitals depended on daily food convoys that war, weather, or blockade could sever in days.	Sieges from Leningrad to Sarajevo: when roads and rails close, urban populations face starvation within weeks regardless of money.	No city on a single thread: minimum two independent supply modes and routes for food and fuel (road + rail, or road + water), urban food reserves of 30 days, and local production floors for fresh essentials. Specification: critical bridges, tunnels, and ports have pre-planned bypass capacity of at least 60% of normal flow; households in isolated regions keep 30 days of staples as the standing norm.	67% (base range 71–86%)	Dual-route supply is classical logistics doctrine; besieged cities with two functioning routes survived, single-route cities starved.	Two independent modes for food and fuel per city; 30-day urban food reserves; bypass capacity of 60% pre-planned for critical links.	Why 1 — single threads everywhere: Cost optimization removed parallel routes and modes. -> Fix: Modal and route caps (40%) for essential flows, written into infrastructure funding conditions. | Why 2 — sprawl locks in fragility: Land use separated homes from needs by design for seventy years. -> Fix: The 15-minute standard for all new development; retrofit priority to districts with no walkable essentials. | Why 3 — fleets homogenize: One vendor, one software stack, one battery chemistry per fleet. -> Fix: Certification requires stack diversity and manual fallback; recycling law recovering 90%+ of critical minerals. | Why 4 — fallbacks are never drilled: Manual procedures exist on paper only. -> Fix: Annual no-network drill days with published results for transit, logistics, and emergency services. | Why 5 — root: Transport is optimized for cost per trip in normal times, not for continuity in bad times. -> Fix: Continuity standards in transport law: every essential flow must demonstrate a degraded-mode operation, tested, before funding.
130	Physical / Material	Transport & Mobility	Tier 2	Now	Household / Personal	Car-dependent sprawl plus fuel and parts chokepoints make daily life brittle; one closure or price spike strands millions.	A single blocked canal halted 12% of world trade; domestically, households without cars in sprawled regions cannot reach work, food, or care at all.	Redundancy and proximity: freight modal split so no single route or mode carries more than 40% of essentials; the 15-minute standard for new development (daily needs reachable on foot or by transit); transit networks funded as essential infrastructure with operating reserves of 6 months; strategic parts and fuel reserves for public fleets. Specification: every settlement over 5,000 has at least two physically separate road connections or one road plus scheduled public transport that runs in emergencies.	76% (base range 60–76%)	Modal-split caps and proximity standards measurably reduce fragility; sprawl already built is the slow constraint.	Cap any single mode/route at 40% of essentials; apply the 15-minute standard to all new development; 6-month operating reserves for transit.	Why 1 — single threads everywhere: Cost optimization removed parallel routes and modes. -> Fix: Modal and route caps (40%) for essential flows, written into infrastructure funding conditions. | Why 2 — sprawl locks in fragility: Land use separated homes from needs by design for seventy years. -> Fix: The 15-minute standard for all new development; retrofit priority to districts with no walkable essentials. | Why 3 — fleets homogenize: One vendor, one software stack, one battery chemistry per fleet. -> Fix: Certification requires stack diversity and manual fallback; recycling law recovering 90%+ of critical minerals. | Why 4 — fallbacks are never drilled: Manual procedures exist on paper only. -> Fix: Annual no-network drill days with published results for transit, logistics, and emergency services. | Why 5 — root: Transport is optimized for cost per trip in normal times, not for continuity in bad times. -> Fix: Continuity standards in transport law: every essential flow must demonstrate a degraded-mode operation, tested, before funding.
131	Physical / Material	Transport & Mobility	Tier 2	Now	Society / National	Car-dependent sprawl plus fuel and parts chokepoints make daily life brittle; one closure or price spike strands millions.	A single blocked canal halted 12% of world trade; domestically, households without cars in sprawled regions cannot reach work, food, or care at all.	Redundancy and proximity: freight modal split so no single route or mode carries more than 40% of essentials; the 15-minute standard for new development (daily needs reachable on foot or by transit); transit networks funded as essential infrastructure with operating reserves of 6 months; strategic parts and fuel reserves for public fleets. Specification: every settlement over 5,000 has at least two physically separate road connections or one road plus scheduled public transport that runs in emergencies.	68% (base range 60–76%)	Modal-split caps and proximity standards measurably reduce fragility; sprawl already built is the slow constraint.	Cap any single mode/route at 40% of essentials; apply the 15-minute standard to all new development; 6-month operating reserves for transit.	Why 1 — single threads everywhere: Cost optimization removed parallel routes and modes. -> Fix: Modal and route caps (40%) for essential flows, written into infrastructure funding conditions. | Why 2 — sprawl locks in fragility: Land use separated homes from needs by design for seventy years. -> Fix: The 15-minute standard for all new development; retrofit priority to districts with no walkable essentials. | Why 3 — fleets homogenize: One vendor, one software stack, one battery chemistry per fleet. -> Fix: Certification requires stack diversity and manual fallback; recycling law recovering 90%+ of critical minerals. | Why 4 — fallbacks are never drilled: Manual procedures exist on paper only. -> Fix: Annual no-network drill days with published results for transit, logistics, and emergency services. | Why 5 — root: Transport is optimized for cost per trip in normal times, not for continuity in bad times. -> Fix: Continuity standards in transport law: every essential flow must demonstrate a degraded-mode operation, tested, before funding.
132	Physical / Material	Transport & Mobility	Tier 2	Now	Global / Civilizational	Car-dependent sprawl plus fuel and parts chokepoints make daily life brittle; one closure or price spike strands millions.	A single blocked canal halted 12% of world trade; domestically, households without cars in sprawled regions cannot reach work, food, or care at all.	Redundancy and proximity: freight modal split so no single route or mode carries more than 40% of essentials; the 15-minute standard for new development (daily needs reachable on foot or by transit); transit networks funded as essential infrastructure with operating reserves of 6 months; strategic parts and fuel reserves for public fleets. Specification: every settlement over 5,000 has at least two physically separate road connections or one road plus scheduled public transport that runs in emergencies.	56% (base range 60–76%)	Modal-split caps and proximity standards measurably reduce fragility; sprawl already built is the slow constraint.	Cap any single mode/route at 40% of essentials; apply the 15-minute standard to all new development; 6-month operating reserves for transit.	Why 1 — single threads everywhere: Cost optimization removed parallel routes and modes. -> Fix: Modal and route caps (40%) for essential flows, written into infrastructure funding conditions. | Why 2 — sprawl locks in fragility: Land use separated homes from needs by design for seventy years. -> Fix: The 15-minute standard for all new development; retrofit priority to districts with no walkable essentials. | Why 3 — fleets homogenize: One vendor, one software stack, one battery chemistry per fleet. -> Fix: Certification requires stack diversity and manual fallback; recycling law recovering 90%+ of critical minerals. | Why 4 — fallbacks are never drilled: Manual procedures exist on paper only. -> Fix: Annual no-network drill days with published results for transit, logistics, and emergency services. | Why 5 — root: Transport is optimized for cost per trip in normal times, not for continuity in bad times. -> Fix: Continuity standards in transport law: every essential flow must demonstrate a degraded-mode operation, tested, before funding.
133	Physical / Material	Transport & Mobility	Tier 2	Future	Household / Personal	Autonomous, electric, platform-controlled mobility concentrates failure: one software fault, cyberattack, or battery-material cutoff grounds a continent.	A corrupted fleet-wide update or a compromised dispatch platform immobilizes autonomous fleets mid-journey, everywhere, at the same hour.	Mandated heterogeneity and manual fallback: fleets certified from at least 2 independent software stacks per region, all autonomous vehicles retaining manual drive capability, transport control systems segmented from public networks with analogue scheduling fallback, and battery-material supply diversified with recycling recovering 90%+ of critical minerals by law. Specification: cities drill a 'no-network day' annually — transit, logistics, and emergency services operating on fallback procedures.	62% (base range 45–63%)	Heterogeneity and manual fallback are cheap relative to fleet value; success depends on certifying them before fleets homogenize.	Two independent software stacks per region by certification rule; manual drive retained in every autonomous vehicle; annual no-network drill day.	Why 1 — single threads everywhere: Cost optimization removed parallel routes and modes. -> Fix: Modal and route caps (40%) for essential flows, written into infrastructure funding conditions. | Why 2 — sprawl locks in fragility: Land use separated homes from needs by design for seventy years. -> Fix: The 15-minute standard for all new development; retrofit priority to districts with no walkable essentials. | Why 3 — fleets homogenize: One vendor, one software stack, one battery chemistry per fleet. -> Fix: Certification requires stack diversity and manual fallback; recycling law recovering 90%+ of critical minerals. | Why 4 — fallbacks are never drilled: Manual procedures exist on paper only. -> Fix: Annual no-network drill days with published results for transit, logistics, and emergency services. | Why 5 — root: Transport is optimized for cost per trip in normal times, not for continuity in bad times. -> Fix: Continuity standards in transport law: every essential flow must demonstrate a degraded-mode operation, tested, before funding.
134	Physical / Material	Transport & Mobility	Tier 2	Future	Society / National	Autonomous, electric, platform-controlled mobility concentrates failure: one software fault, cyberattack, or battery-material cutoff grounds a continent.	A corrupted fleet-wide update or a compromised dispatch platform immobilizes autonomous fleets mid-journey, everywhere, at the same hour.	Mandated heterogeneity and manual fallback: fleets certified from at least 2 independent software stacks per region, all autonomous vehicles retaining manual drive capability, transport control systems segmented from public networks with analogue scheduling fallback, and battery-material supply diversified with recycling recovering 90%+ of critical minerals by law. Specification: cities drill a 'no-network day' annually — transit, logistics, and emergency services operating on fallback procedures.	54% (base range 45–63%)	Heterogeneity and manual fallback are cheap relative to fleet value; success depends on certifying them before fleets homogenize.	Two independent software stacks per region by certification rule; manual drive retained in every autonomous vehicle; annual no-network drill day.	Why 1 — single threads everywhere: Cost optimization removed parallel routes and modes. -> Fix: Modal and route caps (40%) for essential flows, written into infrastructure funding conditions. | Why 2 — sprawl locks in fragility: Land use separated homes from needs by design for seventy years. -> Fix: The 15-minute standard for all new development; retrofit priority to districts with no walkable essentials. | Why 3 — fleets homogenize: One vendor, one software stack, one battery chemistry per fleet. -> Fix: Certification requires stack diversity and manual fallback; recycling law recovering 90%+ of critical minerals. | Why 4 — fallbacks are never drilled: Manual procedures exist on paper only. -> Fix: Annual no-network drill days with published results for transit, logistics, and emergency services. | Why 5 — root: Transport is optimized for cost per trip in normal times, not for continuity in bad times. -> Fix: Continuity standards in transport law: every essential flow must demonstrate a degraded-mode operation, tested, before funding.
135	Physical / Material	Transport & Mobility	Tier 2	Future	Global / Civilizational	Autonomous, electric, platform-controlled mobility concentrates failure: one software fault, cyberattack, or battery-material cutoff grounds a continent.	A corrupted fleet-wide update or a compromised dispatch platform immobilizes autonomous fleets mid-journey, everywhere, at the same hour.	Mandated heterogeneity and manual fallback: fleets certified from at least 2 independent software stacks per region, all autonomous vehicles retaining manual drive capability, transport control systems segmented from public networks with analogue scheduling fallback, and battery-material supply diversified with recycling recovering 90%+ of critical minerals by law. Specification: cities drill a 'no-network day' annually — transit, logistics, and emergency services operating on fallback procedures.	42% (base range 45–63%)	Heterogeneity and manual fallback are cheap relative to fleet value; success depends on certifying them before fleets homogenize.	Two independent software stacks per region by certification rule; manual drive retained in every autonomous vehicle; annual no-network drill day.	Why 1 — single threads everywhere: Cost optimization removed parallel routes and modes. -> Fix: Modal and route caps (40%) for essential flows, written into infrastructure funding conditions. | Why 2 — sprawl locks in fragility: Land use separated homes from needs by design for seventy years. -> Fix: The 15-minute standard for all new development; retrofit priority to districts with no walkable essentials. | Why 3 — fleets homogenize: One vendor, one software stack, one battery chemistry per fleet. -> Fix: Certification requires stack diversity and manual fallback; recycling law recovering 90%+ of critical minerals. | Why 4 — fallbacks are never drilled: Manual procedures exist on paper only. -> Fix: Annual no-network drill days with published results for transit, logistics, and emergency services. | Why 5 — root: Transport is optimized for cost per trip in normal times, not for continuity in bad times. -> Fix: Continuity standards in transport law: every essential flow must demonstrate a degraded-mode operation, tested, before funding.
136	Informational	Communication	Tier 2	Past	Household / Personal	Communication monopoly or collapse isolates populations: empires fell in regions where news took months; rumour governed.	In the 1857 Indian rebellion, cut telegraph lines and rumour cascades moved faster than any verified information, with catastrophic results on all sides.	Multiple independent channels as doctrine: no population dependent on one network, one owner, or one technology for emergency information — broadcast radio, print, courier, and digital maintained in parallel, with amateur and community radio licensed and encouraged as civic infrastructure. Specification: every municipality maintains a battery/solar broadcast capability able to reach all residents within 1 hour without internet or grid power.	83% (base range 67–82%)	Parallel channels historically defeated monopoly and collapse; single-channel populations were governed by rumour.	Maintain broadcast, print, courier, and digital in parallel; municipal emergency broadcast reaching all residents within 1 hour, off-grid capable.	Why 1 — speech moved onto private platforms: Public communication infrastructure was allowed to wither. -> Fix: Maintain postal, broadcast, and municipal networks as real, funded fallbacks — not heritage, but reserve capacity. | Why 2 — exit is costly: Audiences, records, and identities are locked in by design. -> Fix: Portability and interoperability as law for dominant platforms; monthly export as a consumer right. | Why 3 — emergency channels are single-thread: Alerts ride the same networks that fail in disasters. -> Fix: Municipal off-grid broadcast reaching all residents within 1 hour; amateur radio integrated into emergency plans. | Why 4 — privacy and safety are traded falsely: Policy frames encryption and lawful access as absolutes at war. -> Fix: Individualized, judicial, logged, sunsetted access as the legislated middle; blanket access and blanket opacity both prohibited. | Why 5 — root: Communication is treated as a market commodity rather than as civic infrastructure. -> Fix: Charter communication like water and power: universal service duties, resilience standards, and rights protections by design.
137	Informational	Communication	Tier 2	Past	Society / National	Communication monopoly or collapse isolates populations: empires fell in regions where news took months; rumour governed.	In the 1857 Indian rebellion, cut telegraph lines and rumour cascades moved faster than any verified information, with catastrophic results on all sides.	Multiple independent channels as doctrine: no population dependent on one network, one owner, or one technology for emergency information — broadcast radio, print, courier, and digital maintained in parallel, with amateur and community radio licensed and encouraged as civic infrastructure. Specification: every municipality maintains a battery/solar broadcast capability able to reach all residents within 1 hour without internet or grid power.	75% (base range 67–82%)	Parallel channels historically defeated monopoly and collapse; single-channel populations were governed by rumour.	Maintain broadcast, print, courier, and digital in parallel; municipal emergency broadcast reaching all residents within 1 hour, off-grid capable.	Why 1 — speech moved onto private platforms: Public communication infrastructure was allowed to wither. -> Fix: Maintain postal, broadcast, and municipal networks as real, funded fallbacks — not heritage, but reserve capacity. | Why 2 — exit is costly: Audiences, records, and identities are locked in by design. -> Fix: Portability and interoperability as law for dominant platforms; monthly export as a consumer right. | Why 3 — emergency channels are single-thread: Alerts ride the same networks that fail in disasters. -> Fix: Municipal off-grid broadcast reaching all residents within 1 hour; amateur radio integrated into emergency plans. | Why 4 — privacy and safety are traded falsely: Policy frames encryption and lawful access as absolutes at war. -> Fix: Individualized, judicial, logged, sunsetted access as the legislated middle; blanket access and blanket opacity both prohibited. | Why 5 — root: Communication is treated as a market commodity rather than as civic infrastructure. -> Fix: Charter communication like water and power: universal service duties, resilience standards, and rights protections by design.
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139	Informational	Communication	Tier 2	Now	Household / Personal	Platform dependence: a handful of companies carry the world's speech, identity, and emergency communication; outages and bans silence millions instantly.	A single platform outage in 2021 knocked out businesses, emergency lines, and family contact across continents for hours; account bans erase people's records, audiences, and livelihoods without appeal.	Interoperability and exit rights by law: dominant platforms must support data portability and interoperable messaging, public communication infrastructure (postal, broadcast, municipal networks) maintained as a real fallback, and due-process rules for account termination with human appeal. Specification, personal and organizational: own your contact list and records off-platform (exported monthly); maintain one channel you fully control (your own email list, radio net, or phone tree), and never let a platform be the only copy of anything.	77% (base range 61–77%)	Portability and interoperability law is enacted in some jurisdictions and technically straightforward; platform resistance is the constraint.	Own your lists off-platform (monthly exports); keep one fully controlled channel per organization and household; statutory appeal rights for account termination.	Why 1 — speech moved onto private platforms: Public communication infrastructure was allowed to wither. -> Fix: Maintain postal, broadcast, and municipal networks as real, funded fallbacks — not heritage, but reserve capacity. | Why 2 — exit is costly: Audiences, records, and identities are locked in by design. -> Fix: Portability and interoperability as law for dominant platforms; monthly export as a consumer right. | Why 3 — emergency channels are single-thread: Alerts ride the same networks that fail in disasters. -> Fix: Municipal off-grid broadcast reaching all residents within 1 hour; amateur radio integrated into emergency plans. | Why 4 — privacy and safety are traded falsely: Policy frames encryption and lawful access as absolutes at war. -> Fix: Individualized, judicial, logged, sunsetted access as the legislated middle; blanket access and blanket opacity both prohibited. | Why 5 — root: Communication is treated as a market commodity rather than as civic infrastructure. -> Fix: Charter communication like water and power: universal service duties, resilience standards, and rights protections by design.

#	Dimension of Realit	Field of Discipline	Pareto Tier o	Timefram	Scale	Worst Possible Problem	Example	Solution — Full Specification	Probability of Su	Why This Probability	How to Maximize the Chances (detailed)	Root Causes
140	Informational	Communication	Tier 2	Now	Society / National	Platform dependence: a handful of companies carry the world's speech, identity, and emergency communication; outages and bans silence millions instantly.	A single platform outage in 2021 knocked out businesses, emergency lines, and family contact across continents for hours; account bans erase people's records, audiences, and livelihoods without appeal.	Interoperability and exit rights by law: dominant platforms must support data portability and interoperable messaging, public communication infrastructure (postal, broadcast, municipal networks) maintained as a real fallback, and due-process rules for account termination with human appeal. Specification, personal and organizational: own your contact list and records off-platform (exported monthly), maintain one channel you fully control (your own email list, radio net, or phone tree), and never let a platform be the only copy of anything.	69% (base range 61–77%)	Portability and interoperability law is enacted in some jurisdictions and technically straightforward; platform resistance is the constraint.	Own your lists off-platform (monthly exports); keep one fully controlled channel per organization and household; statutory appeal rights for account termination.	Why 1 — speech moved onto private platforms: Public communication infrastructure was allowed to wither. -> Fix: Maintain postal, broadcast, and municipal networks as real, funded fallbacks — not heritage, but reserve capacity.	Why 2 — exit is costly: Audiences, records, and identities are locked in by design. -> Fix: Portability and interoperability as law for dominant platforms; monthly export as a consumer right.	Why 3 — emergency channels are single-thread: Alerts ride the same networks that fail in disasters. -> Fix: Municipal off-grid broadcast reaching all residents within 1 hour; amateur radio integrated into emergency plans.	Why 4 — privacy and safety are traded falsely: Policy frames encryption and lawful access as absolutes at war. -> Fix: Individualized, judicial, logged, sunsetted access as the legislated middle; blanket access and blanket opacity both prohibited.	Why 5 — root: Communication is treated as a market commodity rather than as civic infrastructure. -> Fix: Charter communication like water and power: universal service duties, resilience standards, and rights protections by design.
141	Informational	Communication	Tier 2	Now	Global / Civilizational	Platform dependence: a handful of companies carry the world's speech, identity, and emergency communication; outages and bans silence millions instantly.	A single platform outage in 2021 knocked out businesses, emergency lines, and family contact across continents for hours; account bans erase people's records, audiences, and livelihoods without appeal.	Interoperability and exit rights by law: dominant platforms must support data portability and interoperable messaging, public communication infrastructure (postal, broadcast, municipal networks) maintained as a real fallback, and due-process rules for account termination with human appeal. Specification, personal and organizational: own your contact list and records off-platform (exported monthly), maintain one channel you fully control (your own email list, radio net, or phone tree), and never let a platform be the only copy of anything.	57% (base range 61–77%)	Portability and interoperability law is enacted in some jurisdictions and technically straightforward; platform resistance is the constraint.	Own your lists off-platform (monthly exports); keep one fully controlled channel per organization and household; statutory appeal rights for account termination.	Why 1 — speech moved onto private platforms: Public communication infrastructure was allowed to wither. -> Fix: Maintain postal, broadcast, and municipal networks as real, funded fallbacks — not heritage, but reserve capacity.	Why 2 — exit is costly: Audiences, records, and identities are locked in by design. -> Fix: Portability and interoperability as law for dominant platforms; monthly export as a consumer right.	Why 3 — emergency channels are single-thread: Alerts ride the same networks that fail in disasters. -> Fix: Municipal off-grid broadcast reaching all residents within 1 hour; amateur radio integrated into emergency plans.	Why 4 — privacy and safety are traded falsely: Policy frames encryption and lawful access as absolutes at war. -> Fix: Individualized, judicial, logged, sunsetted access as the legislated middle; blanket access and blanket opacity both prohibited.	Why 5 — root: Communication is treated as a market commodity rather than as civic infrastructure. -> Fix: Charter communication like water and power: universal service duties, resilience standards, and rights protections by design.
142	Informational	Communication	Tier 2	Future	Household / Personal	Total surveillance or total encryption-war makes private communication impossible — or, inversely, untraceable coordination makes law enforcement blind to real conspiracies.	A state that reads everything chills all dissent and error-correction; a world with no lawful access anywhere shelters atrocity planning equally.	Rights-preserving architecture: end-to-end encryption protected in law as the default for private communication, state access only through individualized, judicially authorized, logged, and time-limited orders with after-the-fact notification; anonymous public speech protected while large-scale amplification carries provenance duties. Specification: every surveillance power carries a 5-year sunset and a published annual accounting of use, errors, and outcomes — powers that cannot justify themselves expire.	60% (base range 43–61%)	Encryption with judicially bounded access is a stable equilibrium where legislated; both extremes (total access, total opacity) generate political pressure that eventually breaks them.	Protect end-to-end encryption in law; every surveillance power sunsetted at 5 years with published annual accounting; individualized warrants only.	Why 1 — speech moved onto private platforms: Public communication infrastructure was allowed to wither. -> Fix: Maintain postal, broadcast, and municipal networks as real, funded fallbacks — not heritage, but reserve capacity.	Why 2 — exit is costly: Audiences, records, and identities are locked in by design. -> Fix: Portability and interoperability as law for dominant platforms; monthly export as a consumer right.	Why 3 — emergency channels are single-thread: Alerts ride the same networks that fail in disasters. -> Fix: Municipal off-grid broadcast reaching all residents within 1 hour; amateur radio integrated into emergency plans.	Why 4 — privacy and safety are traded falsely: Policy frames encryption and lawful access as absolutes at war. -> Fix: Individualized, judicial, logged, sunsetted access as the legislated middle; blanket access and blanket opacity both prohibited.	Why 5 — root: Communication is treated as a market commodity rather than as civic infrastructure. -> Fix: Charter communication like water and power: universal service duties, resilience standards, and rights protections by design.
143	Informational	Communication	Tier 2	Future	Society / National	Total surveillance or total encryption-war makes private communication impossible — or, inversely, untraceable coordination makes law enforcement blind to real conspiracies.	A state that reads everything chills all dissent and error-correction; a world with no lawful access anywhere shelters atrocity planning equally.	Rights-preserving architecture: end-to-end encryption protected in law as the default for private communication, state access only through individualized, judicially authorized, logged, and time-limited orders with after-the-fact notification; anonymous public speech protected while large-scale amplification carries provenance duties. Specification: every surveillance power carries a 5-year sunset and a published annual accounting of use, errors, and outcomes — powers that cannot justify themselves expire.	52% (base range 43–61%)	Encryption with judicially bounded access is a stable equilibrium where legislated; both extremes (total access, total opacity) generate political pressure that eventually breaks them.	Protect end-to-end encryption in law; every surveillance power sunsetted at 5 years with published annual accounting; individualized warrants only.	Why 1 — speech moved onto private platforms: Public communication infrastructure was allowed to wither. -> Fix: Maintain postal, broadcast, and municipal networks as real, funded fallbacks — not heritage, but reserve capacity.	Why 2 — exit is costly: Audiences, records, and identities are locked in by design. -> Fix: Portability and interoperability as law for dominant platforms; monthly export as a consumer right.	Why 3 — emergency channels are single-thread: Alerts ride the same networks that fail in disasters. -> Fix: Municipal off-grid broadcast reaching all residents within 1 hour; amateur radio integrated into emergency plans.	Why 4 — privacy and safety are traded falsely: Policy frames encryption and lawful access as absolutes at war. -> Fix: Individualized, judicial, logged, sunsetted access as the legislated middle; blanket access and blanket opacity both prohibited.	Why 5 — root: Communication is treated as a market commodity rather than as civic infrastructure. -> Fix: Charter communication like water and power: universal service duties, resilience standards, and rights protections by design.
144	Informational	Communication	Tier 2	Future	Global / Civilizational	Total surveillance or total encryption-war makes private communication impossible — or, inversely, untraceable coordination makes law enforcement blind to real conspiracies.	A state that reads everything chills all dissent and error-correction; a world with no lawful access anywhere shelters atrocity planning equally.	Rights-preserving architecture: end-to-end encryption protected in law as the default for private communication, state access only through individualized, judicially authorized, logged, and time-limited orders with after-the-fact notification; anonymous public speech protected while large-scale amplification carries provenance duties. Specification: every surveillance power carries a 5-year sunset and a published annual accounting of use, errors, and outcomes — powers that cannot justify themselves expire.	40% (base range 43–61%)	Encryption with judicially bounded access is a stable equilibrium where legislated; both extremes (total access, total opacity) generate political pressure that eventually breaks them.	Protect end-to-end encryption in law; every surveillance power sunsetted at 5 years with published annual accounting; individualized warrants only.	Why 1 — speech moved onto private platforms: Public communication infrastructure was allowed to wither. -> Fix: Maintain postal, broadcast, and municipal networks as real, funded fallbacks — not heritage, but reserve capacity.	Why 2 — exit is costly: Audiences, records, and identities are locked in by design. -> Fix: Portability and interoperability as law for dominant platforms; monthly export as a consumer right.	Why 3 — emergency channels are single-thread: Alerts ride the same networks that fail in disasters. -> Fix: Municipal off-grid broadcast reaching all residents within 1 hour; amateur radio integrated into emergency plans.	Why 4 — privacy and safety are traded falsely: Policy frames encryption and lawful access as absolutes at war. -> Fix: Individualized, judicial, logged, sunsetted access as the legislated middle; blanket access and blanket opacity both prohibited.	Why 5 — root: Communication is treated as a market commodity rather than as civic infrastructure. -> Fix: Charter communication like water and power: universal service duties, resilience standards, and rights protections by design.
145	Biological / Life	Demographics & Population	Tier 2	Past	Household / Personal	Population collapse from plague, war, and famine unmakes societies: fields untilled, cities abandoned, knowledge lineages cut.	The Black Death killed a third to a half of Europe's population; some regions took two centuries to recover their numbers and skills.	Resilience through distribution: population and skills spread across regions rather than concentrated in single cities, craft and knowledge apprenticeship duplicated across at least 2 lineages per critical skill, and post-catastrophe relief law that moves food and people automatically on declared triggers. Specification: no critical occupation (medicine, engineering, agriculture) may have more than 40% of its practitioners in one metropolitan area.	72% (base range 55–72%)	Distributed skills and automatic relief triggers shortened post-catastrophe recoveries where they existed; concentrated populations recovered slowest.	Duplicate every critical skill across 2+ lineages and regions; relief triggers declared in law in advance; no critical profession 40%+ in one metro.	Why 1 — family formation is priced out: Housing and childcare costs peak exactly at family-forming ages. -> Fix: Family housing at the 30%-of-income benchmark and childcare under 10% in binding city and national plans.	Why 2 — pensions promise the unprojected: Formulas were set for pyramids that no longer exist. -> Fix: Automatic balancing mechanisms that adjust gradually by formula, depoliticizing the arithmetic.	Why 3 — migration is improvised: Flows arrive whether planned or not; only the reception differs. -> Fix: Permanent, competent administration sized to the measured gap, with 12-month credential and language completion.	Why 4 — regions hollow silently: Depopulation has no constituency until services collapse. -> Fix: Shrink-smart frameworks funded in advance: consolidated services with guaranteed access standards.	Why 5 — root: Policy is made for the current electorate's age structure while the future population is already born and countable. -> Fix: A 50-year demographic annex, updated every 5 years, binding on infrastructure and pension planning by law.
146	Biological / Life	Demographics & Population	Tier 2	Past	Society / National	Population collapse from plague, war, and famine unmakes societies: fields untilled, cities abandoned, knowledge lineages cut.	The Black Death killed a third to a half of Europe's population; some regions took two centuries to recover their numbers and skills.	Resilience through distribution: population and skills spread across regions rather than concentrated in single cities, craft and knowledge apprenticeship duplicated across at least 2 lineages per critical skill, and post-catastrophe relief law that moves food and people automatically on declared triggers. Specification: no critical occupation (medicine, engineering, agriculture) may have more than 40% of its practitioners in one metropolitan area.	64% (base range 55–72%)	Distributed skills and automatic relief triggers shortened post-catastrophe recoveries where they existed; concentrated populations recovered slowest.	Duplicate every critical skill across 2+ lineages and regions; relief triggers declared in law in advance; no critical profession 40%+ in one metro.	Why 1 — family formation is priced out: Housing and childcare costs peak exactly at family-forming ages. -> Fix: Family housing at the 30%-of-income benchmark and childcare under 10% in binding city and national plans.	Why 2 — pensions promise the unprojected: Formulas were set for pyramids that no longer exist. -> Fix: Automatic balancing mechanisms that adjust gradually by formula, depoliticizing the arithmetic.	Why 3 — migration is improvised: Flows arrive whether planned or not; only the reception differs. -> Fix: Permanent, competent administration sized to the measured gap, with 12-month credential and language completion.	Why 4 — regions hollow silently: Depopulation has no constituency until services collapse. -> Fix: Shrink-smart frameworks funded in advance: consolidated services with guaranteed access standards.	Why 5 — root: Policy is made for the current electorate's age structure while the future population is already born and countable. -> Fix: A 50-year demographic annex, updated every 5 years, binding on infrastructure and pension planning by law.
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148	Biological / Life	Demographics & Population	Tier 2	Now	Household / Personal	Below-replacement fertility plus rapid aging inverts the pyramid: fewer workers support more retirees; towns hollow out.	Countries at fertility rates of 0.7–1.3 face shrinking school cohorts, closing villages, and pension systems whose math no longer closes.	Make family formation affordable where people already want it: housing sized and priced for young families (family units at or below 30% of median income in every major city plan), childcare costing under 10% of median income per child, working-time protection for parents, and pension formulas that adjust transparently and gradually (automatic balancing mechanisms) instead of by crisis. Specification: immigration and integration run as a permanent, competently administered system sized to the measured demographic gap, with language and credential recognition completed within 12 months of arrival.	65% (base range 48–66%)	Housing and childcare cost relief moves fertility modestly where sustained (no country has reversed to replacement by policy alone); pension auto-balancers fully work mechanically.	Family housing at the 30% benchmark in city plans; childcare under 10% of median income; automatic pension balancing so adjustment is gradual, not crisis-driven.	Why 1 — family formation is priced out: Housing and childcare costs peak exactly at family-forming ages. -> Fix: Family housing at the 30%-of-income benchmark and childcare under 10% in binding city and national plans.	Why 2 — pensions promise the unprojected: Formulas were set for pyramids that no longer exist. -> Fix: Automatic balancing mechanisms that adjust gradually by formula, depoliticizing the arithmetic.	Why 3 — migration is improvised: Flows arrive whether planned or not; only the reception differs. -> Fix: Permanent, competent administration sized to the measured gap, with 12-month credential and language completion.	Why 4 — regions hollow silently: Depopulation has no constituency until services collapse. -> Fix: Shrink-smart frameworks funded in advance: consolidated services with guaranteed access standards.	Why 5 — root: Policy is made for the current electorate's age structure while the future population is already born and countable. -> Fix: A 50-year demographic annex, updated every 5 years, binding on infrastructure and pension planning by law.
149	Biological / Life	Demographics & Population	Tier 2	Now	Society / National	Below-replacement fertility plus rapid aging inverts the pyramid: fewer workers support more retirees; towns hollow out.	Countries at fertility rates of 0.7–1.3 face shrinking school cohorts, closing villages, and pension systems whose math no longer closes.	Make family formation affordable where people already want it: housing sized and priced for young families (family units at or below 30% of median income in every major city plan), childcare costing under 10% of median income per child, working-time protection for parents, and pension formulas that adjust transparently and gradually (automatic balancing mechanisms) instead of by crisis. Specification: immigration and integration run as a permanent, competently administered system sized to the measured demographic gap, with language and credential recognition completed within 12 months of arrival.	57% (base range 48–66%)	Housing and childcare cost relief moves fertility modestly where sustained (no country has reversed to replacement by policy alone); pension auto-balancers fully work mechanically.	Family housing at the 30% benchmark in city plans; childcare under 10% of median income; automatic pension balancing so adjustment is gradual, not crisis-driven.	Why 1 — family formation is priced out: Housing and childcare costs peak exactly at family-forming ages. -> Fix: Family housing at the 30%-of-income benchmark and childcare under 10% in binding city and national plans.	Why 2 — pensions promise the unprojected: Formulas were set for pyramids that no longer exist. -> Fix: Automatic balancing mechanisms that adjust gradually by formula, depoliticizing the arithmetic.	Why 3 — migration is improvised: Flows arrive whether planned or not; only the reception differs. -> Fix: Permanent, competent administration sized to the measured gap, with 12-month credential and language completion.	Why 4 — regions hollow silently: Depopulation has no constituency until services collapse. -> Fix: Shrink-smart frameworks funded in advance: consolidated services with guaranteed access standards.	Why 5 — root: Policy is made for the current electorate's age structure while the future population is already born and countable. -> Fix: A 50-year demographic annex, updated every 5 years, binding on infrastructure and pension planning by law.

#	Dimension of Reality	Field of Discipline	Pareto Tier o	Timefram	Scale	Worst Possible Problem	Example	Solution — Full Specification	Probability of Su	Why This Probability	How to Maximize the Chances (detailed)	Root Causes
150	Biological / Life	Demographics & Population	Tier 2	Now	Global / Civilizational	Below-replacement fertility plus rapid aging inverts the pyramid: fewer workers support more retirees, towns hollow out.	Countries at fertility rates of 0.7–1.3 face shrinking school cohorts, closing villages, and pension systems whose math no longer closes.	Make family formation affordable where people already want it: housing sized and priced for young families (family units at or below 30% of median income in every major city plan), childcare costing under 10% of median income per child, working-time protection for parents, and pension formulas that adjust transparently and gradually (automatic balancing mechanisms) instead of by crisis. Specification: immigration and integration run as a permanent, competently administered system sized to the measured demographic gap, with language and credential recognition completed within 12 months of arrival.	45% (base range 48–66%)	Housing and childcare cost relief moves fertility modestly where sustained (no country has reversed to replacement by policy alone); pension auto-balancers fully work mechanically.	Family housing at the 30% benchmark in city plans; childcare under 10% of median income; automatic pension balancing so adjustment is gradual, not crisis-driven.	Why 1 — family formation is priced out: Housing and childcare costs peak exactly at family-forming ages. -> Fix: Family housing at the 30%-of-income benchmark and childcare under 10% in binding city and national plans. | Why 2 — pensions promise the unprojected: Formulas were set for pyramids that no longer exist. -> Fix: Automatic balancing mechanisms that adjust gradually by formula, depoliticizing the arithmetic. | Why 3 — migration is improvised: Flows arrive whether planned or not; only the reception differs. -> Fix: Permanent, competent administration sized to the measured gap, with 12-month credential and language completion. | Why 4 — regions hollow silently: Depopulation has no constituency until services collapse. -> Fix: Shrink-smart frameworks funded in advance: consolidated services with guaranteed access standards. | Why 5 — root: Policy is made for the current electorate's age structure while the future population is already born and countable. -> Fix: A 50-year demographic annex, updated every 5 years, binding on infrastructure and pension planning by law.
151	Biological / Life	Demographics & Population	Tier 2	Future	Household / Personal	Divergent futures collide: some regions depopulate to non-viability while others add billions; unmanaged flows destabilize both.	By 2100 projections, parts of East Asia and Europe halve in population while parts of Africa triple; infrastructure, states, and cultures built for the opposite trend fail.	Plan both directions simultaneously: shrink-smart policy for depopulating regions (consolidated services, maintained dignity and access, land returned to nature or agriculture under stewardship), growth-ready frameworks for expanding regions (land tenure, urban plans, and service grids built ahead of arrival), and pre-negotiated international mobility compacts so movement is governed, not chaotic. Specification: every national plan carries a 50-year demographic annex, updated every 5 years, with infrastructure sized to the projection band — not to today's population.	57% (base range 40–58%)	Both shrink-smart and growth-ready planning are technically known; political systems resist planning for populations that differ from today's voters.	50-year demographic annex in every national plan, updated 5-yearly; mobility compacts negotiated before flows peak; infrastructure sized to the projection band.	Why 1 — family formation is priced out: Housing and childcare costs peak exactly at family-forming ages. -> Fix: Family housing at the 30%-of-income benchmark and childcare under 10% in binding city and national plans. | Why 2 — pensions promise the unprojected: Formulas were set for pyramids that no longer exist. -> Fix: Automatic balancing mechanisms that adjust gradually by formula, depoliticizing the arithmetic. | Why 3 — migration is improvised: Flows arrive whether planned or not; only the reception differs. -> Fix: Permanent, competent administration sized to the measured gap, with 12-month credential and language completion. | Why 4 — regions hollow silently: Depopulation has no constituency until services collapse. -> Fix: Shrink-smart frameworks funded in advance: consolidated services with guaranteed access standards. | Why 5 — root: Policy is made for the current electorate's age structure while the future population is already born and countable. -> Fix: A 50-year demographic annex, updated every 5 years, binding on infrastructure and pension planning by law.
152	Biological / Life	Demographics & Population	Tier 2	Future	Society / National	Divergent futures collide: some regions depopulate to non-viability while others add billions; unmanaged flows destabilize both.	By 2100 projections, parts of East Asia and Europe halve in population while parts of Africa triple; infrastructure, states, and cultures built for the opposite trend fail.	Plan both directions simultaneously: shrink-smart policy for depopulating regions (consolidated services, maintained dignity and access, land returned to nature or agriculture under stewardship), growth-ready frameworks for expanding regions (land tenure, urban plans, and service grids built ahead of arrival), and pre-negotiated international mobility compacts so movement is governed, not chaotic. Specification: every national plan carries a 50-year demographic annex, updated every 5 years, with infrastructure sized to the projection band — not to today's population.	49% (base range 40–58%)	Both shrink-smart and growth-ready planning are technically known; political systems resist planning for populations that differ from today's voters.	50-year demographic annex in every national plan, updated 5-yearly; mobility compacts negotiated before flows peak; infrastructure sized to the projection band.	Why 1 — family formation is priced out: Housing and childcare costs peak exactly at family-forming ages. -> Fix: Family housing at the 30%-of-income benchmark and childcare under 10% in binding city and national plans. | Why 2 — pensions promise the unprojected: Formulas were set for pyramids that no longer exist. -> Fix: Automatic balancing mechanisms that adjust gradually by formula, depoliticizing the arithmetic. | Why 3 — migration is improvised: Flows arrive whether planned or not; only the reception differs. -> Fix: Permanent, competent administration sized to the measured gap, with 12-month credential and language completion. | Why 4 — regions hollow silently: Depopulation has no constituency until services collapse. -> Fix: Shrink-smart frameworks funded in advance: consolidated services with guaranteed access standards. | Why 5 — root: Policy is made for the current electorate's age structure while the future population is already born and countable. -> Fix: A 50-year demographic annex, updated every 5 years, binding on infrastructure and pension planning by law.
153	Biological / Life	Demographics & Population	Tier 2	Future	Global / Civilizational	Divergent futures collide: some regions depopulate to non-viability while others add billions; unmanaged flows destabilize both.	By 2100 projections, parts of East Asia and Europe halve in population while parts of Africa triple; infrastructure, states, and cultures built for the opposite trend fail.	Plan both directions simultaneously: shrink-smart policy for depopulating regions (consolidated services, maintained dignity and access, land returned to nature or agriculture under stewardship), growth-ready frameworks for expanding regions (land tenure, urban plans, and service grids built ahead of arrival), and pre-negotiated international mobility compacts so movement is governed, not chaotic. Specification: every national plan carries a 50-year demographic annex, updated every 5 years, with infrastructure sized to the projection band — not to today's population.	37% (base range 40–58%)	Both shrink-smart and growth-ready planning are technically known; political systems resist planning for populations that differ from today's voters.	50-year demographic annex in every national plan, updated 5-yearly; mobility compacts negotiated before flows peak; infrastructure sized to the projection band.	Why 1 — family formation is priced out: Housing and childcare costs peak exactly at family-forming ages. -> Fix: Family housing at the 30%-of-income benchmark and childcare under 10% in binding city and national plans. | Why 2 — pensions promise the unprojected: Formulas were set for pyramids that no longer exist. -> Fix: Automatic balancing mechanisms that adjust gradually by formula, depoliticizing the arithmetic. | Why 3 — migration is improvised: Flows arrive whether planned or not; only the reception differs. -> Fix: Permanent, competent administration sized to the measured gap, with 12-month credential and language completion. | Why 4 — regions hollow silently: Depopulation has no constituency until services collapse. -> Fix: Shrink-smart frameworks funded in advance: consolidated services with guaranteed access standards. | Why 5 — root: Policy is made for the current electorate's age structure while the future population is already born and countable. -> Fix: A 50-year demographic annex, updated every 5 years, binding on infrastructure and pension planning by law.
154	Social / Order	Justice & Ethics	Tier 2	Past	Household / Personal	Justice systems become weapons: show trials, collective punishment, and law as an instrument of terror.	From Stalin's purges to lynching regimes, courts and mobs delivered punishment by loyalty and identity, not by evidence.	Evidence as the only currency: independent judiciary with security of tenure and salary, prosecution separated from the executive, every criminal conviction reviewable on the evidence record, forensic methods validated scientifically before courtroom use, and absolute bars on collective punishment and retroactive crimes. Specification: wrongful-conviction compensation automatic and generous; every jurisdiction publishes conviction-integrity audits, including exoneration counts, annually.	80% (base range 64–80%)	Independent courts with evidence rules have repeatedly been the last institution standing; where tenure and budgets were protected, terror-by-law was blunted.	Judicial tenure, salary, and budget protected constitutionally; forensic methods validated before courtroom use; wrongful-conviction compensation automatic.	Why 1 — justice is priced: Expertise, delay, and bail convert wealth into outcomes. -> Fix: Legal aid at parity in criminal matters; day-fines; detention only on individualized danger with 48-hour review. | Why 2 — backlogs normalize delay: Court capacity is funded below caseload by habit. -> Fix: Published time standards with automatic capacity triggers (90% of cases within 12 months). | Why 3 — error is unmeasured: Systems that don't count wrongful convictions can't reduce them. -> Fix: Conviction-integrity audits and exoneration statistics published annually; compensation automatic. | Why 4 — algorithms inherit the bias: Historical enforcement data encodes historical disparity. -> Fix: Justice models public, disparity-audited, appealable to a human; no liberty decision on a score alone. | Why 5 — root: Justice is administered as a service to the state rather than as a right held against it. -> Fix: Rights-first charter: every justice metric reported from the citizen's side — time, cost, error, disparity — not only the system's throughput.
155	Social / Order	Justice & Ethics	Tier 2	Past	Society / National	Justice systems become weapons: show trials, collective punishment, and law as an instrument of terror.	From Stalin's purges to lynching regimes, courts and mobs delivered punishment by loyalty and identity, not by evidence.	Evidence as the only currency: independent judiciary with security of tenure and salary, prosecution separated from the executive, every criminal conviction reviewable on the evidence record, forensic methods validated scientifically before courtroom use, and absolute bars on collective punishment and retroactive crimes. Specification: wrongful-conviction compensation automatic and generous; every jurisdiction publishes conviction-integrity audits, including exoneration counts, annually.	72% (base range 64–80%)	Independent courts with evidence rules have repeatedly been the last institution standing; where tenure and budgets were protected, terror-by-law was blunted.	Judicial tenure, salary, and budget protected constitutionally; forensic methods validated before courtroom use; wrongful-conviction compensation automatic.	Why 1 — justice is priced: Expertise, delay, and bail convert wealth into outcomes. -> Fix: Legal aid at parity in criminal matters; day-fines; detention only on individualized danger with 48-hour review. | Why 2 — backlogs normalize delay: Court capacity is funded below caseload by habit. -> Fix: Published time standards with automatic capacity triggers (90% of cases within 12 months). | Why 3 — error is unmeasured: Systems that don't count wrongful convictions can't reduce them. -> Fix: Conviction-integrity audits and exoneration statistics published annually; compensation automatic. | Why 4 — algorithms inherit the bias: Historical enforcement data encodes historical disparity. -> Fix: Justice models public, disparity-audited, appealable to a human; no liberty decision on a score alone. | Why 5 — root: Justice is administered as a service to the state rather than as a right held against it. -> Fix: Rights-first charter: every justice metric reported from the citizen's side — time, cost, error, disparity — not only the system's throughput.
156	Social / Order	Justice & Ethics	Tier 2	Past	Global / Civilizational	Justice systems become weapons: show trials, collective punishment, and law as an instrument of terror.	From Stalin's purges to lynching regimes, courts and mobs delivered punishment by loyalty and identity, not by evidence.	Evidence as the only currency: independent judiciary with security of tenure and salary, prosecution separated from the executive, every criminal conviction reviewable on the evidence record, forensic methods validated scientifically before courtroom use, and absolute bars on collective punishment and retroactive crimes. Specification: wrongful-conviction compensation automatic and generous; every jurisdiction publishes conviction-integrity audits, including exoneration counts, annually.	60% (base range 64–80%)	Independent courts with evidence rules have repeatedly been the last institution standing; where tenure and budgets were protected, terror-by-law was blunted.	Judicial tenure, salary, and budget protected constitutionally; forensic methods validated before courtroom use; wrongful-conviction compensation automatic.	Why 1 — justice is priced: Expertise, delay, and bail convert wealth into outcomes. -> Fix: Legal aid at parity in criminal matters; day-fines; detention only on individualized danger with 48-hour review. | Why 2 — backlogs normalize delay: Court capacity is funded below caseload by habit. -> Fix: Published time standards with automatic capacity triggers (90% of cases within 12 months). | Why 3 — error is unmeasured: Systems that don't count wrongful convictions can't reduce them. -> Fix: Conviction-integrity audits and exoneration statistics published annually; compensation automatic. | Why 4 — algorithms inherit the bias: Historical enforcement data encodes historical disparity. -> Fix: Justice models public, disparity-audited, appealable to a human; no liberty decision on a score alone. | Why 5 — root: Justice is administered as a service to the state rather than as a right held against it. -> Fix: Rights-first charter: every justice metric reported from the citizen's side — time, cost, error, disparity — not only the system's throughput.
157	Social / Order	Justice & Ethics	Tier 2	Now	Household / Personal	Two-tier justice: the wealthy buy delay and expertise; the poor plead guilty to get out of jail; backlogs make 'justice delayed' the norm.	Defendants held pre-trial longer than the sentence for the charge itself; civil cases priced beyond ordinary people's reach; trust in courts falls across democracies.	Equalize the machinery: pre-trial detention only on individualized danger findings with 48-hour review, legal aid funded to parity standards with prosecution in criminal matters, court capacity managed to published time standards (e.g., 90% of cases resolved within 12 months), and fines scaled to income (day-fines) so penalties bite equally. Specification: every court publishes wait times, outcomes by representation type, and backlog monthly; households know their rights in writing — a plain-language rights card is a standard household document.	73% (base range 57–73%)	Detention limits, day-fines, and published time standards work where adopted; the constraint is court capacity funding.	48-hour detention review; day-fines scaled to income; legal aid at parity in criminal matters; monthly publication of waits and outcomes.	Why 1 — justice is priced: Expertise, delay, and bail convert wealth into outcomes. -> Fix: Legal aid at parity in criminal matters; day-fines; detention only on individualized danger with 48-hour review. | Why 2 — backlogs normalize delay: Court capacity is funded below caseload by habit. -> Fix: Published time standards with automatic capacity triggers (90% of cases within 12 months). | Why 3 — error is unmeasured: Systems that don't count wrongful convictions can't reduce them. -> Fix: Conviction-integrity audits and exoneration statistics published annually; compensation automatic. | Why 4 — algorithms inherit the bias: Historical enforcement data encodes historical disparity. -> Fix: Justice models public, disparity-audited, appealable to a human; no liberty decision on a score alone. | Why 5 — root: Justice is administered as a service to the state rather than as a right held against it. -> Fix: Rights-first charter: every justice metric reported from the citizen's side — time, cost, error, disparity — not only the system's throughput.
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160	Social / Order	Justice & Ethics	Tier 2	Future	Household / Personal	Algorithmic justice and predictive policing automate bias at scale; people are punished for probabilities, and no human is answerable.	Risk scores trained on historical enforcement data send patrols and sentences in self-confirming loops; defendants cannot cross-examine a model.	Human answerability as law: no liberty decision (arrest priority, bail, sentencing, parole) may rest solely on an algorithmic score; models used in justice are public, auditable, and tested for error and disparity rates published per demographic group; every automated recommendation names the accountable human who adopted it. Specification: any person scored by a state system can see their inputs, correct errors within 30 days, and receive a human review on request — the same standard as credit records, applied to freedom.	66% (base range 49–67%)	Auditability and human-answerability rules are enforceable today; success depends on adopting them before automated systems entrench.	No liberty decision on algorithmic score alone; models public and disparity-audited; every scored person can see, correct, and appeal to a human within 30 days.	Why 1 — justice is priced: Expertise, delay, and bail convert wealth into outcomes. -> Fix: Legal aid at parity in criminal matters; day-fines; detention only on individualized danger with 48-hour review. | Why 2 — backlogs normalize delay: Court capacity is funded below caseload by habit. -> Fix: Published time standards with automatic capacity triggers (90% of cases within 12 months). | Why 3 — error is unmeasured: Systems that don't count wrongful convictions can't reduce them. -> Fix: Conviction-integrity audits and exoneration statistics published annually; compensation automatic. | Why 4 — algorithms inherit the bias: Historical enforcement data encodes historical disparity. -> Fix: Justice models public, disparity-audited, appealable to a human; no liberty decision on a score alone. | Why 5 — root: Justice is administered as a service to the state rather than as a right held against it. -> Fix: Rights-first charter: every justice metric reported from the citizen's side — time, cost, error, disparity — not only the system's throughput.

#	Dimension of Realit	Field of Discipline	Pareto Tier o	Timefram	Scale	Worst Possible Problem	Example	Solution — Full Specification	Probability of Su	Why This Probability	How to Maximize the Chances (detailed)	Root Causes
161	Social / Order	Justice & Ethics	Tier 2	Future	Society / National	Algorithmic justice and predictive policing automate bias at scale; people are punished for probabilities, and no human is answerable.	Risk scores trained on historical enforcement data send patrols and sentences in self-confirming loops; defendants cannot cross-examine a model.	Human answerability as law: no liberty decision (arrest priority, bail, sentencing, parole) may rest solely on an algorithmic score; models used in justice are public, auditable, and tested for error and disparity rates published per demographic group; every automated recommendation names the accountable human who adopted it. Specification: any person scored by a state system can see their inputs, correct errors within 30 days, and receive a human review on request — the same standard as credit records, applied to freedom.	58% (base range 49–67%)	Auditability and human-answerability rules are enforceable today; success depends on adopting them before automated systems entrench.	No liberty decision on algorithmic score alone; models public and disparity-audited; every scored person can see, correct, and appeal to a human within 30 days.	Why 1 — justice is priced: Expertise, delay, and bail convert wealth into outcomes. -> Fix: Legal aid at parity in criminal matters; day-fines; detention only on individualized danger with 48-hour review. | Why 2 — backlogs normalize delay: Court capacity is funded below caseload by habit. -> Fix: Published time standards with automatic capacity triggers (90% of cases within 12 months). | Why 3 — error is unmeasured: Systems that don't count wrongful convictions can't reduce them. -> Fix: Conviction-integrity audits and exoneration statistics published annually; compensation automatic. | Why 4 — algorithms inherit the bias: Historical enforcement data encodes historical disparity. -> Fix: Justice models public, disparity-audited, appealable to a human; no liberty decision on a score alone. | Why 5 — root: Justice is administered as a service to the state rather than as a right held against it. -> Fix: Rights-first charter: every justice metric reported from the citizen's side — time, cost, error, disparity — not only the system's throughput.
162	Social / Order	Justice & Ethics	Tier 2	Future	Global / Civilizational	Algorithmic justice and predictive policing automate bias at scale; people are punished for probabilities, and no human is answerable.	Risk scores trained on historical enforcement data send patrols and sentences in self-confirming loops; defendants cannot cross-examine a model.	Human answerability as law: no liberty decision (arrest priority, bail, sentencing, parole) may rest solely on an algorithmic score; models used in justice are public, auditable, and tested for error and disparity rates published per demographic group; every automated recommendation names the accountable human who adopted it. Specification: any person scored by a state system can see their inputs, correct errors within 30 days, and receive a human review on request — the same standard as credit records, applied to freedom.	46% (base range 49–67%)	Auditability and human-answerability rules are enforceable today; success depends on adopting them before automated systems entrench.	No liberty decision on algorithmic score alone; models public and disparity-audited; every scored person can see, correct, and appeal to a human within 30 days.	Why 1 — justice is priced: Expertise, delay, and bail convert wealth into outcomes. -> Fix: Legal aid at parity in criminal matters; day-fines; detention only on individualized danger with 48-hour review. | Why 2 — backlogs normalize delay: Court capacity is funded below caseload by habit. -> Fix: Published time standards with automatic capacity triggers (90% of cases within 12 months). | Why 3 — error is unmeasured: Systems that don't count wrongful convictions can't reduce them. -> Fix: Conviction-integrity audits and exoneration statistics published annually; compensation automatic. | Why 4 — algorithms inherit the bias: Historical enforcement data encodes historical disparity. -> Fix: Justice models public, disparity-audited, appealable to a human; no liberty decision on a score alone. | Why 5 — root: Justice is administered as a service to the state rather than as a right held against it. -> Fix: Rights-first charter: every justice metric reported from the citizen's side — time, cost, error, disparity — not only the system's throughput.

Solution #	Field of Discipline	Timefram	Scale	Solution — Full Specification	Probability of S	How to Maximize the Chances (detailed)	Why This Probability
1	Water	Past	Household / Personal	Cap total extraction at measured recharge: meter every well and canal offtake, publish the water balance monthly, and set the legal extraction ceiling at 80% of average recharge so the buffer absorbs dry years. Restore wetlands at river mouths (minimum 10% of historical area) as natural filters and flood buffers. Specification: household storage 4 litres per person per day drinking minimum, 14-day reserve = 56 L/person; community reservoirs sized for 90 days of essential use.	86%	Meter 100% of extraction; publish the balance monthly; set the cap before the crisis decade, not during it.	Extraction caps tied to measured recharge have worked wherever enforced (e.g., managed aquifer districts) because the physics is directly measurable.
2	Water	Past	Society / National	Cap total extraction at measured recharge: meter every well and canal offtake, publish the water balance monthly, and set the legal extraction ceiling at 80% of average recharge so the buffer absorbs dry years. Restore wetlands at river mouths (minimum 10% of historical area) as natural filters and flood buffers. Specification: household storage 4 litres per person per day drinking minimum, 14-day reserve = 56 L/person; community reservoirs sized for 90 days of essential use.	78%	Meter 100% of extraction; publish the balance monthly; set the cap before the crisis decade, not during it.	Extraction caps tied to measured recharge have worked wherever enforced (e.g., managed aquifer districts) because the physics is directly measurable.
3	Water	Past	Global / Civilizational	Cap total extraction at measured recharge: meter every well and canal offtake, publish the water balance monthly, and set the legal extraction ceiling at 80% of average recharge so the buffer absorbs dry years. Restore wetlands at river mouths (minimum 10% of historical area) as natural filters and flood buffers. Specification: household storage 4 litres per person per day drinking minimum, 14-day reserve = 56 L/person; community reservoirs sized for 90 days of essential use.	66%	Meter 100% of extraction; publish the balance monthly; set the cap before the crisis decade, not during it.	Extraction caps tied to measured recharge have worked wherever enforced (e.g., managed aquifer districts) because the physics is directly measurable.
4	Water	Now	Household / Personal	Dual-barrier defence: (1) two independent treatment trains so one failure never reaches taps; (2) point-of-use backup in every household — a certified gravity or pump filter rated to remove bacteria/viruses (0.02–0.2 micron) plus a 72-hour stored reserve (12 L/person drinking). Specification: utilities must hold 5 days of treated-water storage and publish real-time turbidity and chlorine residuals; buildings over 6 storeys keep a 24-hour gravity tank.	93%	Test household filters and stored water quarterly; utilities drill a full plant-loss weekend annually; keep 72-hour reserves rotated, not forgotten.	Dual treatment trains plus household filtration are mature, cheap, tested engineering — failure is almost always maintenance, not technology.
5	Water	Now	Society / National	Dual-barrier defence: (1) two independent treatment trains so one failure never reaches taps; (2) point-of-use backup in every household — a certified gravity or pump filter rated to remove bacteria/viruses (0.02–0.2 micron) plus a 72-hour stored reserve (12 L/person drinking). Specification: utilities must hold 5 days of treated-water storage and publish real-time turbidity and chlorine residuals; buildings over 6 storeys keep a 24-hour gravity tank.	85%	Test household filters and stored water quarterly; utilities drill a full plant-loss weekend annually; keep 72-hour reserves rotated, not forgotten.	Dual treatment trains plus household filtration are mature, cheap, tested engineering — failure is almost always maintenance, not technology.
6	Water	Now	Global / Civilizational	Dual-barrier defence: (1) two independent treatment trains so one failure never reaches taps; (2) point-of-use backup in every household — a certified gravity or pump filter rated to remove bacteria/viruses (0.02–0.2 micron) plus a 72-hour stored reserve (12 L/person drinking). Specification: utilities must hold 5 days of treated-water storage and publish real-time turbidity and chlorine residuals; buildings over 6 storeys keep a 24-hour gravity tank.	73%	Test household filters and stored water quarterly; utilities drill a full plant-loss weekend annually; keep 72-hour reserves rotated, not forgotten.	Dual treatment trains plus household filtration are mature, cheap, tested engineering — failure is almost always maintenance, not technology.
7	Water	Future	Household / Personal	Rebuild the supply stack: wastewater recycling to potable standard (target 30–40% of supply), stormwater capture on all new developments (retain the 25-year storm on site), desalination where coastal, and pricing that guarantees a free lifeline block (first 50 L/person/day) with steep rates above it. Specification: per-capita demand cut to 100 L/day total through fixtures and leakage below 8%; inter-basin agreements signed before crisis, not during.	63%	Start reuse infrastructure 10 years before projected shortfall; lock the lifeline-block pricing in law early; model demand on the worst decade, not the average.	Recycling and demand cuts are proven; the constraint is political will and capital timing against a moving climate baseline.
8	Water	Future	Society / National	Rebuild the supply stack: wastewater recycling to potable standard (target 30–40% of supply), stormwater capture on all new developments (retain the 25-year storm on site), desalination where coastal, and pricing that guarantees a free lifeline block (first 50 L/person/day) with steep rates above it. Specification: per-capita demand cut to 100 L/day total through fixtures and leakage below 8%; inter-basin agreements signed before crisis, not during.	55%	Start reuse infrastructure 10 years before projected shortfall; lock the lifeline-block pricing in law early; model demand on the worst decade, not the average.	Recycling and demand cuts are proven; the constraint is political will and capital timing against a moving climate baseline.
9	Water	Future	Global / Civilizational	Rebuild the supply stack: wastewater recycling to potable standard (target 30–40% of supply), stormwater capture on all new developments (retain the 25-year storm on site), desalination where coastal, and pricing that guarantees a free lifeline block (first 50 L/person/day) with steep rates above it. Specification: per-capita demand cut to 100 L/day total through fixtures and leakage below 8%; inter-basin agreements signed before crisis, not during.	43%	Start reuse infrastructure 10 years before projected shortfall; lock the lifeline-block pricing in law early; model demand on the worst decade, not the average.	Recycling and demand cuts are proven; the constraint is political will and capital timing against a moving climate baseline.
10	Food & Agriculture	Past	Household / Personal	Never let one crop or one supplier exceed 25% of calories. Maintain seed diversity banks with at least 3 climate-matched varieties per staple, rotate crops on a minimum 3-year cycle, and hold strategic grain reserves of 90 days of national consumption. Specification: household level — 90 days of staple calories stored (rice/grain ~ 25 kg per adult per month at 2,000 kcal/day planning), rotated FIFO; regional level — no more than 40% of food imported by calorie.	90%	Enforce the 25% single-source ceiling in law; audit reserves physically twice a year; keep 3 seed varieties minimum per staple, germination-tested.	Diversification and reserves are ancient, repeatedly validated famine preventatives; famines in diversified systems with reserves are historically rare.
11	Food & Agriculture	Past	Society / National	Never let one crop or one supplier exceed 25% of calories. Maintain seed diversity banks with at least 3 climate-matched varieties per staple, rotate crops on a minimum 3-year cycle, and hold strategic grain reserves of 90 days of national consumption. Specification: household level — 90 days of staple calories stored (rice/grain ~ 25 kg per adult per month at 2,000 kcal/day planning), rotated FIFO; regional level — no more than 40% of food imported by calorie.	82%	Enforce the 25% single-source ceiling in law; audit reserves physically twice a year; keep 3 seed varieties minimum per staple, germination-tested.	Diversification and reserves are ancient, repeatedly validated famine preventatives; famines in diversified systems with reserves are historically rare.
12	Food & Agriculture	Past	Global / Civilizational	Never let one crop or one supplier exceed 25% of calories. Maintain seed diversity banks with at least 3 climate-matched varieties per staple, rotate crops on a minimum 3-year cycle, and hold strategic grain reserves of 90 days of national consumption. Specification: household level — 90 days of staple calories stored (rice/grain ~ 25 kg per adult per month at 2,000 kcal/day planning), rotated FIFO; regional level — no more than 40% of food imported by calorie.	70%	Enforce the 25% single-source ceiling in law; audit reserves physically twice a year; keep 3 seed varieties minimum per staple, germination-tested.	Diversification and reserves are ancient, repeatedly validated famine preventatives; famines in diversified systems with reserves are historically rare.
13	Food & Agriculture	Now	Household / Personal	Shorten and parallel the chain: contract at least 2 geographically separate suppliers per staple; local production floor of 30% of fresh food within 100 km; fertilizer independence via green-ammonia plants sized to 50% of national need. Specification: retailers and institutions hold 30 days of non-perishable buffer; households hold 30 days (one month of staples ≈ 25 kg grain-equivalent per adult); price-spike circuit breakers release reserves automatically at +40% price movement.	80%	Dual-source every staple in peacetime contracts; hold the 30-day buffer physically, not as paper claims; pre-sign emergency import routes.	Supplier duplication and buffers are standard logistics; success depends on signing parallel contracts before the shock reprices them.
14	Food & Agriculture	Now	Society / National	Shorten and parallel the chain: contract at least 2 geographically separate suppliers per staple; local production floor of 30% of fresh food within 100 km; fertilizer independence via green-ammonia plants sized to 50% of national need. Specification: retailers and institutions hold 30 days of non-perishable buffer; households hold 30 days (one month of staples ≈ 25 kg grain-equivalent per adult); price-spike circuit breakers release reserves automatically at +40% price movement.	72%	Dual-source every staple in peacetime contracts; hold the 30-day buffer physically, not as paper claims; pre-sign emergency import routes.	Supplier duplication and buffers are standard logistics; success depends on signing parallel contracts before the shock reprices them.
15	Food & Agriculture	Now	Global / Civilizational	Shorten and parallel the chain: contract at least 2 geographically separate suppliers per staple; local production floor of 30% of fresh food within 100 km; fertilizer independence via green-ammonia plants sized to 50% of national need. Specification: retailers and institutions hold 30 days of non-perishable buffer; households hold 30 days (one month of staples ≈ 25 kg grain-equivalent per adult); price-spike circuit breakers release reserves automatically at +40% price movement.	60%	Dual-source every staple in peacetime contracts; hold the 30-day buffer physically, not as paper claims; pre-sign emergency import routes.	Supplier duplication and buffers are standard logistics; success depends on signing parallel contracts before the shock reprices them.
16	Food & Agriculture	Future	Household / Personal	Regenerate the base: rebuild topsoil organic matter to 4%+ through cover cropping and reduced tillage; shift 20% of acreage to drought- and heat-bred varieties; build vertical/greenhouse capacity for greens near cities (target 50% of leafy supply within city limits); protect pollinators with pesticide-free corridors on 10% of farmland. Specification: yield planning assumes the worst decade climate, not the average year.	66%	Begin soil programs a decade early; breed for the 2050 climate now; protect pollinator corridors before pollinator collapse, not after.	Soil regeneration and heat-bred varieties demonstrably work, but on 5–15 year biological timelines that must start before yields fall.
17	Food & Agriculture	Future	Society / National	Regenerate the base: rebuild topsoil organic matter to 4%+ through cover cropping and reduced tillage; shift 20% of acreage to drought- and heat-bred varieties; build vertical/greenhouse capacity for greens near cities (target 50% of leafy supply within city limits); protect pollinators with pesticide-free corridors on 10% of farmland. Specification: yield planning assumes the worst decade climate, not the average year.	58%	Begin soil programs a decade early; breed for the 2050 climate now; protect pollinator corridors before pollinator collapse, not after.	Soil regeneration and heat-bred varieties demonstrably work, but on 5–15 year biological timelines that must start before yields fall.
18	Food & Agriculture	Future	Global / Civilizational	Regenerate the base: rebuild topsoil organic matter to 4%+ through cover cropping and reduced tillage; shift 20% of acreage to drought- and heat-bred varieties; build vertical/greenhouse capacity for greens near cities (target 50% of leafy supply within city limits); protect pollinators with pesticide-free corridors on 10% of farmland. Specification: yield planning assumes the worst decade climate, not the average year.	46%	Begin soil programs a decade early; breed for the 2050 climate now; protect pollinator corridors before pollinator collapse, not after.	Soil regeneration and heat-bred varieties demonstrably work, but on 5–15 year biological timelines that must start before yields fall.
19	Health & Medicine	Past	Household / Personal	Standing pandemic infrastructure, built before it is needed: genomic surveillance of wastewater in every city over 100,000 (weekly), a vaccine platform able to go from sequence to first doses in 100 days, and stockpiled PPE for 90 days of full-hospital use. Specification: hospital surge plans must double ICU capacity in 14 days (pre-mapped wards, cross-trained staff rosters); every household keeps a basic kit — thermometer, pulse oximeter, oral rehydration salts, fever medication for 2 weeks.	88%	Fund the platform in peacetime (idle capacity is the premium, not waste); drill sequence-to-dose annually; wastewater surveillance never switched off.	Surveillance plus platform vaccines compress response from years to months; the 100-day mission is engineering-feasible with standing capacity.
20	Health & Medicine	Past	Society / National	Standing pandemic infrastructure, built before it is needed: genomic surveillance of wastewater in every city over 100,000 (weekly), a vaccine platform able to go from sequence to first doses in 100 days, and stockpiled PPE for 90 days of full-hospital use. Specification: hospital surge plans must double ICU capacity in 14 days (pre-mapped wards, cross-trained staff rosters); every household keeps a basic kit — thermometer, pulse oximeter, oral rehydration salts, fever medication for 2 weeks.	80%	Fund the platform in peacetime (idle capacity is the premium, not waste); drill sequence-to-dose annually; wastewater surveillance never switched off.	Surveillance plus platform vaccines compress response from years to months; the 100-day mission is engineering-feasible with standing capacity.

Solution #	Field of Discipline	Timefram	Scale	Solution — Full Specification	Probability of S	How to Maximize the Chances (detailed)	Why This Probability
21	Health & Medicine	Past	Global / Civilizational	Standing pandemic infrastructure, built before it is needed: genomic surveillance of wastewater in every city over 100,000 (weekly), a vaccine platform able to go from sequence to first doses in 100 days, and stockpiled PPE for 90 days of full-hospital use. Specification: hospital surge plans must double ICU capacity in 14 days (pre-mapped wards, cross-trained staff rosters); every household keeps a basic kit — thermometer, pulse oximeter, oral rehydration salts, fever medication for 2 weeks.	68%	Fund the platform in peacetime (idle capacity is the premium, not waste); drill sequence-to-dose annually; wastewater surveillance never switched off.	Surveillance plus platform vaccines compress response from years to months; the 100-day mission is engineering-feasible with standing capacity.
22	Health & Medicine	Now	Household / Personal	Stewardship with teeth: antibiotics in human medicine by prescription audit only, agricultural antibiotic use cut 75% and banned for growth promotion, hospital infection-control staffing at 1 specialist per 100 beds, and a standing pipeline fund that pays for new antibiotics by subscription (delinked from sales volume). Specification: every hospital publishes resistance rates quarterly; households never stockpile or share antibiotics — leftover courses are returned to pharmacies.	70%	Close the agricultural loophole; tie hospital funding to published resistance rates; fund the subscription antibiotic pipeline internationally.	Stewardship programs cut resistance where enforced, but global leakage (agriculture, travel, unregulated sales) caps success.
23	Health & Medicine	Now	Society / National	Stewardship with teeth: antibiotics in human medicine by prescription audit only, agricultural antibiotic use cut 75% and banned for growth promotion, hospital infection-control staffing at 1 specialist per 100 beds, and a standing pipeline fund that pays for new antibiotics by subscription (delinked from sales volume). Specification: every hospital publishes resistance rates quarterly; households never stockpile or share antibiotics — leftover courses are returned to pharmacies.	62%	Close the agricultural loophole; tie hospital funding to published resistance rates; fund the subscription antibiotic pipeline internationally.	Stewardship programs cut resistance where enforced, but global leakage (agriculture, travel, unregulated sales) caps success.
24	Health & Medicine	Now	Global / Civilizational	Stewardship with teeth: antibiotics in human medicine by prescription audit only, agricultural antibiotic use cut 75% and banned for growth promotion, hospital infection-control staffing at 1 specialist per 100 beds, and a standing pipeline fund that pays for new antibiotics by subscription (delinked from sales volume). Specification: every hospital publishes resistance rates quarterly; households never stockpile or share antibiotics — leftover courses are returned to pharmacies.	50%	Close the agricultural loophole; tie hospital funding to published resistance rates; fund the subscription antibiotic pipeline internationally.	Stewardship programs cut resistance where enforced, but global leakage (agriculture, travel, unregulated sales) caps success.
25	Health & Medicine	Future	Household / Personal	Layered defence: universal indoor air standards in hospitals, schools, and transit (ventilation equivalent to 6 air changes/hour, upper-air UV in high-risk rooms), synthetic-biology screening of all commercial DNA synthesis orders, regional manufacturing of essential medicines (top 100 generics produced on at least 2 continents each), and a reserve clinical workforce (retired and part-time clinicians registered and drilled annually). Specification: no single country supplies more than 40% of any essential medicine's global volume.	56%	Layer at least 3 independent defences (air, screening, surge); keep the reserve clinical workforce registered and drilled; never let occupancy exceed 85% as the normal state.	Air standards and synthesis screening are effective layers, but no layer is complete; success means blunting, not abolishing, the worst case.
26	Health & Medicine	Future	Society / National	Layered defence: universal indoor air standards in hospitals, schools, and transit (ventilation equivalent to 6 air changes/hour, upper-air UV in high-risk rooms), synthetic-biology screening of all commercial DNA synthesis orders, regional manufacturing of essential medicines (top 100 generics produced on at least 2 continents each), and a reserve clinical workforce (retired and part-time clinicians registered and drilled annually). Specification: no single country supplies more than 40% of any essential medicine's global volume.	48%	Layer at least 3 independent defences (air, screening, surge); keep the reserve clinical workforce registered and drilled; never let occupancy exceed 85% as the normal state.	Air standards and synthesis screening are effective layers, but no layer is complete; success means blunting, not abolishing, the worst case.
27	Health & Medicine	Future	Global / Civilizational	Layered defence: universal indoor air standards in hospitals, schools, and transit (ventilation equivalent to 6 air changes/hour, upper-air UV in high-risk rooms), synthetic-biology screening of all commercial DNA synthesis orders, regional manufacturing of essential medicines (top 100 generics produced on at least 2 continents each), and a reserve clinical workforce (retired and part-time clinicians registered and drilled annually). Specification: no single country supplies more than 40% of any essential medicine's global volume.	36%	Layer at least 3 independent defences (air, screening, surge); keep the reserve clinical workforce registered and drilled; never let occupancy exceed 85% as the normal state.	Air standards and synthesis screening are effective layers, but no layer is complete; success means blunting, not abolishing, the worst case.
28	Energy	Past	Household / Personal	Strategic reserves and demand discipline: 90 days of net-import reserves for critical fuels (the standing international standard), efficiency law that cuts vehicle and building consumption 2% per year, and pre-planned rationing tiers so essential services (food logistics, hospitals, water) are served first. Specification: household level — 72 hours of light, heat, and cooking independent of the grid (LED + battery bank of 1 kWh, safe fuel stove); national level — no single supplier above 30% of any critical fuel.	92%	Hold reserves physically audited; pre-write the rationing tiers in law; cut demand 2% per year so the reserve covers more each year.	90-day reserves plus rationing tiers are proven (post-1973 systems survived later shocks without collapse).
29	Energy	Past	Society / National	Strategic reserves and demand discipline: 90 days of net-import reserves for critical fuels (the standing international standard), efficiency law that cuts vehicle and building consumption 2% per year, and pre-planned rationing tiers so essential services (food logistics, hospitals, water) are served first. Specification: household level — 72 hours of light, heat, and cooking independent of the grid (LED + battery bank of 1 kWh, safe fuel stove); national level — no single supplier above 30% of any critical fuel.	84%	Hold reserves physically audited; pre-write the rationing tiers in law; cut demand 2% per year so the reserve covers more each year.	90-day reserves plus rationing tiers are proven (post-1973 systems survived later shocks without collapse).
30	Energy	Past	Global / Civilizational	Strategic reserves and demand discipline: 90 days of net-import reserves for critical fuels (the standing international standard), efficiency law that cuts vehicle and building consumption 2% per year, and pre-planned rationing tiers so essential services (food logistics, hospitals, water) are served first. Specification: household level — 72 hours of light, heat, and cooking independent of the grid (LED + battery bank of 1 kWh, safe fuel stove); national level — no single supplier above 30% of any critical fuel.	72%	Hold reserves physically audited; pre-write the rationing tiers in law; cut demand 2% per year so the reserve covers more each year.	90-day reserves plus rationing tiers are proven (post-1973 systems survived later shocks without collapse).
31	Energy	Now	Household / Personal	Engineer for islanding: grids split into self-sustaining cells that can run 72 hours alone; critical facilities (hospitals, water, 911, data centres) on dual feeds plus on-site generation tested monthly under load; household standard — a 1–3 kWh battery with solar trickle keeps lights, phone, fridge, and medical devices alive 48–72 hours. Specification: black-start capability drilled twice a year; underground or harden the 20% of lines whose failure cascades furthest (Pareto of the grid).	84%	Drill black-start twice yearly under real load; test facility generators monthly; harden the cascading 20% of lines first (Pareto).	Islanding and dual feeds are standard utility engineering; blackouts persist where investment lagged, not where the design was unknown.
32	Energy	Now	Society / National	Engineer for islanding: grids split into self-sustaining cells that can run 72 hours alone; critical facilities (hospitals, water, 911, data centres) on dual feeds plus on-site generation tested monthly under load; household standard — a 1–3 kWh battery with solar trickle keeps lights, phone, fridge, and medical devices alive 48–72 hours. Specification: black-start capability drilled twice a year; underground or harden the 20% of lines whose failure cascades furthest (Pareto of the grid).	76%	Drill black-start twice yearly under real load; test facility generators monthly; harden the cascading 20% of lines first (Pareto).	Islanding and dual feeds are standard utility engineering; blackouts persist where investment lagged, not where the design was unknown.
33	Energy	Now	Global / Civilizational	Engineer for islanding: grids split into self-sustaining cells that can run 72 hours alone; critical facilities (hospitals, water, 911, data centres) on dual feeds plus on-site generation tested monthly under load; household standard — a 1–3 kWh battery with solar trickle keeps lights, phone, fridge, and medical devices alive 48–72 hours. Specification: black-start capability drilled twice a year; underground or harden the 20% of lines whose failure cascades furthest (Pareto of the grid).	64%	Drill black-start twice yearly under real load; test facility generators monthly; harden the cascading 20% of lines first (Pareto).	Islanding and dual feeds are standard utility engineering; blackouts persist where investment lagged, not where the design was unknown.
34	Energy	Future	Household / Personal	Stockpile the long-lead items: a strategic reserve of large transformers and grid components (the items with 12+ month lead times), EMP/GMD-hardened control systems for the top substations, and a designed analogue fallback — paper procedures, manual switching, and local generation — drilled, not just documented. Specification: every region can meet 30% of demand (the survival core: water, food cold-chain, hospitals, comms) on distributed local generation indefinitely.	58%	Buy the long-lead spares in peacetime budgets; keep analogue fallbacks drilled, not documented only; distribute 30% survival-core generation locally.	Transformer stockpiles and hardened controls are feasible; the risk is a once-in-centuries event whose timing cannot be forecast, so readiness must be permanent.
35	Energy	Future	Society / National	Stockpile the long-lead items: a strategic reserve of large transformers and grid components (the items with 12+ month lead times), EMP/GMD-hardened control systems for the top substations, and a designed analogue fallback — paper procedures, manual switching, and local generation — drilled, not just documented. Specification: every region can meet 30% of demand (the survival core: water, food cold-chain, hospitals, comms) on distributed local generation indefinitely.	50%	Buy the long-lead spares in peacetime budgets; keep analogue fallbacks drilled, not documented only; distribute 30% survival-core generation locally.	Transformer stockpiles and hardened controls are feasible; the risk is a once-in-centuries event whose timing cannot be forecast, so readiness must be permanent.
36	Energy	Future	Global / Civilizational	Stockpile the long-lead items: a strategic reserve of large transformers and grid components (the items with 12+ month lead times), EMP/GMD-hardened control systems for the top substations, and a designed analogue fallback — paper procedures, manual switching, and local generation — drilled, not just documented. Specification: every region can meet 30% of demand (the survival core: water, food cold-chain, hospitals, comms) on distributed local generation indefinitely.	38%	Buy the long-lead spares in peacetime budgets; keep analogue fallbacks drilled, not documented only; distribute 30% survival-core generation locally.	Transformer stockpiles and hardened controls are feasible; the risk is a once-in-centuries event whose timing cannot be forecast, so readiness must be permanent.
37	Shelter & Housing	Past	Household / Personal	Build to survive the known local worst: seismic codes to the 2,475-year event for new builds, retrofit of the highest-risk 20% of existing stock first (Pareto: unreinforced masonry and soft-storey buildings cause most deaths), firebreaks and non-combustible roofing in fire zones. Specification: every city holds emergency shelter capacity for 10% of its population for 30 days (pre-sited, pre-supplied), and every household keeps a 72-hour go-kit — documents, medication, water, light.	89%	Retrofit the deadliest 20% of stock first; enforce codes on occupancy, not just construction; shelter capacity pre-sited and pre-supplied.	Seismic and fire codes demonstrably cut deaths by 80–95% where enforced on the highest-risk building classes first.
38	Shelter & Housing	Past	Society / National	Build to survive the known local worst: seismic codes to the 2,475-year event for new builds, retrofit of the highest-risk 20% of existing stock first (Pareto: unreinforced masonry and soft-storey buildings cause most deaths), firebreaks and non-combustible roofing in fire zones. Specification: every city holds emergency shelter capacity for 10% of its population for 30 days (pre-sited, pre-supplied), and every household keeps a 72-hour go-kit — documents, medication, water, light.	81%	Retrofit the deadliest 20% of stock first; enforce codes on occupancy, not just construction; shelter capacity pre-sited and pre-supplied.	Seismic and fire codes demonstrably cut deaths by 80–95% where enforced on the highest-risk building classes first.

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39	Shelter & Housing	Past	Global / Civilizational	Build to survive the known local worst: seismic codes to the 2,475-year event for new builds, retrofit of the highest-risk 20% of existing stock first (Pareto: unreinforced masonry and soft-storey buildings cause most deaths), firebreaks and non-combustible roofing in fire zones. Specification: every city holds emergency shelter capacity for 10% of its population for 30 days (pre-sited, pre-supplied), and every household keeps a 72-hour go-kit — documents, medication, water, light.	69%	Retrofit the deadliest 20% of stock first; enforce codes on occupancy, not just construction; shelter capacity pre-sited and pre-supplied.	Seismic and fire codes demonstrably cut deaths by 80–95% where enforced on the highest-risk building classes first.
40	Shelter & Housing	Now	Household / Personal	Attack the constraint, which is supply near work: legalize mid-density housing on all residential land within 800 m of transit, approvals deemed granted after 12 months, public land leased (not sold) for permanently affordable homes at 30% of income benchmark, and vacancy taxes that make empty investment units unprofitable. Specification: a city is healthy when median rent is at or below 30% of median household income; publish the ratio quarterly and trigger automatic zoning release when it exceeds it for 2 consecutive years.	72%	Automatic triggers (ratio-based zoning release) remove the annual political fight; lease public land instead of selling; measure and publish the rent-to-income ratio quarterly.	Supply-side reforms work where tried (upzoning cities have measurably slowed rent growth), but face political resistance from incumbent owners.
41	Shelter & Housing	Now	Society / National	Attack the constraint, which is supply near work: legalize mid-density housing on all residential land within 800 m of transit, approvals deemed granted after 12 months, public land leased (not sold) for permanently affordable homes at 30% of income benchmark, and vacancy taxes that make empty investment units unprofitable. Specification: a city is healthy when median rent is at or below 30% of median household income; publish the ratio quarterly and trigger automatic zoning release when it exceeds it for 2 consecutive years.	64%	Automatic triggers (ratio-based zoning release) remove the annual political fight; lease public land instead of selling; measure and publish the rent-to-income ratio quarterly.	Supply-side reforms work where tried (upzoning cities have measurably slowed rent growth), but face political resistance from incumbent owners.
42	Shelter & Housing	Now	Global / Civilizational	Attack the constraint, which is supply near work: legalize mid-density housing on all residential land within 800 m of transit, approvals deemed granted after 12 months, public land leased (not sold) for permanently affordable homes at 30% of income benchmark, and vacancy taxes that make empty investment units unprofitable. Specification: a city is healthy when median rent is at or below 30% of median household income; publish the ratio quarterly and trigger automatic zoning release when it exceeds it for 2 consecutive years.	52%	Automatic triggers (ratio-based zoning release) remove the annual political fight; lease public land instead of selling; measure and publish the rent-to-income ratio quarterly.	Supply-side reforms work where tried (upzoning cities have measurably slowed rent growth), but face political resistance from incumbent owners.
43	Shelter & Housing	Future	Household / Personal	Managed transition instead of collapse: pre-mapped receiving cities with pre-built infrastructure capacity, public buyouts at pre-decline values funded by risk-zone insurance levies collected while values remain, and building law that stops new construction in zones projected uninsurable within 30 years. Specification: national housing plans must model the 2050 climate map, not the 2000 one; household level — never hold more than 50% of net worth in a single climate-exposed property.	60%	Collect the risk levy while values remain; sign receiving-city compacts early; stop new building in 30-year uninsurable zones immediately.	Managed retreat succeeds when funded before values collapse; it fails when attempted after, amid litigation and denial.
44	Shelter & Housing	Future	Society / National	Managed transition instead of collapse: pre-mapped receiving cities with pre-built infrastructure capacity, public buyouts at pre-decline values funded by risk-zone insurance levies collected while values remain, and building law that stops new construction in zones projected uninsurable within 30 years. Specification: national housing plans must model the 2050 climate map, not the 2000 one; household level — never hold more than 50% of net worth in a single climate-exposed property.	52%	Collect the risk levy while values remain; sign receiving-city compacts early; stop new building in 30-year uninsurable zones immediately.	Managed retreat succeeds when funded before values collapse; it fails when attempted after, amid litigation and denial.
45	Shelter & Housing	Future	Global / Civilizational	Managed transition instead of collapse: pre-mapped receiving cities with pre-built infrastructure capacity, public buyouts at pre-decline values funded by risk-zone insurance levies collected while values remain, and building law that stops new construction in zones projected uninsurable within 30 years. Specification: national housing plans must model the 2050 climate map, not the 2000 one; household level — never hold more than 50% of net worth in a single climate-exposed property.	40%	Collect the risk levy while values remain; sign receiving-city compacts early; stop new building in 30-year uninsurable zones immediately.	Managed retreat succeeds when funded before values collapse; it fails when attempted after, amid litigation and denial.
46	Security & Defense	Past	Household / Personal	Prevention is the only real solution once killing starts: early-warning indicators (hate-media licensing, militia formation, arms flows) monitored with published thresholds that trigger automatic diplomatic and legal consequences; constitutional hard limits on emergency powers (sunset clauses of 90 days, legislative renewal); and protected humanitarian corridors negotiated in advance frameworks. Specification: civilian layer — neighbourhood mutual-aid rosters, documented evidence habits (photos, records kept off-site), and pre-planned exit routes with 72-hour kits.	53%	Act on the first two indicators, not the tenth; keep exit routes and records as standing household practice in at-risk regions; international triggers automatic, not negotiated mid-crisis.	Once mass violence begins, no intervention reliably stops it quickly; prevention indicators work only if acted on before mobilization completes.
47	Security & Defense	Past	Society / National	Prevention is the only real solution once killing starts: early-warning indicators (hate-media licensing, militia formation, arms flows) monitored with published thresholds that trigger automatic diplomatic and legal consequences; constitutional hard limits on emergency powers (sunset clauses of 90 days, legislative renewal); and protected humanitarian corridors negotiated in advance frameworks. Specification: civilian layer — neighbourhood mutual-aid rosters, documented evidence habits (photos, records kept off-site), and pre-planned exit routes with 72-hour kits.	45%	Act on the first two indicators, not the tenth; keep exit routes and records as standing household practice in at-risk regions; international triggers automatic, not negotiated mid-crisis.	Once mass violence begins, no intervention reliably stops it quickly; prevention indicators work only if acted on before mobilization completes.
48	Security & Defense	Past	Global / Civilizational	Prevention is the only real solution once killing starts: early-warning indicators (hate-media licensing, militia formation, arms flows) monitored with published thresholds that trigger automatic diplomatic and legal consequences; constitutional hard limits on emergency powers (sunset clauses of 90 days, legislative renewal); and protected humanitarian corridors negotiated in advance frameworks. Specification: civilian layer — neighbourhood mutual-aid rosters, documented evidence habits (photos, records kept off-site), and pre-planned exit routes with 72-hour kits.	33%	Act on the first two indicators, not the tenth; keep exit routes and records as standing household practice in at-risk regions; international triggers automatic, not negotiated mid-crisis.	Once mass violence begins, no intervention reliably stops it quickly; prevention indicators work only if acted on before mobilization completes.
49	Security & Defense	Now	Household / Personal	Focused deterrence, the proven Pareto intervention: identify the tiny network driving most violence (typically under 0.5% of a city's population), offer services and exit routes with one hand and certain, swift prosecution with the other; light and maintain the worst micro-locations (lighting, sightlines, occupied storefronts); violence-interrupter teams staffed by credible locals. Specification: measure success by shootings in the target micro-zones, reviewed monthly; every intervention audited for civil-rights compliance.	82%	Keep the target list small and evidence-based; deliver the services offer genuinely; audit civil-rights compliance monthly so legitimacy survives.	Focused deterrence is among the most evaluated interventions in criminology, with violence reductions of 30–60% in target networks across many cities.
50	Security & Defense	Now	Society / National	Focused deterrence, the proven Pareto intervention: identify the tiny network driving most violence (typically under 0.5% of a city's population), offer services and exit routes with one hand and certain, swift prosecution with the other; light and maintain the worst micro-locations (lighting, sightlines, occupied storefronts); violence-interrupter teams staffed by credible locals. Specification: measure success by shootings in the target micro-zones, reviewed monthly; every intervention audited for civil-rights compliance.	74%	Keep the target list small and evidence-based; deliver the services offer genuinely; audit civil-rights compliance monthly so legitimacy survives.	Focused deterrence is among the most evaluated interventions in criminology, with violence reductions of 30–60% in target networks across many cities.
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52	Security & Defense	Future	Household / Personal	Regulate and shield in advance: mandatory remote-identification and geofencing on all drones sold, counter-drone authority and equipment at critical sites (airports, utilities, mass events) with legal rules of engagement written before incidents, export controls on autonomous targeting components, and an international ban framework on fully autonomous lethal systems. Specification: critical infrastructure must pass an annual red-team drone assessment; households and businesses rely on hardened, setback design for the highest-value targets rather than private countermeasures.	64%	Mandate remote ID and geofencing at point of sale now; annual red-team tests for critical sites; write the legal rules of engagement before the first mass incident.	Drone regulation and site hardening are technically ready; the gap is legislative speed versus consumer-technology speed.
53	Security & Defense	Future	Society / National	Regulate and shield in advance: mandatory remote-identification and geofencing on all drones sold, counter-drone authority and equipment at critical sites (airports, utilities, mass events) with legal rules of engagement written before incidents, export controls on autonomous targeting components, and an international ban framework on fully autonomous lethal systems. Specification: critical infrastructure must pass an annual red-team drone assessment; households and businesses rely on hardened, setback design for the highest-value targets rather than private countermeasures.	56%	Mandate remote ID and geofencing at point of sale now; annual red-team tests for critical sites; write the legal rules of engagement before the first mass incident.	Drone regulation and site hardening are technically ready; the gap is legislative speed versus consumer-technology speed.
54	Security & Defense	Future	Global / Civilizational	Regulate and shield in advance: mandatory remote-identification and geofencing on all drones sold, counter-drone authority and equipment at critical sites (airports, utilities, mass events) with legal rules of engagement written before incidents, export controls on autonomous targeting components, and an international ban framework on fully autonomous lethal systems. Specification: critical infrastructure must pass an annual red-team drone assessment; households and businesses rely on hardened, setback design for the highest-value targets rather than private countermeasures.	44%	Mandate remote ID and geofencing at point of sale now; annual red-team tests for critical sites; write the legal rules of engagement before the first mass incident.	Drone regulation and site hardening are technically ready; the gap is legislative speed versus consumer-technology speed.
55	Governance & Law	Past	Household / Personal	Design constitutions against their own abolition: eternity clauses protecting core rights from amendment, emergency powers with hard 90-day sunsets and escalating supermajorities for renewal, judicial appointments split across at least 3 independent bodies, and electoral administration insulated from the executive. Specification: every emergency measure auto-expires; every rights restriction must be re-justified publicly at renewal; citizens keep certified copies of identity and property records in personal custody.	68%	Distribute appointment power across 3+ bodies; keep citizens' own copies of identity and property records; rehearse constitutional defence in civil society, not just courts.	Eternity clauses and sunset rules have preserved some democracies under pressure, but determined majorities with courts captured can still prevail; design raises the cost, it does not make it infinite.

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56	Governance & Law	Past	Society / National	Design constitutions against their own abolition: eternity clauses protecting core rights from amendment, emergency powers with hard 90-day sunsets and escalating supermajorities for renewal, judicial appointments split across at least 3 independent bodies, and electoral administration insulated from the executive. Specification: every emergency measure auto-expires; every rights restriction must be re-justified publicly at renewal; citizens keep certified copies of identity and property records in personal custody.	60%	Distribute appointment power across 3+ bodies; keep citizens' own copies of identity and property records; rehearse constitutional defence in civil society, not just courts.	Eternity clauses and sunset rules have preserved some democracies under pressure, but determined majorities with courts captured can still prevail; design raises the cost, it does not make it infinite.
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58	Governance & Law	Now	Household / Personal	Radical transparency with automatic publication: all public contracts, beneficial ownership of companies bidding, officials' assets, and court statistics published in open data by default — secrecy must be justified line by line, publication is the default state. Independent audit institutions with budgets fixed by formula (e.g., 0.1% of public spending) so governments cannot starve them. Specification: procurement above a low threshold is e-tendered with full bid disclosure; whistleblowers receive statutory protection plus a share of recovered funds.	76%	Make publication the default state of every record; fix audit budgets by formula; protect whistleblowers with recovery shares, not just promises.	Open contracting and asset disclosure measurably reduce corruption where publication is automatic and audited; they fail where disclosure is voluntary.
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61	Governance & Law	Future	Household / Personal	Rebuild trust as infrastructure: citizen assemblies chosen by lottery with real decision power over defined questions, radical localism (decisions at the smallest competent scale, where people can verify outcomes personally), and service-delivery proof — publish wait times, error rates, and fix rates for every public service monthly. Specification: trust is measured (surveyed quarterly, published); agencies falling below threshold trigger automatic independent review.	55%	Start with services people can verify personally (wait times, fix rates); give lottery assemblies real, bounded decisions; publish the trust metrics and act on them visibly.	Trust is rebuilt only by verified performance over years; assemblies and localism help, but no design manufactures legitimacy quickly.
62	Governance & Law	Future	Society / National	Rebuild trust as infrastructure: citizen assemblies chosen by lottery with real decision power over defined questions, radical localism (decisions at the smallest competent scale, where people can verify outcomes personally), and service-delivery proof — publish wait times, error rates, and fix rates for every public service monthly. Specification: trust is measured (surveyed quarterly, published); agencies falling below threshold trigger automatic independent review.	47%	Start with services people can verify personally (wait times, fix rates); give lottery assemblies real, bounded decisions; publish the trust metrics and act on them visibly.	Trust is rebuilt only by verified performance over years; assemblies and localism help, but no design manufactures legitimacy quickly.
63	Governance & Law	Future	Global / Civilizational	Rebuild trust as infrastructure: citizen assemblies chosen by lottery with real decision power over defined questions, radical localism (decisions at the smallest competent scale, where people can verify outcomes personally), and service-delivery proof — publish wait times, error rates, and fix rates for every public service monthly. Specification: trust is measured (surveyed quarterly, published); agencies falling below threshold trigger automatic independent review.	35%	Start with services people can verify personally (wait times, fix rates); give lottery assemblies real, bounded decisions; publish the trust metrics and act on them visibly.	Trust is rebuilt only by verified performance over years; assemblies and localism help, but no design manufactures legitimacy quickly.
64	Economy & Finance	Past	Household / Personal	Circuit breakers built in advance: deposit insurance with pre-funded reserves, central-bank lender-of-last-resort facilities designed before crises, automatic fiscal stabilizers (unemployment benefits that expand by formula, not by vote), and bank resolution law that wipes out shareholders and management — never depositors — first. Specification: systemically important banks hold capital of at least 12% of risk-weighted assets with a leverage cap; households keep 3–6 months of expenses in insured deposits, never in a single institution above the insured limit.	91%	Pre-fund the insurance; write resolution law in peacetime; keep household reserves inside insured limits across institutions.	Deposit insurance ended bank-run dynamics wherever credibly pre-funded; automatic stabilizers are proven counter-cyclical machinery.
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66	Economy & Finance	Past	Global / Civilizational	Circuit breakers built in advance: deposit insurance with pre-funded reserves, central-bank lender-of-last-resort facilities designed before crises, automatic fiscal stabilizers (unemployment benefits that expand by formula, not by vote), and bank resolution law that wipes out shareholders and management — never depositors — first. Specification: systemically important banks hold capital of at least 12% of risk-weighted assets with a leverage cap; households keep 3–6 months of expenses in insured deposits, never in a single institution above the insured limit.	71%	Pre-fund the insurance; write resolution law in peacetime; keep household reserves inside insured limits across institutions.	Deposit insurance ended bank-run dynamics wherever credibly pre-funded; automatic stabilizers are proven counter-cyclical machinery.
67	Economy & Finance	Now	Household / Personal	De-leverage by design: mortgage rules that cap total debt service at 35–40% of verified income under a stressed rate (+2%), automatic refinancing windows when rates fall, public development of non-speculative housing supply, and insolvency law that gives households a fresh start within 3 years instead of permanent debt peonage. Specification: personal rule — total debt payments under 30% of take-home; one income loss must be survivable for 6 months on reserves plus benefits; never hold employer stock plus job plus mortgage in one company town exposure.	71%	Apply the stressed-rate test (+2%) to every loan; open refinancing windows automatically when rates fall; enforce the 30% personal debt-payment ceiling.	Debt-service caps at origination demonstrably reduce default cascades; the obstacle is the credit boom's political popularity.
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70	Economy & Finance	Future	Household / Personal	Share the machine dividend by structure, not charity: sovereign and pension funds holding broad equity stakes so productivity gains flow to all citizens as dividends; reduced standard working hours as productivity rises (the historical pattern, deliberately continued); portable benefits attached to the person, not the job; and public investment in the irreducibly human sectors (care, craft, education, restoration) at the scale of the displacement. Specification: transition planning uses the fastest plausible displacement curve, with retraining accounts funded per worker before displacement, not after.	52%	Capitalize citizen funds from automation rents early; attach benefits to persons in law now; cut standard hours in step with measured productivity, sector by sector.	Ownership-sharing mechanisms are economically sound but politically unbuilt; success depends on constructing them during growth, not after displacement peaks.
71	Economy & Finance	Future	Society / National	Share the machine dividend by structure, not charity: sovereign and pension funds holding broad equity stakes so productivity gains flow to all citizens as dividends; reduced standard working hours as productivity rises (the historical pattern, deliberately continued); portable benefits attached to the person, not the job; and public investment in the irreducibly human sectors (care, craft, education, restoration) at the scale of the displacement. Specification: transition planning uses the fastest plausible displacement curve, with retraining accounts funded per worker before displacement, not after.	44%	Capitalize citizen funds from automation rents early; attach benefits to persons in law now; cut standard hours in step with measured productivity, sector by sector.	Ownership-sharing mechanisms are economically sound but politically unbuilt; success depends on constructing them during growth, not after displacement peaks.
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73	Technology & AI	Past	Household / Personal	Precaution with deadlines: new general-purpose technologies pass staged safety gates (lab, limited, general) with published criteria and automatic review dates; manufacturers carry strict liability for latent harms, funded by mandatory insurance during the entire exposure tail; and a standing technology-assessment office publishes plain-language risk registers. Specification: no product with irreversible population-level exposure ships without a funded 30-year monitoring program; households treat 'new and unstudied' as a reason to delay adoption for non-essentials.	74%	Attach the 30-year monitoring fund to the product, not the taxpayer; give the assessment office subpoena power over safety data; default to delay for irreversible exposures.	Staged gates and strict liability eventually controlled lead and asbestos, but only after long harm tails; earlier gates would have raised success sharply.
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76	Technology & AI	Now	Household / Personal	Assume breach, engineer recovery: immutable offline backups tested by full restoration quarterly (a backup never restored is a rumour), network segmentation so one infection cannot cross from office systems to operations, multi-factor authentication on 100% of remote and privileged access, and 24-hour incident reporting with a national assistance duty. Specification for households and small organizations: automatic updates on, unique passwords in a manager, 3-2-1 backups (3 copies, 2 media, 1 off-site/offline) for irreplaceable data — photos, records, work.	86%	Restore-test backups quarterly — an untested backup is a rumour; segment operations from office networks; 100% MFA on remote and privileged access, no exceptions.	Recovery engineering (immutable backups, segmentation, MFA) defeats the large majority of real-world intrusions; breaches persist where basics are skipped.
77	Technology & AI	Now	Society / National	Assume breach, engineer recovery: immutable offline backups tested by full restoration quarterly (a backup never restored is a rumour), network segmentation so one infection cannot cross from office systems to operations, multi-factor authentication on 100% of remote and privileged access, and 24-hour incident reporting with a national assistance duty. Specification for households and small organizations: automatic updates on, unique passwords in a manager, 3-2-1 backups (3 copies, 2 media, 1 off-site/offline) for irreplaceable data — photos, records, work.	78%	Restore-test backups quarterly — an untested backup is a rumour; segment operations from office networks; 100% MFA on remote and privileged access, no exceptions.	Recovery engineering (immutable backups, segmentation, MFA) defeats the large majority of real-world intrusions; breaches persist where basics are skipped.
78	Technology & AI	Now	Global / Civilizational	Assume breach, engineer recovery: immutable offline backups tested by full restoration quarterly (a backup never restored is a rumour), network segmentation so one infection cannot cross from office systems to operations, multi-factor authentication on 100% of remote and privileged access, and 24-hour incident reporting with a national assistance duty. Specification for households and small organizations: automatic updates on, unique passwords in a manager, 3-2-1 backups (3 copies, 2 media, 1 off-site/offline) for irreplaceable data — photos, records, work.	66%	Restore-test backups quarterly — an untested backup is a rumour; segment operations from office networks; 100% MFA on remote and privileged access, no exceptions.	Recovery engineering (immutable backups, segmentation, MFA) defeats the large majority of real-world intrusions; breaches persist where basics are skipped.
79	Technology & AI	Future	Household / Personal	Governance proportional to capability: frontier training runs above defined compute thresholds are registered and evaluated for dangerous capabilities before release; critical infrastructure keeps human-verified analogue fallbacks and禁止 fully autonomous control of weapons, grid switching, and medical dosing; liability law makes deployers answer for system outputs; and international verification for the largest training facilities, modelled on nuclear safeguards. Specification: every essential service must demonstrate a manual mode, drilled annually; personal rule — no single AI or platform holds your only copy of anything important.	50%	Register and evaluate above compute thresholds before release; drill manual modes of essential services annually; keep deployer liability absolute so caution is priced in.	Capability governance is unproven at frontier scale and races commercial pressure; evaluations and fallbacks reduce risk materially without eliminating it.
80	Technology & AI	Future	Society / National	Governance proportional to capability: frontier training runs above defined compute thresholds are registered and evaluated for dangerous capabilities before release; critical infrastructure keeps human-verified analogue fallbacks and禁止 fully autonomous control of weapons, grid switching, and medical dosing; liability law makes deployers answer for system outputs; and international verification for the largest training facilities, modelled on nuclear safeguards. Specification: every essential service must demonstrate a manual mode, drilled annually; personal rule — no single AI or platform holds your only copy of anything important.	42%	Register and evaluate above compute thresholds before release; drill manual modes of essential services annually; keep deployer liability absolute so caution is priced in.	Capability governance is unproven at frontier scale and races commercial pressure; evaluations and fallbacks reduce risk materially without eliminating it.
81	Technology & AI	Future	Global / Civilizational	Governance proportional to capability: frontier training runs above defined compute thresholds are registered and evaluated for dangerous capabilities before release; critical infrastructure keeps human-verified analogue fallbacks and禁止 fully autonomous control of weapons, grid switching, and medical dosing; liability law makes deployers answer for system outputs; and international verification for the largest training facilities, modelled on nuclear safeguards. Specification: every essential service must demonstrate a manual mode, drilled annually; personal rule — no single AI or platform holds your only copy of anything important.	30%	Register and evaluate above compute thresholds before release; drill manual modes of essential services annually; keep deployer liability absolute so caution is priced in.	Capability governance is unproven at frontier scale and races commercial pressure; evaluations and fallbacks reduce risk materially without eliminating it.
82	Environment & Climate	Past	Household / Personal	Close the loop with law: extraction and pollution permits capped by measured regeneration rates of the specific ecosystem (not national averages), soil-loss limits per hectare enforced like building codes, and full-cost accounting where the price of a product includes its restoration cost. Specification: farmland soil organic matter and erosion measured on a 5-year cycle; land degrading past threshold enters mandatory restoration with cost-sharing.	79%	Measure the specific ecosystem's regeneration, not national averages; put soil and erosion limits in enforceable code; price restoration into products.	Regeneration-capped extraction works where measured per-ecosystem; failures trace to caps set by politics instead of measurement.
83	Environment & Climate	Past	Society / National	Close the loop with law: extraction and pollution permits capped by measured regeneration rates of the specific ecosystem (not national averages), soil-loss limits per hectare enforced like building codes, and full-cost accounting where the price of a product includes its restoration cost. Specification: farmland soil organic matter and erosion measured on a 5-year cycle; land degrading past threshold enters mandatory restoration with cost-sharing.	71%	Measure the specific ecosystem's regeneration, not national averages; put soil and erosion limits in enforceable code; price restoration into products.	Regeneration-capped extraction works where measured per-ecosystem; failures trace to caps set by politics instead of measurement.
84	Environment & Climate	Past	Global / Civilizational	Close the loop with law: extraction and pollution permits capped by measured regeneration rates of the specific ecosystem (not national averages), soil-loss limits per hectare enforced like building codes, and full-cost accounting where the price of a product includes its restoration cost. Specification: farmland soil organic matter and erosion measured on a 5-year cycle; land degrading past threshold enters mandatory restoration with cost-sharing.	59%	Measure the specific ecosystem's regeneration, not national averages; put soil and erosion limits in enforceable code; price restoration into products.	Regeneration-capped extraction works where measured per-ecosystem; failures trace to caps set by politics instead of measurement.
85	Environment & Climate	Now	Household / Personal	Adapt to the climate that exists: building codes and infrastructure designed to the 2050 worst-decile weather, urban canopy targets of 30–40% to cut street-level heat, floodplains returned to water with buyouts instead of repeated rebuilds, and heat action plans with named officials, cooling centres within a 15-minute walk, and check-in systems for isolated elderly. Specification: every home in a risk zone has a written plan — cooling or evacuation, 72-hour supplies, medication continuity; insurers and cities publish address-level risk so decisions are informed.	77%	Design to the 2050 worst decile, not historical weather; canopy and cooling access targets with named owners; address-level risk published so households can act.	Adaptation measures (canopy, codes, heat plans) have strong measured effects on deaths and damage per event.
86	Environment & Climate	Now	Society / National	Adapt to the climate that exists: building codes and infrastructure designed to the 2050 worst-decile weather, urban canopy targets of 30–40% to cut street-level heat, floodplains returned to water with buyouts instead of repeated rebuilds, and heat action plans with named officials, cooling centres within a 15-minute walk, and check-in systems for isolated elderly. Specification: every home in a risk zone has a written plan — cooling or evacuation, 72-hour supplies, medication continuity; insurers and cities publish address-level risk so decisions are informed.	69%	Design to the 2050 worst decile, not historical weather; canopy and cooling access targets with named owners; address-level risk published so households can act.	Adaptation measures (canopy, codes, heat plans) have strong measured effects on deaths and damage per event.
87	Environment & Climate	Now	Global / Civilizational	Adapt to the climate that exists: building codes and infrastructure designed to the 2050 worst-decile weather, urban canopy targets of 30–40% to cut street-level heat, floodplains returned to water with buyouts instead of repeated rebuilds, and heat action plans with named officials, cooling centres within a 15-minute walk, and check-in systems for isolated elderly. Specification: every home in a risk zone has a written plan — cooling or evacuation, 72-hour supplies, medication continuity; insurers and cities publish address-level risk so decisions are informed.	57%	Design to the 2050 worst decile, not historical weather; canopy and cooling access targets with named owners; address-level risk published so households can act.	Adaptation measures (canopy, codes, heat plans) have strong measured effects on deaths and damage per event.
88	Environment & Climate	Future	Household / Personal	Both brakes and lifeboats: emissions cuts on the fastest feasible curve (the only lever on the cause), plus planned adaptation at civilizational scale — protected agricultural relocation zones, freshwater megaprojects only where hydrology closes, migration frameworks negotiated before flows become crises (status, quotas, receiving-city investment), and geoengineering research governed internationally with deployment requiring multinational consent. Specification: all long-lived infrastructure (50+ year life) sited and designed against the high-end sea-level and heat scenarios, not the median.	46%	Cut emissions on the fastest feasible curve — the only causal lever; pre-negotiate migration frameworks; site all 50-year infrastructure against high-end scenarios.	Tipping cascades involve physics already partly committed; success here means slowing and adapting, and it is the lowest-probability row in this file by honesty, not by effort.
89	Environment & Climate	Future	Society / National	Both brakes and lifeboats: emissions cuts on the fastest feasible curve (the only lever on the cause), plus planned adaptation at civilizational scale — protected agricultural relocation zones, freshwater megaprojects only where hydrology closes, migration frameworks negotiated before flows become crises (status, quotas, receiving-city investment), and geoengineering research governed internationally with deployment requiring multinational consent. Specification: all long-lived infrastructure (50+ year life) sited and designed against the high-end sea-level and heat scenarios, not the median.	38%	Cut emissions on the fastest feasible curve — the only causal lever; pre-negotiate migration frameworks; site all 50-year infrastructure against high-end scenarios.	Tipping cascades involve physics already partly committed; success here means slowing and adapting, and it is the lowest-probability row in this file by honesty, not by effort.

Solution #	Field of Discipline	Timefram	Scale	Solution — Full Specification	Probability of S	How to Maximize the Chances (detailed)	Why This Probability
90	Environment & Climate	Future	Global / Civilizational	Both brakes and lifeboats: emissions cuts on the fastest feasible curve (the only lever on the cause), plus planned adaptation at civilizational scale — protected agricultural relocation zones, freshwater megaprojects only where hydrology closes, migration frameworks negotiated before flows become crises (status, quotas, receiving-city investment), and geoengineering research governed internationally with deployment requiring multinational consent. Specification: all long-lived infrastructure (50+ year life) sited and designed against the high-end sea-level and heat scenarios, not the median.	26%	Cut emissions on the fastest feasible curve — the only causal lever; pre-negotiate migration frameworks; site all 50-year infrastructure against high-end scenarios.	Tipping cascades involve physics already partly committed; success here means slowing and adapting, and it is the lowest-probability row in this file by honesty, not by effort.
91	Information & Media	Past	Household / Personal	Pluralism as armour: no single owner or state organ may control a dominant share of any population's information diet (strict cross-media ownership caps), public-interest media funded by formula and governed at arm's length, and school curricula that teach source-tracing — every claim followed to its origin — as a core skill from age 10. Specification: during crises, official information carries verifiable provenance (signed releases, raw data published with summaries) so citizens can check, not just trust.	66%	Enforce ownership caps before capture, not after; teach source-tracing from age 10; sign official crisis information so it can be verified.	Plural ownership and provenance norms blunted propaganda in some societies; where one channel monopolized, they failed — structure decides.
92	Information & Media	Past	Society / National	Pluralism as armour: no single owner or state organ may control a dominant share of any population's information diet (strict cross-media ownership caps), public-interest media funded by formula and governed at arm's length, and school curricula that teach source-tracing — every claim followed to its origin — as a core skill from age 10. Specification: during crises, official information carries verifiable provenance (signed releases, raw data published with summaries) so citizens can check, not just trust.	58%	Enforce ownership caps before capture, not after; teach source-tracing from age 10; sign official crisis information so it can be verified.	Plural ownership and provenance norms blunted propaganda in some societies; where one channel monopolized, they failed — structure decides.
93	Information & Media	Past	Global / Civilizational	Pluralism as armour: no single owner or state organ may control a dominant share of any population's information diet (strict cross-media ownership caps), public-interest media funded by formula and governed at arm's length, and school curricula that teach source-tracing — every claim followed to its origin — as a core skill from age 10. Specification: during crises, official information carries verifiable provenance (signed releases, raw data published with summaries) so citizens can check, not just trust.	46%	Enforce ownership caps before capture, not after; teach source-tracing from age 10; sign official crisis information so it can be verified.	Plural ownership and provenance norms blunted propaganda in some societies; where one channel monopolized, they failed — structure decides.
94	Information & Media	Now	Household / Personal	Change the incentives, not just the content: recommender systems above a scale threshold must offer a chronological/non-optimized feed as a real default option, political advertising carries a public registry of sponsor, spend, and targeting criteria, and proven coordinated inauthentic campaigns trigger platform liability. Personal specification: a two-source rule for any alarming claim (two independent, named, primary sources before acting or forwarding), primary documents over commentary, and one scheduled news window per day instead of continuous feed exposure.	65%	Make the non-optimized feed a true default option; personal rule: two independent primary sources before acting on any alarming claim; one news window daily.	Feed-choice and ad-registry rules are new and partially tested; personal two-source discipline is fully effective at individual scale immediately.
95	Information & Media	Now	Society / National	Change the incentives, not just the content: recommender systems above a scale threshold must offer a chronological/non-optimized feed as a real default option, political advertising carries a public registry of sponsor, spend, and targeting criteria, and proven coordinated inauthentic campaigns trigger platform liability. Personal specification: a two-source rule for any alarming claim (two independent, named, primary sources before acting or forwarding), primary documents over commentary, and one scheduled news window per day instead of continuous feed exposure.	57%	Make the non-optimized feed a true default option; personal rule: two independent primary sources before acting on any alarming claim; one news window daily.	Feed-choice and ad-registry rules are new and partially tested; personal two-source discipline is fully effective at individual scale immediately.
96	Information & Media	Now	Global / Civilizational	Change the incentives, not just the content: recommender systems above a scale threshold must offer a chronological/non-optimized feed as a real default option, political advertising carries a public registry of sponsor, spend, and targeting criteria, and proven coordinated inauthentic campaigns trigger platform liability. Personal specification: a two-source rule for any alarming claim (two independent, named, primary sources before acting or forwarding), primary documents over commentary, and one scheduled news window per day instead of continuous feed exposure.	45%	Make the non-optimized feed a true default option; personal rule: two independent primary sources before acting on any alarming claim; one news window daily.	Feed-choice and ad-registry rules are new and partially tested; personal two-source discipline is fully effective at individual scale immediately.
97	Information & Media	Future	Household / Personal	Provenance infrastructure: cameras and recorders sign content at capture (cryptographic content credentials as the default in professional and official contexts), courts and newsrooms verify by signature chain rather than appearance, and legal process treats unsigned dramatic media as allegation, not evidence. Specification: households and organizations keep original, signed, off-platform copies of identity, property, and credential records — paper originals of the documents that prove who you are and what you own.	69%	Mandate content credentials in official and professional contexts; keep paper originals of identity and property documents; treat unsigned dramatic media as allegation in law.	Capture-signed provenance is mature cryptography; success depends on adoption by courts, newsrooms, and platforms before forgery becomes universal.
98	Information & Media	Future	Society / National	Provenance infrastructure: cameras and recorders sign content at capture (cryptographic content credentials as the default in professional and official contexts), courts and newsrooms verify by signature chain rather than appearance, and legal process treats unsigned dramatic media as allegation, not evidence. Specification: households and organizations keep original, signed, off-platform copies of identity, property, and credential records — paper originals of the documents that prove who you are and what you own.	61%	Mandate content credentials in official and professional contexts; keep paper originals of identity and property documents; treat unsigned dramatic media as allegation in law.	Capture-signed provenance is mature cryptography; success depends on adoption by courts, newsrooms, and platforms before forgery becomes universal.
99	Information & Media	Future	Global / Civilizational	Provenance infrastructure: cameras and recorders sign content at capture (cryptographic content credentials as the default in professional and official contexts), courts and newsrooms verify by signature chain rather than appearance, and legal process treats unsigned dramatic media as allegation, not evidence. Specification: households and organizations keep original, signed, off-platform copies of identity, property, and credential records — paper originals of the documents that prove who you are and what you own.	49%	Mandate content credentials in official and professional contexts; keep paper originals of identity and property documents; treat unsigned dramatic media as allegation in law.	Capture-signed provenance is mature cryptography; success depends on adoption by courts, newsrooms, and platforms before forgery becomes universal.
100	Psychology & Mind	Past	Household / Personal	Treat trauma as a public-health quantity, not a private shame: post-crisis psychological first aid deployed like physical first aid (trained community responders at a ratio of roughly 1 per 500 people), school-based programs after collective events, and long-horizon funding — trauma services budgeted for 25 years after major events, because that is the measured tail. Specification: households know the signs (hypervigilance, numbing, rage, withdrawal) and the rule — connection before correction, routine before analysis, professional help early, not last.	73%	Fund the 25-year tail, not the 6-month headline; train community responders at 1:500 before crises; keep routines and schooling running as treatment infrastructure.	Community-based psychological first aid and school programs measurably reduce long-term disorder after collective trauma where sustained for years.
101	Psychology & Mind	Past	Society / National	Treat trauma as a public-health quantity, not a private shame: post-crisis psychological first aid deployed like physical first aid (trained community responders at a ratio of roughly 1 per 500 people), school-based programs after collective events, and long-horizon funding — trauma services budgeted for 25 years after major events, because that is the measured tail. Specification: households know the signs (hypervigilance, numbing, rage, withdrawal) and the rule — connection before correction, routine before analysis, professional help early, not last.	65%	Fund the 25-year tail, not the 6-month headline; train community responders at 1:500 before crises; keep routines and schooling running as treatment infrastructure.	Community-based psychological first aid and school programs measurably reduce long-term disorder after collective trauma where sustained for years.
102	Psychology & Mind	Past	Global / Civilizational	Treat trauma as a public-health quantity, not a private shame: post-crisis psychological first aid deployed like physical first aid (trained community responders at a ratio of roughly 1 per 500 people), school-based programs after collective events, and long-horizon funding — trauma services budgeted for 25 years after major events, because that is the measured tail. Specification: households know the signs (hypervigilance, numbing, rage, withdrawal) and the rule — connection before correction, routine before analysis, professional help early, not last.	53%	Fund the 25-year tail, not the 6-month headline; train community responders at 1:500 before crises; keep routines and schooling running as treatment infrastructure.	Community-based psychological first aid and school programs measurably reduce long-term disorder after collective trauma where sustained for years.
103	Psychology & Mind	Now	Household / Personal	Rebuild the inputs: sleep protected as infrastructure (school start times aligned to adolescent biology, no-screens wind-down norms), daily physical activity of 60 minutes for youth built into the school day, third places (libraries, clubs, leagues) funded as health infrastructure, and platform design rules for minors (no infinite scroll, no engagement-optimized feeds). Specification, personal: fixed sleep window, morning light exposure, one recurring in-person group activity weekly minimum — belonging is scheduled, not hoped for.	75%	Schedule belonging like medicine (one weekly in-person group minimum); protect the sleep window; build third places as funded health infrastructure.	Sleep, activity, and in-person belonging have large, replicated effects; the constraint is environmental design, not knowledge.
104	Psychology & Mind	Now	Society / National	Rebuild the inputs: sleep protected as infrastructure (school start times aligned to adolescent biology, no-screens wind-down norms), daily physical activity of 60 minutes for youth built into the school day, third places (libraries, clubs, leagues) funded as health infrastructure, and platform design rules for minors (no infinite scroll, no engagement-optimized feeds). Specification, personal: fixed sleep window, morning light exposure, one recurring in-person group activity weekly minimum — belonging is scheduled, not hoped for.	67%	Schedule belonging like medicine (one weekly in-person group minimum); protect the sleep window; build third places as funded health infrastructure.	Sleep, activity, and in-person belonging have large, replicated effects; the constraint is environmental design, not knowledge.
105	Psychology & Mind	Now	Global / Civilizational	Rebuild the inputs: sleep protected as infrastructure (school start times aligned to adolescent biology, no-screens wind-down norms), daily physical activity of 60 minutes for youth built into the school day, third places (libraries, clubs, leagues) funded as health infrastructure, and platform design rules for minors (no infinite scroll, no engagement-optimized feeds). Specification, personal: fixed sleep window, morning light exposure, one recurring in-person group activity weekly minimum — belonging is scheduled, not hoped for.	55%	Schedule belonging like medicine (one weekly in-person group minimum); protect the sleep window; build third places as funded health infrastructure.	Sleep, activity, and in-person belonging have large, replicated effects; the constraint is environmental design, not knowledge.
106	Psychology & Mind	Future	Household / Personal	Protect the real by making it richer, not by banning the synthetic: disclosure law (companions must identify as artificial, always), design rules against dependency engineering (no manufactured jealousy, streaks, or guilt mechanics), public investment in physical gathering — festivals, sport, craft, public space — at a scale that competes, and social norms taught early: the difference between being comforted and being accompanied. Specification: households set explicit budgets — money and hours — for synthetic companionship, reviewed like any other consumption; any product whose retention depends on your loneliness is treated as extractive by definition.	54%	Budget synthetic companionship in hours and money at household level; ban dependency mechanics (jealousy, streaks, guilt) in companion products; invest in physical gathering at competing scale.	Disclosure and anti-dependency design rules can slow synthetic substitution, but they compete against perfectly engineered comfort; the richer-real strategy must actually be funded.

Solution #	Field of Discipline	Timefram	Scale	Solution — Full Specification	Probability of S	How to Maximize the Chances (detailed)	Why This Probability
107	Psychology & Mind	Future	Society / National	Protect the real by making it richer, not by banning the synthetic: disclosure law (companions must identify as artificial, always), design rules against dependency engineering (no manufactured jealousy, streaks, or guilt mechanics), public investment in physical gathering — festivals, sport, craft, public space — at a scale that competes, and social norms taught early: the difference between being comforted and being accompanied. Specification: households set explicit budgets — money and hours — for synthetic companionship, reviewed like any other consumption; any product whose retention depends on your loneliness is treated as extractive by definition.	46%	Budget synthetic companionship in hours and money at household level; ban dependency mechanics (jealousy, streaks, guilt) in companion products; invest in physical gathering at competing scale.	Disclosure and anti-dependency design rules can slow synthetic substitution, but they compete against perfectly engineered comfort; the richer-real strategy must actually be funded.
108	Psychology & Mind	Future	Global / Civilizational	Protect the real by making it richer, not by banning the synthetic: disclosure law (companions must identify as artificial, always), design rules against dependency engineering (no manufactured jealousy, streaks, or guilt mechanics), public investment in physical gathering — festivals, sport, craft, public space — at a scale that competes, and social norms taught early: the difference between being comforted and being accompanied. Specification: households set explicit budgets — money and hours — for synthetic companionship, reviewed like any other consumption; any product whose retention depends on your loneliness is treated as extractive by definition.	34%	Budget synthetic companionship in hours and money at household level; ban dependency mechanics (jealousy, streaks, guilt) in companion products; invest in physical gathering at competing scale.	Disclosure and anti-dependency design rules can slow synthetic substitution, but they compete against perfectly engineered comfort; the richer-real strategy must actually be funded.
109	Education & Knowledge	Past	Household / Personal	Distribute the copies: knowledge survives by redundancy — minimum 3 complete copies of core curricula and records on 3 different media in 3 different jurisdictions (the LOCKSS principle); literacy as a non-derogable right in law; open-licence textbooks for all core subjects so no publisher or regime can withdraw them. Specification: every school and household keeps printed core references (first aid, agriculture, mathematics, local law); digital archives mirrored internationally by default.	94%	Three copies, three media, three jurisdictions for all core knowledge; printed core references in every school and household; open licences so nothing can be withdrawn.	Redundancy is the most reliable preservation strategy in history: texts copied widely survived every fire; texts held once did not.
110	Education & Knowledge	Past	Society / National	Distribute the copies: knowledge survives by redundancy — minimum 3 complete copies of core curricula and records on 3 different media in 3 different jurisdictions (the LOCKSS principle); literacy as a non-derogable right in law; open-licence textbooks for all core subjects so no publisher or regime can withdraw them. Specification: every school and household keeps printed core references (first aid, agriculture, mathematics, local law); digital archives mirrored internationally by default.	86%	Three copies, three media, three jurisdictions for all core knowledge; printed core references in every school and household; open licences so nothing can be withdrawn.	Redundancy is the most reliable preservation strategy in history: texts copied widely survived every fire; texts held once did not.
111	Education & Knowledge	Past	Global / Civilizational	Distribute the copies: knowledge survives by redundancy — minimum 3 complete copies of core curricula and records on 3 different media in 3 different jurisdictions (the LOCKSS principle); literacy as a non-derogable right in law; open-licence textbooks for all core subjects so no publisher or regime can withdraw them. Specification: every school and household keeps printed core references (first aid, agriculture, mathematics, local law); digital archives mirrored internationally by default.	74%	Three copies, three media, three jurisdictions for all core knowledge; printed core references in every school and household; open licences so nothing can be withdrawn.	Redundancy is the most reliable preservation strategy in history: texts copied widely survived every fire; texts held once did not.
112	Education & Knowledge	Now	Household / Personal	Measure learning, not attendance: independent annual assessments of reading and numeracy published per school, funding following measured learning gains, curricula rebuilt around mastery progression (no social promotion without support), and teacher pay and status raised to compete with other professions (top-quartile graduate recruitment). Specification: every child reads fluently by age 9 or triggers automatic intensive support; adult reskilling accounts (funded per citizen) redeemable at accredited providers, including libraries.	78%	Independent published assessment per school; automatic intensive support at the age-9 reading trigger; teacher recruitment from top graduate quartiles with matching pay.	Mastery-based progression and measured-learning funding have strong evidence; systems resist because measurement exposes failure.
113	Education & Knowledge	Now	Society / National	Measure learning, not attendance: independent annual assessments of reading and numeracy published per school, funding following measured learning gains, curricula rebuilt around mastery progression (no social promotion without support), and teacher pay and status raised to compete with other professions (top-quartile graduate recruitment). Specification: every child reads fluently by age 9 or triggers automatic intensive support; adult reskilling accounts (funded per citizen) redeemable at accredited providers, including libraries.	70%	Independent published assessment per school; automatic intensive support at the age-9 reading trigger; teacher recruitment from top graduate quartiles with matching pay.	Mastery-based progression and measured-learning funding have strong evidence; systems resist because measurement exposes failure.
114	Education & Knowledge	Now	Global / Civilizational	Measure learning, not attendance: independent annual assessments of reading and numeracy published per school, funding following measured learning gains, curricula rebuilt around mastery progression (no social promotion without support), and teacher pay and status raised to compete with other professions (top-quartile graduate recruitment). Specification: every child reads fluently by age 9 or triggers automatic intensive support; adult reskilling accounts (funded per citizen) redeemable at accredited providers, including libraries.	58%	Independent published assessment per school; automatic intensive support at the age-9 reading trigger; teacher recruitment from top graduate quartiles with matching pay.	Mastery-based progression and measured-learning funding have strong evidence; systems resist because measurement exposes failure.
115	Education & Knowledge	Future	Household / Personal	Dual-competence standard: every qualification requires demonstrated unaided performance (closed-book, offline, by hand) plus aided performance; curricula teach verification as the core skill — source-tracing, estimation, consistency checking; national knowledge reserves (printed, distributed) maintained as civil infrastructure. Specification: critical professions (medicine, engineering, aviation, law) license on unaided competence first; schools schedule regular no-device weeks, not as punishment but as training.	67%	License critical professions on unaided competence first; regular no-device training weeks; national printed knowledge reserves maintained as infrastructure.	Unaided-competence licensing is fully within institutional control; success depends on examining bodies holding the line against convenience.
116	Education & Knowledge	Future	Society / National	Dual-competence standard: every qualification requires demonstrated unaided performance (closed-book, offline, by hand) plus aided performance; curricula teach verification as the core skill — source-tracing, estimation, consistency checking; national knowledge reserves (printed, distributed) maintained as civil infrastructure. Specification: critical professions (medicine, engineering, aviation, law) license on unaided competence first; schools schedule regular no-device weeks, not as punishment but as training.	59%	License critical professions on unaided competence first; regular no-device training weeks; national printed knowledge reserves maintained as infrastructure.	Unaided-competence licensing is fully within institutional control; success depends on examining bodies holding the line against convenience.
117	Education & Knowledge	Future	Global / Civilizational	Dual-competence standard: every qualification requires demonstrated unaided performance (closed-book, offline, by hand) plus aided performance; curricula teach verification as the core skill — source-tracing, estimation, consistency checking; national knowledge reserves (printed, distributed) maintained as civil infrastructure. Specification: critical professions (medicine, engineering, aviation, law) license on unaided competence first; schools schedule regular no-device weeks, not as punishment but as training.	47%	License critical professions on unaided competence first; regular no-device training weeks; national printed knowledge reserves maintained as infrastructure.	Unaided-competence licensing is fully within institutional control; success depends on examining bodies holding the line against convenience.
118	Work & Livelihood	Past	Household / Personal	Separate the levers by law: wages in legal tender only, housing and retail independent of employers, freedom of movement and association non-waivable, and portable social insurance attached to the person. Specification: no employer may provide more than 2 of {job, housing, food retail, credit} to the same worker; labour inspection funded per worker covered, with unannounced inspection rights.	85%	Enforce the 2-of-4 lever cap (job/housing/food/credit); fund labour inspection per worker; portable insurance attached to the person.	Separating wages, housing, and credit by law ended company-town capture wherever enforced; scrip systems did not survive legal tender rules.
119	Work & Livelihood	Past	Society / National	Separate the levers by law: wages in legal tender only, housing and retail independent of employers, freedom of movement and association non-waivable, and portable social insurance attached to the person. Specification: no employer may provide more than 2 of {job, housing, food retail, credit} to the same worker; labour inspection funded per worker covered, with unannounced inspection rights.	77%	Enforce the 2-of-4 lever cap (job/housing/food/credit); fund labour inspection per worker; portable insurance attached to the person.	Separating wages, housing, and credit by law ended company-town capture wherever enforced; scrip systems did not survive legal tender rules.
120	Work & Livelihood	Past	Global / Civilizational	Separate the levers by law: wages in legal tender only, housing and retail independent of employers, freedom of movement and association non-waivable, and portable social insurance attached to the person. Specification: no employer may provide more than 2 of {job, housing, food retail, credit} to the same worker; labour inspection funded per worker covered, with unannounced inspection rights.	65%	Enforce the 2-of-4 lever cap (job/housing/food/credit); fund labour inspection per worker; portable insurance attached to the person.	Separating wages, housing, and credit by law ended company-town capture wherever enforced; scrip systems did not survive legal tender rules.
121	Work & Livelihood	Now	Household / Personal	Floor under every form of work: benefits (health, pension, injury, unemployment) pro-rated per hour worked regardless of classification, minimum standards applied per task not per contract type, regional diversification funds that trigger automatically when one employer exceeds 20% of local jobs, and wage insurance that tops up displaced workers at 50% of the wage gap for 2 years while retraining. Specification: personal standard — an emergency fund of 3 months, skills documented in portable form, and no single income source treated as permanent.	74%	Pro-rate benefits per hour across all contract types; automatic regional diversification triggers at the 20% single-employer threshold; personal 3-month reserve as the household standard.	Per-hour benefit floors are administratively proven in several systems; classification games shrink when benefits attach to hours, not contracts.
122	Work & Livelihood	Now	Society / National	Floor under every form of work: benefits (health, pension, injury, unemployment) pro-rated per hour worked regardless of classification, minimum standards applied per task not per contract type, regional diversification funds that trigger automatically when one employer exceeds 20% of local jobs, and wage insurance that tops up displaced workers at 50% of the wage gap for 2 years while retraining. Specification: personal standard — an emergency fund of 3 months, skills documented in portable form, and no single income source treated as permanent.	66%	Pro-rate benefits per hour across all contract types; automatic regional diversification triggers at the 20% single-employer threshold; personal 3-month reserve as the household standard.	Per-hour benefit floors are administratively proven in several systems; classification games shrink when benefits attach to hours, not contracts.
123	Work & Livelihood	Now	Global / Civilizational	Floor under every form of work: benefits (health, pension, injury, unemployment) pro-rated per hour worked regardless of classification, minimum standards applied per task not per contract type, regional diversification funds that trigger automatically when one employer exceeds 20% of local jobs, and wage insurance that tops up displaced workers at 50% of the wage gap for 2 years while retraining. Specification: personal standard — an emergency fund of 3 months, skills documented in portable form, and no single income source treated as permanent.	54%	Pro-rate benefits per hour across all contract types; automatic regional diversification triggers at the 20% single-employer threshold; personal 3-month reserve as the household standard.	Per-hour benefit floors are administratively proven in several systems; classification games shrink when benefits attach to hours, not contracts.
124	Work & Livelihood	Future	Household / Personal	Rebuild distribution on ownership: broad-based capital ownership (citizen funds, employee ownership trusts, data and AI dividends), radically shortened standard working time as productivity rises, and public valuation of unpaid contribution (care, community, restoration) through contribution credits convertible to services. Specification: transition is phased by measured displacement rates per sector, with the ownership mechanisms built during the growth phase — financed from automation rents — not improvised after collapse.	51%	Start citizen funds now from automation rents; convert large employers progressively to shared-ownership forms; phase hour reductions by measured sector displacement.	Ownership-based distribution is the only mechanism that scales with full automation, but it must be capitalized during the growth phase; late construction is the main failure mode.

Solution #	Field of Discipline	Timefram	Scale	Solution — Full Specification	Probability of S	How to Maximize the Chances (detailed)	Why This Probability
125	Work & Livelihood	Future	Society / National	Rebuild distribution on ownership: broad-based capital ownership (citizen funds, employee ownership trusts, data and AI dividends), radically shortened standard working time as productivity rises, and public valuation of unpaid contribution (care, community, restoration) through contribution credits convertible to services. Specification: transition is phased by measured displacement rates per sector, with the ownership mechanisms built during the growth phase — financed from automation rents — not improvised after collapse.	43%	Start citizen funds now from automation rents; convert large employers progressively to shared-ownership forms; phase hour reductions by measured sector displacement.	Ownership-based distribution is the only mechanism that scales with full automation, but it must be capitalized during the growth phase; late construction is the main failure mode.
126	Work & Livelihood	Future	Global / Civilizational	Rebuild distribution on ownership: broad-based capital ownership (citizen funds, employee ownership trusts, data and AI dividends), radically shortened standard working time as productivity rises, and public valuation of unpaid contribution (care, community, restoration) through contribution credits convertible to services. Specification: transition is phased by measured displacement rates per sector, with the ownership mechanisms built during the growth phase — financed from automation rents — not improvised after collapse.	31%	Start citizen funds now from automation rents; convert large employers progressively to shared-ownership forms; phase hour reductions by measured sector displacement.	Ownership-based distribution is the only mechanism that scales with full automation, but it must be capitalized during the growth phase; late construction is the main failure mode.
127	Transport & Mobility	Past	Household / Personal	No city on a single thread: minimum two independent supply modes and routes for food and fuel (road + rail, or road + water), urban food reserves of 30 days, and local production floors for fresh essentials. Specification: critical bridges, tunnels, and ports have pre-planned bypass capacity of at least 60% of normal flow; households in isolated regions keep 30 days of staples as the standing norm.	87%	Two independent modes for food and fuel per city; 30-day urban food reserves; bypass capacity of 60% pre-planned for critical links.	Dual-route supply is classical logistics doctrine; besieged cities with two functioning routes survived, single-route cities starved.
128	Transport & Mobility	Past	Society / National	No city on a single thread: minimum two independent supply modes and routes for food and fuel (road + rail, or road + water), urban food reserves of 30 days, and local production floors for fresh essentials. Specification: critical bridges, tunnels, and ports have pre-planned bypass capacity of at least 60% of normal flow; households in isolated regions keep 30 days of staples as the standing norm.	79%	Two independent modes for food and fuel per city; 30-day urban food reserves; bypass capacity of 60% pre-planned for critical links.	Dual-route supply is classical logistics doctrine; besieged cities with two functioning routes survived, single-route cities starved.
129	Transport & Mobility	Past	Global / Civilizational	No city on a single thread: minimum two independent supply modes and routes for food and fuel (road + rail, or road + water), urban food reserves of 30 days, and local production floors for fresh essentials. Specification: critical bridges, tunnels, and ports have pre-planned bypass capacity of at least 60% of normal flow; households in isolated regions keep 30 days of staples as the standing norm.	67%	Two independent modes for food and fuel per city; 30-day urban food reserves; bypass capacity of 60% pre-planned for critical links.	Dual-route supply is classical logistics doctrine; besieged cities with two functioning routes survived, single-route cities starved.
130	Transport & Mobility	Now	Household / Personal	Redundancy and proximity: freight modal split so no single route or mode carries more than 40% of essentials; the 15-minute standard for new development (daily needs reachable on foot or by transit); transit networks funded as essential infrastructure with operating reserves of 6 months; strategic parts and fuel reserves for public fleets. Specification: every settlement over 5,000 has at least two physically separate road connections or one road plus scheduled public transport that runs in emergencies.	76%	Cap any single mode/route at 40% of essentials; apply the 15-minute standard to all new development; 6-month operating reserves for transit.	Modal-split caps and proximity standards measurably reduce fragility; sprawl already built is the slow constraint.
131	Transport & Mobility	Now	Society / National	Redundancy and proximity: freight modal split so no single route or mode carries more than 40% of essentials; the 15-minute standard for new development (daily needs reachable on foot or by transit); transit networks funded as essential infrastructure with operating reserves of 6 months; strategic parts and fuel reserves for public fleets. Specification: every settlement over 5,000 has at least two physically separate road connections or one road plus scheduled public transport that runs in emergencies.	68%	Cap any single mode/route at 40% of essentials; apply the 15-minute standard to all new development; 6-month operating reserves for transit.	Modal-split caps and proximity standards measurably reduce fragility; sprawl already built is the slow constraint.
132	Transport & Mobility	Now	Global / Civilizational	Redundancy and proximity: freight modal split so no single route or mode carries more than 40% of essentials; the 15-minute standard for new development (daily needs reachable on foot or by transit); transit networks funded as essential infrastructure with operating reserves of 6 months; strategic parts and fuel reserves for public fleets. Specification: every settlement over 5,000 has at least two physically separate road connections or one road plus scheduled public transport that runs in emergencies.	56%	Cap any single mode/route at 40% of essentials; apply the 15-minute standard to all new development; 6-month operating reserves for transit.	Modal-split caps and proximity standards measurably reduce fragility; sprawl already built is the slow constraint.
133	Transport & Mobility	Future	Household / Personal	Mandated heterogeneity and manual fallback: fleets certified from at least 2 independent software stacks per region, all autonomous vehicles retaining manual drive capability, transport control systems segmented from public networks with analogue scheduling fallback, and battery-material supply diversified with recycling recovering 90%+ of critical minerals by law. Specification: cities drill a 'no-network day' annually — transit, logistics, and emergency services operating on fallback procedures.	62%	Two independent software stacks per region by certification rule; manual drive retained in every autonomous vehicle; annual no-network drill day.	Heterogeneity and manual fallback are cheap relative to fleet value; success depends on certifying them before fleets homogenize.
134	Transport & Mobility	Future	Society / National	Mandated heterogeneity and manual fallback: fleets certified from at least 2 independent software stacks per region, all autonomous vehicles retaining manual drive capability, transport control systems segmented from public networks with analogue scheduling fallback, and battery-material supply diversified with recycling recovering 90%+ of critical minerals by law. Specification: cities drill a 'no-network day' annually — transit, logistics, and emergency services operating on fallback procedures.	54%	Two independent software stacks per region by certification rule; manual drive retained in every autonomous vehicle; annual no-network drill day.	Heterogeneity and manual fallback are cheap relative to fleet value; success depends on certifying them before fleets homogenize.
135	Transport & Mobility	Future	Global / Civilizational	Mandated heterogeneity and manual fallback: fleets certified from at least 2 independent software stacks per region, all autonomous vehicles retaining manual drive capability, transport control systems segmented from public networks with analogue scheduling fallback, and battery-material supply diversified with recycling recovering 90%+ of critical minerals by law. Specification: cities drill a 'no-network day' annually — transit, logistics, and emergency services operating on fallback procedures.	42%	Two independent software stacks per region by certification rule; manual drive retained in every autonomous vehicle; annual no-network drill day.	Heterogeneity and manual fallback are cheap relative to fleet value; success depends on certifying them before fleets homogenize.
136	Communication	Past	Household / Personal	Multiple independent channels as doctrine: no population dependent on one network, one owner, or one technology for emergency information — broadcast radio, print, courier, and digital maintained in parallel, with amateur and community radio licensed and encouraged as civic infrastructure. Specification: every municipality maintains a battery/solar broadcast capability able to reach all residents within 1 hour without internet or grid power.	83%	Maintain broadcast, print, courier, and digital in parallel; municipal emergency broadcast reaching all residents within 1 hour, off-grid capable.	Parallel channels historically defeated monopoly and collapse; single-channel populations were governed by rumour.
137	Communication	Past	Society / National	Multiple independent channels as doctrine: no population dependent on one network, one owner, or one technology for emergency information — broadcast radio, print, courier, and digital maintained in parallel, with amateur and community radio licensed and encouraged as civic infrastructure. Specification: every municipality maintains a battery/solar broadcast capability able to reach all residents within 1 hour without internet or grid power.	75%	Maintain broadcast, print, courier, and digital in parallel; municipal emergency broadcast reaching all residents within 1 hour, off-grid capable.	Parallel channels historically defeated monopoly and collapse; single-channel populations were governed by rumour.
138	Communication	Past	Global / Civilizational	Multiple independent channels as doctrine: no population dependent on one network, one owner, or one technology for emergency information — broadcast radio, print, courier, and digital maintained in parallel, with amateur and community radio licensed and encouraged as civic infrastructure. Specification: every municipality maintains a battery/solar broadcast capability able to reach all residents within 1 hour without internet or grid power.	63%	Maintain broadcast, print, courier, and digital in parallel; municipal emergency broadcast reaching all residents within 1 hour, off-grid capable.	Parallel channels historically defeated monopoly and collapse; single-channel populations were governed by rumour.
139	Communication	Now	Household / Personal	Interoperability and exit rights by law: dominant platforms must support data portability and interoperable messaging, public communication infrastructure (postal, broadcast, municipal networks) maintained as a real fallback, and due-process rules for account termination with human appeal. Specification, personal and organizational: own your contact list and records off-platform (exported monthly), maintain one channel you fully control (your own email list, radio net, or phone tree), and never let a platform be the only copy of anything.	77%	Own your lists off-platform (monthly exports); keep one fully controlled channel per organization and household; statutory appeal rights for account termination.	Portability and interoperability law is enacted in some jurisdictions and technically straightforward; platform resistance is the constraint.
140	Communication	Now	Society / National	Interoperability and exit rights by law: dominant platforms must support data portability and interoperable messaging, public communication infrastructure (postal, broadcast, municipal networks) maintained as a real fallback, and due-process rules for account termination with human appeal. Specification, personal and organizational: own your contact list and records off-platform (exported monthly), maintain one channel you fully control (your own email list, radio net, or phone tree), and never let a platform be the only copy of anything.	69%	Own your lists off-platform (monthly exports); keep one fully controlled channel per organization and household; statutory appeal rights for account termination.	Portability and interoperability law is enacted in some jurisdictions and technically straightforward; platform resistance is the constraint.
141	Communication	Now	Global / Civilizational	Interoperability and exit rights by law: dominant platforms must support data portability and interoperable messaging, public communication infrastructure (postal, broadcast, municipal networks) maintained as a real fallback, and due-process rules for account termination with human appeal. Specification, personal and organizational: own your contact list and records off-platform (exported monthly), maintain one channel you fully control (your own email list, radio net, or phone tree), and never let a platform be the only copy of anything.	57%	Own your lists off-platform (monthly exports); keep one fully controlled channel per organization and household; statutory appeal rights for account termination.	Portability and interoperability law is enacted in some jurisdictions and technically straightforward; platform resistance is the constraint.
142	Communication	Future	Household / Personal	Rights-preserving architecture: end-to-end encryption protected in law as the default for private communication, state access only through individualized, judicially authorized, logged, and time-limited orders with after-the-fact notification; anonymous public speech protected while large-scale amplification carries provenance duties. Specification: every surveillance power carries a 5-year sunset and a published annual accounting of use, errors, and outcomes — powers that cannot justify themselves expire.	60%	Protect end-to-end encryption in law; every surveillance power sunsetted at 5 years with published annual accounting; individualized warrants only.	Encryption with judicially bounded access is a stable equilibrium where legislated; both extremes (total access, total opacity) generate political pressure that eventually breaks them.
143	Communication	Future	Society / National	Rights-preserving architecture: end-to-end encryption protected in law as the default for private communication, state access only through individualized, judicially authorized, logged, and time-limited orders with after-the-fact notification; anonymous public speech protected while large-scale amplification carries provenance duties. Specification: every surveillance power carries a 5-year sunset and a published annual accounting of use, errors, and outcomes — powers that cannot justify themselves expire.	52%	Protect end-to-end encryption in law; every surveillance power sunsetted at 5 years with published annual accounting; individualized warrants only.	Encryption with judicially bounded access is a stable equilibrium where legislated; both extremes (total access, total opacity) generate political pressure that eventually breaks them.

Solution #	Field of Discipline	Timefram	Scale	Solution — Full Specification	Probability of S	How to Maximize the Chances (detailed)	Why This Probability
144	Communication	Future	Global / Civilizational	Rights-preserving architecture: end-to-end encryption protected in law as the default for private communication, state access only through individualized, judicially authorized, logged, and time-limited orders with after-the-fact notification; anonymous public speech protected while large-scale amplification carries provenance duties. Specification: every surveillance power carries a 5-year sunset and a published annual accounting of use, errors, and outcomes — powers that cannot justify themselves expire.	40%	Protect end-to-end encryption in law; every surveillance power sunsetted at 5 years with published annual accounting; individualized warrants only.	Encryption with judicially bounded access is a stable equilibrium where legislated; both extremes (total access, total opacity) generate political pressure that eventually breaks them.
145	Demographics & Population	Past	Household / Personal	Resilience through distribution: population and skills spread across regions rather than concentrated in single cities, craft and knowledge apprenticeship duplicated across at least 2 lineages per critical skill, and post-catastrophe relief law that moves food and people automatically on declared triggers. Specification: no critical occupation (medicine, engineering, agriculture) may have more than 40% of its practitioners in one metropolitan area.	72%	Duplicate every critical skill across 2+ lineages and regions; relief triggers declared in law in advance; no critical profession 40%+ in one metro.	Distributed skills and automatic relief triggers shortened post-catastrophe recoveries where they existed; concentrated populations recovered slowest.
146	Demographics & Population	Past	Society / National	Resilience through distribution: population and skills spread across regions rather than concentrated in single cities, craft and knowledge apprenticeship duplicated across at least 2 lineages per critical skill, and post-catastrophe relief law that moves food and people automatically on declared triggers. Specification: no critical occupation (medicine, engineering, agriculture) may have more than 40% of its practitioners in one metropolitan area.	64%	Duplicate every critical skill across 2+ lineages and regions; relief triggers declared in law in advance; no critical profession 40%+ in one metro.	Distributed skills and automatic relief triggers shortened post-catastrophe recoveries where they existed; concentrated populations recovered slowest.
147	Demographics & Population	Past	Global / Civilizational	Resilience through distribution: population and skills spread across regions rather than concentrated in single cities, craft and knowledge apprenticeship duplicated across at least 2 lineages per critical skill, and post-catastrophe relief law that moves food and people automatically on declared triggers. Specification: no critical occupation (medicine, engineering, agriculture) may have more than 40% of its practitioners in one metropolitan area.	52%	Duplicate every critical skill across 2+ lineages and regions; relief triggers declared in law in advance; no critical profession 40%+ in one metro.	Distributed skills and automatic relief triggers shortened post-catastrophe recoveries where they existed; concentrated populations recovered slowest.
148	Demographics & Population	Now	Household / Personal	Make family formation affordable where people already want it: housing sized and priced for young families (family units at or below 30% of median income in every major city plan), childcare costing under 10% of median income per child, working-time protection for parents, and pension formulas that adjust transparently and gradually (automatic balancing mechanisms) instead of by crisis. Specification: immigration and integration run as a permanent, competently administered system sized to the measured demographic gap, with language and credential recognition completed within 12 months of arrival.	65%	Family housing at the 30% benchmark in city plans; childcare under 10% of median income; automatic pension balancing so adjustment is gradual, not crisis-driven.	Housing and childcare cost relief moves fertility modestly where sustained (no country has reversed to replacement by policy alone); pension auto-balancers fully work mechanically.
149	Demographics & Population	Now	Society / National	Make family formation affordable where people already want it: housing sized and priced for young families (family units at or below 30% of median income in every major city plan), childcare costing under 10% of median income per child, working-time protection for parents, and pension formulas that adjust transparently and gradually (automatic balancing mechanisms) instead of by crisis. Specification: immigration and integration run as a permanent, competently administered system sized to the measured demographic gap, with language and credential recognition completed within 12 months of arrival.	57%	Family housing at the 30% benchmark in city plans; childcare under 10% of median income; automatic pension balancing so adjustment is gradual, not crisis-driven.	Housing and childcare cost relief moves fertility modestly where sustained (no country has reversed to replacement by policy alone); pension auto-balancers fully work mechanically.
150	Demographics & Population	Now	Global / Civilizational	Make family formation affordable where people already want it: housing sized and priced for young families (family units at or below 30% of median income in every major city plan), childcare costing under 10% of median income per child, working-time protection for parents, and pension formulas that adjust transparently and gradually (automatic balancing mechanisms) instead of by crisis. Specification: immigration and integration run as a permanent, competently administered system sized to the measured demographic gap, with language and credential recognition completed within 12 months of arrival.	45%	Family housing at the 30% benchmark in city plans; childcare under 10% of median income; automatic pension balancing so adjustment is gradual, not crisis-driven.	Housing and childcare cost relief moves fertility modestly where sustained (no country has reversed to replacement by policy alone); pension auto-balancers fully work mechanically.
151	Demographics & Population	Future	Household / Personal	Plan both directions simultaneously: shrink-smart policy for depopulating regions (consolidated services, maintained dignity and access, land returned to nature or agriculture under stewardship), growth-ready frameworks for expanding regions (land tenure, urban plans, and service grids built ahead of arrival), and pre-negotiated international mobility compacts so movement is governed, not chaotic. Specification: every national plan carries a 50-year demographic annex, updated every 5 years, with infrastructure sized to the projection band — not to today's population.	57%	50-year demographic annex in every national plan, updated 5-yearly; mobility compacts negotiated before flows peak; infrastructure sized to the projection band.	Both shrink-smart and growth-ready planning are technically known; political systems resist planning for populations that differ from today's voters.
152	Demographics & Population	Future	Society / National	Plan both directions simultaneously: shrink-smart policy for depopulating regions (consolidated services, maintained dignity and access, land returned to nature or agriculture under stewardship), growth-ready frameworks for expanding regions (land tenure, urban plans, and service grids built ahead of arrival), and pre-negotiated international mobility compacts so movement is governed, not chaotic. Specification: every national plan carries a 50-year demographic annex, updated every 5 years, with infrastructure sized to the projection band — not to today's population.	49%	50-year demographic annex in every national plan, updated 5-yearly; mobility compacts negotiated before flows peak; infrastructure sized to the projection band.	Both shrink-smart and growth-ready planning are technically known; political systems resist planning for populations that differ from today's voters.
153	Demographics & Population	Future	Global / Civilizational	Plan both directions simultaneously: shrink-smart policy for depopulating regions (consolidated services, maintained dignity and access, land returned to nature or agriculture under stewardship), growth-ready frameworks for expanding regions (land tenure, urban plans, and service grids built ahead of arrival), and pre-negotiated international mobility compacts so movement is governed, not chaotic. Specification: every national plan carries a 50-year demographic annex, updated every 5 years, with infrastructure sized to the projection band — not to today's population.	37%	50-year demographic annex in every national plan, updated 5-yearly; mobility compacts negotiated before flows peak; infrastructure sized to the projection band.	Both shrink-smart and growth-ready planning are technically known; political systems resist planning for populations that differ from today's voters.
154	Justice & Ethics	Past	Household / Personal	Evidence as the only currency: independent judiciary with security of tenure and salary, prosecution separated from the executive, every criminal conviction reviewable on the evidence record, forensic methods validated scientifically before courtroom use, and absolute bars on collective punishment and retroactive crimes. Specification: wrongful-conviction compensation automatic and generous; every jurisdiction publishes conviction-integrity audits, including exoneration counts, annually.	80%	Judicial tenure, salary, and budget protected constitutionally; forensic methods validated before courtroom use; wrongful-conviction compensation automatic.	Independent courts with evidence rules have repeatedly been the last institution standing; where tenure and budgets were protected, terror-by-law was blunted.
155	Justice & Ethics	Past	Society / National	Evidence as the only currency: independent judiciary with security of tenure and salary, prosecution separated from the executive, every criminal conviction reviewable on the evidence record, forensic methods validated scientifically before courtroom use, and absolute bars on collective punishment and retroactive crimes. Specification: wrongful-conviction compensation automatic and generous; every jurisdiction publishes conviction-integrity audits, including exoneration counts, annually.	72%	Judicial tenure, salary, and budget protected constitutionally; forensic methods validated before courtroom use; wrongful-conviction compensation automatic.	Independent courts with evidence rules have repeatedly been the last institution standing; where tenure and budgets were protected, terror-by-law was blunted.
156	Justice & Ethics	Past	Global / Civilizational	Evidence as the only currency: independent judiciary with security of tenure and salary, prosecution separated from the executive, every criminal conviction reviewable on the evidence record, forensic methods validated scientifically before courtroom use, and absolute bars on collective punishment and retroactive crimes. Specification: wrongful-conviction compensation automatic and generous; every jurisdiction publishes conviction-integrity audits, including exoneration counts, annually.	60%	Judicial tenure, salary, and budget protected constitutionally; forensic methods validated before courtroom use; wrongful-conviction compensation automatic.	Independent courts with evidence rules have repeatedly been the last institution standing; where tenure and budgets were protected, terror-by-law was blunted.
157	Justice & Ethics	Now	Household / Personal	Equalize the machinery: pre-trial detention only on individualized danger findings with 48-hour review, legal aid funded to parity standards with prosecution in criminal matters, court capacity managed to published time standards (e.g., 90% of cases resolved within 12 months), and fines scaled to income (day-fines) so penalties bite equally. Specification: every court publishes wait times, outcomes by representation type, and backlog monthly; households know their rights in writing — a plain-language rights card is a standard household document.	73%	48-hour detention review; day-fines scaled to income; legal aid at parity in criminal matters; monthly publication of waits and outcomes.	Detention limits, day-fines, and published time standards work where adopted; the constraint is court capacity funding.
158	Justice & Ethics	Now	Society / National	Equalize the machinery: pre-trial detention only on individualized danger findings with 48-hour review, legal aid funded to parity standards with prosecution in criminal matters, court capacity managed to published time standards (e.g., 90% of cases resolved within 12 months), and fines scaled to income (day-fines) so penalties bite equally. Specification: every court publishes wait times, outcomes by representation type, and backlog monthly; households know their rights in writing — a plain-language rights card is a standard household document.	65%	48-hour detention review; day-fines scaled to income; legal aid at parity in criminal matters; monthly publication of waits and outcomes.	Detention limits, day-fines, and published time standards work where adopted; the constraint is court capacity funding.
159	Justice & Ethics	Now	Global / Civilizational	Equalize the machinery: pre-trial detention only on individualized danger findings with 48-hour review, legal aid funded to parity standards with prosecution in criminal matters, court capacity managed to published time standards (e.g., 90% of cases resolved within 12 months), and fines scaled to income (day-fines) so penalties bite equally. Specification: every court publishes wait times, outcomes by representation type, and backlog monthly; households know their rights in writing — a plain-language rights card is a standard household document.	53%	48-hour detention review; day-fines scaled to income; legal aid at parity in criminal matters; monthly publication of waits and outcomes.	Detention limits, day-fines, and published time standards work where adopted; the constraint is court capacity funding.
160	Justice & Ethics	Future	Household / Personal	Human answerability as law: no liberty decision (arrest priority, bail, sentencing, parole) may rest solely on an algorithmic score; models used in justice are public, auditable, and tested for error and disparity rates published per demographic group; every automated recommendation names the accountable human who adopted it. Specification: any person scored by a state system can see their inputs, correct errors within 30 days, and receive a human review on request — the same standard as credit records, applied to freedom.	66%	No liberty decision on algorithmic score alone; models public and disparity-audited; every scored person can see, correct, and appeal to a human within 30 days.	Auditability and human-answerability rules are enforceable today; success depends on adopting them before automated systems entrench.
161	Justice & Ethics	Future	Society / National	Human answerability as law: no liberty decision (arrest priority, bail, sentencing, parole) may rest solely on an algorithmic score; models used in justice are public, auditable, and tested for error and disparity rates published per demographic group; every automated recommendation names the accountable human who adopted it. Specification: any person scored by a state system can see their inputs, correct errors within 30 days, and receive a human review on request — the same standard as credit records, applied to freedom.	58%	No liberty decision on algorithmic score alone; models public and disparity-audited; every scored person can see, correct, and appeal to a human within 30 days.	Auditability and human-answerability rules are enforceable today; success depends on adopting them before automated systems entrench.

Solution #	Field of Discipline	Timefram	Scale	Solution — Full Specification	Probability of S	How to Maximize the Chances (detailed)	Why This Probability
162	Justice & Ethics	Future	Global / Civilizational	Human answerability as law: no liberty decision (arrest priority, bail, sentencing, parole) may rest solely on an algorithmic score; models used in justice are public, auditable, and tested for error and disparity rates published per demographic group; every automated recommendation names the accountable human who adopted it. Specification: any person scored by a state system can see their inputs, correct errors within 30 days, and receive a human review on request — the same standard as credit records, applied to freedom.	46%	No liberty decision on algorithmic score alone; models public and disparity-audited; every scored person can see, correct, and appeal to a human within 30 days.	Auditability and human-answerability rules are enforceable today; success depends on adopting them before automated systems entrench.

Pareto Rank	Solution (short title = its problem)	Field	Timeframe	Scale	Priority Score	Cumulative % of	Probability of Full Solution
1	Municipal supply contamination or infrastructure failure cuts safe drinking water to a city.	Water	Now	Global / Civilizational	27	1.6%	73%	Dual-barrier defence: (1) two independent treatment trains so one failure never reaches taps; (2) point-of-use backup in every household — a certified gravity or pump filter rated to remove bacteria/ viruses (0.02–0.2 micron) plus a 72-hour stored reserve (12 L/person drinking). Specification: utilities must hold 5 days of treated-water storage and publish real-time turbidity and chlorine residuals; buildings over 6 storeys keep a 24-hour gravity tank.
2	Cyber intrusion and ransomware shut down hospitals, pipelines, and city governments; the digital base is fragi	Technology & AI	Now	Global / Civilizational	27	3.3%	66%	Assume breach, engineer recovery: immutable offline backups tested by full restoration quarterly (a backup never restored is a rumour), network segmentation so one infection cannot cross from office systems to operations, multi-factor authentication on 100% of remote and privileged access, and 24-hour incident reporting with a national assistance duty. Specification for households and small organizations: automatic updates on, unique passwords in a manager, 3-2-1 backups (3 copies, 2 media, 1 off-site/offline) for irreplaceable data — photos, records, work.
3	Grid failure — storm, cyberattack, or cascading overload — blacks out regions for days, and everything else fa	Energy	Now	Global / Civilizational	27	4.9%	64%	Engineer for islanding: grids split into self-sustaining cells that can run 72 hours alone; critical facilities (hospitals, water, 911, data centres) on dual feeds plus on-site generation tested monthly under load; household standard — a 1–3 kWh battery with solar trickle keeps lights, phone, fridge, and medical devices alive 48–72 hours. Specification: black-start capability drilled twice a year; underground or harden the 20% of lines whose failure cascades furthest (Pareto of the grid).
4	Organized crime, gang fragmentation, and everyday violence make districts unsafe while institutions lose legit	Security & Defense	Now	Global / Civilizational	27	6.5%	62%	Focused deterrence, the proven Pareto intervention: identify the tiny network driving most violence (typically under 0.5% of a city's population), offer services and exit routes with one hand and certain, swift prosecution with the other; light and maintain the worst micro-locations (lighting, sightlines, occupied storefronts); violence-interrupter teams staffed by credible locals. Specification: measure success by shootings in the target micro-zones, reviewed monthly; every intervention audited for civil-rights compliance.
5	Supply-chain shock (war, fuel, fertilizer, port failure) spikes prices and empties shelves within weeks.	Food & Agriculture	Now	Global / Civilizational	27	8.2%	60%	Shorten and parallel the chain: contract at least 2 geographically separate suppliers per staple; local production floor of 30% of fresh food within 100 km; fertilizer independence via green-ammonia plants sized to 50% of national need. Specification: retailers and institutions hold 30 days of non-perishable buffer; households hold 30 days (one month of staples ≈ 25 kg grain-equivalent per adult); price-spike circuit breakers release reserves automatically at +40% price movement.
6	Extreme heat, flood, and fire exceed the design basis of cities built for a milder century.	Environment & Climate	Now	Global / Civilizational	27	9.8%	57%	Adapt to the climate that exists: building codes and infrastructure designed to the 2050 worst-decile weather, urban canopy targets of 30–40% to cut street-level heat, floodplains returned to water with buyouts instead of repeated rebuilds, and heat action plans with named officials, cooling centres within a 15-minute walk, and check-in systems for isolated elderly. Specification: every home in a risk zone has a written plan — cooling or evacuation, 72-hour supplies, medication continuity; insurers and cities publish address-level risk so decisions are informed.
7	Institutional corruption and state capture hollow out services while keeping the forms of democracy.	Governance & Law	Now	Global / Civilizational	27	11.4%	56%	Radical transparency with automatic publication: all public contracts, beneficial ownership of companies bidding, officials' assets, and court statistics published in open data by default — secrecy must be justified line by line, publication is the default state. Independent audit institutions with budgets fixed by formula (e.g., 0.1% of public spending) so governments cannot starve them. Specification: procurement above a low threshold is e-tendered with full bid disclosure; whistleblowers receive statutory protection plus a share of recovered funds.
8	Housing costs detach from incomes; working households are priced out, homelessness becomes structural.	Shelter & Housing	Now	Global / Civilizational	27	13.0%	52%	Attack the constraint, which is supply near work: legalize mid-density housing on all residential land within 800 m of transit, approvals deemed granted after 12 months, public land leased (not sold) for permanently affordable homes at 30% of income benchmark, and vacancy taxes that make empty investment units unprofitable. Specification: a city is healthy when median rent is at or below 30% of median household income; publish the ratio quarterly and trigger automatic zoning release when it exceeds it for 2 consecutive years.
9	Asset bubbles, household debt, and cost-of-living squeeze hollow out the middle; one rate shock cascades.	Economy & Finance	Now	Global / Civilizational	27	14.7%	51%	De-leverage by design: mortgage rules that cap total debt service at 35–40% of verified income under a stressed rate (+2%), automatic refinancing windows when rates fall, public development of non-speculative housing supply, and insolvency law that gives households a fresh start within 3 years instead of permanent debt peonage. Specification: personal rule — total debt payments under 30% of take-home; one income loss must be survivable for 6 months on reserves plus benefits; never hold employer stock plus job plus mortgage in one company town exposure.
10	Antimicrobial resistance quietly returns routine infections and surgeries to pre-antibiotic risk.	Health & Medicine	Now	Global / Civilizational	27	16.3%	50%	Stewardship with teeth: antibiotics in human medicine by prescription audit only, agricultural antibiotic use cut 75% and banned for growth promotion, hospital infection-control staffing at 1 specialist per 100 beds, and a standing pipeline fund that pays for new antibiotics by subscription (delinked from sales volume). Specification: every hospital publishes resistance rates quarterly; households never stockpile or share antibiotics — leftover courses are returned to pharmacies.
11	Municipal supply contamination or infrastructure failure cuts safe drinking water to a city.	Water	Now	Society / National	18	17.4%	85%	Dual-barrier defence: (1) two independent treatment trains so one failure never reaches taps; (2) point-of-use backup in every household — a certified gravity or pump filter rated to remove bacteria/ viruses (0.02–0.2 micron) plus a 72-hour stored reserve (12 L/person drinking). Specification: utilities must hold 5 days of treated-water storage and publish real-time turbidity and chlorine residuals; buildings over 6 storeys keep a 24-hour gravity tank.
12	Cyber intrusion and ransomware shut down hospitals, pipelines, and city governments; the digital base is fragi	Technology & AI	Now	Society / National	18	18.5%	78%	Assume breach, engineer recovery: immutable offline backups tested by full restoration quarterly (a backup never restored is a rumour), network segmentation so one infection cannot cross from office systems to operations, multi-factor authentication on 100% of remote and privileged access, and 24-hour incident reporting with a national assistance duty. Specification for households and small organizations: automatic updates on, unique passwords in a manager, 3-2-1 backups (3 copies, 2 media, 1 off-site/offline) for irreplaceable data — photos, records, work.
13	Grid failure — storm, cyberattack, or cascading overload — blacks out regions for days, and everything else fa	Energy	Now	Society / National	18	19.6%	76%	Engineer for islanding: grids split into self-sustaining cells that can run 72 hours alone; critical facilities (hospitals, water, 911, data centres) on dual feeds plus on-site generation tested monthly under load; household standard — a 1–3 kWh battery with solar trickle keeps lights, phone, fridge, and medical devices alive 48–72 hours. Specification: black-start capability drilled twice a year; underground or harden the 20% of lines whose failure cascades furthest (Pareto of the grid).
14	Organized crime, gang fragmentation, and everyday violence make districts unsafe while institutions lose legit	Security & Defense	Now	Society / National	18	20.7%	74%	Focused deterrence, the proven Pareto intervention: identify the tiny network driving most violence (typically under 0.5% of a city's population), offer services and exit routes with one hand and certain, swift prosecution with the other; light and maintain the worst micro-locations (lighting, sightlines, occupied storefronts); violence-interrupter teams staffed by credible locals. Specification: measure success by shootings in the target micro-zones, reviewed monthly; every intervention audited for civil-rights compliance.
15	Supply-chain shock (war, fuel, fertilizer, port failure) spikes prices and empties shelves within weeks.	Food & Agriculture	Now	Society / National	18	21.7%	72%	Shorten and parallel the chain: contract at least 2 geographically separate suppliers per staple; local production floor of 30% of fresh food within 100 km; fertilizer independence via green-ammonia plants sized to 50% of national need. Specification: retailers and institutions hold 30 days of non-perishable buffer; households hold 30 days (one month of staples ≈ 25 kg grain-equivalent per adult); price-spike circuit breakers release reserves automatically at +40% price movement.

Pareto Rank	Solution (short title = its problem)	Field	Timeframe	Scale	Priority Score	Cumulative % of	Probability of	Full Solution
16	Extreme heat, flood, and fire exceed the design basis of cities built for a milder century.	Environment & Climate	Now	Society / National	18	22.8%	69%	Adapt to the climate that exists: building codes and infrastructure designed to the 2050 worst-decile weather, urban canopy targets of 30–40% to cut street-level heat, floodplains returned to water with buyouts instead of repeated rebuilds, and heat action plans with named officials, cooling centres within a 15-minute walk, and check-in systems for isolated elderly. Specification: every home in a risk zone has a written plan — cooling or evacuation, 72-hour supplies, medication continuity; insurers and cities publish address-level risk so decisions are informed.
17	Institutional corruption and state capture hollow out services while keeping the forms of democracy.	Governance & Law	Now	Society / National	18	23.9%	68%	Radical transparency with automatic publication: all public contracts, beneficial ownership of companies bidding, officials' assets, and court statistics published in open data by default — secrecy must be justified line by line, publication is the default state. Independent audit institutions with budgets fixed by formula (e.g., 0.1% of public spending) so governments cannot starve them. Specification: procurement above a low threshold is e-tendered with full bid disclosure; whistleblowers receive statutory protection plus a share of recovered funds.
18	Housing costs detach from incomes; working households are priced out, homelessness becomes structural.	Shelter & Housing	Now	Society / National	18	25.0%	64%	Attack the constraint, which is supply near work: legalize mid-density housing on all residential land within 800 m of transit, approvals deemed granted after 12 months, public land leased (not sold) for permanently affordable homes at 30% of income benchmark, and vacancy taxes that make empty investment units unprofitable. Specification: a city is healthy when median rent is at or below 30% of median household income; publish the ratio quarterly and trigger automatic zoning release when it exceeds it for 2 consecutive years.
19	Asset bubbles, household debt, and cost-of-living squeeze hollow out the middle; one rate shock cascades.	Economy & Finance	Now	Society / National	18	26.1%	63%	De-leverage by design: mortgage rules that cap total debt service at 35–40% of verified income under a stressed rate (+2%), automatic refinancing windows when rates fall, public development of non-speculative housing supply, and insolvency law that gives households a fresh start within 3 years instead of permanent debt peonage. Specification: personal rule — total debt payments under 30% of take-home; one income loss must be survivable for 6 months on reserves plus benefits; never hold employer stock plus job plus mortgage in one company town exposure.
20	Antimicrobial resistance quietly returns routine infections and surgeries to pre-antibiotic risk.	Health & Medicine	Now	Society / National	18	27.2%	62%	Stewardship with teeth: antibiotics in human medicine by prescription audit only, agricultural antibiotic use cut 75% and banned for growth promotion, hospital infection-control staffing at 1 specialist per 100 beds, and a standing pipeline fund that pays for new antibiotics by subscription (delinked from sales volume). Specification: every hospital publishes resistance rates quarterly; households never stockpile or share antibiotics — leftover courses are returned to pharmacies.
21	Schooling without learning: credentials inflate while actual literacy and numeracy stagnate or fall; misinform	Education & Knowledge	Now	Global / Civilizational	18	28.3%	58%	Measure learning, not attendance: independent annual assessments of reading and numeracy published per school, funding following measured learning gains, curricula rebuilt around mastery progression (no social promotion without support), and teacher pay and status raised to compete with other professions (top-quartile graduate recruitment). Specification: every child reads fluently by age 9 or triggers automatic intensive support; adult reskilling accounts (funded per citizen) redeemable at accredited providers, including libraries.
22	Platform dependence: a handful of companies carry the world's speech, identity, and emergency communication; o	Communication	Now	Global / Civilizational	18	29.3%	57%	Interoperability and exit rights by law: dominant platforms must support data portability and interoperable messaging, public communication infrastructure (postal, broadcast, municipal networks) maintained as a real fallback, and due-process rules for account termination with human appeal. Specification, personal and organizational: own your contact list and records off-platform (exported monthly), maintain one channel you fully control (your own email list, radio net, or phone tree), and never let a platform be the only copy of anything.
23	Car-dependent sprawl plus fuel and parts chokepoints make daily life brittle; one closure or price spike stran	Transport & Mobility	Now	Global / Civilizational	18	30.4%	56%	Redundancy and proximity: freight modal split so no single route or mode carries more than 40% of essentials; the 15-minute standard for new development (daily needs reachable on foot or by transit); transit networks funded as essential infrastructure with operating reserves of 6 months; strategic parts and fuel reserves for public fleets. Specification: every settlement over 5,000 has at least two physically separate road connections or one road plus scheduled public transport that runs in emergencies.
24	An epidemic of anxiety, depression, loneliness, and attention damage, concentrated in the young and the isolat	Psychology & Mind	Now	Global / Civilizational	18	31.5%	55%	Rebuild the inputs: sleep protected as infrastructure (school start times aligned to adolescent biology, no-screens wind-down norms), daily physical activity of 60 minutes for youth built into the school day, third places (libraries, clubs, leagues) funded as health infrastructure, and platform design rules for minors (no infinite scroll, no engagement-optimized feeds). Specification, personal: fixed sleep window, morning light exposure, one recurring in-person group activity weekly minimum — belonging is scheduled, not hoped for.
25	Precarious gig work, credential inflation, and regional job deserts leave large populations one missed paycheq	Work & Livelihood	Now	Global / Civilizational	18	32.6%	54%	Floor under every form of work: benefits (health, pension, injury, unemployment) pro-rated per hour worked regardless of classification, minimum standards applied per task not per contract type, regional diversification funds that trigger automatically when one employer exceeds 20% of local jobs, and wage insurance that tops up displaced workers at 50% of the wage gap for 2 years while retraining. Specification: personal standard — an emergency fund of 3 months, skills documented in portable form, and no single income source treated as permanent.
26	Two-tier justice: the wealthy buy delay and expertise; the poor plead guilty to get out of jail; backlogs make	Justice & Ethics	Now	Global / Civilizational	18	33.7%	53%	Equalize the machinery: pre-trial detention only on individualized danger findings with 48-hour review, legal aid funded to parity standards with prosecution in criminal matters, court capacity managed to published time standards (e.g., 90% of cases resolved within 12 months), and fines scaled to income (day-fines) so penalties bite equally. Specification: every court publishes wait times, outcomes by representation type, and backlog monthly; households know their rights in writing — a plain-language rights card is a standard household document.
27	Climate-shifted growing zones plus soil loss cut yields across several breadbaskets simultaneously.	Food & Agriculture	Future	Global / Civilizational	18	34.8%	46%	Regenerate the base: rebuild topsoil organic matter to 4%+ through cover cropping and reduced tillage; shift 20% of acreage to drought- and heat-bred varieties; build vertical/greenhouse capacity for greens near cities (target 50% of leafy supply within city limits); protect pollinators with pesticide-free corridors on 10% of farmland. Specification: yield planning assumes the worst decade climate, not the average year.
28	Algorithmic feeds optimize for outrage; shared factual reality fragments; institutions cannot establish basic	Information & Media	Now	Global / Civilizational	18	35.9%	45%	Change the incentives, not just the content: recommender systems above a scale threshold must offer a chronological/non-optimized feed as a real default option, political advertising carries a public registry of sponsor, spend, and targeting criteria, and proven coordinated inauthentic campaigns trigger platform liability. Personal specification: a two-source rule for any alarming claim (two independent, named, primary sources before acting or forwarding), primary documents over commentary, and one scheduled news window per day instead of continuous feed exposure.
29	Below-replacement fertility plus rapid aging inverts the pyramid: fewer workers support more retirees; towns h	Demographics & Population	Now	Global / Civilizational	18	37.0%	45%	Make family formation affordable where people already want it: housing sized and priced for young families (family units at or below 30% of median income in every major city plan), childcare costing under 10% of median income per child, working-time protection for parents, and pension formulas that adjust transparently and gradually (automatic balancing mechanisms) instead of by crisis. Specification: immigration and integration run as a permanent, competently administered system sized to the measured demographic gap, with language and credential recognition completed within 12 months of arrival.

Pareto Rank	Solution (short title = its problem)	Field	Timeframe	Scale	Priority Score	Cumulative % of	Probability of	Full Solution
30	Autonomous weapons and cheap drones make precision violence available to anyone, overwhelming policing built f	Security & Defense	Future	Global / Civilizational	18	38.0%	44%	Regulate and shield in advance: mandatory remote-identification and geofencing on all drones sold, counter-drone authority and equipment at critical sites (airports, utilities, mass events) with legal rules of engagement written before incidents, export controls on autonomous targeting components, and an international ban framework on fully autonomous lethal systems. Specification: critical infrastructure must pass an annual red-team drone assessment; households and businesses rely on hardened, setback design for the highest-value targets rather than private countermeasures.
31	Multi-year megadrought plus population growth makes current cities structurally water-short.	Water	Future	Global / Civilizational	18	39.1%	43%	Rebuild the supply stack: wastewater recycling to potable standard (target 30–40% of supply), stormwater capture on all new developments (retain the 25-year storm on site), desalination where coastal, and pricing that guarantees a free lifeline block (first 50 L/person/day) with steep rates above it. Specification: per-capita demand cut to 100 L/day total through fixtures and leakage below 8%; inter-basin agreements signed before crisis, not during.
32	Climate migration plus uninsurable regions strand populations in housing that cannot be sold, insured, or safe	Shelter & Housing	Future	Global / Civilizational	18	40.2%	40%	Managed transition instead of collapse: pre-mapped receiving cities with pre-built infrastructure capacity, public buyouts at pre-decline values funded by risk-zone insurance levies collected while values remain, and building law that stops new construction in zones projected uninsurable within 30 years. Specification: national housing plans must model the 2050 climate map, not the 2000 one; household level — never hold more than 50% of net worth in a single climate-exposed property.
33	Electrification of everything concentrates all risk on one system; a solar storm or systemic cyber event takes	Energy	Future	Global / Civilizational	18	41.3%	38%	Stockpile the long-lead items: a strategic reserve of large transformers and grid components (the items with 12+ month lead times), EMP/GMD-hardened control systems for the top substations, and a designed analogue fallback — paper procedures, manual switching, and local generation — drilled, not just documented. Specification: every region can meet 30% of demand (the survival core: water, food cold-chain, hospitals, comms) on distributed local generation indefinitely.
34	An engineered or novel pathogen, or total care-system overload from aging plus chronic disease, breaks medicin	Health & Medicine	Future	Global / Civilizational	18	42.4%	36%	Layered defence: universal indoor air standards in hospitals, schools, and transit (ventilation equivalent to 6 air changes/hour, upper-air UV in high-risk rooms), synthetic-biology screening of all commercial DNA synthesis orders, regional manufacturing of essential medicines (top 100 generics produced on at least 2 continents each), and a reserve clinical workforce (retired and part-time clinicians registered and drilled annually). Specification: no single country supplies more than 40% of any essential medicine's global volume.
35	Legitimacy collapse: populations stop believing any institution, and coordination itself fails even without a	Governance & Law	Future	Global / Civilizational	18	43.5%	35%	Rebuild trust as infrastructure: citizen assemblies chosen by lottery with real decision power over defined questions, radical localism (decisions at the smallest competent scale, where people can verify outcomes personally), and service-delivery proof — publish wait times, error rates, and fix rates for every public service monthly. Specification: trust is measured (surveyed quarterly, published); agencies falling below threshold trigger automatic independent review.
36	Automation displaces labour faster than new livelihoods form, while wealth concentrates in owners of machines	Economy & Finance	Future	Global / Civilizational	18	44.6%	32%	Share the machine dividend by structure, not charity: sovereign and pension funds holding broad equity stakes so productivity gains flow to all citizens as dividends; reduced standard working hours as productivity rises (the historical pattern, deliberately continued); portable benefits attached to the person, not the job; and public investment in the irreducibly human sectors (care, craft, education, restoration) at the scale of the displacement. Specification: transition planning uses the fastest plausible displacement curve, with retraining accounts funded per worker before displacement, not after.
37	A misaligned or misused advanced AI, or total dependence on opaque systems, produces civilizational-scale fail	Technology & AI	Future	Global / Civilizational	18	45.7%	30%	Governance proportional to capability: frontier training runs above defined compute thresholds are registered and evaluated for dangerous capabilities before release; critical infrastructure keeps human-verified analogue fallbacks and禁止 fully autonomous control of weapons, grid switching, and medical dosing; liability law makes deployers answer for system outputs; and international verification for the largest training facilities, modelled on nuclear safeguards. Specification: every essential service must demonstrate a manual mode, drilled annually; personal rule — no single AI or platform holds your only copy of anything important.
38	Tipping points cascade: ice loss, permafrost methane, ocean-circulation shifts lock in metres of sea-level ris	Environment & Climate	Future	Global / Civilizational	18	46.7%	26%	Both brakes and lifeboats: emissions cuts on the fastest feasible curve (the only lever on the cause), plus planned adaptation at civilizational scale — protected agricultural relocation zones, freshwater megaprojects only where hydrology closes, migration frameworks negotiated before flows become crises (status, quotas, receiving-city investment), and geoengineering research governed internationally with deployment requiring multinational consent. Specification: all long-lived infrastructure (50+ year life) sited and designed against the high-end sea-level and heat scenarios, not the median.
39	Schooling without learning: credentials inflate while actual literacy and numeracy stagnate or fall; misinform	Education & Knowledge	Now	Society / National	12	47.5%	70%	Measure learning, not attendance: independent annual assessments of reading and numeracy published per school, funding following measured learning gains, curricula rebuilt around mastery progression (no social promotion without support), and teacher pay and status raised to compete with other professions (top-quartile graduate recruitment). Specification: every child reads fluently by age 9 or triggers automatic intensive support; adult reskilling accounts (funded per citizen) redeemable at accredited providers, including libraries.
40	Platform dependence: a handful of companies carry the world's speech, identity, and emergency communication; o	Communication	Now	Society / National	12	48.2%	69%	Interoperability and exit rights by law: dominant platforms must support data portability and interoperable messaging, public communication infrastructure (postal, broadcast, municipal networks) maintained as a real fallback, and due-process rules for account termination with human appeal. Specification, personal and organizational: own your contact list and records off-platform (exported monthly), maintain one channel you fully control (your own email list, radio net, or phone tree), and never let a platform be the only copy of anything.
41	Car-dependent sprawl plus fuel and parts chokepoints make daily life brittle; one closure or price spike stran	Transport & Mobility	Now	Society / National	12	48.9%	68%	Redundancy and proximity: freight modal split so no single route or mode carries more than 40% of essentials; the 15-minute standard for new development (daily needs reachable on foot or by transit); transit networks funded as essential infrastructure with operating reserves of 6 months; strategic parts and fuel reserves for public fleets. Specification: every settlement over 5,000 has at least two physically separate road connections or one road plus scheduled public transport that runs in emergencies.
42	An epidemic of anxiety, depression, loneliness, and attention damage, concentrated in the young and the isolat	Psychology & Mind	Now	Society / National	12	49.6%	67%	Rebuild the inputs: sleep protected as infrastructure (school start times aligned to adolescent biology, no-screens wind-down norms), daily physical activity of 60 minutes for youth built into the school day, third places (libraries, clubs, leagues) funded as health infrastructure, and platform design rules for minors (no infinite scroll, no engagement-optimized feeds). Specification, personal: fixed sleep window, morning light exposure, one recurring in-person group activity weekly minimum — belonging is scheduled, not hoped for.
43	Precarious gig work, credential inflation, and regional job deserts leave large populations one missed paycheq	Work & Livelihood	Now	Society / National	12	50.4%	66%	Floor under every form of work: benefits (health, pension, injury, unemployment) pro-rated per hour worked regardless of classification, minimum standards applied per task not per contract type, regional diversification funds that trigger automatically when one employer exceeds 20% of local jobs, and wage insurance that tops up displaced workers at 50% of the wage gap for 2 years while retraining. Specification: personal standard — an emergency fund of 3 months, skills documented in portable form, and no single income source treated as permanent.

Pareto Rank	Solution (short title = its problem)	Field	Timeframe	Scale	Priority Score	Cumulative % of	Probability of	Full Solution
44	Two-tier justice: the wealthy buy delay and expertise; the poor plead guilty to get out of jail; backlogs make	Justice & Ethics	Now	Society / National	12	51.1%	65%	Equalize the machinery: pre-trial detention only on individualized danger findings with 48-hour review, legal aid funded to parity standards with prosecution in criminal matters, court capacity managed to published time standards (e.g., 90% of cases resolved within 12 months), and fines scaled to income (day-fines) so penalties bite equally. Specification: every court publishes wait times, outcomes by representation type, and backlog monthly; households know their rights in writing — a plain-language rights card is a standard household document.
45	Climate-shifted growing zones plus soil loss cut yields across several breadbaskets simultaneously.	Food & Agriculture	Future	Society / National	12	51.8%	58%	Regenerate the base: rebuild topsoil organic matter to 4%+ through cover cropping and reduced tillage; shift 20% of acreage to drought- and heat-bred varieties; build vertical/greenhouse capacity for greens near cities (target 50% of leafy supply within city limits); protect pollinators with pesticide-free corridors on 10% of farmland. Specification: yield planning assumes the worst decade climate, not the average year.
46	Algorithmic feeds optimize for outrage; shared factual reality fragments; institutions cannot establish basic	Information & Media	Now	Society / National	12	52.5%	57%	Change the incentives, not just the content: recommender systems above a scale threshold must offer a chronological/non-optimized feed as a real default option, political advertising carries a public registry of sponsor, spend, and targeting criteria, and proven coordinated inauthentic campaigns trigger platform liability. Personal specification: a two-source rule for any alarming claim (two independent, named, primary sources before acting or forwarding), primary documents over commentary, and one scheduled news window per day instead of continuous feed exposure.
47	Below-replacement fertility plus rapid aging inverts the pyramid: fewer workers support more retirees; towns h	Demographics & Population	Now	Society / National	12	53.3%	57%	Make family formation affordable where people already want it: housing sized and priced for young families (family units at or below 30% of median income in every major city plan), childcare costing under 10% of median income per child, working-time protection for parents, and pension formulas that adjust transparently and gradually (automatic balancing mechanisms) instead of by crisis. Specification: immigration and integration run as a permanent, competently administered system sized to the measured demographic gap, with language and credential recognition completed within 12 months of arrival.
48	Autonomous weapons and cheap drones make precision violence available to anyone, overwhelming policing built f	Security & Defense	Future	Society / National	12	54.0%	56%	Regulate and shield in advance: mandatory remote-identification and geofencing on all drones sold, counter-drone authority and equipment at critical sites (airports, utilities, mass events) with legal rules of engagement written before incidents, export controls on autonomous targeting components, and an international ban framework on fully autonomous lethal systems. Specification: critical infrastructure must pass an annual red-team drone assessment; households and businesses rely on hardened, setback design for the highest-value targets rather than private countermeasures.
49	Multi-year megadrought plus population growth makes current cities structurally water-short.	Water	Future	Society / National	12	54.7%	55%	Rebuild the supply stack: wastewater recycling to potable standard (target 30–40% of supply), stormwater capture on all new developments (retain the 25-year storm on site), desalination where coastal, and pricing that guarantees a free lifeline block (first 50 L/person/day) with steep rates above it. Specification: per-capita demand cut to 100 L/day total through fixtures and leakage below 8%; inter-basin agreements signed before crisis, not during.
50	Climate migration plus uninsurable regions strand populations in housing that cannot be sold, insured, or safe	Shelter & Housing	Future	Society / National	12	55.4%	52%	Managed transition instead of collapse: pre-mapped receiving cities with pre-built infrastructure capacity, public buyouts at pre-decline values funded by risk-zone insurance levies collected while values remain, and building law that stops new construction in zones projected uninsurable within 30 years. Specification: national housing plans must model the 2050 climate map, not the 2000 one; household level — never hold more than 50% of net worth in a single climate-exposed property.
51	Electrification of everything concentrates all risk on one system; a solar storm or systemic cyber event takes	Energy	Future	Society / National	12	56.2%	50%	Stockpile the long-lead items: a strategic reserve of large transformers and grid components (the items with 12+ month lead times), EMP/GMD-hardened control systems for the top substations, and a designed analogue fallback — paper procedures, manual switching, and local generation — drilled, not just documented. Specification: every region can meet 30% of demand (the survival core: water, food cold-chain, hospitals, comms) on distributed local generation indefinitely.
52	Synthetic media makes any audio, video, or document forgeable at zero cost; 'seeing' stops being believing, an	Information & Media	Future	Global / Civilizational	12	56.9%	49%	Provenance infrastructure: cameras and recorders sign content at capture (cryptographic content credentials as the default in professional and official contexts), courts and newsrooms verify by signature chain rather than appearance, and legal process treats unsigned dramatic media as allegation, not evidence. Specification: households and organizations keep original, signed, off-platform copies of identity, property, and credential records — paper originals of the documents that prove who you are and what you own.
53	An engineered or novel pathogen, or total care-system overload from aging plus chronic disease, breaks medicin	Health & Medicine	Future	Society / National	12	57.6%	48%	Layered defence: universal indoor air standards in hospitals, schools, and transit (ventilation equivalent to 6 air changes/hour, upper-air UV in high-risk rooms), synthetic-biology screening of all commercial DNA synthesis orders, regional manufacturing of essential medicines (top 100 generics produced on at least 2 continents each), and a reserve clinical workforce (retired and part-time clinicians registered and drilled annually). Specification: no single country supplies more than 40% of any essential medicine's global volume.
54	Legitimacy collapse: populations stop believing any institution, and coordination itself fails even without a	Governance & Law	Future	Society / National	12	58.3%	47%	Rebuild trust as infrastructure: citizen assemblies chosen by lottery with real decision power over defined questions, radical localism (decisions at the smallest competent scale, where people can verify outcomes personally), and service-delivery proof — publish wait times, error rates, and fix rates for every public service monthly. Specification: trust is measured (surveyed quarterly, published); agencies falling below threshold trigger automatic independent review.
55	AI tutors and synthetic content make human knowledge optional; populations lose the ability to verify, calcula	Education & Knowledge	Future	Global / Civilizational	12	59.1%	47%	Dual-competence standard: every qualification requires demonstrated unaided performance (closed-book, offline, by hand) plus aided performance; curricula teach verification as the core skill — source-tracing, estimation, consistency checking; national knowledge reserves (printed, distributed) maintained as civil infrastructure. Specification: critical professions (medicine, engineering, aviation, law) license on unaided competence first; schools schedule regular no-device weeks, not as punishment but as training.
56	Algorithmic justice and predictive policing automate bias at scale; people are punished for probabilities, and	Justice & Ethics	Future	Global / Civilizational	12	59.8%	46%	Human answerability as law: no liberty decision (arrest priority, bail, sentencing, parole) may rest solely on an algorithmic score; models used in justice are public, auditable, and tested for error and disparity rates published per demographic group; every automated recommendation names the accountable human who adopted it. Specification: any person scored by a state system can see their inputs, correct errors within 30 days, and receive a human review on request — the same standard as credit records, applied to freedom.
57	Automation displaces labour faster than new livelihoods form, while wealth concentrates in owners of machines	Economy & Finance	Future	Society / National	12	60.5%	44%	Share the machine dividend by structure, not charity: sovereign and pension funds holding broad equity stakes so productivity gains flow to all citizens as dividends; reduced standard working hours as productivity rises (the historical pattern, deliberately continued); portable benefits attached to the person, not the job; and public investment in the irreducibly human sectors (care, craft, education, restoration) at the scale of the displacement. Specification: transition planning uses the fastest plausible displacement curve, with retraining accounts funded per worker before displacement, not after.

Pareto Rank	Solution (short title = its problem)	Field	Timeframe	Scale	Priority Score	Cumulative % of	Probability of	Full Solution
58	A misaligned or misused advanced AI, or total dependence on opaque systems, produces civilizational-scale fail	Technology & AI	Future	Society / National	12	61.2%	42%	Governance proportional to capability: frontier training runs above defined compute thresholds are registered and evaluated for dangerous capabilities before release; critical infrastructure keeps human-verified analogue fallbacks and禁止 fully autonomous control of weapons, grid switching, and medical dosing; liability law makes deployers answer for system outputs; and international verification for the largest training facilities, modelled on nuclear safeguards. Specification: every essential service must demonstrate a manual mode, drilled annually; personal rule — no single AI or platform holds your only copy of anything important.
59	Autonomous, electric, platform-controlled mobility concentrates failure: one software fault, cyberattack, or b	Transport & Mobility	Future	Global / Civilizational	12	62.0%	42%	Mandated heterogeneity and manual fallback: fleets certified from at least 2 independent software stacks per region, all autonomous vehicles retaining manual drive capability, transport control systems segmented from public networks with analogue scheduling fallback, and battery-material supply diversified with recycling recovering 90%+ of critical minerals by law. Specification: cities drill a 'no-network day' annually — transit, logistics, and emergency services operating on fallback procedures.
60	Total surveillance or total encryption-war makes private communication impossible — or, inversely, untraceable	Communication	Future	Global / Civilizational	12	62.7%	40%	Rights-preserving architecture: end-to-end encryption protected in law as the default for private communication, state access only through individualized, judicially authorized, logged, and time-limited orders with after-the-fact notification; anonymous public speech protected while large-scale amplification carries provenance duties. Specification: every surveillance power carries a 5-year sunset and a published annual accounting of use, errors, and outcomes — powers that cannot justify themselves expire.
61	Tipping points cascade: ice loss, permafrost methane, ocean-circulation shifts lock in metres of sea-level ris	Environment & Climate	Future	Society / National	12	63.4%	38%	Both brakes and lifeboats: emissions cuts on the fastest feasible curve (the only lever on the cause), plus planned adaptation at civilizational scale — protected agricultural relocation zones, freshwater megaprojects only where hydrology closes, migration frameworks negotiated before flows become crises (status, quotas, receiving-city investment), and geoengineering research governed internationally with deployment requiring multinational consent. Specification: all long-lived infrastructure (50+ year life) sited and designed against the high-end sea-level and heat scenarios, not the median.
62	Divergent futures collide: some regions depopulate to non-viability while others add billions; unmanaged flows	Demographics & Population	Future	Global / Civilizational	12	64.1%	37%	Plan both directions simultaneously: shrink-smart policy for depopulating regions (consolidated services, maintained dignity and access, land returned to nature or agriculture under stewardship), growth-ready frameworks for expanding regions (land tenure, urban plans, and service grids built ahead of arrival), and pre-negotiated international mobility compacts so movement is governed, not chaotic. Specification: every national plan carries a 50-year demographic annex, updated every 5 years, with infrastructure sized to the projection band — not to today's population.
63	Synthetic companionship and immersive virtual worlds outcompete reality; populations withdraw from human conta	Psychology & Mind	Future	Global / Civilizational	12	64.9%	34%	Protect the real by making it richer, not by banning the synthetic: disclosure law (companions must identify as artificial, always), design rules against dependency engineering (no manufactured jealousy, streaks, or guilt mechanics), public investment in physical gathering — festivals, sport, craft, public space — at a scale that competes, and social norms taught early: the difference between being comforted and being accompanied. Specification: households set explicit budgets — money and hours — for synthetic companionship, reviewed like any other consumption; any product whose retention depends on your loneliness is treated as extractive by definition.
64	Machines take the majority of paid tasks; the link between contribution and income breaks entirely.	Work & Livelihood	Future	Global / Civilizational	12	65.6%	31%	Rebuild distribution on ownership: broad-based capital ownership (citizen funds, employee ownership trusts, data and AI dividends), radically shortened standard working time as productivity rises, and public valuation of unpaid contribution (care, community, restoration) through contribution credits convertible to services. Specification: transition is phased by measured displacement rates per sector, with the ownership mechanisms built during the growth phase — financed from automation rents — not improvised after collapse.
65	Municipal supply contamination or infrastructure failure cuts safe drinking water to a city.	Water	Now	Household / Personal	9	66.1%	93%	Dual-barrier defence: (1) two independent treatment trains so one failure never reaches taps; (2) point-of-use backup in every household — a certified gravity or pump filter rated to remove bacteria/viruses (0.02–0.2 micron) plus a 72-hour stored reserve (12 L/person drinking). Specification: utilities must hold 5 days of treated-water storage and publish real-time turbidity and chlorine residuals; buildings over 6 storeys keep a 24-hour gravity tank.
66	Cyber intrusion and ransomware shut down hospitals, pipelines, and city governments; the digital base is fragi	Technology & AI	Now	Household / Personal	9	66.7%	86%	Assume breach, engineer recovery: immutable offline backups tested by full restoration quarterly (a backup never restored is a rumour), network segmentation so one infection cannot cross from office systems to operations, multi-factor authentication on 100% of remote and privileged access, and 24-hour incident reporting with a national assistance duty. Specification for households and small organizations: automatic updates on, unique passwords in a manager, 3-2-1 backups (3 copies, 2 media, 1 off-site/offline) for irreplaceable data — photos, records, work.
67	Grid failure — storm, cyberattack, or cascading overload — blacks out regions for days, and everything else fa	Energy	Now	Household / Personal	9	67.2%	84%	Engineer for islanding: grids split into self-sustaining cells that can run 72 hours alone; critical facilities (hospitals, water, 911, data centres) on dual feeds plus on-site generation tested monthly under load; household standard — a 1–3 kWh battery with solar trickle keeps lights, phone, fridge, and medical devices alive 48–72 hours. Specification: black-start capability drilled twice a year; underground or harden the 20% of lines whose failure cascades furthest (Pareto of the grid).
68	Organized crime, gang fragmentation, and everyday violence make districts unsafe while institutions lose legit	Security & Defense	Now	Household / Personal	9	67.8%	82%	Focused deterrence, the proven Pareto intervention: identify the tiny network driving most violence (typically under 0.5% of a city's population), offer services and exit routes with one hand and certain, swift prosecution with the other; light and maintain the worst micro-locations (lighting, sightlines, occupied storefronts); violence-interrupter teams staffed by credible locals. Specification: measure success by shootings in the target micro-zones, reviewed monthly; every intervention audited for civil-rights compliance.
69	Supply-chain shock (war, fuel, fertilizer, port failure) spikes prices and empties shelves within weeks.	Food & Agriculture	Now	Household / Personal	9	68.3%	80%	Shorten and parallel the chain: contract at least 2 geographically separate suppliers per staple; local production floor of 30% of fresh food within 100 km; fertilizer independence via green-ammonia plants sized to 50% of national need. Specification: retailers and institutions hold 30 days of non-perishable buffer; households hold 30 days (one month of staples ≈ 25 kg grain-equivalent per adult); price-spike circuit breakers release reserves automatically at +40% price movement.
70	Extreme heat, flood, and fire exceed the design basis of cities built for a milder century.	Environment & Climate	Now	Household / Personal	9	68.8%	77%	Adapt to the climate that exists: building codes and infrastructure designed to the 2050 worst-decile weather, urban canopy targets of 30–40% to cut street-level heat, floodplains returned to water with buyouts instead of repeated rebuilds, and heat action plans with named officials, cooling centres within a 15-minute walk, and check-in systems for isolated elderly. Specification: every home in a risk zone has a written plan — cooling or evacuation, 72-hour supplies, medication continuity; insurers and cities publish address-level risk so decisions are informed.
71	Institutional corruption and state capture hollow out services while keeping the forms of democracy.	Governance & Law	Now	Household / Personal	9	69.4%	76%	Radical transparency with automatic publication: all public contracts, beneficial ownership of companies bidding, officials' assets, and court statistics published in open data by default — secrecy must be justified line by line, publication is the default state. Independent audit institutions with budgets fixed by formula (e.g., 0.1% of public spending) so governments cannot starve them. Specification: procurement above a low threshold is e-tendered with full bid disclosure; whistleblowers receive statutory protection plus a share of recovered funds.

Pareto Rank	Solution (short title = its problem)	Field	Timeframe	Scale	Priority Score	Cumulative % of	Probability of	Full Solution
72	Fuel embargo or supply cutoff paralyzes transport, heating, and industry in dependent nations.	Energy	Past	Global / Civilizational	9	69.9%	72%	Strategic reserves and demand discipline: 90 days of net-import reserves for critical fuels (the standing international standard), efficiency law that cuts vehicle and building consumption 2% per year, and pre-planned rationing tiers so essential services (food logistics, hospitals, water) are served first. Specification: household level — 72 hours of light, heat, and cooking independent of the grid (LED + battery bank of 1 kWh, safe fuel stove); national level — no single supplier above 30% of any critical fuel.
73	Housing costs detach from incomes; working households are priced out, homelessness becomes structural.	Shelter & Housing	Now	Household / Personal	9	70.5%	72%	Attack the constraint, which is supply near work: legalize mid-density housing on all residential land within 800 m of transit, approvals deemed granted after 12 months, public land leased (not sold) for permanently affordable homes at 30% of income benchmark, and vacancy taxes that make empty investment units unprofitable. Specification: a city is healthy when median rent is at or below 30% of median household income; publish the ratio quarterly and trigger automatic zoning release when it exceeds it for 2 consecutive years.
74	Banking panic and deflation spiral convert a market crash into a decade of mass unemployment.	Economy & Finance	Past	Global / Civilizational	9	71.0%	71%	Circuit breakers built in advance: deposit insurance with pre-funded reserves, central-bank lender-of-last-resort facilities designed before crises, automatic fiscal stabilizers (unemployment benefits that expand by formula, not by vote), and bank resolution law that wipes out shareholders and management — never depositors — first. Specification: systemically important banks hold capital of at least 12% of risk-weighted assets with a leverage cap; households keep 3–6 months of expenses in insured deposits, never in a single institution above the insured limit.
75	Asset bubbles, household debt, and cost-of-living squeeze hollow out the middle; one rate shock cascades.	Economy & Finance	Now	Household / Personal	9	71.6%	71%	De-leverage by design: mortgage rules that cap total debt service at 35–40% of verified income under a stressed rate (+2%), automatic refinancing windows when rates fall, public development of non-speculative housing supply, and insolvency law that gives households a fresh start within 3 years instead of permanent debt peonage. Specification: personal rule — total debt payments under 30% of take-home; one income loss must be survivable for 6 months on reserves plus benefits; never hold employer stock plus job plus mortgage in one company town exposure.
76	Single-crop dependence plus blight or drought produces famine at national scale.	Food & Agriculture	Past	Global / Civilizational	9	72.1%	70%	Never let one crop or one supplier exceed 25% of calories. Maintain seed diversity banks with at least 3 climate-matched varieties per staple, rotate crops on a minimum 3-year cycle, and hold strategic grain reserves of 90 days of national consumption. Specification: household level — 90 days of staple calories stored (rice/grain ~ 25 kg per adult per month at 2,000 kcal/day planning), rotated FIFO; regional level — no more than 40% of food imported by calorie.
77	Antimicrobial resistance quietly returns routine infections and surgeries to pre-antibiotic risk.	Health & Medicine	Now	Household / Personal	9	72.6%	70%	Stewardship with teeth: antibiotics in human medicine by prescription audit only, agricultural antibiotic use cut 75% and banned for growth promotion, hospital infection-control staffing at 1 specialist per 100 beds, and a standing pipeline fund that pays for new antibiotics by subscription (delinked from sales volume). Specification: every hospital publishes resistance rates quarterly; households never stockpile or share antibiotics — leftover courses are returned to pharmacies.
78	Fire, earthquake, or war destroys a city's housing stock in days, leaving populations in the open.	Shelter & Housing	Past	Global / Civilizational	9	73.2%	69%	Build to survive the known local worst: seismic codes to the 2,475-year event for new builds, retrofit of the highest-risk 20% of existing stock first (Pareto: unreinforced masonry and soft-storey buildings cause most deaths), firebreaks and non-combustible roofing in fire zones. Specification: every city holds emergency shelter capacity for 10% of its population for 30 days (pre-sited, pre-supplied), and every household keeps a 72-hour go-kit — documents, medication, water, light.
79	Pandemic disease sweeps a population with no immunity, no germ theory, and no surge capacity.	Health & Medicine	Past	Global / Civilizational	9	73.7%	68%	Standing pandemic infrastructure, built before it is needed: genomic surveillance of wastewater in every city over 100,000 (weekly), a vaccine platform able to go from sequence to first doses in 100 days, and stockpiled PPE for 90 days of full-hospital use. Specification: hospital surge plans must double ICU capacity in 14 days (pre-mapped wards, cross-trained staff rosters); every household keeps a basic kit — thermometer, pulse oximeter, oral rehydration salts, fever medication for 2 weeks.
80	Regional aquifer and river systems collapse from overuse and contamination, making whole regions uninhabitable	Water	Past	Global / Civilizational	9	74.3%	66%	Cap total extraction at measured recharge: meter every well and canal offtake, publish the water balance monthly, and set the legal extraction ceiling at 80% of average recharge so the buffer absorbs dry years. Restore wetlands at river mouths (minimum 10% of historical area) as natural filters and flood buffers. Specification: household storage 4 litres per person per day drinking minimum, 14-day reserve = 56 L/person; community reservoirs sized for 90 days of essential use.
81	Civilizations degrade their own ecological base — soil, forests, water — until the base fails and the society	Environment & Climate	Past	Global / Civilizational	9	74.8%	59%	Close the loop with law: extraction and pollution permits capped by measured regeneration rates of the specific ecosystem (not national averages), soil-loss limits per hectare enforced like building codes, and full-cost accounting where the price of a product includes its restoration cost. Specification: farmland soil organic matter and erosion measured on a 5-year cycle; land degrading past threshold enters mandatory restoration with cost-sharing.
82	A general-purpose technology spreads faster than safety norms, and its harms compound for decades before corre	Technology & AI	Past	Global / Civilizational	9	75.4%	54%	Precaution with deadlines: new general-purpose technologies pass staged safety gates (lab, limited, general) with published criteria and automatic review dates; manufacturers carry strict liability for latent harms, funded by mandatory insurance during the entire exposure tail; and a standing technology-assessment office publishes plain-language risk registers. Specification: no product with irreversible population-level exposure ships without a funded 30-year monitoring program; households treat 'new and unstudied' as a reason to delay adoption for non-essentials.
83	Constitutional order collapses into dictatorship through legal steps, emergency decrees, and captured courts.	Governance & Law	Past	Global / Civilizational	9	75.9%	48%	Design constitutions against their own abolition: eternity clauses protecting core rights from amendment, emergency powers with hard 90-day sunsets and escalating supermajorities for renewal, judicial appointments split across at least 3 independent bodies, and electoral administration insulated from the executive. Specification: every emergency measure auto-expires; every rights restriction must be re-justified publicly at renewal; citizens keep certified copies of identity and property records in personal custody.
84	Invasion or civil war destroys the state's monopoly on force; civilians face violence with no protection.	Security & Defense	Past	Global / Civilizational	9	76.4%	33%	Prevention is the only real solution once killing starts: early-warning indicators (hate-media licensing, militia formation, arms flows) monitored with published thresholds that trigger automatic diplomatic and legal consequences; constitutional hard limits on emergency powers (sunset clauses of 90 days, legislative renewal); and protected humanitarian corridors negotiated in advance frameworks. Specification: civilian layer — neighbourhood mutual-aid rosters, documented evidence habits (photos, records kept off-site), and pre-planned exit routes with 72-hour kits.
85	Synthetic media makes any audio, video, or document forgeable at zero cost; 'seeing' stops being believing, an	Information & Media	Future	Society / National	8	76.9%	61%	Provenance infrastructure: cameras and recorders sign content at capture (cryptographic content credentials as the default in professional and official contexts), courts and newsrooms verify by signature chain rather than appearance, and legal process treats unsigned dramatic media as allegation, not evidence. Specification: households and organizations keep original, signed, off-platform copies of identity, property, and credential records — paper originals of the documents that prove who you are and what you own.

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86	AI tutors and synthetic content make human knowledge optional; populations lose the ability to verify, calcula	Education & Knowledge	Future	Society / National	8	77.4%	59%	Dual-competence standard: every qualification requires demonstrated unaided performance (closed-book, offline, by hand) plus aided performance; curricula teach verification as the core skill — source-tracing, estimation, consistency checking; national knowledge reserves (printed, distributed) maintained as civil infrastructure. Specification: critical professions (medicine, engineering, aviation, law) license on unaided competence first; schools schedule regular no-device weeks, not as punishment but as training.
87	Algorithmic justice and predictive policing automate bias at scale; people are punished for probabilities, and	Justice & Ethics	Future	Society / National	8	77.9%	58%	Human answerability as law: no liberty decision (arrest priority, bail, sentencing, parole) may rest solely on an algorithmic score; models used in justice are public, auditable, and tested for error and disparity rates published per demographic group; every automated recommendation names the accountable human who adopted it. Specification: any person scored by a state system can see their inputs, correct errors within 30 days, and receive a human review on request — the same standard as credit records, applied to freedom.
88	Autonomous, electric, platform-controlled mobility concentrates failure: one software fault, cyberattack, or b	Transport & Mobility	Future	Society / National	8	78.4%	54%	Mandated heterogeneity and manual fallback: fleets certified from at least 2 independent software stacks per region, all autonomous vehicles retaining manual drive capability, transport control systems segmented from public networks with analogue scheduling fallback, and battery-material supply diversified with recycling recovering 90%+ of critical minerals by law. Specification: cities drill a 'no-network day' annually — transit, logistics, and emergency services operating on fallback procedures.
89	Total surveillance or total encryption-war makes private communication impossible — or, inversely, untraceable	Communication	Future	Society / National	8	78.9%	52%	Rights-preserving architecture: end-to-end encryption protected in law as the default for private communication, state access only through individualized, judicially authorized, logged, and time-limited orders with after-the-fact notification; anonymous public speech protected while large-scale amplification carries provenance duties. Specification: every surveillance power carries a 5-year sunset and a published annual accounting of use, errors, and outcomes — powers that cannot justify themselves expire.
90	Divergent futures collide: some regions depopulate to non-viability while others add billions; unmanaged flows	Demographics & Population	Future	Society / National	8	79.3%	49%	Plan both directions simultaneously: shrink-smart policy for depopulating regions (consolidated services, maintained dignity and access, land returned to nature or agriculture under stewardship), growth-ready frameworks for expanding regions (land tenure, urban plans, and service grids built ahead of arrival), and pre-negotiated international mobility compacts so movement is governed, not chaotic. Specification: every national plan carries a 50-year demographic annex, updated every 5 years, with infrastructure sized to the projection band — not to today's population.
91	Synthetic companionship and immersive virtual worlds outcompete reality; populations withdraw from human conta	Psychology & Mind	Future	Society / National	8	79.8%	46%	Protect the real by making it richer, not by banning the synthetic: disclosure law (companions must identify as artificial, always), design rules against dependency engineering (no manufactured jealousy, streaks, or guilt mechanics), public investment in physical gathering — festivals, sport, craft, public space — at a scale that competes, and social norms taught early: the difference between being comforted and being accompanied. Specification: households set explicit budgets — money and hours — for synthetic companionship, reviewed like any other consumption; any product whose retention depends on your loneliness is treated as extractive by definition.
92	Machines take the majority of paid tasks; the link between contribution and income breaks entirely.	Work & Livelihood	Future	Society / National	8	80.3%	43%	Rebuild distribution on ownership: broad-based capital ownership (citizen funds, employee ownership trusts, data and AI dividends), radically shortened standard working time as productivity rises, and public valuation of unpaid contribution (care, community, restoration) through contribution credits convertible to services. Specification: transition is phased by measured displacement rates per sector, with the ownership mechanisms built during the growth phase — financed from automation rents — not improvised after collapse.
93	Fuel embargo or supply cutoff paralyzes transport, heating, and industry in dependent nations.	Energy	Past	Society / National	6	80.7%	84%	Strategic reserves and demand discipline: 90 days of net-import reserves for critical fuels (the standing international standard), efficiency law that cuts vehicle and building consumption 2% per year, and pre-planned rationing tiers so essential services (food logistics, hospitals, water) are served first. Specification: household level — 72 hours of light, heat, and cooking independent of the grid (LED + battery bank of 1 kWh, safe fuel stove); national level — no single supplier above 30% of any critical fuel.
94	Banking panic and deflation spiral convert a market crash into a decade of mass unemployment.	Economy & Finance	Past	Society / National	6	81.0%	83%	Circuit breakers built in advance: deposit insurance with pre-funded reserves, central-bank lender-of-last-resort facilities designed before crises, automatic fiscal stabilizers (unemployment benefits that expand by formula, not by vote), and bank resolution law that wipes out shareholders and management — never depositors — first. Specification: systemically important banks hold capital of at least 12% of risk-weighted assets with a leverage cap; households keep 3–6 months of expenses in insured deposits, never in a single institution above the insured limit.
95	Single-crop dependence plus blight or drought produces famine at national scale.	Food & Agriculture	Past	Society / National	6	81.4%	82%	Never let one crop or one supplier exceed 25% of calories. Maintain seed diversity banks with at least 3 climate-matched varieties per staple, rotate crops on a minimum 3-year cycle, and hold strategic grain reserves of 90 days of national consumption. Specification: household level — 90 days of staple calories stored (rice/grain ~ 25 kg per adult per month at 2,000 kcal/day planning), rotated FIFO; regional level — no more than 40% of food imported by calorie.
96	Fire, earthquake, or war destroys a city's housing stock in days, leaving populations in the open.	Shelter & Housing	Past	Society / National	6	81.8%	81%	Build to survive the known local worst: seismic codes to the 2,475-year event for new builds, retrofit of the highest-risk 20% of existing stock first (Pareto: unreinforced masonry and soft-storey buildings cause most deaths), firebreaks and non-combustible roofing in fire zones. Specification: every city holds emergency shelter capacity for 10% of its population for 30 days (pre-sited, pre-supplied), and every household keeps a 72-hour go-kit — documents, medication, water, light.
97	Pandemic disease sweeps a population with no immunity, no germ theory, and no surge capacity.	Health & Medicine	Past	Society / National	6	82.1%	80%	Standing pandemic infrastructure, built before it is needed: genomic surveillance of wastewater in every city over 100,000 (weekly), a vaccine platform able to go from sequence to first doses in 100 days, and stockpiled PPE for 90 days of full-hospital use. Specification: hospital surge plans must double ICU capacity in 14 days (pre-mapped wards, cross-trained staff rosters); every household keeps a basic kit — thermometer, pulse oximeter, oral rehydration salts, fever medication for 2 weeks.
98	Regional aquifer and river systems collapse from overuse and contamination, making whole regions uninhabitable	Water	Past	Society / National	6	82.5%	78%	Cap total extraction at measured recharge: meter every well and canal offtake, publish the water balance monthly, and set the legal extraction ceiling at 80% of average recharge so the buffer absorbs dry years. Restore wetlands at river mouths (minimum 10% of historical area) as natural filters and flood buffers. Specification: household storage 4 litres per person per day drinking minimum, 14-day reserve = 56 L/person; community reservoirs sized for 90 days of essential use.
99	Schooling without learning: credentials inflate while actual literacy and numeracy stagnate or fall; misinform	Education & Knowledge	Now	Household / Personal	6	82.9%	78%	Measure learning, not attendance: independent annual assessments of reading and numeracy published per school, funding following measured learning gains, curricula rebuilt around mastery progression (no social promotion without support), and teacher pay and status raised to compete with other professions (top-quartile graduate recruitment). Specification: every child reads fluently by age 9 or triggers automatic intensive support; adult reskilling accounts (funded per citizen) redeemable at accredited providers, including libraries.

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100	Platform dependence: a handful of companies carry the world's speech, identity, and emergency communication; o	Communication	Now	Household / Personal	6	83.2%	77%	Interoperability and exit rights by law: dominant platforms must support data portability and interoperable messaging, public communication infrastructure (postal, broadcast, municipal networks) maintained as a real fallback, and due-process rules for account termination with human appeal. Specification, personal and organizational: own your contact list and records off-platform (exported monthly), maintain one channel you fully control (your own email list, radio net, or phone tree), and never let a platform be the only copy of anything.
101	Car-dependent sprawl plus fuel and parts chokepoints make daily life brittle; one closure or price spike stran	Transport & Mobility	Now	Household / Personal	6	83.6%	76%	Redundancy and proximity: freight modal split so no single route or mode carries more than 40% of essentials; the 15-minute standard for new development (daily needs reachable on foot or by transit); transit networks funded as essential infrastructure with operating reserves of 6 months; strategic parts and fuel reserves for public fleets. Specification: every settlement over 5,000 has at least two physically separate road connections or one road plus scheduled public transport that runs in emergencies.
102	An epidemic of anxiety, depression, loneliness, and attention damage, concentrated in the young and the isolat	Psychology & Mind	Now	Household / Personal	6	83.9%	75%	Rebuild the inputs: sleep protected as infrastructure (school start times aligned to adolescent biology, no-screens wind-down norms), daily physical activity of 60 minutes for youth built into the school day, third places (libraries, clubs, leagues) funded as health infrastructure, and platform design rules for minors (no infinite scroll, no engagement-optimized feeds). Specification, personal: fixed sleep window, morning light exposure, one recurring in-person group activity weekly minimum — belonging is scheduled, not hoped for.
103	Knowledge monopolized or destroyed: libraries burned, literacy forbidden to classes and genders, learning dies	Education & Knowledge	Past	Global / Civilizational	6	84.3%	74%	Distribute the copies: knowledge survives by redundancy — minimum 3 complete copies of core curricula and records on 3 different media in 3 different jurisdictions (the LOCKSS principle); literacy as a non-derogable right in law; open-licence textbooks for all core subjects so no publisher or regime can withdraw them. Specification: every school and household keeps printed core references (first aid, agriculture, mathematics, local law); digital archives mirrored internationally by default.
104	Precarious gig work, credential inflation, and regional job deserts leave large populations one missed paycheq	Work & Livelihood	Now	Household / Personal	6	84.7%	74%	Floor under every form of work: benefits (health, pension, injury, unemployment) pro-rated per hour worked regardless of classification, minimum standards applied per task not per contract type, regional diversification funds that trigger automatically when one employer exceeds 20% of local jobs, and wage insurance that tops up displaced workers at 50% of the wage gap for 2 years while retraining. Specification: personal standard — an emergency fund of 3 months, skills documented in portable form, and no single income source treated as permanent.
105	Two-tier justice: the wealthy buy delay and expertise; the poor plead guilty to get out of jail; backlogs make	Justice & Ethics	Now	Household / Personal	6	85.0%	73%	Equalize the machinery: pre-trial detention only on individualized danger findings with 48-hour review, legal aid funded to parity standards with prosecution in criminal matters, court capacity managed to published time standards (e.g., 90% of cases resolved within 12 months), and fines scaled to income (day-fines) so penalties bite equally. Specification: every court publishes wait times, outcomes by representation type, and backlog monthly; households know their rights in writing — a plain-language rights card is a standard household document.
106	Civilizations degrade their own ecological base — soil, forests, water — until the base fails and the society	Environment & Climate	Past	Society / National	6	85.4%	71%	Close the loop with law: extraction and pollution permits capped by measured regeneration rates of the specific ecosystem (not national averages), soil-loss limits per hectare enforced like building codes, and full-cost accounting where the price of a product includes its restoration cost. Specification: farmland soil organic matter and erosion measured on a 5-year cycle; land degrading past threshold enters mandatory restoration with cost-sharing.
107	Transport failure starves cities: pre-industrial capitals depended on daily food convoys that war, weather, or	Transport & Mobility	Past	Global / Civilizational	6	85.7%	67%	No city on a single thread: minimum two independent supply modes and routes for food and fuel (road + rail, or road + water), urban food reserves of 30 days, and local production floors for fresh essentials. Specification: critical bridges, tunnels, and ports have pre-planned bypass capacity of at least 60% of normal flow; households in isolated regions keep 30 days of staples as the standing norm.
108	Climate-shifted growing zones plus soil loss cut yields across several breadbaskets simultaneously.	Food & Agriculture	Future	Household / Personal	6	86.1%	66%	Regenerate the base: rebuild topsoil organic matter to 4%+ through cover cropping and reduced tillage; shift 20% of acreage to drought- and heat-bred varieties; build vertical/greenhouse capacity for greens near cities (target 50% of leafy supply within city limits); protect pollinators with pesticide-free corridors on 10% of farmland. Specification: yield planning assumes the worst decade climate, not the average year.
109	A general-purpose technology spreads faster than safety norms, and its harms compound for decades before corre	Technology & AI	Past	Society / National	6	86.5%	66%	Precaution with deadlines: new general-purpose technologies pass staged safety gates (lab, limited, general) with published criteria and automatic review dates; manufacturers carry strict liability for latent harms, funded by mandatory insurance during the entire exposure tail; and a standing technology-assessment office publishes plain-language risk registers. Specification: no product with irreversible population-level exposure ships without a funded 30-year monitoring program; households treat 'new and unstudied' as a reason to delay adoption for non-essentials.
110	Algorithmic feeds optimize for outrage; shared factual reality fragments; institutions cannot establish basic	Information & Media	Now	Household / Personal	6	86.8%	65%	Change the incentives, not just the content: recommender systems above a scale threshold must offer a chronological/non-optimized feed as a real default option, political advertising carries a public registry of sponsor, spend, and targeting criteria, and proven coordinated inauthentic campaigns trigger platform liability. Personal specification: a two-source rule for any alarming claim (two independent, named, primary sources before acting or forwarding), primary documents over commentary, and one scheduled news window per day instead of continuous feed exposure.
111	Livelihood systems collapse into serfdom, slavery, or company-town dependence: one master controls work, housi	Work & Livelihood	Past	Global / Civilizational	6	87.2%	65%	Separate the levers by law: wages in legal tender only, housing and retail independent of employers, freedom of movement and association non-waivable, and portable social insurance attached to the person. Specification: no employer may provide more than 2 of {job, housing, food retail, credit} to the same worker; labour inspection funded per worker covered, with unannounced inspection rights.
112	Below-replacement fertility plus rapid aging inverts the pyramid: fewer workers support more retirees; towns h	Demographics & Population	Now	Household / Personal	6	87.6%	65%	Make family formation affordable where people already want it: housing sized and priced for young families (family units at or below 30% of median income in every major city plan), childcare costing under 10% of median income per child, working-time protection for parents, and pension formulas that adjust transparently and gradually (automatic balancing mechanisms) instead of by crisis. Specification: immigration and integration run as a permanent, competently administered system sized to the measured demographic gap, with language and credential recognition completed within 12 months of arrival.
113	Autonomous weapons and cheap drones make precision violence available to anyone, overwhelming policing built f	Security & Defense	Future	Household / Personal	6	87.9%	64%	Regulate and shield in advance: mandatory remote-identification and geofencing on all drones sold, counter-drone authority and equipment at critical sites (airports, utilities, mass events) with legal rules of engagement written before incidents, export controls on autonomous targeting components, and an international ban framework on fully autonomous lethal systems. Specification: critical infrastructure must pass an annual red-team drone assessment; households and businesses rely on hardened, setback design for the highest-value targets rather than private countermeasures.
114	Multi-year megadrought plus population growth makes current cities structurally water-short.	Water	Future	Household / Personal	6	88.3%	63%	Rebuild the supply stack: wastewater recycling to potable standard (target 30–40% of supply), stormwater capture on all new developments (retain the 25-year storm on site), desalination where coastal, and pricing that guarantees a free lifeline block (first 50 L/person/day) with steep rates above it. Specification: per-capita demand cut to 100 L/day total through fixtures and leakage below 8%; inter-basin agreements signed before crisis, not during.

Pareto Rank	Solution (short title = its problem)	Field	Timeframe	Scale	Priority Score	Cumulative % of	Probability of Full Solution
115	Communication monopoly or collapse isolates populations: empires fell in regions where news took months; rumou	Communication	Past	Global / Civilizational	6	88.6%	63%	Multiple independent channels as doctrine: no population dependent on one network, one owner, or one technology for emergency information — broadcast radio, print, courier, and digital maintained in parallel, with amateur and community radio licensed and encouraged as civic infrastructure. Specification: every municipality maintains a battery/solar broadcast capability able to reach all residents within 1 hour without internet or grid power.
116	Climate migration plus uninsurable regions strand populations in housing that cannot be sold, insured, or safe	Shelter & Housing	Future	Household / Personal	6	89.0%	60%	Managed transition instead of collapse: pre-mapped receiving cities with pre-built infrastructure capacity, public buyouts at pre-decline values funded by risk-zone insurance levies collected while values remain, and building law that stops new construction in zones projected uninsurable within 30 years. Specification: national housing plans must model the 2050 climate map, not the 2000 one; household level — never hold more than 50% of net worth in a single climate-exposed property.
117	Constitutional order collapses into dictatorship through legal steps, emergency decrees, and captured courts.	Governance & Law	Past	Society / National	6	89.4%	60%	Design constitutions against their own abolition: eternity clauses protecting core rights from amendment, emergency powers with hard 90-day sunsets and escalating supermajorities for renewal, judicial appointments split across at least 3 independent bodies, and electoral administration insulated from the executive. Specification: every emergency measure auto-expires; every rights restriction must be re-justified publicly at renewal; citizens keep certified copies of identity and property records in personal custody.
118	Justice systems become weapons: show trials, collective punishment, and law as an instrument of terror.	Justice & Ethics	Past	Global / Civilizational	6	89.7%	60%	Evidence as the only currency: independent judiciary with security of tenure and salary, prosecution separated from the executive, every criminal conviction reviewable on the evidence record, forensic methods validated scientifically before courtroom use, and absolute bars on collective punishment and retroactive crimes. Specification: wrongful-conviction compensation automatic and generous; every jurisdiction publishes conviction-integrity audits, including exoneration counts, annually.
119	Electrification of everything concentrates all risk on one system; a solar storm or systemic cyber event takes	Energy	Future	Household / Personal	6	90.1%	58%	Stockpile the long-lead items: a strategic reserve of large transformers and grid components (the items with 12+ month lead times), EMP/GMD-hardened control systems for the top substations, and a designed analogue fallback — paper procedures, manual switching, and local generation — drilled, not just documented. Specification: every region can meet 30% of demand (the survival core: water, food cold-chain, hospitals, comms) on distributed local generation indefinitely.
120	An engineered or novel pathogen, or total care-system overload from aging plus chronic disease, breaks medicin	Health & Medicine	Future	Household / Personal	6	90.5%	56%	Layered defence: universal indoor air standards in hospitals, schools, and transit (ventilation equivalent to 6 air changes/hour, upper-air UV in high-risk rooms), synthetic-biology screening of all commercial DNA synthesis orders, regional manufacturing of essential medicines (top 100 generics produced on at least 2 continents each), and a reserve clinical workforce (retired and part-time clinicians registered and drilled annually). Specification: no single country supplies more than 40% of any essential medicine's global volume.
121	Legitimacy collapse: populations stop believing any institution, and coordination itself fails even without a	Governance & Law	Future	Household / Personal	6	90.8%	55%	Rebuild trust as infrastructure: citizen assemblies chosen by lottery with real decision power over defined questions, radical localism (decisions at the smallest competent scale, where people can verify outcomes personally), and service-delivery proof — publish wait times, error rates, and fix rates for every public service monthly. Specification: trust is measured (surveyed quarterly, published); agencies falling below threshold trigger automatic independent review.
122	Mass psychological trauma — war, displacement, famine — goes untreated and transmits across generations.	Psychology & Mind	Past	Global / Civilizational	6	91.2%	53%	Treat trauma as a public-health quantity, not a private shame: post-crisis psychological first aid deployed like physical first aid (trained community responders at a ratio of roughly 1 per 500 people), school-based programs after collective events, and long-horizon funding — trauma services budgeted for 25 years after major events, because that is the measured tail. Specification: households know the signs (hypervigilance, numbing, rage, withdrawal) and the rule — connection before correction, routine before analysis, professional help early, not last.
123	Automation displaces labour faster than new livelihoods form, while wealth concentrates in owners of machines	Economy & Finance	Future	Household / Personal	6	91.5%	52%	Share the machine dividend by structure, not charity: sovereign and pension funds holding broad equity stakes so productivity gains flow to all citizens as dividends; reduced standard working hours as productivity rises (the historical pattern, deliberately continued); portable benefits attached to the person, not the job; and public investment in the irreducibly human sectors (care, craft, education, restoration) at the scale of the displacement. Specification: transition planning uses the fastest plausible displacement curve, with retraining accounts funded per worker before displacement, not after.
124	Population collapse from plague, war, and famine unmakes societies: fields untilled, cities abandoned, knowled	Demographics & Population	Past	Global / Civilizational	6	91.9%	52%	Resilience through distribution: population and skills spread across regions rather than concentrated in single cities, craft and knowledge apprenticeship duplicated across at least 2 lineages per critical skill, and post-catastrophe relief law that moves food and people automatically on declared triggers. Specification: no critical occupation (medicine, engineering, agriculture) may have more than 40% of its practitioners in one metropolitan area.
125	A misaligned or misused advanced AI, or total dependence on opaque systems, produces civilizational-scale fail	Technology & AI	Future	Household / Personal	6	92.3%	50%	Governance proportional to capability: frontier training runs above defined compute thresholds are registered and evaluated for dangerous capabilities before release; critical infrastructure keeps human-verified analogue fallbacks and禁止 fully autonomous control of weapons, grid switching, and medical dosing; liability law makes deployers answer for system outputs; and international verification for the largest training facilities, modelled on nuclear safeguards. Specification: every essential service must demonstrate a manual mode, drilled annually; personal rule — no single AI or platform holds your only copy of anything important.
126	Tipping points cascade: ice loss, permafrost methane, ocean-circulation shifts lock in metres of sea-level ris	Environment & Climate	Future	Household / Personal	6	92.6%	46%	Both brakes and lifeboats: emissions cuts on the fastest feasible curve (the only lever on the cause), plus planned adaptation at civilizational scale — protected agricultural relocation zones, freshwater megaprojects only where hydrology closes, migration frameworks negotiated before flows become crises (status, quotas, receiving-city investment), and geoengineering research governed internationally with deployment requiring multinational consent. Specification: all long-lived infrastructure (50+ year life) sited and designed against the high-end sea-level and heat scenarios, not the median.
127	Propaganda systems capture whole populations; lies with state amplification prepare and excuse mass violence.	Information & Media	Past	Global / Civilizational	6	93.0%	46%	Pluralism as armour: no single owner or state organ may control a dominant share of any population's information diet (strict cross-media ownership caps), public-interest media funded by formula and governed at arm's length, and school curricula that teach source-tracing — every claim followed to its origin — as a core skill from age 10. Specification: during crises, official information carries verifiable provenance (signed releases, raw data published with summaries) so citizens can check, not just trust.
128	Invasion or civil war destroys the state's monopoly on force; civilians face violence with no protection.	Security & Defense	Past	Society / National	6	93.4%	45%	Prevention is the only real solution once killing starts: early-warning indicators (hate-media licensing, militia formation, arms flows) monitored with published thresholds that trigger automatic diplomatic and legal consequences; constitutional hard limits on emergency powers (sunset clauses of 90 days, legislative renewal); and protected humanitarian corridors negotiated in advance frameworks. Specification: civilian layer — neighbourhood mutual-aid rosters, documented evidence habits (photos, records kept off-site), and pre-planned exit routes with 72-hour kits.

Pareto Rank	Solution (short title = its problem)	Field	Timeframe	Scale	Priority Score	Cumulative % of	Probability of Success	Full Solution
129	Knowledge monopolized or destroyed: libraries burned, literacy forbidden to classes and genders, learning dies	Education & Knowledge	Past	Society / National	4	93.6%	86%	Distribute the copies: knowledge survives by redundancy — minimum 3 complete copies of core curricula and records on 3 different media in 3 different jurisdictions (the LOCKSS principle); literacy as a non-derogable right in law; open-licence textbooks for all core subjects so no publisher or regime can withdraw them. Specification: every school and household keeps printed core references (first aid, agriculture, mathematics, local law); digital archives mirrored internationally by default.
130	Transport failure starves cities: pre-industrial capitals depended on daily food convoys that war, weather, or	Transport & Mobility	Past	Society / National	4	93.8%	79%	No city on a single thread: minimum two independent supply modes and routes for food and fuel (road + rail, or road + water), urban food reserves of 30 days, and local production floors for fresh essentials. Specification: critical bridges, tunnels, and ports have pre-planned bypass capacity of at least 60% of normal flow; households in isolated regions keep 30 days of staples as the standing norm.
131	Livelihood systems collapse into serfdom, slavery, or company-town dependence: one master controls work, housi	Work & Livelihood	Past	Society / National	4	94.1%	77%	Separate the levers by law: wages in legal tender only, housing and retail independent of employers, freedom of movement and association non-waivable, and portable social insurance attached to the person. Specification: no employer may provide more than 2 of {job, housing, food retail, credit} to the same worker; labour inspection funded per worker covered, with unannounced inspection rights.
132	Communication monopoly or collapse isolates populations: empires fell in regions where news took months; rumou	Communication	Past	Society / National	4	94.3%	75%	Multiple independent channels as doctrine: no population dependent on one network, one owner, or one technology for emergency information — broadcast radio, print, courier, and digital maintained in parallel, with amateur and community radio licensed and encouraged as civic infrastructure. Specification: every municipality maintains a battery/solar broadcast capability able to reach all residents within 1 hour without internet or grid power.
133	Justice systems become weapons: show trials, collective punishment, and law as an instrument of terror.	Justice & Ethics	Past	Society / National	4	94.6%	72%	Evidence as the only currency: independent judiciary with security of tenure and salary, prosecution separated from the executive, every criminal conviction reviewable on the evidence record, forensic methods validated scientifically before courtroom use, and absolute bars on collective punishment and retroactive crimes. Specification: wrongful-conviction compensation automatic and generous; every jurisdiction publishes conviction-integrity audits, including exoneration counts, annually.
134	Synthetic media makes any audio, video, or document forgeable at zero cost; 'seeing' stops being believing, an	Information & Media	Future	Household / Personal	4	94.8%	69%	Provenance infrastructure: cameras and recorders sign content at capture (cryptographic content credentials as the default in professional and official contexts), courts and newsrooms verify by signature chain rather than appearance, and legal process treats unsigned dramatic media as allegation, not evidence. Specification: households and organizations keep original, signed, off-platform copies of identity, property, and credential records — paper originals of the documents that prove who you are and what you own.
135	AI tutors and synthetic content make human knowledge optional; populations lose the ability to verify, calcula	Education & Knowledge	Future	Household / Personal	4	95.0%	67%	Dual-competence standard: every qualification requires demonstrated unaided performance (closed-book, offline, by hand) plus aided performance; curricula teach verification as the core skill — source-tracing, estimation, consistency checking; national knowledge reserves (printed, distributed) maintained as civil infrastructure. Specification: critical professions (medicine, engineering, aviation, law) license on unaided competence first; schools schedule regular no-device weeks, not as punishment but as training.
136	Algorithmic justice and predictive policing automate bias at scale; people are punished for probabilities, and	Justice & Ethics	Future	Household / Personal	4	95.3%	66%	Human answerability as law: no liberty decision (arrest priority, bail, sentencing, parole) may rest solely on an algorithmic score; models used in justice are public, auditable, and tested for error and disparity rates published per demographic group; every automated recommendation names the accountable human who adopted it. Specification: any person scored by a state system can see their inputs, correct errors within 30 days, and receive a human review on request — the same standard as credit records, applied to freedom.
137	Mass psychological trauma — war, displacement, famine — goes untreated and transmits across generations.	Psychology & Mind	Past	Society / National	4	95.5%	65%	Treat trauma as a public-health quantity, not a private shame: post-crisis psychological first aid deployed like physical first aid (trained community responders at a ratio of roughly 1 per 500 people), school-based programs after collective events, and long-horizon funding — trauma services budgeted for 25 years after major events, because that is the measured tail. Specification: households know the signs (hypervigilance, numbing, rage, withdrawal) and the rule — connection before correction, routine before analysis, professional help early, not last.
138	Population collapse from plague, war, and famine unmakes societies: fields untilled, cities abandoned, knowled	Demographics & Population	Past	Society / National	4	95.8%	64%	Resilience through distribution: population and skills spread across regions rather than concentrated in single cities, craft and knowledge apprenticeship duplicated across at least 2 lineages per critical skill, and post-catastrophe relief law that moves food and people automatically on declared triggers. Specification: no critical occupation (medicine, engineering, agriculture) may have more than 40% of its practitioners in one metropolitan area.
139	Autonomous, electric, platform-controlled mobility concentrates failure: one software fault, cyberattack, or b	Transport & Mobility	Future	Household / Personal	4	96.0%	62%	Mandated heterogeneity and manual fallback: fleets certified from at least 2 independent software stacks per region, all autonomous vehicles retaining manual drive capability, transport control systems segmented from public networks with analogue scheduling fallback, and battery-material supply diversified with recycling recovering 90%+ of critical minerals by law. Specification: cities drill a 'no-network day' annually — transit, logistics, and emergency services operating on fallback procedures.
140	Total surveillance or total encryption-war makes private communication impossible — or, inversely, untraceable	Communication	Future	Household / Personal	4	96.3%	60%	Rights-preserving architecture: end-to-end encryption protected in law as the default for private communication, state access only through individualized, judicially authorized, logged, and time-limited orders with after-the-fact notification; anonymous public speech protected while large-scale amplification carries provenance duties. Specification: every surveillance power carries a 5-year sunset and a published annual accounting of use, errors, and outcomes — powers that cannot justify themselves expire.
141	Propaganda systems capture whole populations; lies with state amplification prepare and excuse mass violence.	Information & Media	Past	Society / National	4	96.5%	58%	Pluralism as armour: no single owner or state organ may control a dominant share of any population's information diet (strict cross-media ownership caps), public-interest media funded by formula and governed at arm's length, and school curricula that teach source-tracing — every claim followed to its origin — as a core skill from age 10. Specification: during crises, official information carries verifiable provenance (signed releases, raw data published with summaries) so citizens can check, not just trust.
142	Divergent futures collide: some regions depopulate to non-viability while others add billions; unmanaged flows	Demographics & Population	Future	Household / Personal	4	96.7%	57%	Plan both directions simultaneously: shrink-smart policy for depopulating regions (consolidated services, maintained dignity and access, land returned to nature or agriculture under stewardship), growth-ready frameworks for expanding regions (land tenure, urban plans, and service grids built ahead of arrival), and pre-negotiated international mobility compacts so movement is governed, not chaotic. Specification: every national plan carries a 50-year demographic annex, updated every 5 years, with infrastructure sized to the projection band — not to today's population.
143	Synthetic companionship and immersive virtual worlds outcompete reality; populations withdraw from human conta	Psychology & Mind	Future	Household / Personal	4	97.0%	54%	Protect the real by making it richer, not by banning the synthetic: disclosure law (companions must identify as artificial, always), design rules against dependency engineering (no manufactured jealousy, streaks, or guilt mechanics), public investment in physical gathering — festivals, sport, craft, public space — at a scale that competes, and social norms taught early: the difference between being comforted and being accompanied. Specification: households set explicit budgets — money and hours — for synthetic companionship, reviewed like any other consumption; any product whose retention depends on your loneliness is treated as extractive by definition.

Pareto Rank	Solution (short title = its problem)	Field	Timeframe	Scale	Priority Score	Cumulative % of	Probability of	Full Solution
144	Machines take the majority of paid tasks; the link between contribution and income breaks entirely.	Work & Livelihood	Future	Household / Personal	4	97.2%	51%	Rebuild distribution on ownership: broad-based capital ownership (citizen funds, employee ownership trusts, data and AI dividends), radically shortened standard working time as productivity rises, and public valuation of unpaid contribution (care, community, restoration) through contribution credits convertible to services. Specification: transition is phased by measured displacement rates per sector, with the ownership mechanisms built during the growth phase — financed from automation rents — not improvised after collapse.
145	Fuel embargo or supply cutoff paralyzes transport, heating, and industry in dependent nations.	Energy	Past	Household / Personal	3	97.4%	92%	Strategic reserves and demand discipline: 90 days of net-import reserves for critical fuels (the standing international standard), efficiency law that cuts vehicle and building consumption 2% per year, and pre-planned rationing tiers so essential services (food logistics, hospitals, water) are served first. Specification: household level — 72 hours of light, heat, and cooking independent of the grid (LED + battery bank of 1 kWh, safe fuel stove); national level — no single supplier above 30% of any critical fuel.
146	Banking panic and deflation spiral convert a market crash into a decade of mass unemployment.	Economy & Finance	Past	Household / Personal	3	97.6%	91%	Circuit breakers built in advance: deposit insurance with pre-funded reserves, central-bank lender-of-last-resort facilities designed before crises, automatic fiscal stabilizers (unemployment benefits that expand by formula, not by vote), and bank resolution law that wipes out shareholders and management — never depositors — first. Specification: systemically important banks hold capital of at least 12% of risk-weighted assets with a leverage cap; households keep 3–6 months of expenses in insured deposits, never in a single institution above the insured limit.
147	Single-crop dependence plus blight or drought produces famine at national scale.	Food & Agriculture	Past	Household / Personal	3	97.8%	90%	Never let one crop or one supplier exceed 25% of calories. Maintain seed diversity banks with at least 3 climate-matched varieties per staple, rotate crops on a minimum 3-year cycle, and hold strategic grain reserves of 90 days of national consumption. Specification: household level — 90 days of staple calories stored (rice/grain ~ 25 kg per adult per month at 2,000 kcal/day planning), rotated FIFO; regional level — no more than 40% of food imported by calorie.
148	Fire, earthquake, or war destroys a city's housing stock in days, leaving populations in the open.	Shelter & Housing	Past	Household / Personal	3	97.9%	89%	Build to survive the known local worst: seismic codes to the 2,475-year event for new builds, retrofit of the highest-risk 20% of existing stock first (Pareto: unreinforced masonry and soft-storey buildings cause most deaths), firebreaks and non-combustible roofing in fire zones. Specification: every city holds emergency shelter capacity for 10% of its population for 30 days (pre-sited, pre-supplied), and every household keeps a 72-hour go-kit — documents, medication, water, light.
149	Pandemic disease sweeps a population with no immunity, no germ theory, and no surge capacity.	Health & Medicine	Past	Household / Personal	3	98.1%	88%	Standing pandemic infrastructure, built before it is needed: genomic surveillance of wastewater in every city over 100,000 (weekly), a vaccine platform able to go from sequence to first doses in 100 days, and stockpiled PPE for 90 days of full-hospital use. Specification: hospital surge plans must double ICU capacity in 14 days (pre-mapped wards, cross-trained staff rosters); every household keeps a basic kit — thermometer, pulse oximeter, oral rehydration salts, fever medication for 2 weeks.
150	Regional aquifer and river systems collapse from overuse and contamination, making whole regions uninhabitable	Water	Past	Household / Personal	3	98.3%	86%	Cap total extraction at measured recharge: meter every well and canal offtake, publish the water balance monthly, and set the legal extraction ceiling at 80% of average recharge so the buffer absorbs dry years. Restore wetlands at river mouths (minimum 10% of historical area) as natural filters and flood buffers. Specification: household storage 4 litres per person per day drinking minimum, 14-day reserve = 56 L/person; community reservoirs sized for 90 days of essential use.
151	Civilizations degrade their own ecological base — soil, forests, water — until the base fails and the society	Environment & Climate	Past	Household / Personal	3	98.5%	79%	Close the loop with law: extraction and pollution permits capped by measured regeneration rates of the specific ecosystem (not national averages), soil-loss limits per hectare enforced like building codes, and full-cost accounting where the price of a product includes its restoration cost. Specification: farmland soil organic matter and erosion measured on a 5-year cycle; land degrading past threshold enters mandatory restoration with cost-sharing.
152	A general-purpose technology spreads faster than safety norms, and its harms compound for decades before corre	Technology & AI	Past	Household / Personal	3	98.7%	74%	Precaution with deadlines: new general-purpose technologies pass staged safety gates (lab, limited, general) with published criteria and automatic review dates; manufacturers carry strict liability for latent harms, funded by mandatory insurance during the entire exposure tail; and a standing technology-assessment office publishes plain-language risk registers. Specification: no product with irreversible population-level exposure ships without a funded 30-year monitoring program; households treat 'new and unstudied' as a reason to delay adoption for non-essentials.
153	Constitutional order collapses into dictatorship through legal steps, emergency decrees, and captured courts.	Governance & Law	Past	Household / Personal	3	98.9%	68%	Design constitutions against their own abolition: eternity clauses protecting core rights from amendment, emergency powers with hard 90-day sunsets and escalating supermajorities for renewal, judicial appointments split across at least 3 independent bodies, and electoral administration insulated from the executive. Specification: every emergency measure auto-expires; every rights restriction must be re-justified publicly at renewal; citizens keep certified copies of identity and property records in personal custody.
154	Invasion or civil war destroys the state's monopoly on force; civilians face violence with no protection.	Security & Defense	Past	Household / Personal	3	99.0%	53%	Prevention is the only real solution once killing starts: early-warning indicators (hate-media licensing, militia formation, arms flows) monitored with published thresholds that trigger automatic diplomatic and legal consequences; constitutional hard limits on emergency powers (sunset clauses of 90 days, legislative renewal); and protected humanitarian corridors negotiated in advance frameworks. Specification: civilian layer — neighbourhood mutual-aid rosters, documented evidence habits (photos, records kept off-site), and pre-planned exit routes with 72-hour kits.
155	Knowledge monopolized or destroyed: libraries burned, literacy forbidden to classes and genders, learning dies	Education & Knowledge	Past	Household / Personal	2	99.2%	94%	Distribute the copies: knowledge survives by redundancy — minimum 3 complete copies of core curricula and records on 3 different media in 3 different jurisdictions (the LOCKSS principle); literacy as a non-derogable right in law; open-licence textbooks for all core subjects so no publisher or regime can withdraw them. Specification: every school and household keeps printed core references (first aid, agriculture, mathematics, local law); digital archives mirrored internationally by default.
156	Transport failure starves cities: pre-industrial capitals depended on daily food convoys that war, weather, or	Transport & Mobility	Past	Household / Personal	2	99.3%	87%	No city on a single thread: minimum two independent supply modes and routes for food and fuel (road + rail, or road + water), urban food reserves of 30 days, and local production floors for fresh essentials. Specification: critical bridges, tunnels, and ports have pre-planned bypass capacity of at least 60% of normal flow; households in isolated regions keep 30 days of staples as the standing norm.
157	Livelihood systems collapse into serfdom, slavery, or company-town dependence: one master controls work, housi	Work & Livelihood	Past	Household / Personal	2	99.4%	85%	Separate the levers by law: wages in legal tender only, housing and retail independent of employers, freedom of movement and association non-waivable, and portable social insurance attached to the person. Specification: no employer may provide more than 2 of {job, housing, food retail, credit} to the same worker; labour inspection funded per worker covered, with unannounced inspection rights.
158	Communication monopoly or collapse isolates populations: empires fell in regions where news took months; rumou	Communication	Past	Household / Personal	2	99.5%	83%	Multiple independent channels as doctrine: no population dependent on one network, one owner, or one technology for emergency information — broadcast radio, print, courier, and digital maintained in parallel, with amateur and community radio licensed and encouraged as civic infrastructure. Specification: every municipality maintains a battery/solar broadcast capability able to reach all residents within 1 hour without internet or grid power.

Pareto Rank	Solution (short title = its problem)	Field	Timeframe	Scale	Priority Score	Cumulative % of	Probability of Full Solution	
159	Justice systems become weapons: show trials, collective punishment, and law as an instrument of terror.	Justice & Ethics	Past	Household / Personal	2	99.6%	80%	Evidence as the only currency: independent judiciary with security of tenure and salary, prosecution separated from the executive, every criminal conviction reviewable on the evidence record, forensic methods validated scientifically before courtroom use, and absolute bars on collective punishment and retroactive crimes. Specification: wrongful-conviction compensation automatic and generous; every jurisdiction publishes conviction-integrity audits, including exoneration counts, annually.
160	Mass psychological trauma — war, displacement, famine — goes untreated and transmits across generations.	Psychology & Mind	Past	Household / Personal	2	99.8%	73%	Treat trauma as a public-health quantity, not a private shame: post-crisis psychological first aid deployed like physical first aid (trained community responders at a ratio of roughly 1 per 500 people), school-based programs after collective events, and long-horizon funding — trauma services budgeted for 25 years after major events, because that is the measured tail. Specification: households know the signs (hypervigilance, numbing, rage, withdrawal) and the rule — connection before correction, routine before analysis, professional help early, not last.
161	Population collapse from plague, war, and famine unmakes societies: fields untilled, cities abandoned, knowled	Demographics & Population	Past	Household / Personal	2	99.9%	72%	Resilience through distribution: population and skills spread across regions rather than concentrated in single cities, craft and knowledge apprenticeship duplicated across at least 2 lineages per critical skill, and post-catastrophe relief law that moves food and people automatically on declared triggers. Specification: no critical occupation (medicine, engineering, agriculture) may have more than 40% of its practitioners in one metropolitan area.
162	Propaganda systems capture whole populations; lies with state amplification prepare and excuse mass violence.	Information & Media	Past	Household / Personal	2	100.0%	66%	Pluralism as armour: no single owner or state organ may control a dominant share of any population's information diet (strict cross-media ownership caps), public-interest media funded by formula and governed at arm's length, and school curricula that teach source-tracing — every claim followed to its origin — as a core skill from age 10. Specification: during crises, official information carries verifiable provenance (signed releases, raw data published with summaries) so citizens can check, not just trust.

Field of Discipline	Why Level	Root Cause of Possible Failure	Solution to This Cause
Water	Why 1 — supplies fail	Infrastructure ages and maintenance is deferred because water is priced below its cost.	Price water at full lifecycle cost with a protected free lifeline block, so revenue funds maintenance without denying anyone the minimum.
Water	Why 2 — deferral is invisible	Underground assets fail out of sight; politicians gain nothing from pipes nobody sees.	Publish asset-condition grades and failure statistics annually, like financial audits; tie executive bonuses in utilities to condition grades.
Water	Why 3 — no one owns the number	Responsibility is split across municipalities, regions, and private operators.	One accountable water authority per basin with the full balance — extraction, quality, losses — published monthly under one signature.
Water	Why 4 — basins cross borders	Upstream and downstream jurisdictions optimize against each other.	Basin treaties with pre-agreed drought-sharing formulas signed in wet years, when generosity is cheap.
Water	Why 5 — root	Water is treated as a free gift of nature rather than a managed stock with a measurable balance.	Adopt the stock-and-flow account as law: every basin keeps a public ledger, and extraction rights are issued only against it.
Food & Agriculture	Why 1 — shelves empty fast	Just-in-time logistics hold almost no buffer by design.	Mandate buffer stocks at each level (household 30 days, retail 30 days, national 90 days) so the chain has slack by law, not by hope.
Food & Agriculture	Why 2 — buffers cost money	Private actors are punished by markets for holding 'inefficient' stock.	Pay for buffers as a public service: reserve-holding contracts with guaranteed purchase, like capacity payments in electricity.
Food & Agriculture	Why 3 — efficiency outbids resilience	Quarterly metrics reward cost per unit, never survival per decade.	Add resilience metrics to procurement law: supplier diversity and buffer days scored alongside price in every public food contract.
Food & Agriculture	Why 4 — soil is mined, not farmed	Yield this year is priced; soil next decade is not.	Soil organic matter measured per farm on a 5-year cycle, with payments for building it and limits on losing it.
Food & Agriculture	Why 5 — root	Food is treated as a commodity first and a survival system second.	Constitutional or statutory food-security duty: the state must maintain the measured capacity to feed the population through a 1-year supply shock.
Health & Medicine	Why 1 — systems run at 95%+ occupancy	Efficiency targets eliminate the slack that crises consume.	Fund surge slack explicitly: an 85% occupancy ceiling as the regulated normal, with the 15% paid for as readiness.
Health & Medicine	Why 2 — readiness looks like waste	Idle beds and stockpiles photograph badly in budget debates.	Account for readiness as insurance with published expected-loss figures, reviewed by auditors, not by headlines.
Health & Medicine	Why 3 — prevention is underpaid	Payment follows procedures performed, not illness avoided.	Shift a fixed share (target 15–20%) of health budgets to prevention with outcome contracts: paid on measured population results.
Health & Medicine	Why 4 — workforce pipelines are long	Clinicians take a decade to train; shortages cannot be improvised.	Train against the 10-year demographic projection, maintain a drilled reserve register, and recognize foreign credentials within 12 months.
Health & Medicine	Why 5 — root	Health is budgeted as a consumption expense instead of survival infrastructure.	Classify core health capacity with infrastructure law: multi-year protected funding, like defence, insulated from annual cuts.
Energy	Why 1 — grids fail in cascade	Systems are optimized for average load and cost, with single points of failure.	Engineer for islanding and N-2 redundancy on critical paths; publish cascade maps and harden the worst 20% of nodes first.
Energy	Why 2 — redundancy is cut in procurement	Lowest-bid rules select the design with the least slack.	Score bids on failure behaviour (tested black-start, islanding) with the same weight as price.
Energy	Why 3 — long-lead parts are not stocked	Transformers and breakers are treated as order-when-needed items.	A strategic spares pool, jointly owned by utilities, with the 12+ month-lead items physically in warehouses.
Energy	Why 4 — demand only grows	Every new load (vehicles, data centres, heating) is connected first and planned later.	Connection approvals require a supply plan: new large loads bring generation or storage with them by rule.
Energy	Why 5 — root	Energy is sold as a commodity while being used as a life-support system.	Regulate a defined survival core (water, food cold chain, hospitals, communications) with life-support reliability standards above all commercial tiers.
Shelter & Housing	Why 1 — not enough homes near work	Zoning forbids the density the population needs where it needs it.	Legalize mid-density near transit by right; approvals deemed granted after 12 months of silence.
Shelter & Housing	Why 2 — incumbents block supply	Homeowners gain from scarcity and hold local veto power.	Move housing targets to regional level with automatic triggers (rent-to-income ratio) that override local refusal.
Shelter & Housing	Why 3 — risk is mispriced	Insurance and mortgages ignore the 2050 risk map, so building continues in doomed zones.	Publish address-level risk and price it in; stop new construction in zones uninsurable within 30 years.
Shelter & Housing	Why 4 — construction capacity is thin	Trades pipelines and factory-built capacity were allowed to shrink.	Fund trades training against the housing target and permit factory-built housing at national standard.
Shelter & Housing	Why 5 — root	Housing is treated primarily as an investment asset and only secondarily as shelter.	Dual mandate in law: every housing policy scored on both asset stability and the shelter benchmark (median rent at or below 30% of median income).
Security & Defense	Why 1 — violence concentrates	A tiny network and a few places generate most harm, while responses spread thin everywhere.	Fund focused deterrence and place-based fixes at the measured concentration points; audit monthly.
Security & Defense	Why 2 — legitimacy collapses first	Abusive or absent policing destroys the cooperation that produces information.	Procedural-justice standards with published complaint and outcome data; violence interrupters as credible messengers.
Security & Defense	Why 3 — weapons get cheaper	Each decade lowers the cost of precision harm.	Regulate at the component and platform layer (drone ID, geofencing, synthesis screening) before incidents, not after.
Security & Defense	Why 4 — early warnings are ignored	Acting early looks like overreaction until it looks like negligence.	Pre-commit triggers in law: named indicators compel named responses, removing the discretion to wait.
Security & Defense	Why 5 — root	Security is funded as reaction — after events — rather than as prevention with measured returns.	Budget prevention like insurance with published expected-harm reductions per program, renewed only on measured results.
Governance & Law	Why 1 — institutions are captured	Appointment, money, and media levers concentrate in one network.	Split every lever across independent bodies; publish ownership, contracts, and assets by default.
Governance & Law	Why 2 — capture is gradual and legal	Each step is small enough to seem normal.	Democracy audits on a fixed cycle with published capture indicators (court independence, media concentration, procurement openness).
Governance & Law	Why 3 — citizens cannot verify	Records are secret or unusable, so accountability is theatrical.	Open data as the default state of every public record; secrecy justified line by line, expiring automatically.
Governance & Law	Why 4 — emergencies become permanent	Temporary powers are never handed back voluntarily.	Hard sunsets (90 days) with escalating supermajorities; every renewal re-justified in public.

Field of Discipline	Why Level	Root Cause of Possible Failure	Solution to This Cause
Governance & Law	Why 5 — root	Constitutions are written for good-faith actors and stress-tested by bad-faith ones only in real time.	Design against the worst plausible officeholder: eternity clauses, distributed appointment power, and rehearsed civic defence.
Economy & Finance	Why 1 — crises cascade through debt	Leverage links every balance sheet to every other.	Capital and leverage caps that bind in booms, when they are politically hardest and economically cheapest.
Economy & Finance	Why 2 — booms silence regulators	Warnings during bubbles are punished as pessimism.	Automatic stabilizers and counter-cyclical rules set by formula in law, not by discretion in the moment.
Economy & Finance	Why 3 — gains privatize, losses socialize	Institutions too big to fail keep the upside.	Resolution law that wipes shareholders and management first, rehearsed with living wills tested annually.
Economy & Finance	Why 4 — households carry the fragility	Wages stagnate while housing and credit costs compound.	Debt-service caps at origination under stressed rates; fresh-start insolvency within 3 years; public supply of non-speculative housing.
Economy & Finance	Why 5 — root	Finance is treated as an end in itself rather than as plumbing for the real economy.	Charter test: financial activity that does not demonstrably serve production, housing, or risk-sharing carries higher capital and tax weight.
Technology & AI	Why 1 — harms arrive before rules	Development cycles are months; legislation is years.	Staged gates with automatic review dates, so oversight moves at the technology's speed by default.
Technology & AI	Why 2 — builders don't pay for harm	Latent and externalized costs fall on the public.	Strict liability plus mandatory harm insurance across the exposure tail, priced by insurers who will demand safety data.
Technology & AI	Why 3 — basics are skipped	Backups, segmentation, and MFA are known but unglamorous.	Make the basics conditions of insurance and procurement: no contract, no coverage without restore-tested backups and MFA.
Technology & AI	Why 4 — dependence concentrates	One platform, one model, one vendor becomes a single point of civilizational failure.	Interoperability, portability, and manual fallbacks as certification requirements for essential services.
Technology & AI	Why 5 — root	Speed is rewarded and safety is unpriced, so the race always outruns the guardrails.	Price safety into the race: liability, insurance, and gate criteria make the safe path the fast path.
Environment & Climate	Why 1 — emissions continue	The cost of carbon falls on everyone except the emitter.	Price emissions at the measured damage cost with the revenue returned visibly (dividends or bill credits) so the policy survives politics.
Environment & Climate	Why 2 — adaptation lags	Infrastructure is built to historical weather by code and habit.	Design codes keyed to the 2050 worst decile, updated on a fixed 5-year cycle from published projections.
Environment & Climate	Why 3 — risk is hidden	Buyers and builders cannot see address-level future risk.	Mandatory published risk maps (flood, heat, fire, sea level) attached to every property transaction and building permit.
Environment & Climate	Why 4 — feedback is slow	Damage accumulates silently for decades before it is undeniable.	Instrument the thresholds: ice, ocean circulation, and permafrost monitored with published tripwires tied to pre-agreed actions.
Environment & Climate	Why 5 — root	Nature is accounted as an externality — a free input and a free dump.	Natural capital on the national balance sheet: soil, water, canopy, and climate stability carried as assets whose depreciation must be reported and offset.
Information & Media	Why 1 — outrage outcompetes fact	Engagement optimization pays for emotion, not accuracy.	Require a non-optimized feed as a real default for large platforms; political ad registries with sponsor, spend, and targeting public.
Information & Media	Why 2 — corrections don't travel	A lie reaches millions before its correction reaches thousands.	Provenance at capture (signed content credentials) so authenticity is checkable in seconds, not debunkable in days.
Information & Media	Why 3 — ownership concentrates	A few owners set a population's information diet.	Cross-media ownership caps enforced before capture; arm's-length public media funded by formula.
Information & Media	Why 4 — citizens lack verification habits	Schools teach content, not source-tracing.	Source-tracing as a core skill from age 10, tested like literacy: every claim followed to its origin.
Information & Media	Why 5 — root	Attention is treated as a free resource to be mined rather than a commons to be governed.	Attention-commons rules: the information environment regulated like the physical one — pollution (coordinated deception) carries liability.
Psychology & Mind	Why 1 — distress is treated late	Help arrives at crisis, years after onset.	Psychological first aid and early screening embedded in schools, workplaces, and primary care — first contact within days, not months.
Psychology & Mind	Why 2 — environments manufacture distress	Sleep loss, isolation, and engineered attention damage are built into daily design.	Regulate the inputs: school start times, minor-platform design rules, and third places funded as health infrastructure.
Psychology & Mind	Why 3 — belonging collapsed	Clubs, congregations, leagues, and unions thinned for fifty years.	Fund recurring in-person association directly (venues, leagues, choirs, workshops) with the same seriousness as clinics.
Psychology & Mind	Why 4 — synthetic comfort outcompetes people	Machines offer frictionless company; humans are comparatively costly.	Anti-dependency design law for companions, plus public investment that makes real gathering richer and easier.
Psychology & Mind	Why 5 — root	Mental health is framed as individual chemistry rather than as the output of a designed environment.	Assess every major policy for its psychological externalities (sleep, belonging, attention, agency) in a published impact statement.
Education & Knowledge	Why 1 — systems measure attendance, not learning	Enrolment and graduation are countable; learning is not, so it is not managed.	Independent annual learning assessments published per school; funding follows measured gains.
Education & Knowledge	Why 2 — teachers are undervalued	Pay and status lose the competition for strong graduates.	Top-quartile recruitment with matching pay and professional autonomy; teaching as a selective, honoured profession.
Education & Knowledge	Why 3 — knowledge is centralized	Single publishers, platforms, and archives are single points of failure.	Redundancy law for knowledge: 3 copies, 3 media, 3 jurisdictions; open-licence core curricula.
Education & Knowledge	Why 4 — verification is untaught	Curricula deliver content while the environment demands checking.	Verification (source-tracing, estimation, consistency) as a tested core skill at every level.
Education & Knowledge	Why 5 — root	Education is run as a sorting machine for employers rather than as the formation of capable citizens.	Dual charter in law: every education system reports both employability and unaided competence/civic capability, and is funded on both.
Work & Livelihood	Why 1 — protections attach to job titles	Benefits were designed for one employer, one career, one contract type.	Attach all social insurance to the person, pro-rated per hour worked in any form.
Work & Livelihood	Why 2 — employers arbitrage the categories	Misclassification is cheaper than compliance where penalties are rare.	Per-task standards with automatic penalties and joint liability up the contracting chain.

Field of Discipline	Why Level	Root Cause of Possible Failure	Solution to This Cause
Work & Livelihood	Why 3 — regions depend on single employers	Local economies were allowed to become monocultures.	Automatic diversification triggers at the 20% threshold, with funds released by formula.
Work & Livelihood	Why 4 — transitions are unfunded	Retraining is offered after displacement, from general revenue, too late.	Transition accounts funded per worker during employment, from automation and payroll levies, before displacement.
Work & Livelihood	Why 5 — root	Income is tied exclusively to selling labour in a world where labour's share is structurally falling.	Diversify the income base by design: ownership shares, citizen funds, and contribution credits alongside wages.
Transport & Mobility	Why 1 — single threads everywhere	Cost optimization removed parallel routes and modes.	Modal and route caps (40%) for essential flows, written into infrastructure funding conditions.
Transport & Mobility	Why 2 — sprawl locks in fragility	Land use separated homes from needs by design for seventy years.	The 15-minute standard for all new development; retrofit priority to districts with no walkable essentials.
Transport & Mobility	Why 3 — fleets homogenize	One vendor, one software stack, one battery chemistry per fleet.	Certification requires stack diversity and manual fallback; recycling law recovering 90%+ of critical minerals.
Transport & Mobility	Why 4 — fallbacks are never drilled	Manual procedures exist on paper only.	Annual no-network drill days with published results for transit, logistics, and emergency services.
Transport & Mobility	Why 5 — root	Transport is optimized for cost per trip in normal times, not for continuity in bad times.	Continuity standards in transport law: every essential flow must demonstrate a degraded-mode operation, tested, before funding.
Communication	Why 1 — speech moved onto private platforms	Public communication infrastructure was allowed to wither.	Maintain postal, broadcast, and municipal networks as real, funded fallbacks — not heritage, but reserve capacity.
Communication	Why 2 — exit is costly	Audiences, records, and identities are locked in by design.	Portability and interoperability as law for dominant platforms; monthly export as a consumer right.
Communication	Why 3 — emergency channels are single-thread	Alerts ride the same networks that fail in disasters.	Municipal off-grid broadcast reaching all residents within 1 hour; amateur radio integrated into emergency plans.
Communication	Why 4 — privacy and safety are traded falsely	Policy frames encryption and lawful access as absolutes at war.	Individualized, judicial, logged, sunsetted access as the legislated middle; blanket access and blanket opacity both prohibited.
Communication	Why 5 — root	Communication is treated as a market commodity rather than as civic infrastructure.	Charter communication like water and power: universal service duties, resilience standards, and rights protections by design.
Demographics & Population	Why 1 — family formation is priced out	Housing and childcare costs peak exactly at family-forming ages.	Family housing at the 30%-of-income benchmark and childcare under 10% in binding city and national plans.
Demographics & Population	Why 2 — pensions promise the unprojected	Formulas were set for pyramids that no longer exist.	Automatic balancing mechanisms that adjust gradually by formula, depoliticizing the arithmetic.
Demographics & Population	Why 3 — migration is improvised	Flows arrive whether planned or not; only the reception differs.	Permanent, competent administration sized to the measured gap, with 12-month credential and language completion.
Demographics & Population	Why 4 — regions hollow silently	Depopulation has no constituency until services collapse.	Shrink-smart frameworks funded in advance: consolidated services with guaranteed access standards.
Demographics & Population	Why 5 — root	Policy is made for the current electorate's age structure while the future population is already born and countable.	A 50-year demographic annex, updated every 5 years, binding on infrastructure and pension planning by law.
Justice & Ethics	Why 1 — justice is priced	Expertise, delay, and bail convert wealth into outcomes.	Legal aid at parity in criminal matters; day-fines; detention only on individualized danger with 48-hour review.
Justice & Ethics	Why 2 — backlogs normalize delay	Court capacity is funded below caseload by habit.	Published time standards with automatic capacity triggers (90% of cases within 12 months).
Justice & Ethics	Why 3 — error is unmeasured	Systems that don't count wrongful convictions can't reduce them.	Conviction-integrity audits and exoneration statistics published annually; compensation automatic.
Justice & Ethics	Why 4 — algorithms inherit the bias	Historical enforcement data encodes historical disparity.	Justice models public, disparity-audited, appealable to a human; no liberty decision on a score alone.
Justice & Ethics	Why 5 — root	Justice is administered as a service to the state rather than as a right held against it.	Rights-first charter: every justice metric reported from the citizen's side — time, cost, error, disparity — not only the system's throughput.

#	Coinciding Scenario	Why They Amplify Each Other	Action Sequence — First 24 Hours	First Week	First Month	Joint Probabi	Why This Joint Probability	How to Maximize (detailed)
1	Grid failure + extreme heatwave + hospital surge	Power loss kills cooling and medical devices exactly when heat casualties peak; hospitals lose power as admissions triple; water pressure falls as pumps stop.	Open cooling centres on generator power first (before restoring general load); hospitals to emergency generation with fuel convoys escorted; publish the restoration order publicly: hospitals, water, cooling, then neighbourhoods.	Rolling restoration by survival-core priority; mutual-aid crews from unaffected regions; daily public accounting of deaths, capacity, and restoration percentages.	After-action rebuild: islanding cells, transformer spares, canopy and cooling standards — the scenario becomes a funded program, not a report.	62%	Each element alone scores 69–76%, but they share one cause (the grid) and one defence (generation), so fixing generation lifts all three at once; the joint figure sits below the best single because heat casualties already incurred cannot be undone.	Pre-position generation at hospitals and cooling sites BEFORE heat season; drill the compound scenario (not single failures) annually; hold 72 hours of fuel at every critical site; check in on isolated elderly by roster within the first 6 hours — most heat deaths are solitary and early.
2	Pandemic + supply-chain break + misinformation wave	Disease raises demand for medicine and food exactly as imports fail; false cures and denial delay every effective behaviour; panic buying empties shelves faster than the virus empties beds.	Publish the verified facts with signed provenance and the raw data; release strategic reserves visibly to break panic buying; protect the two highest-risk settings (care homes, hospitals) with air and testing barriers first.	Stand up local production of the top shortage items; daily briefings with error corrections stated openly; targeted support (delivery, income) so isolation is economically possible.	Institutionalize: wastewater surveillance permanent, platform provenance duties enforced, reserve levels restored to 90 days.	55%	Health measures alone score ~80% in the best case, but the coinciding information failure degrades compliance — the binding constraint is behaviour, not medicine; joint success is set by the weakest layer.	Pre-build the trust infrastructure in peacetime (citizen assemblies, published service metrics) — it cannot be manufactured mid-crisis; pre-sign local production contracts; communicate uncertainty honestly from hour one, because the first correction costs more trust than the first admission.
3	Household triple: job loss + serious illness + housing loss	Income stops as medical costs start; housing depends on the income that stopped; illness removes the capacity to job-hunt; each problem removes the resource needed to solve the others.	Freeze the bleeding in order: (1) housing — apply for eviction protection/assistance the same day (deadlines are short); (2) income — file unemployment and disability claims day one; (3) health — confirm coverage continuity before any treatment gap.	Stack every available support simultaneously (benefits, charity care, food programs, utility arrears plans); one case worker or one organized folder with every claim number; creditors notified in writing with the plan.	Rebuild in sequence: stabilized housing, treated health, then re-employment at sustainable load; rebuild the 3-month reserve before any lifestyle restoration.	71%	Household-scale problems are the most controllable in this file (+8 scale rule) and the solutions are administrative, not technological; the joint penalty comes from deadline collisions, which sequencing removes.	Act inside the first 72 hours while every program is still open; keep documents in one folder (ID, lease, medical, employment) — claim speed is a documents problem; tell every creditor and agency the whole picture in writing: coordinated honesty unlocks discretion that silence forfeits.
4	Cyberattack on payments + bank panic + synthetic-media hoax	Payment failure feels like insolvency; a forged official video 'confirming' collapse converts a technical outage into a bank run within hours; cash demand then creates the real shortage.	Banks and the central bank publish signed, verifiable statements through pre-registered channels within 1 hour; cash distribution points announced with visible stock; deposit-insurance reminder with the exact coverage figure.	Forensic restoration in stages with public status percentages; platforms label and throttle the forged media under the provenance rules; prosecutions announced for the hoax's amplifiers if coordinated.	Provenance infrastructure completed (capture-signing default for official media); payment systems segmented with tested analogue fallback (offline card limits, cash logistics).	58%	Technical recovery is high-probability (~78%), but panic is a belief event, not a technical one; success is capped by how much verifiable trust infrastructure existed before the attack.	Pre-register the official channels and teach the public where they are in peacetime; hold the cash-logistics reserve (physical notes, distribution points) as seriously as the digital one; the first statement must be fast, signed, and numerically specific — vagueness is what the hoax feeds on.
5	Multi-year drought + food price spike + civil unrest	Drought cuts supply; prices transfer the pain to the urban poor; unrest disrupts the transport that moves food; governments fall, which removes the coordinator.	Release reserves and open subsidized staple channels in the cities first (unrest is urban and price-driven); protect transport corridors as critical infrastructure; publish the reserve position and import schedule to kill rumour pricing.	Rationing by lifeline entitlement (guaranteed staple quantity per person at fixed price) rather than by price; emergency employment on water infrastructure (wells, canals, soil works) paying partly in food.	Structural: extraction caps to recharge, soil regeneration, diversified import contracts, and the 90-day reserve rebuilt — with the drought plan written into law.	49%	The physics (drought) is not solvable by policy, only blunted; success means preventing famine and state collapse, not preventing hardship — hence below 50% for the full compound, while famine prevention alone rates ~75%.	Spend on the cities' staple price first — that is where the political cascade starts; never let the reserve position be a secret (secrecy converts shortage into panic); begin the structural water program during the crisis, with crisis legitimacy, instead of after it, when attention dies.
6	Major earthquake + urban fire + communications outage	Shaking ignites fires as it breaks the water mains needed to fight them; comms loss prevents coordination; aftershocks hit responders during rescue.	Search and rescue in the pre-mapped highest-risk building classes first (Pareto: the deadliest 20% of stock holds most victims); fire crews to water-independent tactics (firebreaks, demolitions) where mains are dead; emergency broadcast by battery radio and vehicle loudspeaker.	Shelter capacity for 10% of population activated; mutual-aid and military logistics under civilian coordination; damage map published daily so aid follows need, not cameras.	Retrofit program for the deadliest building classes with deadlines; water system segmented so breaks cannot drain a district; comms fallback (municipal broadcast, amateur radio) made permanent.	66%	Rescue and shelter are drilled, logistical problems with high success rates; the coinciding fire-without-water element is the irreducible loss driver, which is why the joint figure stays below the ~81% for the shelter solution alone.	The work is 90% before the event: retrofits, pre-sited shelters, mapped risk. In the event: one coordination authority, one public map, one broadcast channel — parallel uncoordinated heroics waste the scarcest resource, which is organized hours.
7	Automation displacement wave + fiscal crisis + legitimacy collapse	Displacement cuts income tax as demand for support rises; deficits force austerity into the downturn; both parties of the social contract (work pays, state protects) appear broken simultaneously.	Automatic stabilizers fire by formula (no emergency vote needed): benefits expand, wage insurance starts, public works rosters open in the hardest-hit sectors and regions first.	Citizen-fund dividends declared from automation rents; standard hours reduced in the most displaced sectors; a published transition dashboard (displacement, re-employment, wage recovery) updated monthly.	Ownership structures deepened while the economy recovers (the funds buy the dip); education rebuilt around unaided competence plus AI collaboration; legitimacy rebuilt through verified local service performance.	41%	This is the slowest compound in the file: every lever works on a 5–20 year lag, and legitimacy — the binding constraint — cannot be purchased; the honest figure reflects generational, not electoral, timescales.	Build the ownership and benefit mechanisms during the preceding growth years — starting them inside the crisis halves their effect; protect the stabilizers' automaticity against emergency politics; measure and publish relentlessly, because in a legitimacy crisis only verified performance compounds.
8	River flood + power loss + drinking-water contamination	Floodwater carries sewage and chemicals into wells and treatment intakes while power loss stops pumps and treatment; the disaster manufactures the contamination that outlasts it.	Boil-water and no-drink orders within the first hour via every channel including door-to-door in the flooded zone; bottled and tanker water to pre-mapped distribution points; power restoration sequenced water-plant-first.	Well testing across the flooded area with published results per street; temporary treatment (mobile units) until plants are certified; mould and contamination remediation standards enforced before reoccupation.	Floodplain buyouts executed while memory and funding last; treatment plants hardened and dual-powered; address-level risk attached to every property record.	68%	Each element has a mature playbook (water 85%, energy 76%, adaptation 69% at best) and they share logistics rather than competing for them; the deduction is for the contamination tail, which is slow and unevenly distributed.	Sequence publicly and keep the sequence: water before general power, testing before reassurance, buyouts before rebuilding-in-place; never declare water safe on a schedule — declare it per tested zone, with the numbers published.

Statistic	Value
Total problems	162
Total solutions listed	162
Fields of discipline	18
Tier 1 (survival fields) solutions	90
Tier 2 solutions	72
Past problems	54
Now problems	54
Future problems	54
Root causes with solutions (5 Whys x 18 fields)	90
Coinciding scenarios analysed	8
Average success probability — all	62.5%
Average — Past	71.2%
Average — Now	67.0%
Average — Future	49.3%
Top 20% of solutions (32) — share of weighted ha	40.2%

Field	Solutions	Average Success %	AvgProbNum
Water	9	71.3%	71.3
Food & Agriculture	9	69.3%	69.3
Health & Medicine	9	62.0%	62
Energy	9	68.7%	68.7
Shelter & Housing	9	64.3%	64.3
Security & Defense	9	57.0%	57
Governance & Law	9	57.0%	57
Economy & Finance	9	62.0%	62
Technology & AI	9	60.7%	60.7
Environment & Climate	9	58.0%	58
Information & Media	9	57.3%	57.3
Psychology & Mind	9	58.0%	58
Education & Knowledge	9	70.3%	70.3
Work & Livelihood	9	60.7%	60.7
Transport & Mobility	9	65.7%	65.7

Communication	9	64.0%		64				
Demographics & Population	9	55.3%		55.3				
Justice & Ethics	9	63.7%		63.7				

